

Advanced Statistics for Physics Analysis

Bayesian Inference — Lecture Notes

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1. About these notes

1.1. Scope

These lecture notes accompany the course *Advanced Statistics for Physics Analysis* (Bayesian inference, second-semester elective, Master in Physics of Data, University of Padova). They cover everything taught from the first lecture (Introduction to R, 2 March) to the last (Generalised Linear Models, 22 May), in roughly the order the material was presented.

1.2. Reading the notes

Each chapter is built around one lecture (or a short run of consecutive lectures on the same topic). Definitions, theorems, propositions, and worked examples live inside the boxed environments defined in the project preamble — they are visually distinct from the surrounding prose so that a reader can find a result quickly.

Code is treated in two different ways:

- **Chapter 1 — *Introduction to R*.** Listings appear inline, with annotated comments of the form `# <1>`, `# <2>`, ... that match a numbered explanation in the body. This is the chapter you read with a console open next to you.
- **All later chapters.** Listings are moved automatically to *Appendix: Source code* by a small Pandoc filter (`move-code.lua`). In the body, the reader sees a brief reference like “*See code \ref{lst:beta-update} in Appendix for the implementation.*” This keeps the mathematical exposition uncluttered while preserving the full computational record.

Every figure produced from code is also reproducible; the plot-generating chunk is the one that gets relocated to the appendix.

1.3. Notation

- Random variables in upper-case (X, Y); realisations in lower-case (x, y).
- Probabilities and densities share the symbol $p(\cdot)$; the argument tells you whether it is a PMF, a PDF, or a posterior.
- Vectors and matrices in bold ($x, \theta, \mathbf{X}$).
- Conditional dependence is read right-to-left: $p(\theta | y)$ is the density of θ given y .
- Background information common to every term in an expression is suppressed except where its inclusion changes the discussion.

1.4. Sources

The notes draw on three textbooks and one set of lecture notes throughout:

- Sivia, D.S. & Skilling, J. *Data Analysis: A Bayesian Tutorial*, 2nd ed., Oxford University Press, 2006.

1. About these notes

- von der Linden, W., Dose, V. & von Toussaint, U. *Bayesian Probability Theory: Applications in the Physical Sciences*, Cambridge University Press, 2014.
- Robert, C.P. & Casella, G. *Monte Carlo Statistical Methods*, 2nd ed., Springer, 2004.
- Falasco, G. *Advanced Statistics for Physics Analysis — lecture notes*, University of Padova, 2026.

Every chapter ends with a *Sources* section naming the slide deck and the book chapters from which its content was drawn.

2. Introduction to R for Bayesian inference

This chapter introduces the R language and the small set of facilities that the rest of the book relies on: vectors and data frames, basic plotting, sampling from probability distributions, simple Monte-Carlo estimates, and a first look at the workflow we follow when running a Bayesian analysis. Code listings in this chapter are kept **inline** with explanatory annotations of the form `# <1>` so the reader can match each line to the surrounding text. From Chapter 2 onward, code is hidden in the body and reproduced in the *Appendix: Source code* at the back of the book.

Remark. Why R. Built-in statistical functions, excellent plotting (`base`, `ggplot2`), and first-class Bayesian packages (`rstan`, `rjags`, `brms`, `coda`) make R the default language of the course. Everything in this chapter runs on a vanilla R installation; the Bayesian packages are introduced in their own chapters.

2.1. A first session

Open R (or RStudio) and type the expression at the prompt; the result is printed below.

```
1 + 1
exp(1)
sqrt(2)
```

①
②
③

- ① Arithmetic at the prompt — R behaves as a calculator.
- ② The constant $e = 2.71828 \dots$ is computed by `exp(1)`.
- ③ Square roots, logarithms, trigonometric functions and so on are all built in.

```
[1] 2
[1] 2.718282
[1] 1.414214
```

Assigning a value uses the arrow `<-`. The result is stored but not displayed; print it explicitly by typing the name.

```
x <- 1:5
x
length(x)
mean(x); sd(x)
```

①
②
③
④

- ① `:` builds an integer sequence; here `x = (1, 2, 3, 4, 5)`.
- ② Typing the name at the prompt prints the value.
- ③ `length()` returns the number of entries — fundamental for sample sizes.
- ④ Two statements on the same line separated by `;`. The sample mean and standard deviation are the workhorses of frequentist summaries.

```
[1] 1 2 3 4 5
[1] 5
[1] 3
[1] 1.581139
```

2.2. Vectors, indexing and arithmetic

R is a *vectorised* language: arithmetic and most functions operate element-wise.

```
x <- c(2.1, 3.4, 5.0, 7.8, 11.2)
x + 1
log(x)
x[2]
x[x > 5]
sum(x); cumsum(x)
```

①
②
③
④
⑤
⑥

- ① `c()` concatenates values into a numeric vector.
- ② `Scalar + 1` is added to every entry — no loop necessary.
- ③ `log` is the natural logarithm. `log10`, `exp`, `sin`, ... all behave element-wise.
- ④ Square brackets index from 1 (not 0). Negative indices *exclude* entries.
- ⑤ A logical vector is a valid index — this is how we filter data.
- ⑥ `sum` collapses a vector to a scalar; `cumsum` returns the running total. We will use these constantly in Monte-Carlo work.

```
[1] 3.1 4.4 6.0 8.8 12.2
[1] 0.7419373 1.2237754 1.6094379 2.0541237 2.4159138
[1] 3.4
[1] 7.8 11.2
[1] 29.5
[1] 2.1 5.5 10.5 18.3 29.5
```

2.3. Data frames

A data frame is a list of equal-length vectors — the table-of-data abstraction that statisticians reach for first.

```
df <- data.frame(
  x = c(1.0, 1.5, 2.2, 3.4, 4.1),
  y = c(2.1, 3.0, 4.2, 6.5, 7.4)
)
df
df$y
df[df$x > 2, ]
```

①

②
③
④

- ① Create a data frame from named columns.
- ② Print the whole table.
- ③ `$` extracts a column as a vector.
- ④ Row-subset using a logical vector on the rows; the trailing comma keeps all columns.

```
      x y
1 1.0 2.1
2 1.5 3.0
3 2.2 4.2
4 3.4 6.5
5 4.1 7.4
[1] 2.1 3.0 4.2 6.5 7.4
      x y
3 2.2 4.2
4 3.4 6.5
5 4.1 7.4
```

2.4. Probability distributions

Every standard distribution in R follows the `d / p / q / r` convention:

- `d...`(*x*, ...) — **d**ensity (PDF) or mass function (PMF)
- `p...`(*q*, ...) — **p**robability $\Pr(X \leq q)$ (CDF)
- `q...`(*p*, ...) — **q**uantile (inverse CDF)
- `r...`(*n*, ...) — generate *n* random draws

```
dnorm(0)
pnorm(1.96)
qnorm(0.975)
set.seed(1)
rnorm(5, mean = 10, sd = 2)
```

①
②
③
④
⑤

- ① Density of the standard normal at 0; equals $1/\sqrt{2\pi} \approx 0.3989$.
- ② The CDF at 1.96 — the familiar 97.5%-point of the standard normal.
- ③ The inverse of (2): $z_{0.975} = 1.96$.
- ④ **Always** set a seed before sampling so the analysis is reproducible.
- ⑤ Draw five values from $\mathcal{N}(10, 2^2)$.

```
[1] 0.3989423
[1] 0.9750021
[1] 1.959964
[1] 8.747092 10.367287 8.328743 13.190562 10.659016
```

Note (Discrete distributions in R). The same convention works for discrete laws: `dbinom`, `dpois`, `dgeom`, `dnbinom`, `dhyper`, `dmultinom`. For these, `d` returns the *probability mass*; otherwise the semantics are identical.

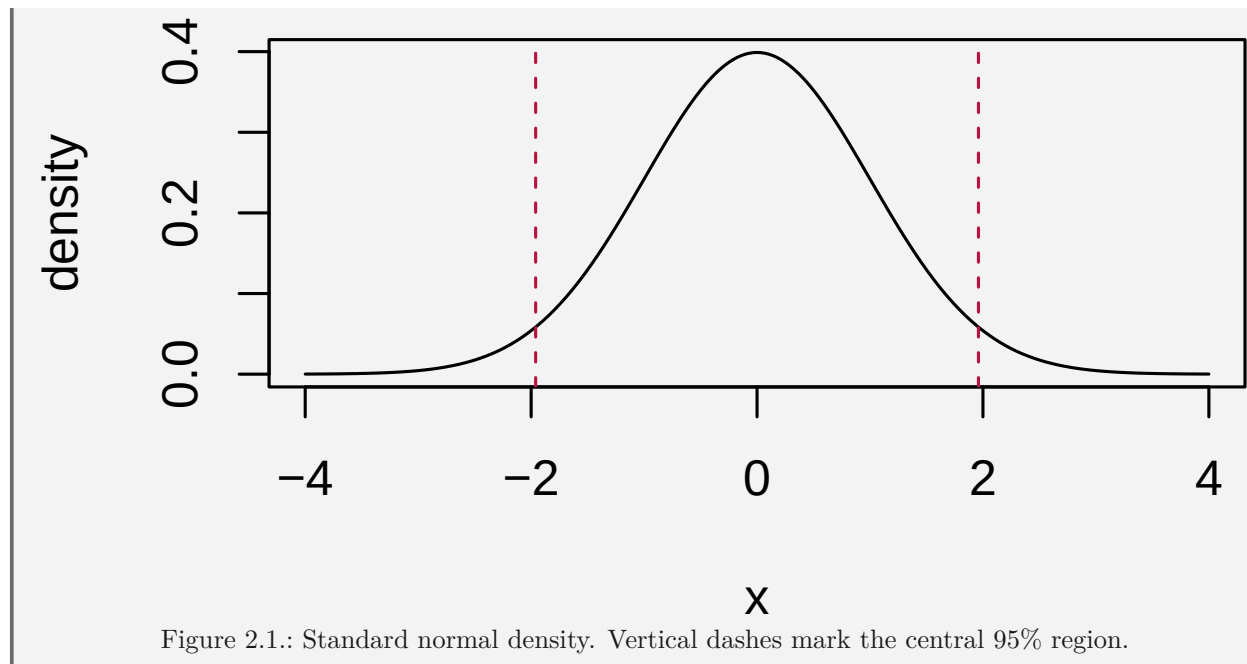
2.5. Plotting

We use base R graphics throughout the book; the defaults set in `.Rprofile` (serif font, thin lines, transparent backgrounds, legends outside the panel) keep figures consistent.

```
x <- seq(-4, 4, length.out = 400)
plot(x, dnorm(x), type = "l", lwd = 1,
      xlab = "x", ylab = "density")
abline(v = c(-1.96, 1.96),
       lty = 2, col = "#b8123e")
```

①
②
③

- ① A dense grid on the *x*-axis.
- ② `type = "l"` joins points with a line; the rest of the call labels the axes.
- ③ The 2.5% and 97.5% quantiles of the standard normal mark the conventional 95% interval. We will revisit this picture (with proper Bayesian intervals) in Chapter 5.



2.6. A first Monte-Carlo estimate

Monte-Carlo replaces a deterministic integral with the average of a function over a random sample. Here we estimate $\mathbb{E}[X^2]$ for $X \sim \mathcal{N}(0, 1)$, which we know equals 1.

Example 2.1 (Monte-Carlo estimate of a Gaussian second moment)

```
set.seed(42)
N <- 10000
x <- rnorm(N)
mean(x^2)
sd(x^2) / sqrt(N)
```

①
②
③
④
⑤

- ① Reproducibility.
- ② Number of draws.
- ③ Sample $X_1, \dots, X_N \sim \mathcal{N}(0, 1)$.
- ④ The Monte-Carlo estimator $\hat{\mathbb{E}}[X^2] = \frac{1}{N} \sum_i X_i^2$. It should be close to 1.
- ⑤ The standard error of the estimator — how much we trust the result.

```
[1] 1.012334
[1] 0.0142808
```

The estimator converges at rate $1/\sqrt{N}$ regardless of dimension; this is the property that makes MC the workhorse of Bayesian computation (Chapter 7).

2.7. A first Bayesian update

The simplest update we will meet in the book is the *Beta-Binomial* model: if the prior on the success probability is $\theta \sim \text{Beta}(\alpha, \beta)$ and we observe k successes out of n trials, the posterior is $\theta \mid k \sim \text{Beta}(\alpha + k, \beta + n - k)$.

Example 2.2 (*Beta–Binomial conjugate update*)

```

alpha <- 2
beta_param <- 2
n <- 10; k <- 7
theta <- seq(0, 1, length.out = 400)
prior <- dbeta(theta, alpha, beta_param)
posterior <- dbeta(theta, alpha + k, beta_param + n - k)

plot(theta, prior, type = "l", lwd = 1, lty = 2,
      xlab = expression(theta), ylab = "density",
      ylim = c(0, max(posterior) * 1.05))
lines(theta, posterior, lwd = 1.2, col = "#b8123e")
legend("topleft", c("Prior Beta(2,2)", "Posterior Beta(9,5)"),
      lty = c(2, 1), col = c("black", "#b8123e"),
      bty = "n", cex = 0.8)

```

- ① The prior shape parameter α . Beta(2, 2) is a mild prior centred at 1/2.
- ② The prior shape parameter β (renamed to avoid clashing with R's `beta` function).
- ③ The data: 7 successes out of 10 trials.
- ④ The posterior is also Beta — this is **conjugacy** (formally introduced in Chapter 4).
- ⑤ Plot the two densities together; the data have shifted the mass to the right and concentrated it.

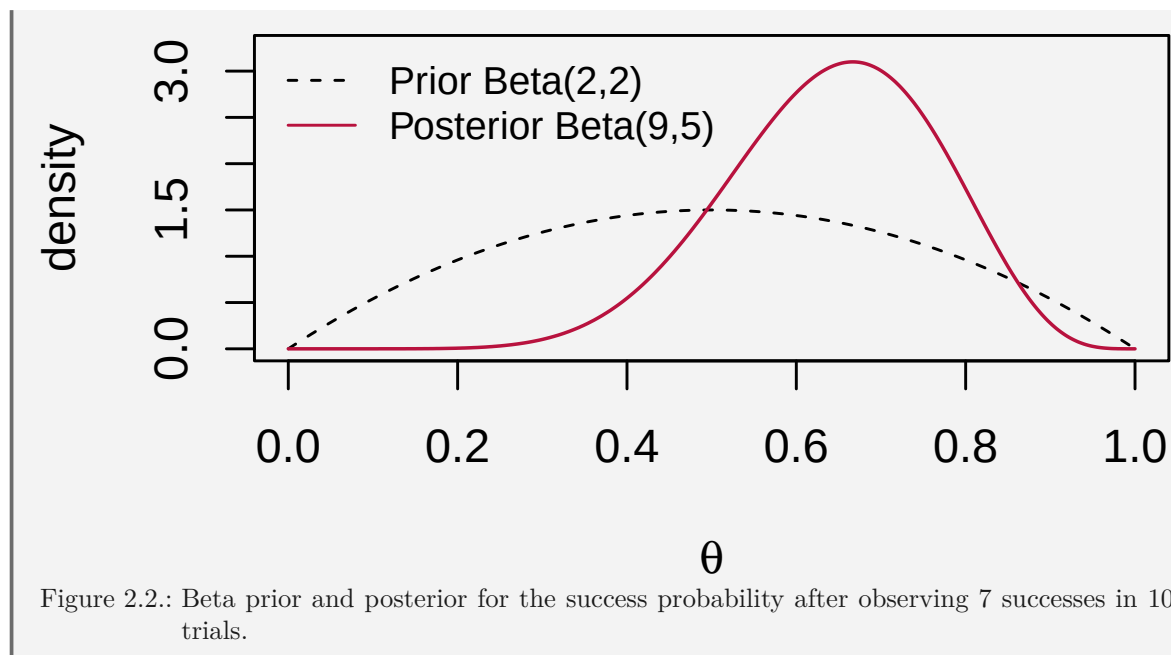


Figure 2.2.: Beta prior and posterior for the success probability after observing 7 successes in 10 trials.

2.8. A reading map

Every later chapter builds on a small part of what you have just seen:

- sampling with `r...` underlies all of Chapters 7–10;
- the `d/p/q` convention is the language for likelihoods and posteriors in Chapters 4–6;
- the Beta–Binomial update is the simplest instance of the general machinery developed in Chapter 5;
- data frames return when we move to regression (Chapter 11) and GLMs (Chapter 13);
- `rstan` / `rjags` calls in Chapter 12 wrap the same workflow we used here, just with a posterior specified by code.

2. Introduction to R for Bayesian inference

From now on, code blocks no longer appear inline. Each listing is moved to the *Appendix: Source code* with a label of the form `lst:something`, and a placeholder in the body reads “*See code \ref{lst:something} in Appendix for the implementation.*” — this keeps the body of the book uncluttered while preserving the full computational record.

Part I.

PART I: PROBABILITY RECAP

3. Discrete probability

3.1. Foundations: sample spaces and axioms

A probability space is the formal mathematical foundation for quantifying uncertainty. It consists of three components: a sample space Ω (the set of all possible outcomes of an experiment), a sigma-algebra $\mathcal{F}$ of subsets of Ω (the collection of events to which we can assign probabilities), and a probability measure P that assigns a number in $[0, 1]$ to each event.

Definition 3.1 (Kolmogorov axioms). A probability measure $P : \mathcal{F} \rightarrow [0, 1]$ satisfies:

1. **Non-negativity:** $P(A) \geq 0$ for all $A \in \mathcal{F}$.
2. **Normalization:** $P(\Omega) = 1$.
3. **Countable additivity:** For any countable collection of disjoint events $\{A_i\}_{i=1}^{\infty}$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

These axioms ensure that probability theory is internally consistent and forms a measure-theoretic foundation for all probability calculations.

Events are combined using set operations. For any two events A and B :

- **Union** $A \cup B$: at least one of A or B occurs. By inclusion-exclusion, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. If A and B are disjoint, $P(A \cup B) = P(A) + P(B)$.
- **Intersection** $A \cap B$: both A and B occur. We have $P(A \cap B) = P(A | B)P(B) = P(B | A)P(A)$.
- **Complement** $\bar{A}$: A does not occur. Then $P(\bar{A}) = 1 - P(A)$.

A **partition** of the sample space is a collection of disjoint events $\{A_i\}_{i=1}^n$ such that $\bigcup_{i=1}^n A_i = \Omega$. Partitions arise naturally in inference, where they often represent mutually exclusive hypotheses or models. For any partition, $\sum_{i=1}^n P(A_i) = 1$.

3.2. Conditional probability and Bayes' theorem

Having fixed the language of events and partitions, we now ask how observing one event updates the probability of another. The conditional probability of event B given that event A has occurred is

$$P(B | A) = \frac{P(A \cap B)}{P(A)},$$

provided $P(A) > 0$. Intuitively, we restrict our sample space to A and renormalize. Rearranging gives the **product rule**:

$$P(A \cap B) = P(B | A)P(A) = P(A | B)P(B).$$

Definition 3.2 (Independence). Two events A and B are **independent** if $P(A \cap B) = P(A)P(B)$, equivalently $P(B | A) = P(B)$.

Bayes' theorem is the cornerstone of Bayesian reasoning. For any partition $\{H_i\}_{i=1}^n$ of hypotheses and observed data D :

$$P(H_i | D) = \frac{P(D | H_i)P(H_i)}{\sum_{j=1}^n P(D | H_j)P(H_j)} = \frac{P(D | H_i)P(H_i)}{P(D)}.$$

The posterior probability $P(H_i | D)$ quantifies belief in hypothesis H_i after observing D . It is proportional to the **likelihood** $P(D | H_i)$ (the probability of the data given the hypothesis) times the **prior** $P(H_i)$ (our pre-data belief). The denominator $P(D) = \sum_j P(D | H_j)P(H_j)$ is the marginal likelihood, ensuring normalization.

Example 3.1 (*Signal vs background (exam-style)*)

A source emits either signal particles or background particles. The prior probability of signal is $P(S) = 0.05$. A selection cut keeps 80% of signals but only 10% of backgrounds. An event passes the cut. What is the posterior probability that it is signal?

Solution. Let C be the event *passes the cut*. By Bayes' theorem,

$$P(S | C) = \frac{P(C | S)P(S)}{P(C | S)P(S) + P(C | B)P(B)} = \frac{0.8 \cdot 0.05}{0.8 \cdot 0.05 + 0.1 \cdot 0.95} = \frac{0.04}{0.135} \approx 0.296.$$

Even a fairly efficient cut does not produce a signal-dominated sample when the prior signal rate is small. This is the typical low base-rate situation in particle physics.

3.3. Combinatorics and counting

To assign concrete probabilities — and later to write down the PMFs of the standard distributions — we need to count outcomes. Many probability distributions arise directly from counting the number of ways events can occur. This connection is fundamental to entropy and information theory.

Definition 3.3 (Permutations). A **permutation** is an ordered arrangement of objects. The number of permutations of n distinct objects is

$$n! = n(n-1)(n-2) \cdots 1.$$

The number of permutations of k objects chosen from n without replacement is

$$P(n, k) = \frac{n!}{(n-k)!} = n(n-1) \cdots (n-k+1).$$

With replacement, it is n^k .

Definition 3.4 (Combinations). A **combination** counts selections where order does not matter. The number of ways to choose k objects from n without replacement is

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

This is the binomial coefficient, which appears in $(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$.

With replacement, the number of ways to choose k objects from n is $\binom{n+k-1}{k}$.

Definition 3.5 (Pairings and matchings). From n objects, the number of ways to select:

- One unordered pair: $\binom{n}{2} = \frac{n(n-1)}{2}$.
- k unordered pairs (partitioning $2k$ elements): $\frac{n!}{(n-2k)!2^k k!}$.
- k ordered pairs: $\frac{n!}{(n-2k)!k!}$.
- A complete matching of $n = 2m$ elements into pairs (the double factorial):

$$(2m-1)!! = \frac{(2m)!}{2^m m!}.$$

These combinatorial objects structure the probability mass functions of the discrete distributions we now encounter.

3.4. Discrete random variables and distributions

A **discrete random variable** X takes values in a countable set, typically $\mathbb{N} = \{0, 1, 2, \dots\}$. Its distribution is described by the **probability mass function** (PMF):

$$p_X(x) = P(X = x), \quad \sum_x p_X(x) = 1.$$

The **cumulative distribution function** (CDF) is $F_X(x) = P(X \leq x) = \sum_{x_i \leq x} p_X(x_i)$.

For a discrete random variable, the expectation (mean) and variance are

$$E[X] = \sum_x x p_X(x), \quad \text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2.$$

The “statistical zoo” of common discrete laws can be organised around a single building block — the **Bernoulli trial**, with two outcomes (success with probability p , failure with probability $q = 1 - p$) — and the different questions one can ask about a sequence of such trials. A coin toss is the canonical example, with $p = 1/2$ for a fair coin.

Binomial — **how many successes in n trials?** If we perform n independent Bernoulli trials and count the number of successes, we obtain a binomial random variable.

Definition 3.6 (Binomial distribution).

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n.$$

Parameters: n (number of trials), p (success probability).

Mean: $E[X] = np$.

Variance: $\text{Var}(X) = np(1-p)$.

Special case: $n = 1$ recovers the Bernoulli distribution, $P(X = 1) = p$, $P(X = 0) = 1 - p$.

Example 3.2 (*Restaurant entrees*)

In a restaurant, 12 beef, 8 fish, and 10 poultry entrees are served. If 4 customers order independently (with replacement), what is the probability that exactly 2 order fish?

The probability that a customer orders fish is $p = 8/30 = 4/15$. The number of fish orders X follows $\text{Binomial}(n = 4, p = 4/15)$:

$$P(X = 2) = \binom{4}{2} \left(\frac{4}{15}\right)^2 \left(\frac{11}{15}\right)^2 = 6 \cdot \frac{16}{225} \cdot \frac{121}{225} \approx 0.23.$$

Multinomial — **how many counts in each of k categories?** The binomial describes outcomes with two categories. The multinomial generalizes to k categories.

Definition 3.7 (Multinomial distribution). With k outcomes having probabilities $p_1, \dots, p_k$ (and $\sum_i p_i = 1$), the joint PMF of counts $(x_1, \dots, x_k)$ with $\sum_i x_i = n$ is

$$p(x_1, \dots, x_k) = \frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}.$$

Special case: $k = 2$ recovers the binomial distribution.

Geometric — **how many trials until the first success?** The geometric distribution counts the number of trials until the first success.

Definition 3.8 (Geometric distribution).

$$P(N = n) = (1 - p)^{n-1} p, \quad n = 1, 2, 3, \dots$$

Mean: $E[N] = \frac{1}{p}$.

Variance: $\text{Var}(N) = \frac{1-p}{p^2}$.

Memoryless property: $P(N > n + m \mid N > n) = P(N > m)$. The waiting time does not depend on how long we have already waited.

Note. In R, the function `dgeom(k, prob=p)` computes the PMF of the number of *failures* before the first success, so $P(N_{\text{fail}} = k) = (1 - p)^k p$. The expected value in R is thus $(1 - p)/p = E[N] - 1$.

Negative binomial — **how many failures before the r -th success?** The negative binomial distribution counts the number of failures before the r -th success.

Definition 3.9 (Negative binomial distribution).

$$P(X = k) = \binom{k + r - 1}{k} (1 - p)^k p^r, \quad k = 0, 1, 2, \dots$$

Parameters: r (number of successes), p (success probability).

Special case: $r = 1$ recovers the geometric distribution.

Hypergeometric — **how many successes when sampling without replacement?** The hypergeometric distribution arises when sampling *without replacement*. It is used when the population size and composition are finite.

Definition 3.10 (Hypergeometric distribution). Consider a population of N objects, of which K are of interest (e.g., success). Drawing n objects without replacement, the number of successes is

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}, \quad k = 0, 1, \dots, \min(n, K).$$

Mean: $E[X] = n \frac{K}{N}$.

Variance: $\text{Var}(X) = n \frac{K}{N} \left(1 - \frac{K}{N}\right) \frac{N-n}{N-1}$.

The hypergeometric generalizes the Bernoulli case (sampling with replacement). The success probability becomes history-dependent: the probability changes as objects are drawn.

Example 3.3 (Urn problem)

An urn contains 40 white balls and 60 black balls. Drawing 20 balls at random without replacement, what is the probability that at least 50 black balls remain (i.e., at most 10 black balls are drawn)?

Let X be the number of black balls drawn. Then $X \sim \text{Hypergeometric}(N = 100, K = 60, n = 20)$:

$$P(X \leq 10) = \sum_{k=0}^{10} \frac{\binom{60}{k} \binom{40}{20-k}}{\binom{100}{20}} \approx 0.22.$$

Example 3.4 (Hypergeometric vs binomial variance (exam-style))

An urn contains K black balls and $N - K$ white balls. You draw n balls without replacement and observe X black balls. Compare $\text{Var}(X)$ with the variance of a binomial model with $p = K/N$. Why is one smaller?

Solution. For the hypergeometric model,

$$\text{Var}(X) = n \frac{K}{N} \left(1 - \frac{K}{N}\right) \frac{N-n}{N-1}.$$

The binomial variance with the same success probability is $n(K/N)(1 - K/N)$. The hypergeometric variance is smaller by the finite-population factor $(N-n)/(N-1)$. Drawing without replacement creates negative dependence between successive draws — drawing a black ball lowers the chance of another — which reduces fluctuations relative to the independent binomial case.

Poisson — how many rare events in a fixed interval? The Poisson distribution describes the count of rare, independent events occurring at a constant average rate λ over a fixed time or space interval.

Definition 3.11 (Poisson distribution).

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

Parameter: λ (rate).

Mean: $E[X] = \lambda$.

Variance: $\text{Var}(X) = \lambda$.

A remarkable property: all cumulants equal λ .

Example 3.5 (*Radioactive decay*)

Suppose particles from a radioactive source arrive at an average rate of $\lambda = 3$ per second. The number of arrivals in a 1-second window follows $\text{Poisson}(\lambda = 3)$. The probability of observing exactly 2 particles is

$$P(X = 2) = \frac{3^2 e^{-3}}{2!} = \frac{9e^{-3}}{2} \approx 0.224.$$

3.5. The rare-event limit: Poisson as a limit of binomial

A deep insight: the Poisson distribution emerges as the limit of the binomial distribution when $n \rightarrow \infty$, $p \rightarrow 0$, with $np \rightarrow \lambda$ held constant. This explains why rare events follow a Poisson distribution.

Theorem 3.1 (Poisson limit). *Let $X_n \sim \text{Binomial}(n, p_n)$ with $np_n \rightarrow \lambda$ as $n \rightarrow \infty$. Then*

$$\lim_{n \rightarrow \infty} P(X_n = k) = \frac{\lambda^k e^{-\lambda}}{k!}.$$

Proof sketch: Start from the binomial PMF:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}.$$

Let $p = \lambda/n$. Then:

$$P(X = k) = \frac{n(n-1)\cdots(n-k+1)}{k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}.$$

As $n \rightarrow \infty$: - $\frac{n(n-1)\cdots(n-k+1)}{n^k} \rightarrow 1$. - $\left(\frac{\lambda}{n}\right)^k \rightarrow \frac{\lambda^k}{k!}$ (times the normalization). - $\left(1 - \frac{\lambda}{n}\right)^n \rightarrow e^{-\lambda}$.

Thus $P(X = k) \rightarrow \frac{\lambda^k e^{-\lambda}}{k!}$.

This limit is exact for large n and small p with np moderate. In practice, the Poisson approximation is accurate when $n \geq 20$ and $p \leq 0.05$.

Example 3.6 (*When the Poisson approximation breaks (exam-style)*)

A bin receives $X \sim \text{Binomial}(n, p)$ events. The usual Poisson approximation is $X \sim \text{Poisson}(np)$. For $n = 20$ and $p = 0.2$, compare the exact variance with the Poisson approximate variance. What does the comparison reveal?

Solution. The binomial mean is $np = 4$ and the exact variance is

$$\text{Var}(X) = np(1-p) = 20 \cdot 0.2 \cdot 0.8 = 3.2.$$

The Poisson approximation would give variance $np = 4$. The Poisson approximation **overestimates** the variance here because $p = 0.2$ is not very small — the rare-event condition $p \rightarrow 0$ is not well satisfied. The relative correction is $(1-p) = 0.8$.

3.6. Quantiles and percentiles

Mean and variance summarise location and scale, but they hide the shape of the distribution. When reporting or analyzing data, we often describe how probability mass is distributed without computing the full PMF. A **quantile** captures this information.

Definition 3.12 (Quantile). The q -quantile x_q of a random variable X satisfies

$$P(X \leq x_q) = F_X(x_q) = q,$$

where $0 \leq q \leq 1$. The quantile function is the inverse of the CDF: $x_q = F_X^{-1}(q)$.

Common quantiles include: - **Median** ($q = 0.5$): the middle value. - **Lower quartile** ($q = 0.25$) and **upper quartile** ($q = 0.75$).

Percentiles express quantiles as percentages: the p -th percentile satisfies $P(X \leq x_p) = p/100$.

3.7. Entropy and information

Information theory provides a bridge between combinatorics, probability, and statistical inference. The **Shannon entropy** quantifies the average information content (or uncertainty) of a distribution.

Definition 3.13 (Shannon entropy). For a discrete random variable X with PMF $p_X(x)$, the entropy is

$$H(X) = - \sum_x p_X(x) \log p_X(x),$$

where the logarithm is typically base 2 (bits) or natural log (nats). By convention, $0 \log 0 = 0$.

- Note.*
- Entropy is maximized when the distribution is uniform: $H(X) = \log N$ for N equally likely outcomes.
 - Entropy is minimized (zero) when one outcome has probability 1.
 - For a Bernoulli trial, $H = -p \log p - (1-p) \log(1-p)$, which is symmetric around $p = 0.5$ and peaks at $H_{\max} = 1$ bit.

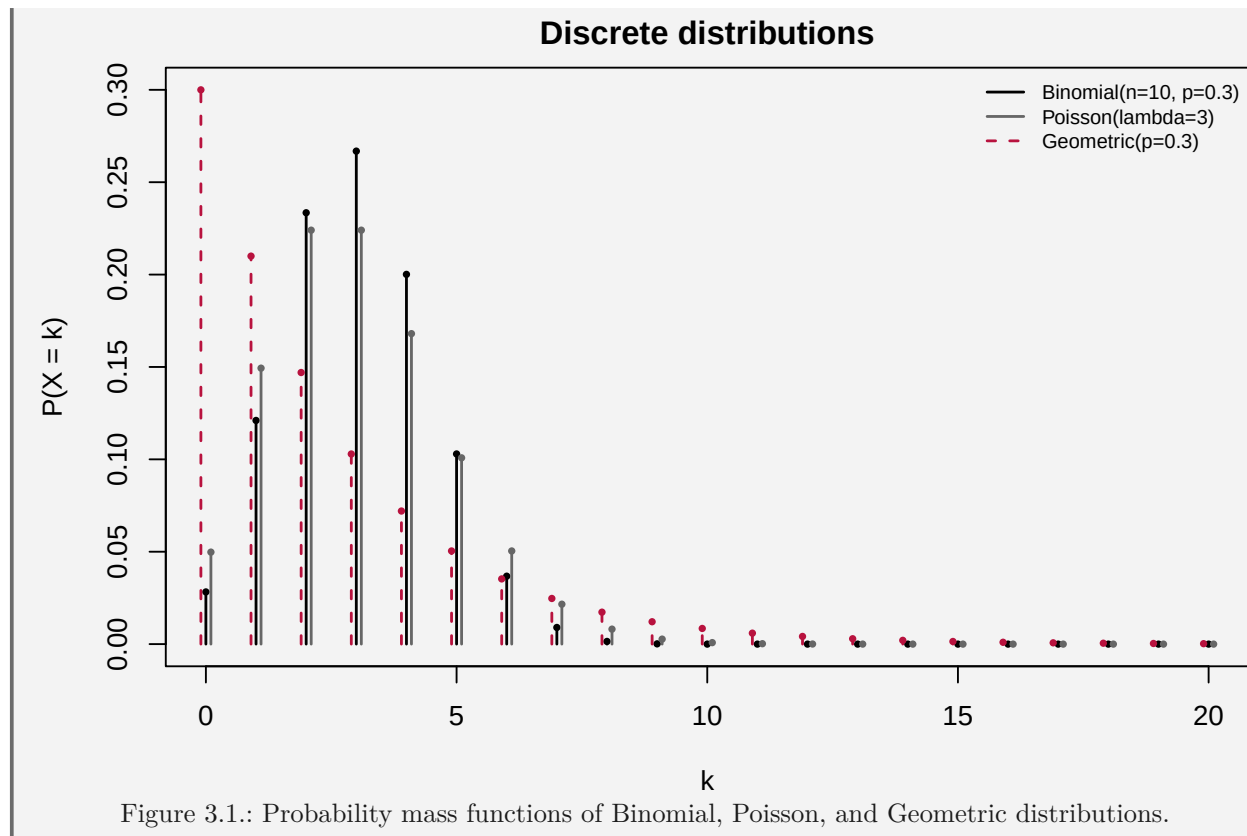
Entropy appears in many guises throughout Bayesian inference: in the definition of divergence measures (Kullback–Leibler), in maximum-entropy priors, and in information-theoretic model selection criteria.

3.8. Summary of discrete distributions

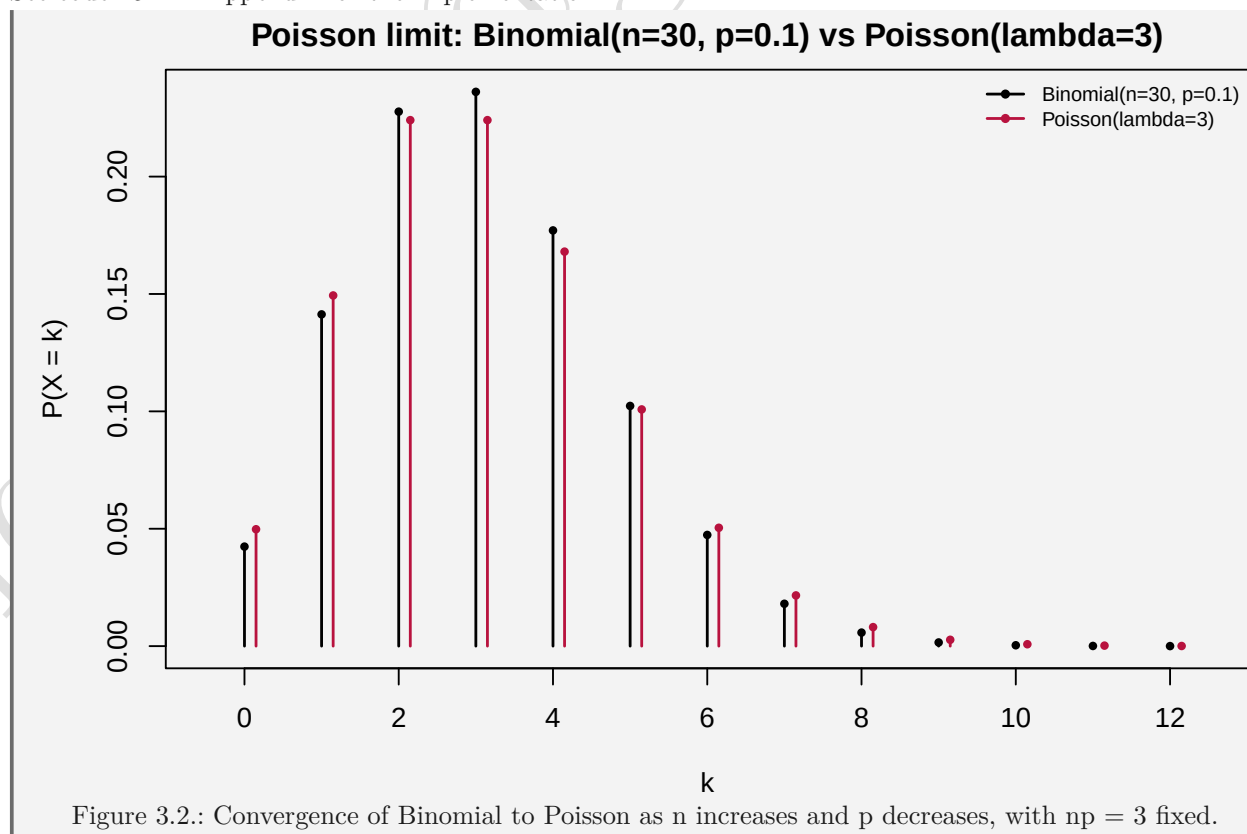
The “statistical zoo” of discrete distributions can be organized by the question they answer:

- **Bernoulli** ($n = 1$): Success or failure? $P(X = 1) = p$.
- **Binomial** ($n \geq 1$): How many successes in n trials?
- **Multinomial**: How many counts in each of k categories?
- **Geometric**: How many trials until the first success?
- **Negative binomial**: How many failures before the r -th success?
- **Hypergeometric**: How many successes when sampling without replacement?
- **Poisson**: How many rare events in a fixed interval?

See code 15.1 in Appendix for the implementation.



See code 15.2 in Appendix for the implementation.



3.9. Generating functions and moments (discrete)

The distributions above were introduced one at a time, with means and variances derived case by case. Generating functions package this work into a single computational object. The **probability generating function** and related tools provide powerful methods to compute moments, analyze distributions, and understand the behavior of sums of independent random variables. This section develops the theory and applies it to the key discrete distributions.

Definition 3.14 (Probability generating function). For a non-negative integer-valued random variable X , the **probability generating function** (PGF) is

$$G_X(s) = \mathbb{E}[s^X] = \sum_{k \geq 0} s^k P(X = k) = \sum_{k \geq 0} s^k p_k,$$

defined for $|s| \leq 1$. The PGF converges for all $|s| \leq 1$ (since it is a power series with radius of convergence at least 1). Crucially, the PGF uniquely determines the distribution: if two random variables have the same PGF, they have the same PMF.

Moments from the PGF

The PGF is named for its ability to generate moments. Differentiating term-by-term:

Proposition 3.2 (Factorial moments from the PGF). *The r -th factorial moment of X is*

$$\mathbb{E}[X(X-1) \cdots (X-r+1)] = G_X^{(r)}(1),$$

where $G_X^{(r)}$ denotes the r -th derivative of G_X .

Proof sketch: By term-by-term differentiation,

$$G'_X(s) = \sum_{k \geq 1} k s^{k-1} p_k, \quad G''_X(s) = \sum_{k \geq 2} k(k-1) s^{k-2} p_k, \quad \text{etc.}$$

Evaluating at $s = 1$ extracts the factorial moments.

In particular: $\mathbb{E}[X] = G'(1)$ - $\mathbb{E}[X(X-1)] = G''(1)$, so $\text{Var}(X) = G''(1) + G'(1) - [G'(1)]^2$

PGFs of sums of independent random variables

A key advantage of the PGF is its multiplicative property:

Proposition 3.3 (PGF of an independent sum). *If X and Y are independent random variables, then*

$$G_{X+Y}(s) = G_X(s) G_Y(s).$$

Proof sketch: By independence,

$$G_{X+Y}(s) = \mathbb{E}[s^{X+Y}] = \mathbb{E}[s^X s^Y] = \mathbb{E}[s^X] \mathbb{E}[s^Y] = G_X(s) G_Y(s).$$

More generally, if $X_1, \dots, X_n$ are independent, $G_{\sum X_i}(s) = \prod_i G_{X_i}(s)$.

Worked examples

3. Discrete probability

Example 3.7 (Bernoulli distribution)

For $X \sim \text{Bernoulli}(p)$ with $P(X = 1) = p$ and $P(X = 0) = 1 - p$:

$$G(s) = (1 - p) \cdot 1 + p \cdot s = 1 - p + ps.$$

Then $G'(1) = p$, so $\mathbb{E}[X] = p$. Also $G''(1) = 0$, so $\text{Var}(X) = 0 + p - p^2 = p(1 - p)$.

Example 3.8 (Binomial distribution)

For $X \sim \text{Binomial}(n, p)$:

$$G(s) = \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} s^k = [(1-p) + ps]^n.$$

Differentiating: $G'(s) = n[(1-p) + ps]^{n-1} \cdot p$, so $G'(1) = np$. Thus $\mathbb{E}[X] = np$. Also, $G''(s) = n(n-1)p^2[(1-p) + ps]^{n-2}$, giving $G''(1) = n(n-1)p^2$. Therefore,

$$\text{Var}(X) = n(n-1)p^2 + np - (np)^2 = np(1-p).$$

Example 3.9 (Geometric distribution)

For $X \sim \text{Geometric}(p)$ on $\{1, 2, 3, \dots\}$ with $P(X = n) = (1-p)^{n-1}p$:

$$G(s) = \sum_{n=1}^{\infty} (1-p)^{n-1} p s^n = ps \sum_{n=1}^{\infty} [(1-p)s]^{n-1} = \frac{ps}{1 - (1-p)s}.$$

Then $G'(s) = \frac{p[1-(1-p)s] + ps(1-p)}{[1-(1-p)s]^2} = \frac{p}{[1-(1-p)s]^2}$, so $G'(1) = \frac{p}{p^2} = \frac{1}{p}$. Thus $\mathbb{E}[X] = 1/p$.

Example 3.10 (Poisson distribution)

For $X \sim \text{Poisson}(\lambda)$:

$$G(s) = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} s^k = e^{-\lambda} e^{\lambda s} = e^{\lambda(s-1)}.$$

All derivatives have the form $G^{(r)}(s) = \lambda^r e^{\lambda(s-1)}$, so $G^{(r)}(1) = \lambda^r$. In particular, all raw and factorial moments are equal to λ , and the distribution has the remarkable property that $\mathbb{E}[X] = \text{Var}(X) = \lambda$.

Moment generating function for discrete random variables

The PGF is specific to non-negative integer-valued random variables. For a more general approach, we introduce the **moment generating function**:

Definition 3.15 (Moment generating function (discrete)). The **moment generating function** (MGF) of a discrete random variable X is

$$M_X(t) = \mathbb{E}[e^{tX}] = \sum_k e^{tk} p_k,$$

defined for $|t|$ in a neighborhood of 0. The n -th moment is $\mathbb{E}[X^n] = M_X^{(n)}(0)$.

Note that for non-negative integer-valued X , $M_X(t) = G_X(e^t)$: the MGF is the PGF evaluated at $s = e^t$. This connection makes the MGF the bridge to continuous random variables, where the PGF does not apply.

3.10. Exercises

Exercise 3.1

A diagnostic test has sensitivity $P(+ | D) = 0.95$ and specificity $P(- | \bar{D}) = 0.98$. The disease has prevalence $P(D) = 0.01$. Compute the posterior probability of disease given a positive test, $P(D | +)$. Why is this so much lower than the sensitivity?

Solution

By Bayes' theorem with $P(+ | \bar{D}) = 1 - 0.98 = 0.02$,

$$P(D | +) = \frac{P(+ | D) P(D)}{P(+ | D) P(D) + P(+ | \bar{D}) P(\bar{D})} = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.02 \cdot 0.99} = \frac{0.0095}{0.0293} \approx 0.324.$$

The posterior is far below the 0.95 sensitivity because the prior $P(D) = 0.01$ is so small that the false-positive mass $0.02 \cdot 0.99$ dominates the true-positive mass $0.95 \cdot 0.01$ — the classic low base-rate effect.

Exercise 3.2

Three urns contain (2R, 8B), (5R, 5B), (8R, 2B) balls respectively. An urn is chosen uniformly at random and a single red ball is drawn. Compute the posterior probability that the chosen urn was the third one.

Solution

With uniform prior $P(U_i) = 1/3$ and likelihoods $P(R | U_1) = 2/10$, $P(R | U_2) = 5/10$, $P(R | U_3) = 8/10$, Bayes' theorem gives

$$P(U_3 | R) = \frac{(1/3)(8/10)}{(1/3)(2/10) + (1/3)(5/10) + (1/3)(8/10)} = \frac{8}{2 + 5 + 8} = \frac{8}{15}.$$

Exercise 3.3

For $X \sim \text{Binomial}(n, p)$, derive $E[X] = np$ and $\text{Var}(X) = np(1 - p)$ by writing $X = \sum_{i=1}^n Y_i$ as a sum of i.i.d. Bernoulli indicators.

Solution

Each $Y_i \sim \text{Bernoulli}(p)$ has $E[Y_i] = p$ and $\text{Var}(Y_i) = p(1 - p)$. By linearity of expectation and, for the variance, independence of the Y_i ,

$$E[X] = \sum_{i=1}^n E[Y_i] = np, \quad \text{Var}(X) = \sum_{i=1}^n \text{Var}(Y_i) = np(1 - p).$$

Exercise 3.4

A radioactive source emits at average rate λ counts per second. Compute the probability of observing exactly k counts in a window of length t in closed form. Compute $P(X \geq 1)$ for $\lambda t = 2$ exactly (no calculator — leave e^{-2} symbolic).

3. Discrete probability

Solution

The count X in a window of length t is $X \sim \text{Poisson}(\lambda t)$, so

$$P(X = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}.$$

Then $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-\lambda t}$, and for $\lambda t = 2$ this is $1 - e^{-2}$.

Exercise 3.5

Compute $E[X]$ for $X \sim \text{Geometric}(p)$ on $\{1, 2, 3, \dots\}$ using the PGF $G(s) = ps/[1 - (1 - p)s]$. Then derive $\text{Var}(X) = (1 - p)/p^2$.

Solution

Let $q = 1 - p$. Differentiating $G(s) = ps/(1 - qs)$ gives

$$G'(s) = \frac{p}{(1 - qs)^2}, \quad G''(s) = \frac{2pq}{(1 - qs)^3},$$

so $E[X] = G'(1) = p/p^2 = 1/p$ and $E[X(X - 1)] = G''(1) = 2q/p^2$. Hence

$$\text{Var}(X) = G''(1) + G'(1) - [G'(1)]^2 = \frac{2q}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{1 - p}{p^2}.$$

Exercise 3.6

Show that the PGF of $X \sim \text{Binomial}(n, p)$ is $G_X(s) = (1 - p + ps)^n$. Use it to compute $E[X]$ and $E[X(X - 1)]$, and hence $\text{Var}(X)$.

Solution

By the binomial theorem,

$$G_X(s) = \sum_{k=0}^n \binom{n}{k} p^k (1 - p)^{n-k} s^k = (1 - p + ps)^n.$$

Differentiating, $G'_X(s) = np(1 - p + ps)^{n-1}$ and $G''_X(s) = n(n - 1)p^2(1 - p + ps)^{n-2}$, so $E[X] = G'_X(1) = np$ and $E[X(X - 1)] = G''_X(1) = n(n - 1)p^2$. Therefore

$$\text{Var}(X) = n(n - 1)p^2 + np - (np)^2 = np(1 - p).$$

Exercise 3.7

Two independent Poisson processes have rates λ_1 and λ_2 . Show that the total number of events in time t is Poisson with rate $\lambda_1 + \lambda_2$, and that, given the total count, the split into the two processes is binomial.

Solution

Let $X_i \sim \text{Poisson}(\lambda_i t)$ be independent. Using PGFs, $G_{X_1+X_2}(s) = e^{\lambda_1 t(s-1)} e^{\lambda_2 t(s-1)} = e^{(\lambda_1 + \lambda_2)t(s-1)}$,

so $X_1 + X_2 \sim \text{Poisson}((\lambda_1 + \lambda_2)t)$. For the conditional split, with $N = X_1 + X_2 = n$,

$$P(X_1 = k \mid N = n) = \frac{P(X_1 = k)P(X_2 = n - k)}{P(N = n)} = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k},$$

i.e. $X_1 \mid N = n \sim \text{Binomial}(n, \lambda_1/(\lambda_1 + \lambda_2))$.

Exercise 3.8

Negative binomial. X is the number of failures before the r -th success in i.i.d. Bernoulli trials with success probability p . Give the PMF, the PGF, and use the PGF to compute $E[X]$ and $\text{Var}(X)$.

Solution

Writing $X = \sum_{i=1}^r X_i$ as a sum of r i.i.d. geometric (on $\{0, 1, \dots\}$) variables with $G_{X_i}(s) = p/[1 - (1 - p)s]$, the PMF and PGF are

$$P(X = k) = \binom{k + r - 1}{k} (1 - p)^k p^r, \quad G_X(s) = \left(\frac{p}{1 - (1 - p)s} \right)^r.$$

Differentiating with $q = 1 - p$ gives $G'_X(1) = rq/p$ and $G''_X(1) = r(r + 1)q^2/p^2$, so

$$E[X] = \frac{r(1 - p)}{p}, \quad \text{Var}(X) = \frac{r(1 - p)}{p^2}.$$

Exercise 3.9

For a Bernoulli random variable X with success probability p , compute $E[X^k]$ for every integer $k \geq 1$. Comment briefly on why all moments of X are equal.

Solution

Since $X \in \{0, 1\}$, $X^k = X$ for every integer $k \geq 1$, so

$$E[X^k] = 0^k(1 - p) + 1^k p = p.$$

All powers of X coincide with X itself, hence all moments equal p .

Exercise 3.10

A multinomial trial has n draws over three categories with probabilities (p_1, p_2, p_3) . Compute $\text{Cov}(N_1, N_2)$. Explain why the covariance is negative.

Solution

Writing $N_j = \sum_{i=1}^n \mathbb{1}\{\text{draw } i \in \text{category } j\}$, the draws are independent and within a single draw $\mathbb{1}_1 \mathbb{1}_2 = 0$, so $E[\mathbb{1}_1 \mathbb{1}_2] = 0$ and the per-draw covariance is $-p_1 p_2$. Summing over n independent draws,

$$\text{Cov}(N_1, N_2) = -n p_1 p_2.$$

The covariance is negative because the total count is fixed at n : an extra draw in category 1 leaves one fewer draw available for category 2.

3. Discrete probability

Exercise 3.11

An urn contains $K = 5$ black balls and $N - K = 15$ white balls. You draw $n = 4$ balls without replacement. Compute the probability of drawing exactly 2 black balls using the hypergeometric PMF. Then compare with the binomial approximation $p = K/N$.

Solution

With $X \sim \text{Hypergeometric}(N = 20, K = 5, n = 4)$,

$$P(X = 2) = \frac{\binom{5}{2}\binom{15}{2}}{\binom{20}{4}} = \frac{10 \cdot 105}{4845} = \frac{1050}{4845} = \frac{70}{323}.$$

The binomial approximation with $p = 5/20 = 1/4$ gives $\binom{4}{2}(1/4)^2(3/4)^2 = 6 \cdot 9/256 = 54/256 \approx 0.211$, slightly below the exact $70/323 \approx 0.217$. The sampling-without-replacement model has slightly less variance, concentrating mass near the mean.

Exercise 3.12

Compare the exact probability $P(X \geq 1)$ for $X \sim \text{Binomial}(10, 0.1)$ with the Poisson approximation $X \sim \text{Poisson}(1)$. Compute both expressions in closed form (no calculator) and comment on the gap.

Solution

For the exact binomial,

$$P(X \geq 1) = 1 - (1 - 0.1)^{10} = 1 - (9/10)^{10}.$$

The Poisson approximation with $\lambda = np = 1$ gives $P(X \geq 1) = 1 - e^{-1}$. The gap is $(9/10)^{10} - e^{-1}$, which is small because $(1 - 1/n)^n \rightarrow e^{-1}$, but positive: the binomial assigns slightly more mass to $X = 0$ than the Poisson does.

Exercise 3.13

Two coin tosses. Let X be the number of heads in two independent tosses with success probability p , and let $Y = \mathbb{1}\{X \geq 1\}$. Compute $P(X = k | Y = 1)$ for $k = 1, 2$. Verify your answer sums to one.

Solution

$X \sim \text{Binomial}(2, p)$ gives $P(X = 1) = 2p(1 - p)$ and $P(X = 2) = p^2$, with $P(Y = 1) = 1 - (1 - p)^2 = p(2 - p)$. Hence

$$P(X = 1 | Y = 1) = \frac{2p(1 - p)}{p(2 - p)} = \frac{2(1 - p)}{2 - p}, \quad P(X = 2 | Y = 1) = \frac{p^2}{p(2 - p)} = \frac{p}{2 - p}.$$

Their sum is $[2(1 - p) + p]/(2 - p) = (2 - p)/(2 - p) = 1$.

Exercise 3.14

Show by direct calculation that $E[X] = K/N \cdot n$ for $X \sim \text{Hypergeometric}(N, K, n)$, using $X = \sum_{i=1}^n \mathbb{1}\{\text{draw } i \text{ is black}\}$.

Solution

By exchangeability of the draws, each draw is marginally a uniform pick from the urn, so $P(\text{draw } i \text{ is black}) = K/N$ for every i , independently of the order. Linearity of expectation gives

$$E[X] = \sum_{i=1}^n P(\text{draw } i \text{ is black}) = n \cdot \frac{K}{N}.$$

Linearity does not require independence, so the without-replacement structure is not an obstacle.

Exercise 3.15

For $X \sim \text{Poisson}(\lambda)$, show using the PGF that the sum of n independent $\text{Poisson}(\lambda)$ variables is $\text{Poisson}(n\lambda)$. Identify the corresponding result via the MGF and explain why both routes agree.

Solution

Each X_i has PGF $G_{X_i}(s) = e^{\lambda(s-1)}$. By independence,

$$G_{\sum_i X_i}(s) = \prod_{i=1}^n e^{\lambda(s-1)} = e^{n\lambda(s-1)},$$

which is the PGF of $\text{Poisson}(n\lambda)$. The MGF route gives $M_{\sum X_i}(t) = e^{n\lambda(e^t-1)}$, also a $\text{Poisson}(n\lambda)$ MGF. The two routes agree because $M_X(t) = G_X(e^t)$ for non-negative integer-valued X , so the multiplicative-under-independence property and the uniqueness theorem hold identically in both.

3.11. Sources

These notes draw on the course slides *Advanced Statistics: Discrete Probability* (Gianmaria Falasco, 2026) and material from the following references:

- Falasco, G. (2026). *Advanced Statistics for Physics Analysis: Bayesian inference*. Lecture notes, University of Padova.
- von der Linde, A. & von Toussaint, U. (2011). *Bayesian probability theory: Applications in the physical sciences*. Cambridge University Press.
- Sivia, D. & Skilling, J. (2006). *Data analysis: A Bayesian tutorial*. Oxford University Press, 2nd ed.

4. Continuous probability and limit theorems

4.1. Introduction

This chapter covers continuous random variables and the fundamental limit theorems that govern the behavior of sums of independent samples. These results form the mathematical backbone of Bayesian statistics: they explain why posteriors concentrate, why Gaussian approximations emerge, and how rare events scale exponentially with sample size. Together, they provide a unified picture of how randomness resolves into determinism and structure as data accumulates.

4.2. Continuous random variables

A continuous random variable X is characterized by a probability density function (pdf) $f(x)$ defined over some interval (often $\mathbb{R}$). Unlike discrete random variables, X cannot take specific values with positive probability; instead, we specify probabilities over intervals:

$$P(a \leq X \leq b) = \int_a^b f(x) dx,$$

with the normalization $\int_{-\infty}^{\infty} f(x) dx = 1$. At any single point, $P(X = x) = 0$.

Definition 4.1 (Cumulative distribution function). The **cumulative distribution function** (CDF) of a continuous random variable X is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt.$$

The pdf is recovered by differentiation: $f(x) = \frac{d}{dx} F(x)$.

For multivariate distributions, the concepts extend naturally. Given random variables X and Y : - The **joint pdf** $f(x, y)$ satisfies $\iint f(x, y) dx dy = 1$. - The **marginal pdf** of X is $f(x) = \int f(x, y) dy$. - The **conditional pdf** of X given $Y = y$ is $f(x | y) = \frac{f(x, y)}{f(y)}$ (when $f(y) > 0$).

4.3. Expectation and moments

For a continuous random variable X with pdf $f(x)$, the **expectation** is

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx.$$

More generally, for any function $g(X)$,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx.$$

4. Continuous probability and limit theorems

The n -th **raw moment** is $\mu'_n = E[X^n]$, and the n -th **central moment** is $\mu_n = E[(X - E[X])^n]$. The second central moment is the **variance**:

$$\text{Var}(X) = \mu_2 = E[(X - E[X])^2] = E[X^2] - (E[X])^2.$$

The **standard deviation** is $\sigma = \sqrt{\text{Var}(X)}$.

Definition 4.2 (Moment generating function). The **moment generating function** (MGF) of X is

$$M(t) = E[e^{tX}],$$

defined in a neighborhood of $t = 0$. The n -th moment is recovered via

$$E[X^n] = \left. \frac{d^n M}{dt^n} \right|_{t=0}.$$

The MGF uniquely characterizes the distribution (when it exists) and is particularly useful for analyzing sums of independent random variables.

Definition 4.3 (Covariance and correlation). For random variables X and Y , the **covariance** is

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])].$$

The **correlation coefficient** is

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y},$$

which lies in $[-1, 1]$ and measures linear association.

4.4. Standard continuous distributions

The distributions below recur throughout Bayesian inference, either as likelihoods, as conjugate priors, or as limit shapes that emerge asymptotically. For each, we record the density, the first two moments, the role in conjugate analysis, and the physical setting in which it typically arises.

- **Uniform.** The **uniform distribution** on $[a, b]$ has constant density $f(x) = 1/(b - a)$ for $x \in [a, b]$. For $X \sim U(a, b)$, the mean is $E[X] = (a + b)/2$ and the variance is $\text{Var}(X) = (b - a)^2/12$. It is the canonical non-informative prior on a bounded interval, encoding complete ignorance within fixed bounds (the classic example is a draw from a random lottery).
- **Normal (Gaussian).** The **normal distribution** with mean μ and variance σ^2 has pdf

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

For $X \sim N(\mu, \sigma^2)$, $E[X] = \mu$ and $\text{Var}(X) = \sigma^2$. It is **stable** under linear combinations: if $X_i \sim N(\mu_i, \sigma_i^2)$ are independent, then $Y = \sum_i a_i X_i \sim N(\sum_i a_i \mu_i, \sum_i a_i^2 \sigma_i^2)$. It is the conjugate prior for normal likelihoods, emerges via the Central Limit Theorem, and dominates applied work as a model for errors, noise, and aggregates of many small effects (e.g. heights in a population approximate $N(170, 100)$ in cm when averaged over many groups).

Example 4.1 (*Two Gaussian distributions*)

Consider two measurements with uncertainties: $X_1 \sim N(0, 1)$ and $X_2 \sim N(2, 0.5^2)$. A comparison is easily visualized by overlaying their pdfs. The first is centered at 0 with width ~ 2 ; the second is centered at 2 with narrower width ~ 1 . A posterior combining both observations would be pulled toward the more precise measurement.

See code 15.3 in Appendix for the implementation.

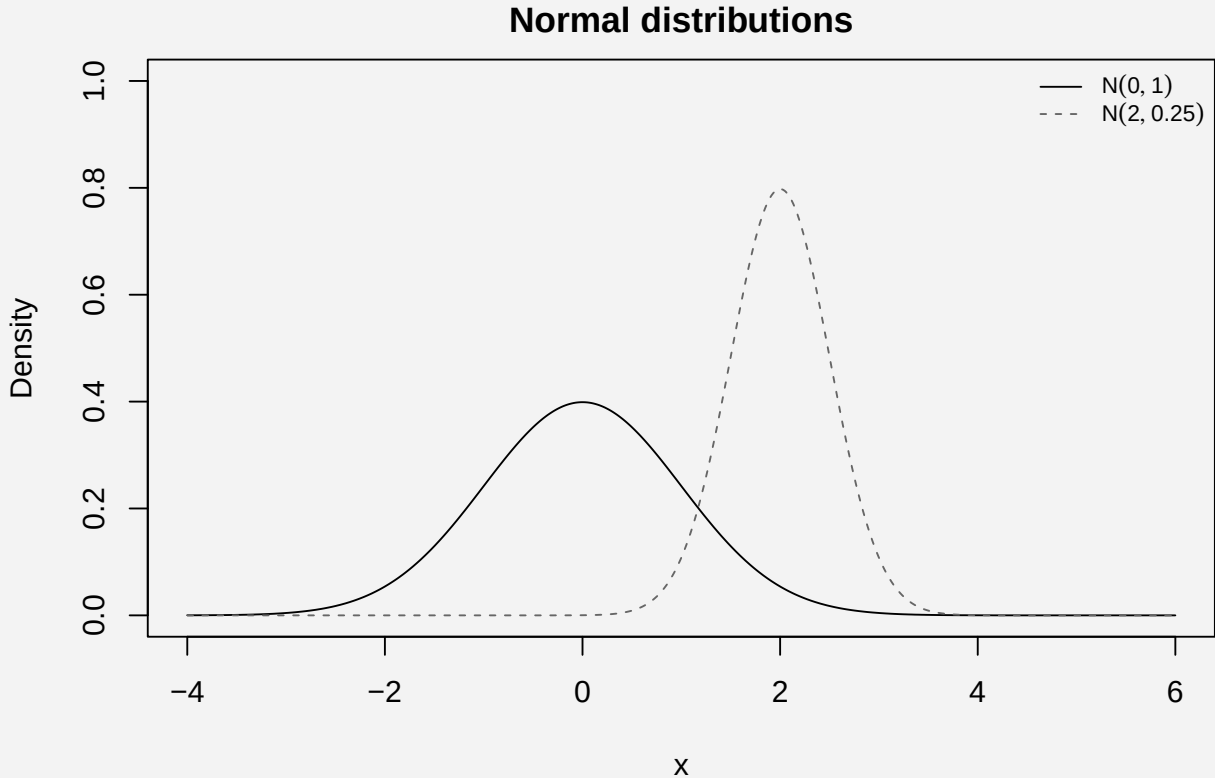


Figure 4.1.: Two normal distributions with different parameters

- **Exponential.** The **exponential distribution** with rate $\lambda > 0$ has pdf $f(t) = \lambda e^{-\lambda t}$ for $t \geq 0$. For $X \sim \text{Exp}(\lambda)$, $E[X] = 1/\lambda$ and $\text{Var}(X) = 1/\lambda^2$. It is conjugate to itself and arises as the waiting time between successive events in a Poisson process, in radioactive decay, and in reliability lifetimes (e.g. if calls arrive at rate $\lambda = 2$ per minute, the time until the next call is $\text{Exp}(2)$ with expected value 0.5 minutes). Its defining feature is the **memoryless property** $P(X > t + s \mid X > t) = P(X > s)$ for all $s, t \geq 0$: the residual waiting time does not depend on how long we have already waited, and the exponential is the unique continuous distribution with this property.
- **Erlang and Gamma.** The **Erlang distribution** describes the sum of n independent exponentials with the same rate λ and has pdf $f(t) = \lambda^n t^{n-1} e^{-\lambda t} / (n-1)!$ for $t \geq 0$. The **Gamma distribution** generalises Erlang by allowing the shape parameter $n \rightarrow \alpha \in \mathbb{R}^+$,

$$f(t) = \frac{\lambda^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\lambda t}, \quad t \geq 0,$$

where $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$ extends the factorial via $\Gamma(n) = (n-1)!$ for positive integers. For $X \sim \text{Gamma}(\alpha, \lambda)$, $E[X] = \alpha/\lambda$ and $\text{Var}(X) = \alpha/\lambda^2$; it scales as $cX \sim \text{Gamma}(\alpha, \lambda/c)$ for $c > 0$. It is the conjugate prior for Poisson and exponential likelihoods and arises as sums of exponentials, occurrence times in Poisson processes, rainfall accumulation, and insurance claim sizes (e.g. the time until the 5th customer at a store with arrival rate $\lambda = 1$ per minute follows $\text{Gamma}(5, 1)$). A special case is the **chi-squared distribution** χ_k^2 : with $\alpha = k/2$ and $\lambda = 1/2$ we recover the distribution of

4. Continuous probability and limit theorems

$\sum_{i=1}^k Z_i^2$ for independent $Z_i \sim N(0, 1)$, which underlies goodness-of-fit testing.

- **Beta.** The **Beta distribution** on $[0, 1]$ with shape parameters $\alpha, \beta > 0$ has pdf

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad x \in [0, 1],$$

where $B(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1-t)^{\beta-1} dt = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$ is the **Beta function**. For $X \sim \text{Beta}(\alpha, \beta)$, $E[X] = \alpha/(\alpha+\beta)$ and $\text{Var}(X) = \alpha\beta/[(\alpha+\beta)^2(\alpha+\beta+1)]$. The shape is unimodal when $\alpha, \beta > 1$, uniform when $\alpha = \beta = 1$, and U-shaped when $\alpha, \beta < 1$. It is the conjugate prior for Bernoulli and binomial likelihoods and is the natural model for probabilities, proportions constrained to $[0, 1]$, and order statistics (e.g. for inference on a coin bias p , a $\text{Beta}(2, 2)$ prior — symmetric, peaked near 0.5 — is more informative than $\text{Beta}(1, 1)$, which is uniform). Multimodal distributions on $[0, 1]$ are easily obtained as convex combinations of Betas, e.g. $\frac{1}{2}\text{Beta}(\alpha_1, \beta_1) + \frac{1}{2}\text{Beta}(\alpha_2, \beta_2)$.

See code 15.4 in Appendix for the implementation.

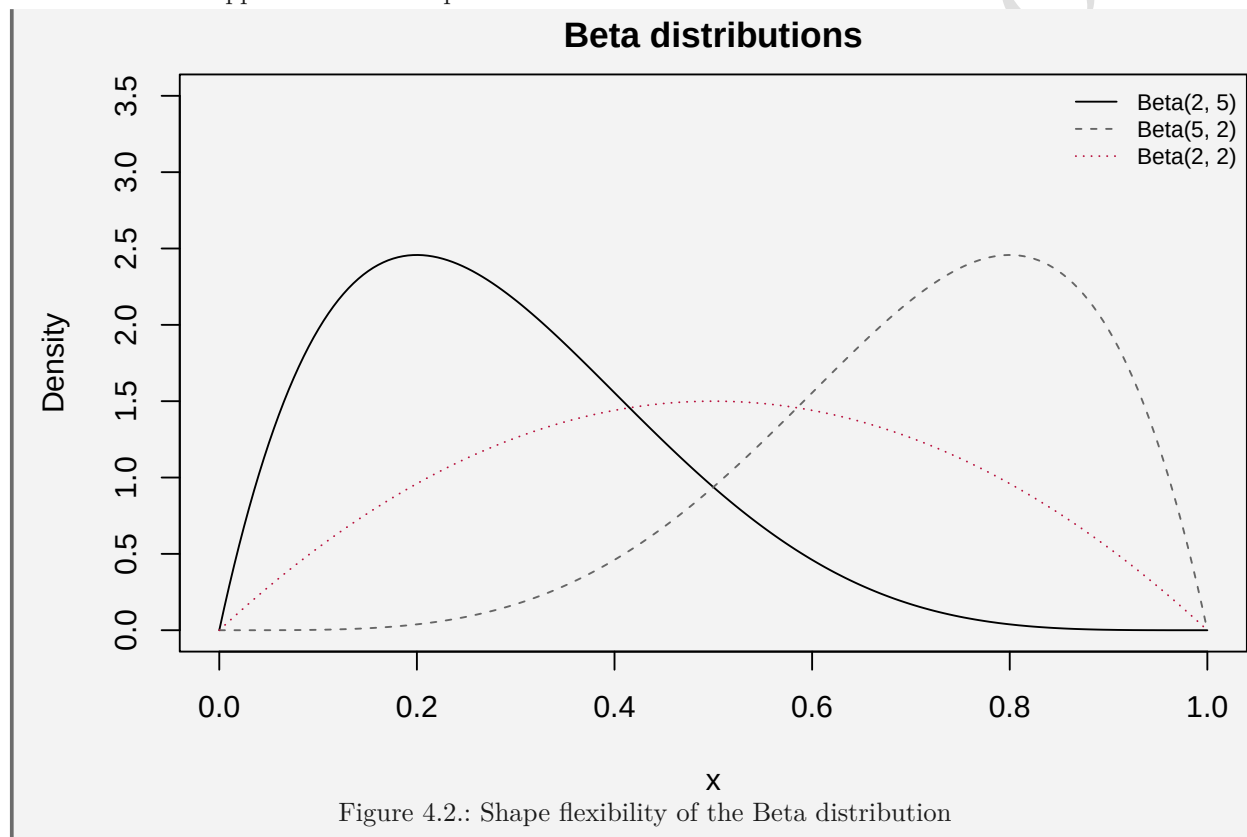


Figure 4.2.: Shape flexibility of the Beta distribution

Several other continuous distributions appear repeatedly as priors, sampling distributions, or generalisations of the families above:

- **Student's t .** With ν degrees of freedom, its pdf is

$$f(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi\nu}\Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}.$$

It has mean 0 (for $\nu > 1$) and variance $\nu/(\nu-2)$ (for $\nu > 2$), with tails heavier than the normal — especially for small ν — and converges to $N(0, 1)$ as $\nu \rightarrow \infty$. It arises as the distribution of $\sqrt{n}(\bar{X}_n - \mu)/S_n$ where S_n is the sample standard deviation, making it fundamental for frequentist confidence intervals.

- **Inverse Gamma.** With shape α and rate λ , its pdf is

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} \exp\left(-\frac{\lambda}{x}\right), \quad x > 0,$$

with mean $\lambda/(\alpha - 1)$ for $\alpha > 1$. It is conjugate to the variance parameter of a normal likelihood and is commonly used as a prior for precision and variance parameters.

- **Dirichlet.** On the $(k - 1)$ -simplex (vectors $\mathbf{x} = (x_1, \dots, x_k)$ with $x_i \geq 0$ and $\sum_i x_i = 1$), with parameters $\alpha = (\alpha_1, \dots, \alpha_k)$,

$$f(\mathbf{x}) = \frac{\Gamma(\sum_i \alpha_i)}{\prod_i \Gamma(\alpha_i)} \prod_{i=1}^k x_i^{\alpha_i-1}.$$

It is conjugate to the multinomial likelihood and is the natural extension of the Beta to multiple categories.

- **Cauchy.** With location μ and scale σ , $f(x) = (1/\pi\sigma) \cdot 1/(1 + ((x - \mu)/\sigma)^2)$. The Cauchy has no finite mean or variance and its tails decay as $1/x^2$, making it useful as a prior that admits extreme values while remaining non-informative in a heavy-tailed sense.

4.5. Transformation of variables and sums of random variables

Often we need the distribution of a transformed random variable or a sum. Two key techniques are the Jacobian and the moment-generating function.

Proposition 4.1 (Transformation via Jacobian). *If $Y = g(X)$ where g is invertible and differentiable, then*

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dg^{-1}}{dy} \right|.$$

For multivariate transformations $\mathbf{Y} = \mathbf{g}(\mathbf{X})$ with Jacobian matrix $\mathbf{J}$,

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(\mathbf{g}^{-1}(\mathbf{y})) |\det \mathbf{J}^{-1}(\mathbf{y})|.$$

Example 4.2 (Sum of uniform random variables)

Let X_1, X_2 be independent $U(0, 1)$ random variables. We seek the distribution of $Y = X_1 + X_2$.

By definition, $f_Y(y) = \int_{-\infty}^{\infty} f_{X_1}(x_1) f_{X_2}(y - x_1) dx_1$. Since $f_{X_1} = f_{X_2} = 1$ on $[0, 1]$, we need both $0 \leq x_1 \leq 1$ and $0 \leq y - x_1 \leq 1$, which constrains $x_1 \in [\max(0, y - 1), \min(1, y)]$.

For $y \in [0, 2]$:

$$f_Y(y) = \int_{\max(0, y-1)}^{\min(1, y)} 1 dx_1 = \min(1, y) - \max(0, y - 1) = \begin{cases} y & 0 \leq y \leq 1 \\ 2 - y & 1 < y \leq 2 \end{cases}$$

This is the triangular distribution—a classic result showing how convolution of uniform pdfs yields a piecewise linear shape. Generalizing to n uniforms gives the Irwin–Hall distribution.

Example 4.3 (Sum of two independent exponentials (exam-style))

Let X and Y be independent and identically distributed exponential random variables with rate $\lambda > 0$, i.e. $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Find the density of $Z = X + Y$.

Solution. Since X and Y are independent, the density of the sum is the convolution

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x) dx = \int_{-\infty}^{+\infty} f_X(x) f_X(z-x) dx.$$

The integrand is non-zero only for $0 \leq x \leq z$, so for $z \geq 0$,

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx = \lambda^2 e^{-\lambda z} \int_0^z dx = \lambda^2 z e^{-\lambda z}.$$

Hence

$$f_Z(z) = \begin{cases} \lambda^2 z e^{-\lambda z}, & z \geq 0, \\ 0, & z < 0. \end{cases}$$

This is the Erlang / Gamma distribution with shape 2 and rate λ .

Example 4.4 (*Detector dead time plus exponential waiting (exam-style)*)

A detector has a deterministic dead time $t_0 > 0$. After this dead time, the waiting time to the next event is exponential with rate λ . If $Z = t_0 + T$ with $T \sim \text{Exp}(\lambda)$, find the density and mean of Z .

Solution. A deterministic shift just translates the support. For $z \geq t_0$,

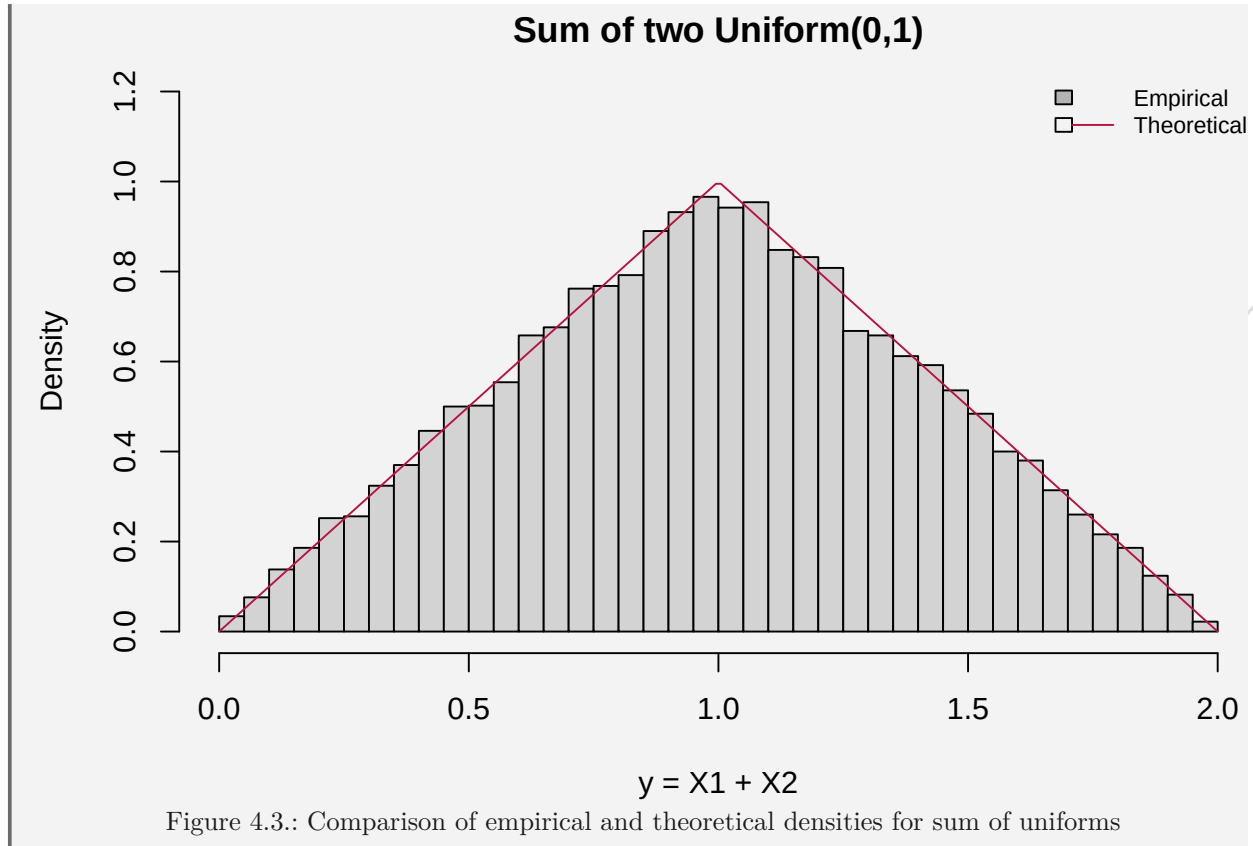
$$f_Z(z) = \lambda e^{-\lambda(z-t_0)} \mathbb{1}_{\{z \geq t_0\}}.$$

The mean shifts by the dead time:

$$E[Z] = t_0 + E[T] = t_0 + \frac{1}{\lambda}.$$

The variance is unchanged: $\text{Var}(Z) = 1/\lambda^2$. A deterministic shift does not add variability.

See code 15.5 in Appendix for the implementation.



Proposition 4.2 (MGF of sums). *If $X_1, \dots, X_n$ are independent with MGFs $M_1(t), \dots, M_n(t)$, then the MGF of $S_n = \sum_i X_i$ is*

$$M_{S_n}(t) = \prod_{i=1}^n M_i(t).$$

This provides a direct route to the mean and variance of sums via derivatives.

4.6. Law of Large Numbers

The **Law of Large Numbers** (LLN) states that as we accumulate more samples, their average converges to the population mean. This is the mathematical foundation for why empirical frequencies approach probabilities.

Theorem 4.3 (Weak Law of Large Numbers). *Let $\{X_i\}_{i=1}^\infty$ be i.i.d. random variables with $E[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Then for any $\epsilon > 0$,*

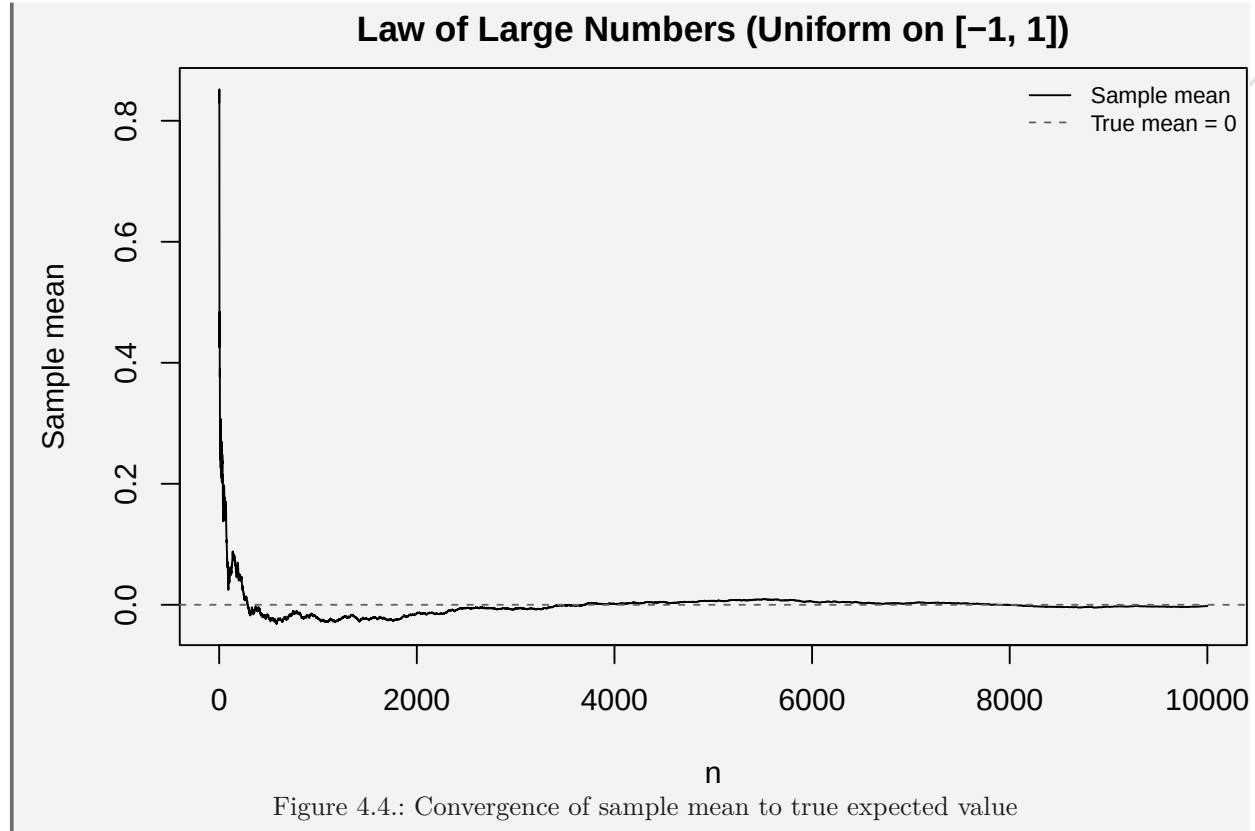
$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| > \epsilon) = 0,$$

where $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. We write $\bar{X}_n \xrightarrow{P} \mu$ (convergence in probability).

The weak LLN says that the sample mean concentrates around its expected value, with deviations becoming arbitrarily unlikely as n grows. There is also a **strong LLN**, which states almost-sure convergence: $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$ with probability 1.

Note. In Bayesian inference, the LLN underlies **posterior consistency**: as data accumulate, the posterior concentrates near the true parameter value, and the prior becomes negligible. This is why Bayesian and frequentist estimators often agree asymptotically, despite their philosophies differing.

See code 15.6 in Appendix for the implementation.



4.7. Central Limit Theorem

The **Central Limit Theorem** (CLT) is the most important result in probability theory for statistics. It says that sample means converge in distribution to a normal, regardless of the underlying distribution.

Theorem 4.4 (Central Limit Theorem). *Let $\{X_i\}_{i=1}^{\infty}$ be i.i.d. random variables with $E[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 > 0$. Then*

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

as $n \rightarrow \infty$, where $\xrightarrow{d}$ denotes convergence in distribution. Equivalently,

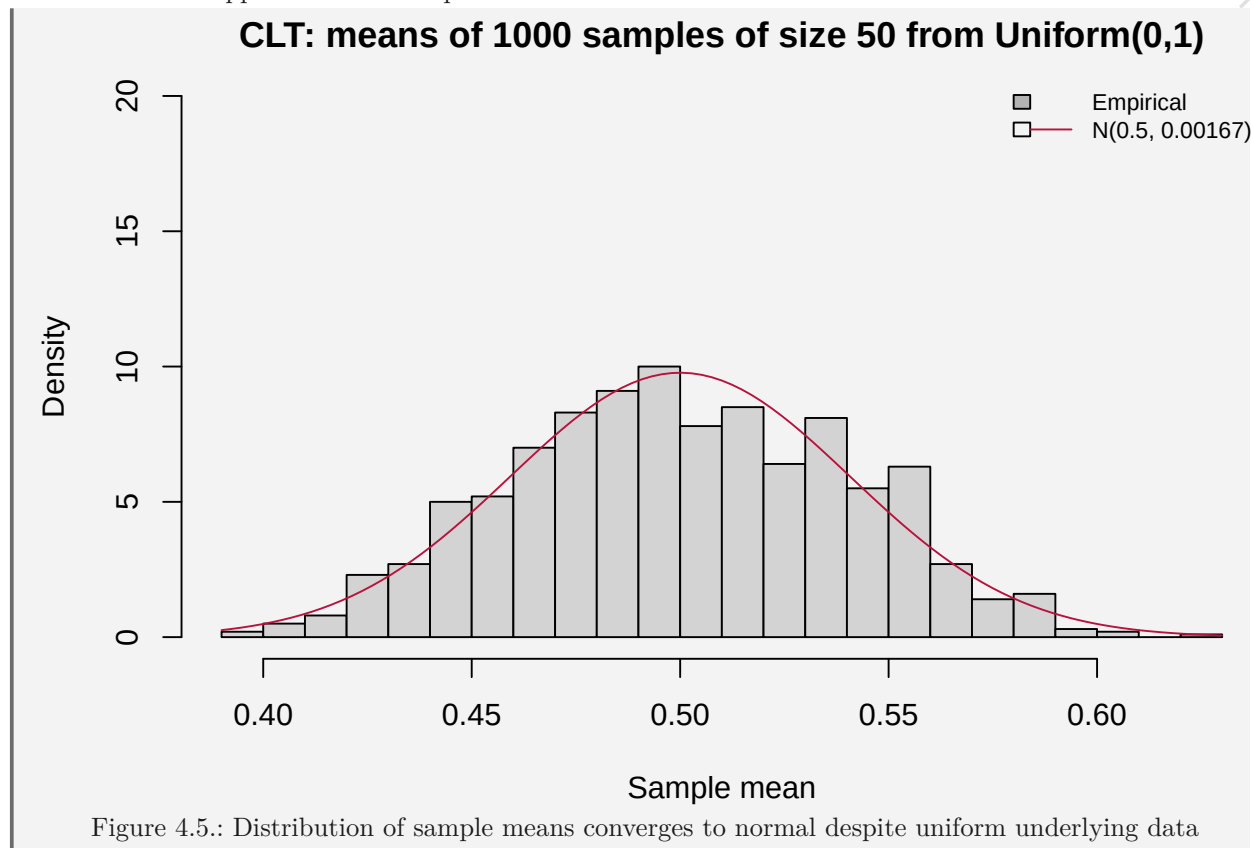
$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{for large } n.$$

The CLT is **universal**: the form of the limiting normal depends only on the mean and variance, not on higher moments or the original shape of the distribution. This explains why the normal distribution is so ubiquitous in practice.

Example 4.5 (*CLT with non-normal data*)

Suppose we draw $n = 50$ samples from $U(0, 1)$, which is decidedly non-normal (uniform and bounded). The mean $\bar{X}_{50}$ is approximately $N(0.5, 1/(12 \cdot 50)) = N(0.5, 0.00167)$, so $\bar{X}_{50}$ has standard deviation ≈ 0.04 . Repeating this experiment 1000 times and histogramming the sample means shows a bell curve, despite the underlying distribution being uniform. As n increases, the histogram becomes ever more Gaussian.

See code 15.7 in Appendix for the implementation.



The CLT extends naturally to **multivariate settings**: if $\mathbf{X}_1, \dots, \mathbf{X}_n$ are i.i.d. d -dimensional random vectors with mean μ and covariance matrix Σ , then

$$\sqrt{n}(\bar{\mathbf{X}}_n - \mu) \xrightarrow{d} N_d(\mathbf{0}, \Sigma).$$

This is crucial for Bayesian inference on multiple parameters, since it justifies Gaussian posterior approximations via Laplace's method.

Note. In Bayesian inference, the CLT explains why: - Posteriors are approximately Gaussian for large n (Bernstein–von Mises theorem) - Fisher information quantifies curvature of the log-likelihood at its mode - Credible intervals around the posterior mean approximate confidence intervals

4.8. Large Deviation Principle

The CLT describes typical fluctuations (of order $\sqrt{n}$), but it does not capture rare, atypical events. The **Large Deviation Principle** (LDP) describes how the probability of observing a sample mean far from the true mean decays exponentially with sample size.

Theorem 4.5 (Large Deviation Principle (Cramér)). Let $\{X_i\}_{i=1}^\infty$ be i.i.d. random variables with MGF $M(t) = E[e^{tX}]$ existing in a neighborhood of zero. Then for any x in the interior of the support,

$$P(\bar{X}_n \approx x) \asymp e^{-nI(x)},$$

where the **rate function** is

$$I(x) = \sup_t [tx - \log M(t)].$$

This is the **Fenchel–Legendre transform** of the log-MGF.

The rate function has the properties: - $I(\mu) = 0$ (typical events at the mean have zero cost) - $I(x) > 0$ for $x \neq \mu$ (atypical events carry exponential cost) - Expanding $I(x)$ to second order around $x = \mu$ recovers the Gaussian behavior of the CLT

Example 4.6 (LDP for normal samples)

If $X_i \sim N(\mu, \sigma^2)$, then $M(t) = \exp(\mu t + \sigma^2 t^2/2)$, so

$$\log M(t) = \mu t + \frac{\sigma^2 t^2}{2}.$$

The rate function is

$$I(x) = \sup_t \left[tx - \mu t - \frac{\sigma^2 t^2}{2} \right].$$

Optimizing: $\frac{d}{dt}[\dots] = x - \mu - \sigma^2 t = 0 \Rightarrow t^* = (x - \mu)/\sigma^2$, yielding

$$I(x) = \frac{(x - \mu)^2}{2\sigma^2}.$$

Thus, $P(\bar{X}_n \approx x) \sim \exp\left(-\frac{n(x-\mu)^2}{2\sigma^2}\right)$, which is consistent with the Gaussian CLT: rare events decay as a Gaussian tail.

4.9. Unified picture of asymptotics

These three results form a hierarchy of limiting behavior as $n \rightarrow \infty$:

1. **Law of Large Numbers:** The sample mean becomes deterministic. Fluctuations scale as $O(n^{-1})$ relative to the mean.
2. **Central Limit Theorem:** Gaussian fluctuations of order $\sqrt{n}^{-1}$ emerge around the LLN limit.
3. **Large Deviation Principle:** Rare events with $\bar{X}_n \approx x$ occur with probability $\sim e^{-nI(x)}$. For events beyond a few standard deviations, this exponential suppression dominates.

Remark. Conceptually, the picture is: **Randomness = Determinism + Fluctuations + Rare Events**. As data accumulate, the deterministic part grows, fluctuations remain Gaussian and relatively shrink, and rare events become exponentially suppressed. This is why Bayesian posteriors concentrate, why frequentist intervals shrink, and why model misspecification becomes detectable: the data force all inferences toward the truth under the asymptotic regime.

4.10. Exercises

Exercise 4.1

Find the CDF and PDF of the half-normal random variable $Y = |X|$ when $X \sim \mathcal{N}(0, 1)$.

Solution

For $y \geq 0$, $F_Y(y) = P(|X| \leq y) = P(-y \leq X \leq y) = 2\Phi(y) - 1$, where Φ is the standard normal CDF. Differentiating,

$$f_Y(y) = 2\varphi(y) = \sqrt{\frac{2}{\pi}} e^{-y^2/2}, \quad y \geq 0,$$

and $f_Y(y) = 0$ for $y < 0$.

Exercise 4.2

Let $X \sim \mathcal{N}(0, 1)$ and $Y = X^2$. Derive the density of Y using the change-of-variables formula. Identify the resulting distribution by name.

Solution

The map $y = x^2$ is two-to-one, with inverse branches $x = \pm\sqrt{y}$ and $|dx/dy| = 1/(2\sqrt{y})$. Summing both branches,

$$f_Y(y) = 2 \cdot \frac{1}{\sqrt{2\pi}} e^{-y/2} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi y}} e^{-y/2}, \quad y > 0.$$

This is the χ_1^2 distribution, i.e. $\text{Gamma}(\frac{1}{2}, \frac{1}{2})$.

Exercise 4.3

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = e^X$. Derive the density of Y (log-normal). Compute $E[Y]$ in closed form using the MGF of X .

Solution

With $x = \ln y$ and $|dx/dy| = 1/y$,

$$f_Y(y) = \frac{1}{y\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\ln y - \mu)^2}{2\sigma^2}\right), \quad y > 0.$$

By definition $E[Y] = E[e^X] = M_X(1)$, and since $M_X(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$,

$$E[Y] = \exp(\mu + \frac{1}{2}\sigma^2).$$

Exercise 4.4

Let X, Y be independent with $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$. Use MGFs to show that $X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Solution

By independence, $M_{X+Y}(t) = M_X(t) M_Y(t)$. Substituting the Gaussian MGFs,

$$M_{X+Y}(t) = e^{\mu_1 t + \frac{1}{2}\sigma_1^2 t^2} e^{\mu_2 t + \frac{1}{2}\sigma_2^2 t^2} = \exp((\mu_1 + \mu_2)t + \frac{1}{2}(\sigma_1^2 + \sigma_2^2)t^2),$$

4. Continuous probability and limit theorems

which is the MGF of $\mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. By uniqueness of the MGF, $X + Y$ has that distribution.

Exercise 4.5

Two independent exponentials with rate λ . Find the density of $M = \min(X, Y)$. Identify it by name.

Solution

Using the survival function for the minimum of independents,

$$P(M > m) = P(X > m) P(Y > m) = e^{-\lambda m} e^{-\lambda m} = e^{-2\lambda m}, \quad m \geq 0.$$

Hence $f_M(m) = 2\lambda e^{-2\lambda m}$ for $m \geq 0$, so $M \sim \text{Exp}(2\lambda)$.

Exercise 4.6

Two independent Uniform(0, 1) variables. Find the density of $M = \max(X, Y)$ and compute $E[M]$.

Solution

For $m \in [0, 1]$, $F_M(m) = P(X \leq m) P(Y \leq m) = m^2$, so $f_M(m) = 2m$. The mean is

$$E[M] = \int_0^1 m \cdot 2m \, dm = \frac{2}{3}.$$

Exercise 4.7

For $X \sim \text{Uniform}(a, b)$, compute $E[X]$ and $\text{Var}(X)$ directly from the PDF.

Solution

With $f(x) = 1/(b-a)$ on $[a, b]$,

$$E[X] = \frac{1}{b-a} \int_a^b x \, dx = \frac{a+b}{2}, \quad E[X^2] = \frac{1}{b-a} \int_a^b x^2 \, dx = \frac{a^2 + ab + b^2}{3}.$$

Subtracting gives

$$\text{Var}(X) = E[X^2] - E[X]^2 = \frac{(b-a)^2}{12}.$$

Exercise 4.8

For $X \sim \text{Gamma}(\alpha, \beta)$ with rate β , derive $E[X] = \alpha/\beta$ and $\text{Var}(X) = \alpha/\beta^2$ via the MGF. Then derive the same results via the cumulant generating function.

Solution

The MGF is $M(t) = (1 - t/\beta)^{-\alpha}$ for $t < \beta$. Then $M'(0) = \alpha/\beta$ and $M''(0) = \alpha(\alpha + 1)/\beta^2$, so $E[X] = \alpha/\beta$ and $\text{Var}(X) = M''(0) - M'(0)^2 = \alpha/\beta^2$. Via the CGF

$$K(t) = -\alpha \log(1 - t/\beta),$$

we get $K'(0) = \alpha/\beta = E[X]$ and $K''(0) = \alpha/\beta^2 = \text{Var}(X)$ directly.

Exercise 4.9

Let X, Y be i.i.d. $\text{Exp}(1)$. Find the density of $Z = X/(X + Y)$. Identify it by name.

Solution

This is a standard Gamma ratio: if $X \sim \text{Gamma}(\alpha_1, \lambda)$ and $Y \sim \text{Gamma}(\alpha_2, \lambda)$ are independent, then $X/(X + Y) \sim \text{Beta}(\alpha_1, \alpha_2)$, independent of $X + Y$. Here $\alpha_1 = \alpha_2 = 1$, giving

$$f_Z(z) = 1, \quad z \in [0, 1],$$

i.e. $Z \sim \text{Uniform}(0, 1) = \text{Beta}(1, 1)$.

Exercise 4.10

Compute the MGF of $X \sim \text{Beta}(\alpha, \beta)$ as a series (no closed form is required). Use the series to derive $E[X] = \alpha/(\alpha + \beta)$.

Solution

Expanding $e^{tx} = \sum_{k=0}^{\infty} t^k x^k / k!$ and integrating term-by-term against the Beta density,

$$M(t) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E[X^k] = \sum_{k=0}^{\infty} \frac{t^k}{k!} \frac{B(\alpha + k, \beta)}{B(\alpha, \beta)} = \sum_{k=0}^{\infty} \frac{t^k}{k!} \prod_{j=0}^{k-1} \frac{\alpha + j}{\alpha + \beta + j}.$$

The coefficient of t is $E[X] = B(\alpha + 1, \beta)/B(\alpha, \beta) = \alpha/(\alpha + \beta)$, using $\Gamma(\alpha + 1) = \alpha\Gamma(\alpha)$.

Exercise 4.11

Central limit theorem. Let $X_1, \dots, X_n \sim \text{Exp}(\lambda)$ i.i.d. Give the approximate distribution of $\bar{X}_n$ for large n in terms of λ and n .

Solution

Here $\mu = 1/\lambda$ and $\sigma^2 = 1/\lambda^2$, so the CLT gives

$$\bar{X}_n \approx \mathcal{N}\left(\frac{1}{\lambda}, \frac{1}{n\lambda^2}\right) \quad \text{for large } n.$$

Exercise 4.12

Inverse-CDF for the Cauchy distribution. The Cauchy CDF is $F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan x$. Derive an explicit sampler $X = g(U)$ with $U \sim \text{Uniform}(0, 1)$.

Solution

Setting $U = F(X)$ and solving, $\arctan X = \pi(U - \frac{1}{2})$, hence

$$X = \tan\left(\pi\left(U - \frac{1}{2}\right)\right), \quad U \sim \text{Uniform}(0, 1).$$

Exercise 4.13

Three independent Gaussians with the same mean μ and variances σ_i^2 . Compute the variance of the precision-weighted mean $\hat{\mu} = \sum_i x_i / \sigma_i^2 / \sum_i 1 / \sigma_i^2$. Compare with the unweighted mean.

Solution

Write $\hat{\mu} = \sum_i w_i x_i$ with $w_i = (1/\sigma_i^2)/S$ and $S = \sum_j 1/\sigma_j^2$. By independence,

$$\text{Var}(\hat{\mu}) = \sum_i w_i^2 \sigma_i^2 = \frac{1}{S^2} \sum_i \frac{1}{\sigma_i^2} = \frac{1}{\sum_i 1/\sigma_i^2}.$$

The unweighted mean $\bar{X} = \frac{1}{3} \sum_i x_i$ has $\text{Var}(\bar{X}) = \frac{1}{9} \sum_i \sigma_i^2$. By the Cauchy–Schwarz / AM–HM inequality $\text{Var}(\hat{\mu}) \leq \text{Var}(\bar{X})$, with equality iff all σ_i are equal.

Exercise 4.14

Sum of two independent Gammas with the same rate. Show, using MGFs, that $\text{Gamma}(\alpha_1, \beta) + \text{Gamma}(\alpha_2, \beta) \sim \text{Gamma}(\alpha_1 + \alpha_2, \beta)$.

Solution

With $M_i(t) = (1 - t/\beta)^{-\alpha_i}$ for $t < \beta$, independence gives

$$M_{X_1+X_2}(t) = (1 - t/\beta)^{-\alpha_1} (1 - t/\beta)^{-\alpha_2} = (1 - t/\beta)^{-(\alpha_1+\alpha_2)},$$

which is the MGF of $\text{Gamma}(\alpha_1 + \alpha_2, \beta)$. Uniqueness of the MGF yields the identification.

Exercise 4.15

A random variable has CGF $K(t) = \mu t + \frac{1}{2} \sigma^2 t^2$. What is its distribution? Give a one-line justification using the uniqueness of the MGF.

Solution

Exponentiating, the MGF is

$$M(t) = e^{K(t)} = \exp\left(\mu t + \frac{1}{2} \sigma^2 t^2\right),$$

which is the MGF of $\mathcal{N}(\mu, \sigma^2)$; by uniqueness of the MGF, $X \sim \mathcal{N}(\mu, \sigma^2)$.

4.11. Sources

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- van Dyk, D. A., & Gehrels, N. (2023). *Bayesian Probability Theory: Applications in the Physical Sciences*. Cambridge University Press.
- Sivia, D. S., & Skilling, J. (2006). *Data Analysis: A Bayesian Tutorial*. Oxford University Press, 2nd ed.

Part II.

PART II: BAYESIAN INFERENCE

5. Model comparison and Bayes factors

5.1. Introduction

When faced with multiple competing hypotheses or models, scientists need a principled way to compare them in light of observed data. The Bayesian framework provides an elegant and automatic way to do this: the posterior odds ratio, built from the Bayes factor and the prior odds. Unlike frequentist hypothesis tests that focus on rejecting a single null hypothesis, Bayesian model comparison directly assigns probabilities to models and penalizes unjustified complexity.

The key insight—Occam’s razor—emerges automatically from the mathematics. Complex models that require fine-tuning of their parameters to fit data are penalized, while simple models that make robust predictions across a wide range of parameter values are favored. This chapter develops the theory and practice of Bayesian model selection, from the basic posterior odds formula to sophisticated applications in scientific inference.

Example 5.1 (*Bayes’ theorem and the role of evidence (exam-style)*)

State Bayes’ theorem for a continuous parameter θ and data D . Explain the meaning of likelihood, prior, posterior, and evidence. Why is the evidence often omitted in parameter estimation but essential in model comparison?

Solution. Bayes’ theorem reads

$$p(\theta | D, H) = \frac{p(D | \theta, H) p(\theta | H)}{p(D | H)}, \quad p(D | H) = \int p(D | \theta, H) p(\theta | H) d\theta.$$

Here $p(D | \theta, H)$ is the **likelihood**, $p(\theta | H)$ is the **prior**, $p(\theta | D, H)$ is the **posterior**, and $p(D | H)$ is the **evidence** (or marginal likelihood).

In parameter estimation within a fixed model H , the evidence is independent of θ and acts only as a normalisation constant, so one writes $p(\theta | D, H) \propto p(D | \theta, H) p(\theta | H)$. In model comparison, however, different models have different evidences, and the ratio of evidences is precisely the Bayes factor — so $p(D | H)$ cannot be dropped.

5.2. Posterior odds and the Bayes factor

Suppose we have two mutually exclusive and exhaustive hypotheses H_0 and H_1 , and we observe data D . Bayes’ theorem tells us:

$$P(H_i | D) = \frac{P(D | H_i) P(H_i)}{P(D)}$$

where $P(D) = P(D | H_0) P(H_0) + P(D | H_1) P(H_1)$ is the total probability of the data, obtained by marginalizing over both hypotheses.

Definition 5.1 (Posterior odds ratio). The posterior odds ratio is the ratio of the posterior probabilities:

$$R = \frac{P(H_1|D)}{P(H_0|D)}$$

This directly quantifies which hypothesis is more likely given the data. Values $R > 1$ favor H_1 , while $R < 1$ favor H_0 .

Now divide the posterior for H_1 by the posterior for H_0 :

$$R = \frac{P(H_1|D)}{P(H_0|D)} = \frac{P(D|H_1)}{P(D|H_0)} \cdot \frac{P(H_1)}{P(H_0)}$$

The first ratio is the Bayes factor; the second is the prior odds.

Definition 5.2 (Bayes factor). The Bayes factor is:

$$\text{BF}_{10} = \frac{P(D|H_1)}{P(D|H_0)}$$

It expresses how much the data shifts the odds between the two models, independent of prior beliefs.

Thus we have the fundamental equation of Bayesian model comparison:

$$R = \text{BF}_{10} \times \frac{P(H_1)}{P(H_0)}$$

Note. Under uniform priors—when $P(H_1) = P(H_0) = 1/2$ —the posterior odds equal the Bayes factor. This is why some practitioners use $\text{BF} > 1$ as a criterion for favoring H_1 “objectively,” removing apparent subjectivity of the prior. However, this merely replaces one prior assumption (about models) with another (equal prior probability).

Example 5.2 (Bayes factor — definition and role (exam-style))

Consider two models H_0 and H_1 . Define the Bayes factor in favour of H_1 against H_0 , both for parameter-free models and for parametric models. Show how it relates prior odds and posterior odds, and explain the role of the marginal likelihood.

Solution. The Bayes factor is

$$\text{BF}_{10} = \frac{p(D | H_1)}{p(D | H_0)}.$$

By Bayes’ theorem, the posterior odds satisfy

$$\frac{p(H_1 | D)}{p(H_0 | D)} = \frac{p(D | H_1) p(H_1)}{p(D | H_0) p(H_0)} = \text{BF}_{10} \frac{p(H_1)}{p(H_0)}.$$

The Bayes factor is therefore the multiplicative factor by which the data update prior odds into posterior odds. If $\text{BF}_{10} > 1$, the data favour H_1 over H_0 relative to the prior odds; if $\text{BF}_{10} < 1$, they favour H_0 .

For parametric models H_j with parameters θ_j and prior $p(\theta_j | H_j)$,

$$p(D | H_j) = \int p(D | \theta_j, H_j) p(\theta_j | H_j) d\theta_j,$$

so the evidence averages the likelihood over the prior. This integral automatically embeds an Occam penalty: models that spread prior mass over regions not supported by the data are penalised.

Two quick consequences round off the basic machinery before we move to the parametric case.

- **Posterior probabilities from the odds.** Once R is known, the model probabilities are recovered from $P(H_0 | D) = 1/(1 + R)$ and $P(H_1 | D) = R/(1 + R)$.
- **Jeffreys' interpretation scale.** Jeffreys (1961) proposed a rough scale for the magnitude of a Bayes factor: BF_{10} in $(1, 3]$ is weak evidence for H_1 , $(3, 10]$ moderate, $(10, 30]$ strong, $(30, 100]$ very strong, and above 100 decisive (reverse the ratios for evidence against H_1). The cutoffs are conventional and context-dependent; report the magnitude alongside its uncertainty and the cost of being wrong rather than treating any threshold as binding.

Note. Jeffreys' scale is a communication device, not a decision rule. Combine it with a sensitivity analysis (see below) before reporting "strong" or "decisive" evidence.

5.3. The marginal likelihood and Occam's razor

The Bayes factor depends on the marginal likelihood (also called the model evidence), the probability of the data averaged over all possible parameter values under that model:

$$P(D|H_i) = \int P(D|\theta, H_i) \pi(\theta|H_i) d\theta$$

Here $\pi(\theta|H_i)$ is the prior distribution over parameters under hypothesis H_i , and $P(D|\theta, H_i)$ is the likelihood.

Definition 5.3 (Occam's razor in Bayesian inference). Marginal likelihoods automatically encode a penalty for model complexity. A complex model with many parameters typically spreads its prior mass over a large region of parameter space. Only a small region of this parameter space will fit the data well. When we integrate over all parameters, the contribution from the region that fits data is small relative to the total prior volume, yielding a low marginal likelihood—even if the model achieves a higher best-fit likelihood than a simpler competitor.

To see this concretely, consider two models: - H_0 : a simple model with a fixed parameter value θ_0 . - H_1 : a complex model with parameter θ uniformly distributed over the range $[a, b]$.

If both models fit the data equally well at their best-fit parameter values, then $P(D|\theta_0, H_0) \approx P(D|\hat{\theta}, H_1)$ where $\hat{\theta}$ is the MLE under H_1 .

But:

$$P(D|H_0) = P(D|\theta_0, H_0)$$

$$P(D|H_1) \approx \frac{1}{b-a} \int_a^b P(D|\theta, H_1) d\theta \approx P(D|\hat{\theta}, H_1) \cdot \frac{\Delta\theta_{\text{eff}}}{b-a}$$

where $\Delta\theta_{\text{eff}}$ is the effective width of the likelihood peak. If $\Delta\theta_{\text{eff}} \ll b-a$, then $P(D|H_1) \ll P(D|H_0)$ despite H_1 fitting the data equally well locally. The Bayes factor penalizes H_1 for having a large prior parameter space most of which is inconsistent with the data.

This intuition becomes quantitative through Laplace's approximation, which isolates the penalty as a distinct *Occam factor*.

- **Laplace approximation of the evidence.** For continuous parameters, assume $P(D|\theta, H)$ is sharply peaked around its maximum $\hat{\theta}$ and the prior is smooth on the scale of that peak. Then $P(D|H) \approx P(D|\hat{\theta}, H) \cdot \pi(\hat{\theta}) \cdot \Delta\theta_{\text{eff}}$, factoring the evidence into a best-fit likelihood and a prior-volume term.

- **Occam factor.** The combination $\text{Occam} = \pi(\hat{\theta}) \cdot \Delta\theta_{\text{eff}}$ is the prior probability of the region of parameter space that actually fits the data. For a d -dimensional model with roughly uniform prior of range L_i in each direction, $\text{Occam} \approx \prod_i (\Delta\theta_{\text{eff},i}/L_i)$: narrow likelihood peaks inside wide priors are heavily penalised.
- **Decomposed Bayes factor.** Combining the two ingredients gives

$$\text{BF}_{10} = \frac{P(D|\hat{\theta}_1, H_1)}{P(D|\hat{\theta}_0, H_0)} \cdot \frac{\text{Occam}_1}{\text{Occam}_0}.$$

The first ratio favours the more flexible model; the second penalises it for complexity. The balance — arising purely from integration, not from any ad-hoc penalty — determines the winner.

5.4. Example: Testing coin fairness

This classic example from Sivia (2006) illustrates the full machinery. Suppose we flip a coin $n = 250$ times and observe $h = 140$ heads. We compare:

- H_0 : The coin is fair ($\theta = 0.5$).
- H_1 : The coin has unknown bias $\theta \sim \text{Beta}(1, 1)$ (uniform on $[0, 1]$).

The likelihood for h heads in n flips is:

$$P(D|\theta) = \binom{n}{h} \theta^h (1 - \theta)^{n-h}$$

Under H_0 :

$$P(D|H_0) = \binom{n}{h} (0.5)^n$$

Under H_1 , with the $\text{Beta}(1,1)$ prior:

$$P(D|H_1) = \int_0^1 \binom{n}{h} \theta^h (1 - \theta)^{n-h} d\theta = \binom{n}{h} \cdot \frac{h!(n-h)!}{(n+1)!}$$

Therefore:

$$\text{BF}_{10} = \frac{P(D|H_1)}{P(D|H_0)} = \frac{h!(n-h)!}{(n+1)!} \cdot 2^n$$

With $n = 250$ and $h = 140$:

$$\text{BF}_{10} = \frac{140! \cdot 110!}{251!} \cdot 2^{250} \approx 0.044$$

This strong evidence against the coin being fair makes intuitive sense: observing 140 heads in 250 flips is notably different from 50-50 randomness.

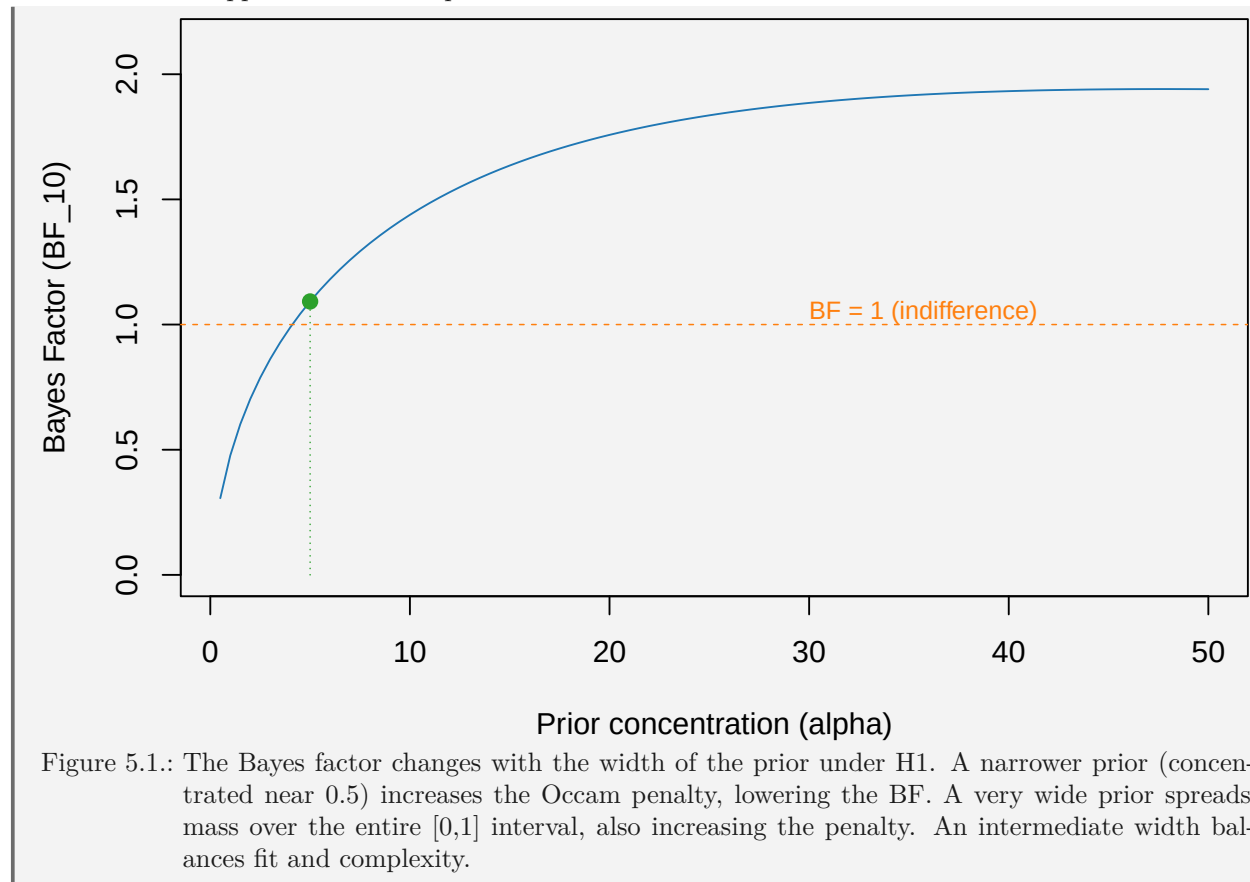
Example 5.3 (*Sensitivity to prior width*)

Now suppose we use a wider prior for H_1 : $\theta \sim \text{Beta}(\alpha, \alpha)$ for some $\alpha > 1$. This concentrates the prior more around $\theta = 0.5$. The Occam factor shrinks, raising the Bayes factor. Conversely, using $\alpha < 1$ (a U-shaped prior favoring extreme values) lowers it.

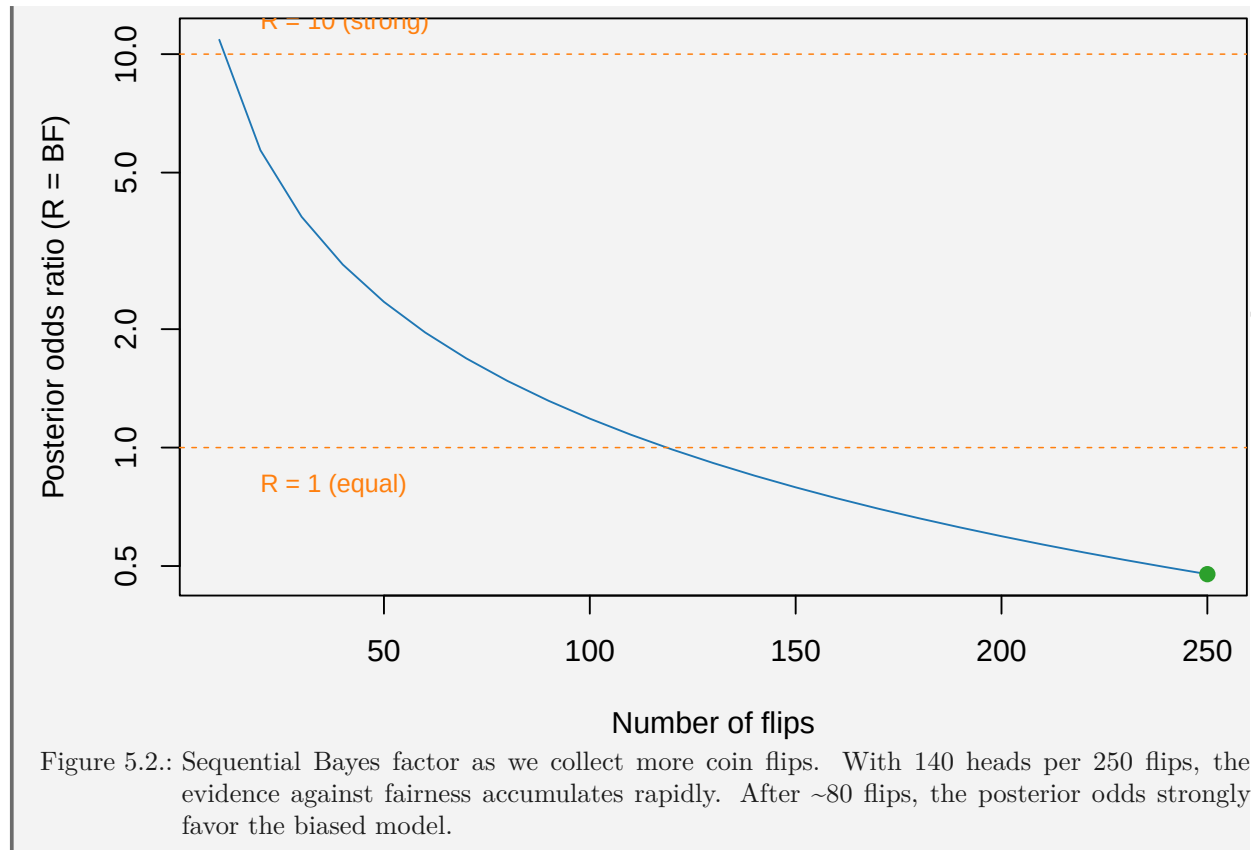
This illustrates a key insight: the Bayes factor is not prior-independent. However, it responds *sensibly*: wider priors (more agnostic) increase the Occam penalty, favoring simpler models; narrower priors (more peaked) decrease the penalty. A practitioner should explore this sensitivity and report it alongside point estimates.

The next three figures unpack the coin example numerically: how the Bayes factor depends on the prior concentration α , how the posterior odds grow as data accumulates, and a stylised picture of how the Occam factor penalises wide priors.

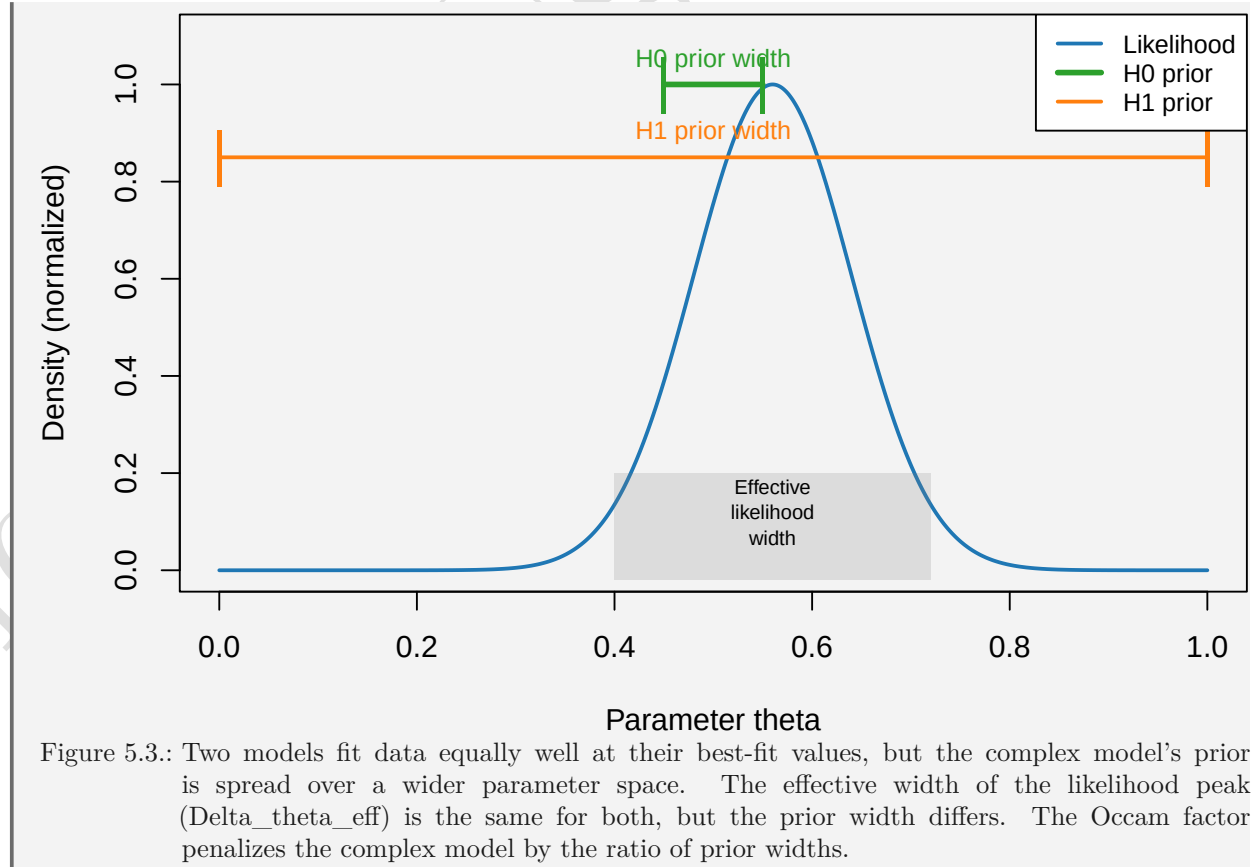
See code 15.8 in Appendix for the implementation.



See code 15.9 in Appendix for the implementation.



See code 15.10 in Appendix for the implementation.



5.5. Lindley's paradox

A famous puzzle arises when comparing a point null hypothesis (e.g., $\theta = 0$) to a broad alternative. As sample size increases, frequentist p-values eventually fall below any threshold (e.g., 0.05), calling for rejection of the null. Yet the Bayes factor may simultaneously favor the null, if the alternative prior is very diffuse.

Example 5.4 (Lindley's paradox)

Suppose we test $H_0 : \mu = 0$ versus $H_1 : \mu \sim N(0, \sigma_\mu^2)$ on data $x_1, \dots, x_n \sim N(\mu, 1)$. With a very large sample and $\bar{x}$ only slightly away from zero, a frequentist test will reject H_0 (small p-value), but if σ_μ is very large (vague prior on μ), the Occam penalty under H_1 keeps the Bayes factor favoring H_0 . The resolution is that the “paradox” reflects a real difference in how the two frameworks treat vague alternatives. Frequentist tests focus on the likelihood under H_0 . Bayesian tests also account for where the alternative prior was placed. This is not an error but a feature: it discourages chasing phantom effects in vague parameter spaces.

Example 5.5 (Closed-form Bayes factor for a Gaussian mean (exam-style))

Let $D = \{x_1, \dots, x_n\}$ be i.i.d. observations from $\mathcal{N}(\mu, \sigma^2)$ with known variance σ^2 . Compare

$$H_0 : \mu = 0, \quad H_1 : \mu \sim \mathcal{N}(0, \tau^2).$$

Compute BF_{10} in terms of the sample mean $\bar{x}$.

Solution. The sample mean $\bar{x}$ is a sufficient statistic for μ with $\bar{x} \mid \mu \sim \mathcal{N}(\mu, \sigma^2/n)$. Under H_0 ,

$$\bar{x} \mid H_0 \sim \mathcal{N}\left(0, \frac{\sigma^2}{n}\right).$$

Under H_1 , marginalising over the Gaussian prior on μ gives

$$\bar{x} \mid H_1 \sim \mathcal{N}\left(0, \tau^2 + \frac{\sigma^2}{n}\right).$$

Therefore

$$\text{BF}_{10} = \frac{p(\bar{x} \mid H_1)}{p(\bar{x} \mid H_0)} = \sqrt{\frac{\sigma^2/n}{\tau^2 + \sigma^2/n}} \exp\left[\frac{\bar{x}^2}{2} \left(\frac{n}{\sigma^2} - \frac{1}{\tau^2 + \sigma^2/n}\right)\right].$$

Large $|\bar{x}|$ favours the model with a free mean, while the prefactor $\sqrt{(\sigma^2/n)/(\tau^2 + \sigma^2/n)} \ll 1$ penalises the broader prior-predictive distribution under H_1 — the Occam factor. Letting $\tau \rightarrow \infty$ at fixed $\bar{x}$ drives the Bayes factor to favour H_0 : this is Lindley's paradox.

5.6. Sensitivity analysis and recommendations

Because the Bayes factor depends on prior specification under each model, responsible practice demands a sensitivity analysis:

1. **Vary the prior width** under the complex model within a reasonable range.
2. **Plot or tabulate** the Bayes factor as a function of prior width.
3. **Report the range** of Bayes factors across plausible priors.
4. **Use multiple Bayes factor values** (e.g., the median and bounds) rather than a single point estimate.

If the Bayes factor remains in the same evidential category across a wide range of prior specifications, the conclusion is robust. If it changes materially, the inference is prior-sensitive and deserves careful discussion.

Note. Some software implementations use automatic prior calibration rules (e.g., based on effect size or information content in the data) to reduce arbitrariness. While convenient, these methods still embed assumptions. Transparency about prior choice is always preferable to the illusion of objectivity.

5.7. Brief comparison with frequentist hypothesis testing

The frequentist approach to hypothesis testing is conceptually different. It treats parameters as fixed but unknown and focuses on controlling error rates across repeated hypothetical samples:

Definition 5.4 (Frequentist hypothesis test). A test statistic $T(D)$ is compared to a threshold derived from the null distribution $P(T|H_0)$. The p-value is $P(T(D) \geq t_{\text{obs}}|H_0)$, the probability of observing data as or more extreme than observed, if H_0 were true. A p-value below α (e.g., 0.05) triggers rejection of H_0 .

Key differences:

Feature	Bayesian	Frequentist
Assigns probability to hypotheses	Yes	No; tests hypothesis rejection rates
Uses prior information	Yes; explicitly	No; aims for objectivity through error control
Penalizes complexity	Automatically via Occam factor	Only through significance level (weak link)
Interpretation of p-value	N/A	Probability of data under H_0 , not probability H_0 is true
Sample size effects	Accounted for naturally	Strong: small effects are rejected in large samples, regardless of effect size
Binary decision	Unnecessary; provides probabilities	Often yes/no; less flexible

Both frameworks are useful and address different questions. Frequentist tests excel at controlling false-positive rates in exploratory settings. Bayesian methods excel when you want to rank competing theories and make decisions balancing evidence and consequences. Modern practice often combines both: use frequentist tests for initial screening, then Bayesian analysis for focused inference.

5.8. Computing the marginal likelihood

Having seen *what* the marginal likelihood does for model comparison, we turn to *how* to compute it. The integral $p(D | M) = \int p(D | \theta, M) p(\theta | M) d\theta$ is the primary computational bottleneck in Bayesian model comparison. For simple conjugate models — where the prior and likelihood are conjugate families — analytical solutions exist (the coin example above being one). In all other settings we must resort to numerical approximation. The four families of methods below cover the practical landscape; each is presented with its idea, strengths, and pitfalls.

- **Laplace approximation.** The cheapest analytic estimator: assume the posterior is sharply peaked around its mode and expand the log-posterior to second order around the MAP estimate $\hat{\theta}$. *Strengths:*

requires only the MAP and Hessian; often accurate when the posterior is well-approximated by a high-dimensional Gaussian. *Weaknesses:* poor for multimodal or heavy-tailed posteriors and degrades in high dimensions.

Definition 5.5 (Laplace approximation for marginal likelihood).

$$p(D | M) \approx p(D | \hat{\theta}, M) p(\hat{\theta} | M) (2\pi)^{d/2} |\mathcal{H}|^{-1/2}$$

where d is the dimension of the parameter space and $\mathcal{H}$ is the Hessian of the negative log-posterior evaluated at $\hat{\theta}$.

Note. For many applied problems, the Laplace approximation is a good starting point. Use it as a baseline and compare with more robust methods if possible.

- **Harmonic mean estimator.** If posterior samples $\theta^{(s)}$, $s = 1, \dots, S$ are already available from MCMC, one can write

$$\hat{Z}_{\text{HM}} = \left[\frac{1}{S} \sum_{s=1}^S \frac{1}{p(D | \theta^{(s)}, M)} \right]^{-1}.$$

This arises as an importance-sampling estimator with the prior as proposal: posterior samples are reweighted by the reciprocal of the likelihood, inverting the usual likelihood weighting. Tempting in its simplicity, it is unreliable for the reasons made precise below.

Remark. Pitfall: catastrophic instability. The harmonic mean estimator has infinite variance whenever the prior has lighter tails than the posterior, which occurs in nearly every realistic setting. A single posterior sample in a region of low prior density can dominate the sum, causing enormous variance and unreliable estimates. Newton & Raftery (1994) and Wolpert & Schmidler (2012) have thoroughly documented this pathology. **We strongly discourage the use of the harmonic mean estimator in production analyses.** Its tempting simplicity belies a fundamental flaw.

- **Bridge sampling.** Meng & Wong (1996) corrected the importance-sampling bias of the harmonic mean using the identity

$$\frac{Z_1}{Z_2} = \frac{\mathbb{E}_{q_2}[\alpha(\theta) q_1(\theta)]}{\mathbb{E}_{q_1}[\alpha(\theta) q_2(\theta)]},$$

for any two densities q_1, q_2 with tractable normalising constants Z_1, Z_2 and any “bridge” function α . The optimal (minimum-variance) α has the geometric-mean form; setting q_1 to the posterior and q_2 to the prior yields a marginal-likelihood estimate. *Strengths:* much lower variance than the harmonic mean; scales to moderate-to-high dimensions; widely implemented (e.g., in the R package `bridgesampling` of Gronau et al. 2017). *Weaknesses:* requires posterior samples and careful selection of the bridge function and proposal.

Note. Bridge sampling is the recommended default when posterior samples are already in hand (e.g., from an MCMC run) and model comparison is a secondary goal. The `bridgesampling` R package provides a robust, production-ready implementation.

- **Nested sampling.** Skilling (2006) reformulated the evidence integral as a one-dimensional integration over enclosed prior mass. With $X(\lambda) = \int_{\{L(\theta) > \lambda\}} p(\theta | M) d\theta$,

$$Z = \int p(D | \theta, M) p(\theta | M) d\theta = \int_0^1 L(X) dX,$$

5. Model comparison and Bayes factors

where $L(X)$ is the likelihood as a function of enclosed prior mass X . The algorithm maintains a set of K “live points” drawn from the prior; at each iteration the lowest-likelihood point is removed (becoming a “dead point”) and replaced by a new draw from the prior constrained to higher likelihood. The evidence is accumulated as $Z \approx \sum_i L_i \Delta X_i$, where L_i and ΔX_i are the likelihoods and prior-mass increments of the dead points. *Strengths*: handles multimodal posteriors naturally; delivers both Z and (importance-reweighted) posterior samples; popular in astrophysics and cosmology. *Weaknesses*: more expensive than Laplace; requires tuning of live-point count and the constrained sampler.

Note. Nested sampling is implemented in **MultiNest** (Feroz et al.), **dynesty** (Speagle), and **PolyChord** (Handley et al.). Choose nested sampling when the posterior is complex, multimodal, or when you need both evidence and posterior samples with high confidence.

- **Thermodynamic integration and annealed importance sampling.** A fourth family interpolates between prior and posterior through the tempered family

$$\pi_\beta(\theta) \propto p(D | \theta, M)^\beta p(\theta | M), \quad \beta \in [0, 1],$$

with $\beta = 0$ the prior and $\beta = 1$ the posterior. The marginal likelihood then satisfies

$$\log Z = \int_0^1 \mathbb{E}_{\pi_\beta}[\log p(D | \theta, M)] d\beta,$$

the foundation of thermodynamic integration (Gelman & Meng 1998) and annealed importance sampling (Neal 2001). *Strengths*: robust in high dimensions and on multimodal problems; sits naturally inside parallel-tempering MCMC (Chapter 10). *Weaknesses*: requires careful choice of temperature schedule and numerical integration scheme.

The trade-offs are summarised in the table below.

Method	When to use	Strengths	Weaknesses	Cost
Laplace	Quick baseline with smooth, unimodal posterior	Fast; only MAP and Hessian needed	Poor for multimodal or heavy-tailed posteriors	Very low
Harmonic mean	<i>Avoid</i>	Simplicity; uses existing MCMC samples	Catastrophic instability; infinite variance	Very low (but unreliable)
Bridge sampling	MCMC samples in hand; moderate-to-high dimensions	Low variance; theoretically principled; well-implemented	Requires tuning; sensitive to proposal	Low-to-moderate
Nested sampling	Multimodal or complex posteriors; want both Z and posterior	Handles multimodality naturally; provides full uncertainty; robust	Computationally expensive; requires tuning	Moderate-to-high
Thermodynamic integration	High-dimensional problems; parallel tempering available	Robust; scalable to high dimensions	Needs temperature schedule; numerical integration error	Moderate-to-high

Practical recommendations. For rapid Bayes-factor assessment with a well-conditioned, unimodal posterior, compute the Laplace approximation as a baseline and supplement it with uncertainty bounds. If posterior samples from MCMC are in hand and a robust point estimate is needed, **use bridge sampling** — the practical sweet spot of accuracy, interpretability, and implementation maturity. For complex, multimodal posteriors or when maximum confidence in the evidence is essential, **employ nested sampling** despite its higher cost. **Never rely on the harmonic mean estimator** in production. In all cases, report not just a point estimate but also an uncertainty band, and pair it with the sensitivity analysis introduced above whenever feasible.

5.9. Exercises

Exercise 5.1

(Written test, 15 June 2026 — Q1.) A detector can operate in two mutually exclusive regimes, H_0 and H_1 . Before seeing the data, the calibration team assigns $P(H_1) = 0.35$. For the observed data D , the likelihood ratio is

$$\frac{P(D | H_1)}{P(D | H_0)} = 4.$$

Compute the posterior probability $P(H_1 | D)$. Is the Bayes factor alone sufficient to determine this posterior probability? Briefly explain.

Solution

The prior odds are $\pi(H_1)/\pi(H_0) = 0.35/0.65 = 7/13$, so the posterior odds are

$$\frac{P(H_1 | D)}{P(H_0 | D)} = \text{BF}_{10} \frac{\pi(H_1)}{\pi(H_0)} = 4 \cdot \frac{7}{13} = \frac{28}{13}.$$

Therefore $P(H_1 | D) = (28/13)/(1 + 28/13) = 28/41 \approx 0.683$. The Bayes factor alone is not sufficient: it only quantifies how the data update the odds, so one also needs the prior odds (here $\pi(H_1) = 0.35$) to recover a posterior probability.

Exercise 5.2

Two mutually exclusive hypotheses H_0, H_1 are assigned equal prior probability. The Bayes factor in favor of H_1 is $\text{BF}_{10} = 9$. Compute the posterior odds and $P(H_1 | D)$.

Solution

With equal priors, $\pi(H_1)/\pi(H_0) = 1$, so the posterior odds equal the Bayes factor:

$$\frac{P(H_1 | D)}{P(H_0 | D)} = \text{BF}_{10} = 9.$$

Hence $P(H_1 | D) = 9/(1 + 9) = 9/10 = 0.9$.

Exercise 5.3

Three mutually exclusive models M_1, M_2, M_3 have prior probabilities $(0.5, 0.3, 0.2)$. The likelihoods of the observed data are in the ratio $P(D | M_1) : P(D | M_2) : P(D | M_3) = 2 : 1 : 4$. Compute the posterior probability of each model.

5. Model comparison and Bayes factors

Solution

The unnormalised posterior weights are $\pi(M_i) P(D | M_i)$:

$$(0.5)(2) : (0.3)(1) : (0.2)(4) = 1.0 : 0.3 : 0.8.$$

Normalising by $1.0 + 0.3 + 0.8 = 2.1$ gives

$$P(M_1 | D) = \frac{10}{21}, \quad P(M_2 | D) = \frac{3}{21} = \frac{1}{7}, \quad P(M_3 | D) = \frac{8}{21}.$$

Exercise 5.4

A coin is flipped $n = 5$ times and four heads are observed. Compare two simple hypotheses with equal prior probability: H_0 : the coin is fair ($p = 1/2$); H_1 : the coin has $p = 3/4$. Compute BF_{10} exactly (no calculator — leave powers of 2, 3, 4 unsimplified) and give the posterior odds.

Solution

The binomial coefficient $\binom{5}{4}$ cancels in the ratio, leaving

$$\text{BF}_{10} = \frac{(3/4)^4 (1/4)^1}{(1/2)^5} = \frac{3^4/4^5}{1/2^5} = \frac{3^4}{4^5} \cdot 2^5 = \frac{3^4}{2^5} = \frac{81}{32}.$$

With equal priors the posterior odds equal $\text{BF}_{10} = 81/32 \approx 2.53$, so $P(H_1 | D) = 81/113$.

Exercise 5.5

Two Poisson models are compared on a single count $k = 2$: M_0 with rate $\lambda_0 = 1$, M_1 with rate $\lambda_1 = 3$. Write BF_{10} in closed form using the Poisson PMF. Express it in terms of e (do not evaluate numerically).

Solution

The Poisson PMF is $P(k | \lambda) = e^{-\lambda} \lambda^k / k!$. The factorial $k! = 2!$ cancels:

$$\text{BF}_{10} = \frac{e^{-3} 3^2 / 2!}{e^{-1} 1^2 / 2!} = 9 e^{-2}.$$

Exercise 5.6

A Gaussian observation x has known variance σ^2 . Compare M_0 : $\mu = 0$ versus M_1 : $\mu = \mu_1$ (a fixed non-zero value). Derive

$$\text{BF}_{10}(x) = \exp\left(\frac{2\mu_1 x - \mu_1^2}{2\sigma^2}\right).$$

For which x does the data favor M_1 ?

Solution

The Gaussian densities at x give

$$\text{BF}_{10}(x) = \frac{\exp[-(x - \mu_1)^2 / (2\sigma^2)]}{\exp[-x^2 / (2\sigma^2)]} = \exp\left(\frac{2\mu_1 x - \mu_1^2}{2\sigma^2}\right).$$

The data favour M_1 (i.e. $\text{BF}_{10} > 1$) iff $2\mu_1 x - \mu_1^2 > 0$, i.e. $x > \mu_1/2$ when $\mu_1 > 0$ (and $x < \mu_1/2$ when

$\mu_1 < 0$): the cross-over is exactly the midpoint between the two competing means.

Exercise 5.7

The prior odds against H_1 are 1 : 9. What Bayes factor BF_{10} is required to push the posterior probability $P(H_1 | D)$ above (a) 1/2 and (b) 0.95?

Solution

Posterior odds $R = P/(1 - P)$ are obtained from prior odds 1/9 via $R = \text{BF}_{10} \cdot (1/9)$, so $\text{BF}_{10} = 9R$. (a) $P = 1/2 \Rightarrow R = 1 \Rightarrow \text{BF}_{10} = 9$. (b) $P = 0.95 \Rightarrow R = 0.95/0.05 = 19 \Rightarrow \text{BF}_{10} = 171$.

Exercise 5.8

Repeat the calculation of Exercise 1 with the prior changed first to $P(H_1) = 0.10$ and then to $P(H_1) = 0.70$, keeping the same likelihood ratio. Comment on the sensitivity of the conclusion to the prior.

Solution

With $\text{BF}_{10} = 4$ the posterior odds are $R = 4\pi(H_1)/(1 - \pi(H_1))$ and $P(H_1 | D) = R/(1 + R)$. For $\pi(H_1) = 0.10$: $R = 4 \cdot (1/9) = 4/9$, so $P(H_1 | D) = 4/13 \approx 0.308$. For $\pi(H_1) = 0.70$: $R = 4 \cdot (7/3) = 28/3$, so $P(H_1 | D) = 28/31 \approx 0.903$. The qualitative conclusion flips: a moderate likelihood ratio of 4 is not enough to overturn a sceptical prior, but easily reinforces an already-favourable one. The Bayes factor must always be reported with the prior used.

Exercise 5.9

Show that when the two hypotheses are assigned equal prior probability, the posterior odds equal the Bayes factor. State the converse and discuss its practical meaning.

Solution

By definition,

$$\frac{P(H_1 | D)}{P(H_0 | D)} = \text{BF}_{10} \frac{\pi(H_1)}{\pi(H_0)}.$$

If $\pi(H_1) = \pi(H_0) = 1/2$ the prior-odds factor is 1, so the posterior odds equal BF_{10} . Converse: if the posterior odds equal the Bayes factor, then $\pi(H_1) = \pi(H_0)$. Practically, quoting BF_{10} as if it were a probability tacitly assumes a 50/50 prior — perfectly fine if disclosed, misleading if not.

Exercise 5.10

A binary outcome $X | \theta \sim \text{Bernoulli}(\theta)$ is observed n times with s successes. Under the prior $\theta \sim \text{Uniform}(0, 1)$, show that the marginal likelihood is

$$p(\text{data}) = \frac{s!(n-s)!}{(n+1)!}.$$

5. Model comparison and Bayes factors

Solution

The likelihood for an ordered sequence with s successes is $\theta^s(1-\theta)^{n-s}$. With $\pi(\theta) = 1$ on $[0, 1]$,

$$p(\text{data}) = \int_0^1 \theta^s(1-\theta)^{n-s} d\theta = B(s+1, n-s+1) = \frac{s!(n-s)!}{(n+1)!},$$

using $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ and $\Gamma(k+1) = k!$.

Exercise 5.11

Same likelihood as in the previous exercise but with prior $\theta \sim \text{Beta}(\alpha, \beta)$. Express the marginal likelihood in terms of the Beta function $B(\cdot, \cdot)$. Use it to compute the Bayes factor between $\text{Beta}(\alpha_1, \beta_1)$ and $\text{Beta}(\alpha_0, \beta_0)$ priors.

Solution

With $\pi(\theta) = \theta^{\alpha-1}(1-\theta)^{\beta-1}/B(\alpha, \beta)$, the marginal likelihood is

$$p(\text{data} \mid \alpha, \beta) = \frac{1}{B(\alpha, \beta)} \int_0^1 \theta^{s+\alpha-1}(1-\theta)^{n-s+\beta-1} d\theta = \frac{B(s+\alpha, n-s+\beta)}{B(\alpha, \beta)}.$$

The Bayes factor between two Beta priors is therefore

$$\text{BF}_{10} = \frac{B(s+\alpha_1, n-s+\beta_1)/B(\alpha_1, \beta_1)}{B(s+\alpha_0, n-s+\beta_0)/B(\alpha_0, \beta_0)}.$$

Exercise 5.12

A Gaussian observation $x \mid \theta \sim \mathcal{N}(\theta, 1)$ is compared under two models: $M_0: \theta = 0$; $M_1: \theta \sim \mathcal{N}(0, \tau^2)$ with prior width τ . Derive BF_{10} as a function of x and τ . What happens to BF_{10} as $\tau \rightarrow \infty$ with x fixed? Comment on Lindley's paradox.

Solution

Marginalising the Gaussian likelihood $x \mid \theta \sim \mathcal{N}(\theta, 1)$ over the $\mathcal{N}(0, \tau^2)$ prior on θ gives $x \mid M_1 \sim \mathcal{N}(0, 1+\tau^2)$, while $x \mid M_0 \sim \mathcal{N}(0, 1)$. Hence

$$\text{BF}_{10}(x, \tau) = \frac{1}{\sqrt{1+\tau^2}} \exp\left[\frac{x^2}{2} \cdot \frac{\tau^2}{1+\tau^2}\right].$$

As $\tau \rightarrow \infty$ with x fixed, the exponential tends to $e^{x^2/2}$ (finite) while the prefactor $1/\sqrt{1+\tau^2} \rightarrow 0$, so $\text{BF}_{10} \rightarrow 0$ and M_0 always wins. This is Lindley's paradox: an arbitrarily diffuse alternative is penalised by an unbounded Occam factor, so a single observation can never overturn the null no matter how far from zero.

Exercise 5.13

Classify the strength of evidence on the (Jeffreys / Kass–Raftery) scale for each of the Bayes factors $\text{BF}_{10} \in \{1, 3, 12, 150\}$. State the cut-offs you use.

Solution

Using Jeffreys' cut-offs (1, 3] weak, (3, 10] moderate (substantial), (10, 30] strong, (30, 100] very strong, > 100 decisive: $\text{BF}_{10} = 1$: no evidence (indifference between H_0 and H_1). $\text{BF}_{10} = 3$: at the boundary of weak/moderate — “bare mention” or just-substantial evidence for H_1 . $\text{BF}_{10} = 12$: strong evidence for H_1 . $\text{BF}_{10} = 150$: decisive evidence for H_1 .

Exercise 5.14

Two models are retained after data analysis with posterior probabilities $P(M_1 | D) = 0.4$, $P(M_2 | D) = 0.6$. Their posterior-predictive densities for a future observation y are $\mathcal{N}(0, 1)$ and $\mathcal{N}(2, 1)$ respectively. Write the model-averaged predictive density of y and compute its mean and variance.

Solution

The model-averaged predictive is a mixture:

$$p(y | D) = 0.4 \phi(y; 0, 1) + 0.6 \phi(y; 2, 1).$$

Mean: $\mathbb{E}[y] = 0.4 \cdot 0 + 0.6 \cdot 2 = 1.2$. Variance (law of total variance): $\text{Var}(y) = \mathbb{E}[\text{Var}(y | M)] + \text{Var}(\mathbb{E}[y | M]) = 1 + (0.4 \cdot 0^2 + 0.6 \cdot 2^2 - 1.2^2) = 1 + (2.4 - 1.44) = 1.96$.

Exercise 5.15

For a Gaussian likelihood with known variance and a Gaussian prior of width $\sigma_p \gg \sigma_{\text{post}}$, show that the marginal likelihood factorises as

$$p(D) \approx p(D | \hat{\theta}_{\text{MAP}}) \cdot \frac{\sigma_{\text{post}}}{\sigma_p}.$$

Identify the Occam factor and explain why broader priors are automatically penalised.

Solution

Around the MAP the likelihood is approximately $p(D | \theta) \approx p(D | \hat{\theta}_{\text{MAP}}) \exp[-(\theta - \hat{\theta}_{\text{MAP}})^2 / (2\sigma_{\text{post}}^2)]$, while the prior on the scale of the peak is essentially flat: $\pi(\theta) \approx 1/(\sqrt{2\pi} \sigma_p)$. Marginalising,

$$p(D) \approx p(D | \hat{\theta}_{\text{MAP}}) \cdot \frac{1}{\sqrt{2\pi} \sigma_p} \cdot \sqrt{2\pi} \sigma_{\text{post}} = p(D | \hat{\theta}_{\text{MAP}}) \cdot \frac{\sigma_{\text{post}}}{\sigma_p}.$$

The Occam factor is $\sigma_{\text{post}}/\sigma_p < 1$: a broader prior σ_p spreads the prior over a region most of which is inconsistent with the data, multiplicatively shrinking the evidence and so penalising the model.

Exercise 5.16

(Define and explain.) Define the Bayes factor in favour of H_1 against H_0 , both for parameter-free models and for parametric models. Show how it relates prior odds and posterior odds, and explain the role of the marginal likelihood. (Brief; no calculation needed.)

Solution

The Bayes factor is the ratio of marginal likelihoods, $\text{BF}_{10} = p(D | H_1)/p(D | H_0)$. For parameter-free models this is just the ratio of two ordinary likelihoods; for parametric models H_j with parameters

$\theta_j,$

$$p(D | H_j) = \int p(D | \theta_j, H_j) \pi(\theta_j | H_j) d\theta_j,$$

the prior-averaged likelihood. It multiplies prior odds into posterior odds,

$$\frac{P(H_1 | D)}{P(H_0 | D)} = \text{BF}_{10} \frac{\pi(H_1)}{\pi(H_0)}.$$

The marginal likelihood quantifies how well a model predicted the data *before* fitting; the integral automatically penalises models whose prior wastes mass on regions inconsistent with the data (Occam's razor).

Exercise 5.17

(Compare and contrast.) State two practical differences between a Bayesian Bayes-factor analysis and a frequentist p -value. Pick a concrete example (e.g. testing $\mu = 0$ from n Gaussian samples) and describe the qualitative conclusion each framework draws when n is moderately large.

Solution

Two key differences: (i) the Bayes factor compares two models symmetrically and can support either H_0 or H_1 , whereas a p -value only quantifies tail evidence against H_0 ; (ii) the Bayes factor automatically embeds an Occam penalty against diffuse alternatives, while a p -value is independent of the alternative. For $\mu = 0$ vs. $\mu \sim \mathcal{N}(0, \tau^2)$ on n samples with sample mean $\bar{x}$, the p -value is $2(1 - \Phi(\sqrt{n}|\bar{x}|/\sigma))$ and shrinks like $1/\sqrt{n}$, so any non-zero $\bar{x}$ is eventually “significant”. The Bayes factor $\text{BF}_{10} \propto \sqrt{\sigma^2/(n\tau^2)} \exp(\dots)$ instead carries a $\sqrt{1/n}$ Occam prefactor, so for moderately large n and small $\bar{x}$ it may favour H_0 — Lindley's paradox.

Exercise 5.18

(Define and explain.) Define the Occam factor and explain why broader priors over a complex model are automatically penalised in the Bayes factor. Give a sketch using a one-dimensional parameter with width σ_p and posterior width σ_{post} .

Solution

The Occam factor is the multiplicative term that appears in the Laplace decomposition of the evidence,

$$p(D | H) \approx p(D | \hat{\theta}) \cdot \underbrace{\pi(\hat{\theta}) \cdot \sqrt{2\pi} \sigma_{\text{post}}}_{\text{Occam factor}} \approx p(D | \hat{\theta}) \cdot \frac{\sigma_{\text{post}}}{\sigma_p},$$

i.e. the fraction of the prior volume that the posterior actually occupies. Doubling the prior width σ_p halves the Occam factor and hence the marginal likelihood, even though the best-fit likelihood does not change. Broader priors waste mass on regions the data exclude, so the Bayes factor penalises them automatically — no ad-hoc complexity term needed.

Exercise 5.19

(Show that.) Show that the Bayes factor for two parametric models is invariant under bijective reparametrisation of either model's parameters. (Hint: the marginal likelihood is an integral against the prior, so any Jacobian cancels.)

Solution

Let $\phi = g(\theta)$ be a bijection with $\theta = g^{-1}(\phi)$ and Jacobian $|d\theta/d\phi|$. The transformed prior density is $\tilde{\pi}(\phi) = \pi(g^{-1}(\phi)) |d\theta/d\phi|$ and the likelihood becomes $\tilde{p}(D | \phi) = p(D | g^{-1}(\phi))$. Then

$$\int \tilde{p}(D | \phi) \tilde{\pi}(\phi) d\phi = \int p(D | g^{-1}(\phi)) \pi(g^{-1}(\phi)) \left| \frac{d\theta}{d\phi} \right| d\phi = \int p(D | \theta) \pi(\theta) d\theta,$$

by change of variables. So each model's marginal likelihood is unchanged under reparametrisation, and hence so is BF_{10} .

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6. Bayesian parameter estimation, intervals and prediction

6.1. Introduction

Parameter estimation lies at the heart of Bayesian inference. Once we observe data, we want to infer the unknown parameters that generated it. Unlike frequentist approaches that treat parameters as fixed but unknown, the Bayesian framework treats parameters as random variables described by probability distributions. This chapter synthesizes three core topics: how to estimate parameters using loss functions, how to quantify uncertainty via credible intervals, and how to make predictions about future observations.

The structure follows a natural progression: we begin with the Bayesian estimation problem and loss functions, explore how to characterize posterior shapes and construct intervals, then move to prediction and compare with frequentist confidence intervals.

6.2. The Bayesian estimation problem

Definition 6.1. Bayesian inference proceeds through Bayes' theorem:

$$p(\theta \mid D, H) = \frac{p(D \mid \theta, H) p(\theta \mid H)}{p(D \mid H)},$$

where - $p(D \mid \theta, H)$ is the **likelihood**: the probability of observing data D under parameter θ and model H , - $p(\theta \mid H)$ is the **prior**: our belief about θ before seeing data, - $p(\theta \mid D, H)$ is the **posterior**: our updated belief after data, - $p(D \mid H) = \int p(D \mid \theta, H) p(\theta \mid H) d\theta$ is the **evidence** (normalizing constant).

The posterior distribution contains all information about θ given the data. Once we have it, we face a practical question: should we quote a point estimate or a full distribution?

6.3. Point estimation under loss functions

The Bayesian answer to point estimation is principled: choose an estimate by **minimizing expected loss given the data**. Different loss functions lead to different estimators.

Definition 6.2. Given data D , the **Bayesian estimator** is

$$\hat{\theta}(D) = \arg \min_a E[L(a, \theta) \mid D],$$

where $L(a, \theta)$ is a loss function penalizing how far the estimate a is from the true parameter θ .

Three canonical loss functions cover most practical needs:

- **Squared loss and the posterior mean.** $L(a, \theta) = (a - \theta)^2$ yields $\hat{\theta}_{\text{LMS}} = E[\theta | D]$, the **posterior mean** (Least Mean Square estimator). Completing the square,

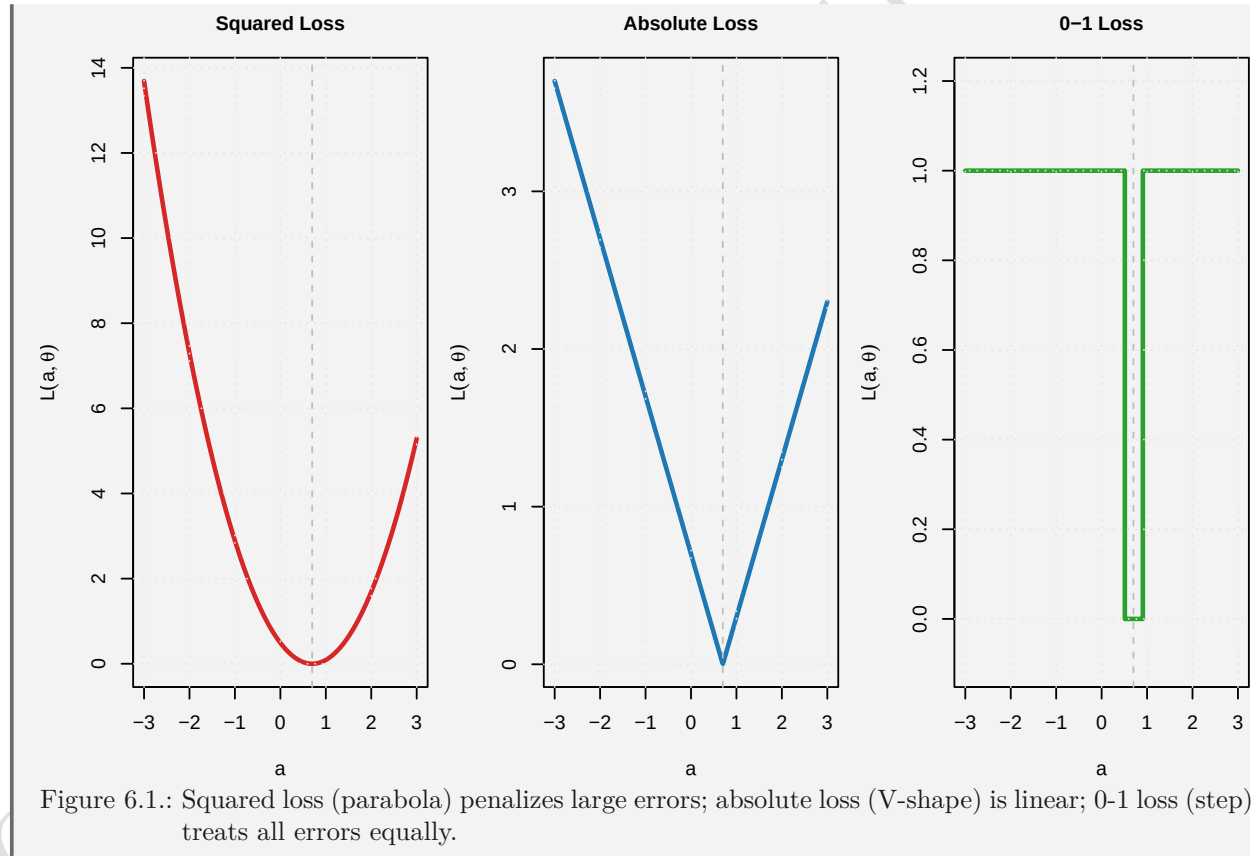
$$E[(a - \theta)^2 | D] = (a - E[\theta | D])^2 + \text{Var}(\theta | D).$$

The posterior mean uses the full posterior, is invariant under reparametrization, but can be pulled by heavy tails.

- **Absolute loss and the posterior median.** $L(a, \theta) = |a - \theta|$ yields $\hat{\theta}_{\text{med}} = \text{median}(\theta | D)$. The posterior median is more robust to outliers than the mean and divides the posterior into equal probability masses.
- **0-1 loss and the maximum a posteriori (MAP).** $L(a, \theta) = 1\{a \neq \theta\}$ yields $\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta | D)$, the **posterior mode** — computationally convenient but uses only local information around the peak. **Critically, MAP is not invariant under reparametrization:** if we change variables $\phi = g(\theta)$, the mode of $p(\phi | D)$ generally does not equal $g(\hat{\theta}_{\text{MAP}})$.

The three losses have distinct geometries: the parabola of squared loss penalises large errors most heavily, the V-shape of absolute loss is linear, and the step of 0-1 loss treats every nonzero error identically.

See code 15.11 in Appendix for the implementation.



6.4. Asymmetric posteriors: MAP vs mean vs median

When the posterior is sharply peaked and symmetric (e.g., Gaussian), the MAP, mean, and median coincide. However, realistic posteriors are often skewed or multimodal.

Example 6.1**Bernoulli parameter estimation with small samples**

Suppose we toss a coin $n = 10$ times and observe $k = 3$ heads. Using a uniform prior $\text{Beta}(1, 1)$, the posterior is $\text{Beta}(1 + 3, 1 + 7) = \text{Beta}(4, 8)$.

See code 15.12 in Appendix for the implementation.

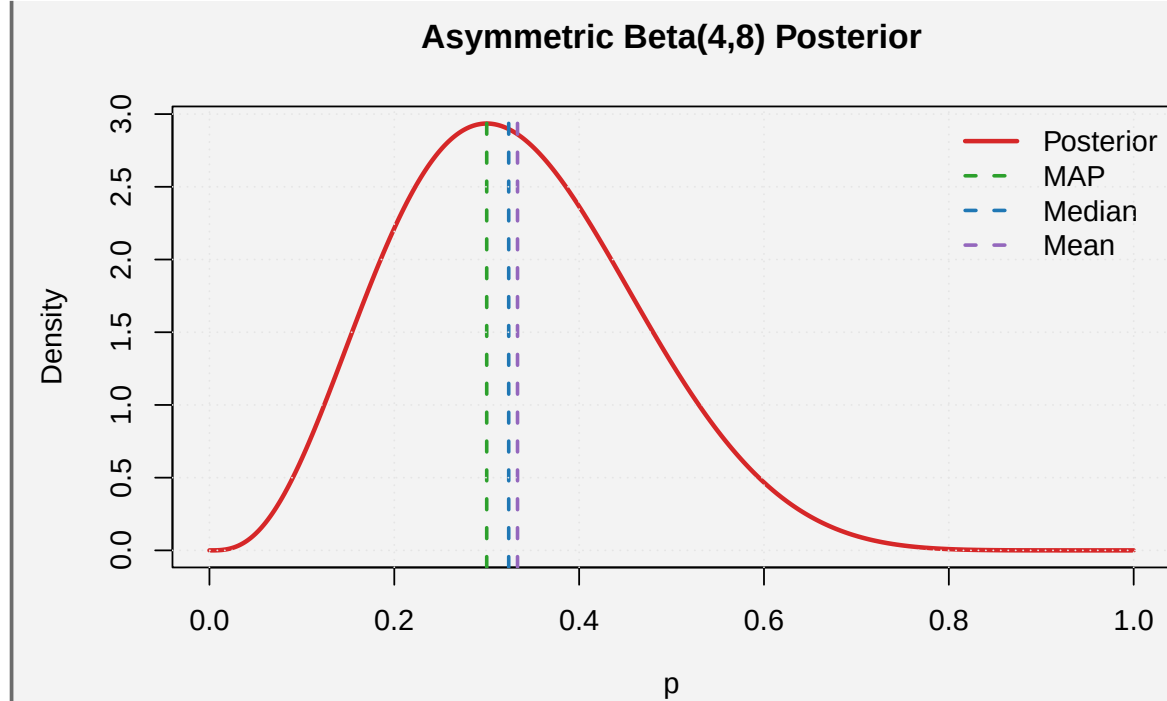


Figure 6.2.: Beta(4,8) posterior (red) is skewed right. MAP (green), median (blue), and mean (purple) differ substantially.

Here, $\hat{p}_{\text{MAP}} \approx 0.27$, median ≈ 0.31 , and $E[p \mid D] \approx 0.33$. The mean is pulled toward 0.5 (the neutral Beta(1,1) prior), while the MAP stays closer to the data.

Example 6.2 (MAP at the boundary — Beta(1/2,1/2) prior (exam-style))

A Bernoulli parameter has prior $p \sim \text{Beta}(1/2, 1/2)$. You observe $n = 1$ trial and it is a success. Compute the posterior mean and the MAP, if it exists. Why are they different?

Solution. Conjugate updating gives

$$p \mid D \sim \text{Beta}\left(\frac{1}{2} + 1, \frac{1}{2}\right) = \text{Beta}\left(\frac{3}{2}, \frac{1}{2}\right).$$

The posterior mean is the standard Beta mean,

$$E[p \mid D] = \frac{3/2}{3/2 + 1/2} = \frac{3}{4}.$$

The interior MAP formula $\hat{p}_{\text{MAP}} = (\alpha - 1)/(\alpha + \beta - 2)$ requires *both* shape parameters greater than 1. Here the second shape is $1/2 < 1$, so the posterior density $p^{1/2}(1-p)^{-1/2}$ diverges as $p \rightarrow 1$ and the MAP sits at the boundary $\hat{p}_{\text{MAP}} = 1$. The mean averages over the full posterior, while the MAP just tracks the local maximum — and in this case the maximum is a boundary singularity. This is a classic case where the MAP is a misleading point summary.

The reparametrization invariance problem. This is a critical issue for MAP estimation. Consider the logit transformation $\phi = \log \frac{p}{1-p}$, so $p = \frac{e^\phi}{1+e^\phi}$.

Note. When we change variables, the Jacobian enters the posterior density:

$$p(\phi \mid D) = p(p \mid D) \left| \frac{dp}{d\phi} \right| = p(p \mid D) \cdot p(1-p).$$

This Jacobian modifies the exponents in the likelihood, so $\arg \max_{\phi} p(\phi \mid D) \neq g(\arg \max_p p(p \mid D))$ unless the Jacobian is constant.

The posterior mean and median are invariant under reparametrization because they are defined via integration, not pointwise optimization. This makes them more principled choices.

6.5. Reliability: Gaussian approximation and Laplace's method

For large samples or sharply peaked posteriors, we can approximate the posterior as Gaussian using a Taylor expansion around the MAP.

Theorem 6.1. Laplace's method: *If the posterior is sharply unimodal around $\hat{\theta}_{MAP}$, then*

$$p(\theta \mid D, H) \approx N(\hat{\theta}_{MAP}, C),$$

where the covariance matrix is

$$C = H^{-1}, \quad H_{ij} = - \left. \frac{\partial^2 \ln p(\theta \mid D)}{\partial \theta_i \partial \theta_j} \right|_{\theta = \hat{\theta}_{MAP}}$$

is the negative Hessian of the log-posterior.

The uncertainty in each parameter is $\sigma_i = \sqrt{C_{ii}}$.

When is Laplace reliable? The Gaussian approximation works well when the posterior is unimodal and symmetric, the sample size is large (CLT effect), and parameters stay away from natural boundaries. It breaks down when the posterior is broad, skewed, or multimodal, when data are sparse (small n , far from the CLT regime), or when parameters are bounded (e.g., $p \in [0, 1]$, $\sigma^2 > 0$).

Example 6.3

Bernoulli reliability

For the Bernoulli likelihood with a Beta prior, the log-posterior is

$$L(p) = (a-1) \ln p + (b-1) \ln(1-p),$$

where $a = \alpha + k$ and $b = \beta + n - k$.

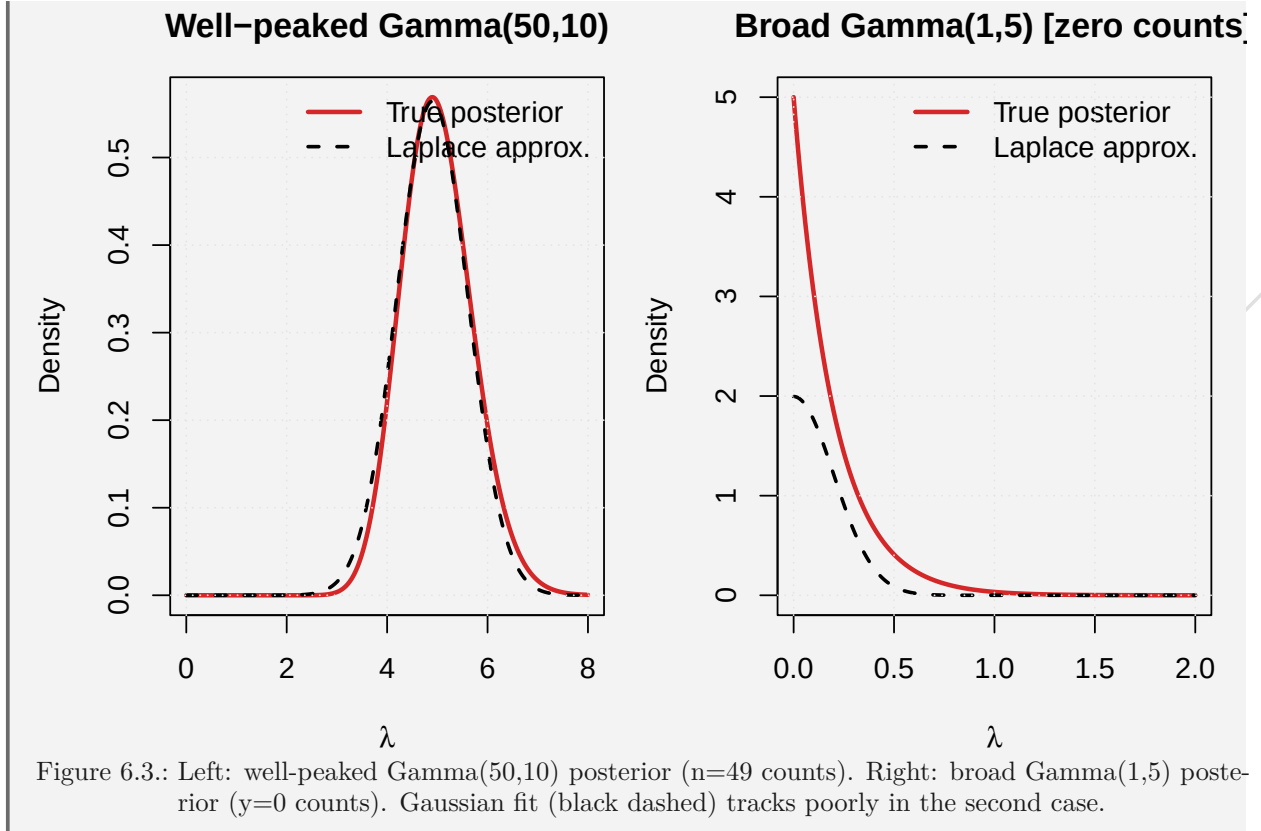
The second derivative at the MAP is

$$L''(\hat{p}_{MAP}) = - \frac{(a+b-2)^2}{(a-1)(b-1)}.$$

The uncertainty scales as $\sigma \sim (n + \alpha + \beta)^{-1/2}$. For small n or extreme counts, the quadratic approximation is poor because the Beta posterior has heavy tails.

The next figure contrasts a well-peaked posterior, where the Gaussian fit is excellent, with a broad zero-count posterior, where it fails.

See code 15.13 in Appendix for the implementation.



6.6. Credible intervals

When the posterior is broad or asymmetric, a single point estimate with symmetric error bars is misleading. Instead, we summarize the posterior with **credible intervals**: regions containing a specified probability mass.

Definition 6.3. A credible interval $[\theta^-, \theta^+]$ at **credibility level** γ satisfies

$$P(\theta \in [\theta^-, \theta^+] \mid D, H) = \gamma.$$

The posterior probability that θ lies in this interval is γ , **conditioned on the observed data**.

Two natural constructions are widely used:

- **Central (equal-tail) credible interval.** Places equal probability mass in each tail:

$$\int_{-\infty}^{\theta^-} p(\theta \mid D) d\theta = \int_{\theta^+}^{\infty} p(\theta \mid D) d\theta = \frac{1-\gamma}{2}.$$

The endpoints are quantiles: $\theta^\pm = F^{-1}\left(\frac{1\pm\gamma}{2}\right)$, where F is the CDF of the posterior. For a Gaussian posterior with mean μ and std σ , the 68%, 95%, and 99.7% central intervals are the familiar $\mu \pm 1\sigma$, $\mu \pm 1.96\sigma$, and $\mu \pm 3\sigma$.

- **Highest Posterior Density (HPD) interval.** The smallest-width region containing probability mass γ .

Definition 6.4. A **Highest Posterior Density (HPD) interval** at credibility level γ is the smallest-width region containing probability mass γ with the highest posterior density. Equivalently,

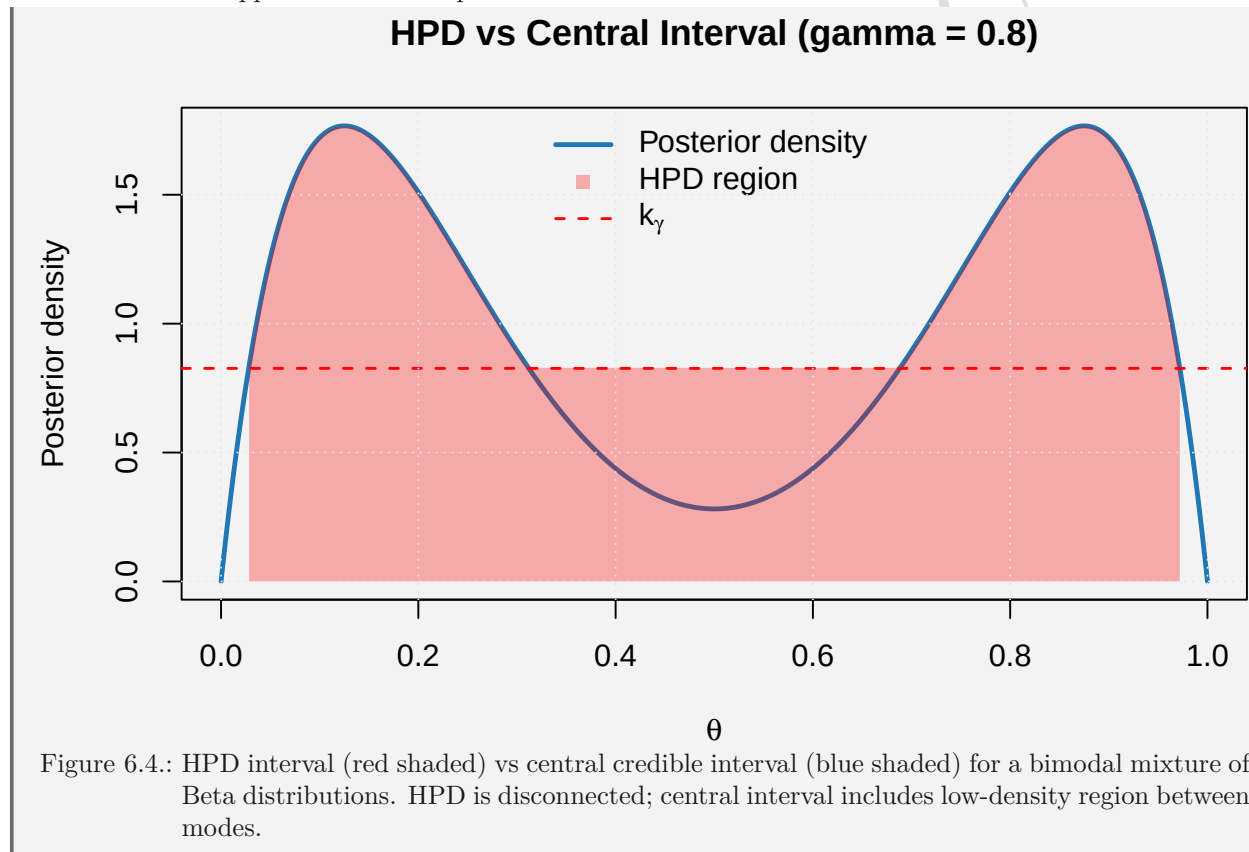
$$\text{HPD}_\gamma = \{\theta : p(\theta | D) \geq k_\gamma\},$$

where k_γ is the largest constant such that $\int_{\text{HPD}_\gamma} p(\theta | D) d\theta = \gamma$.

Properties: - Every point inside the HPD has higher density than every point outside. - All points in the HPD have posterior density $\geq k_\gamma$. - For symmetric distributions, HPD coincides with the central interval. - For asymmetric or multimodal posteriors, the HPD can be disconnected. - HPD is the minimum-width interval.

HPD construction by threshold. The practical algorithm is simple: sort posterior density values in descending order, accumulate probability mass until reaching γ , and find the density threshold k_γ where this occurs. Then the HPD is the set of all θ with $p(\theta | D) \geq k_\gamma$.

See code 15.14 in Appendix for the implementation.



Worked example: Beta posterior with equal-tail and HPD intervals. Suppose we observe $k = 5$ heads in $n = 20$ coin flips with a uniform prior $\text{Beta}(1,1)$. The posterior is $\text{Beta}(6, 16)$.

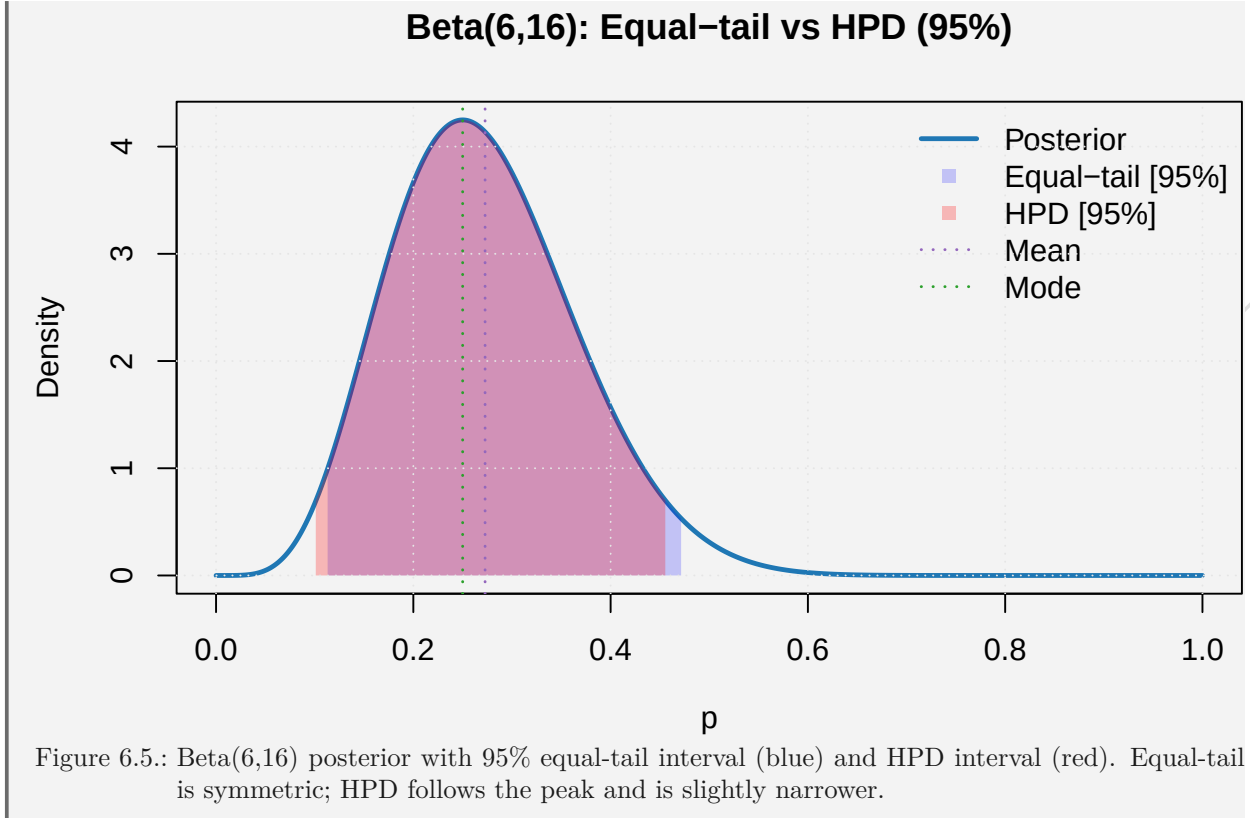
See code 15.15 in Appendix for the implementation.

Equal-tail: [0.1128, 0.4717]

See code 15.16 in Appendix for the implementation.

HPD: [0.1011, 0.4555]

See code 15.17 in Appendix for the implementation.



The equal-tail interval is $[0.1368, 0.3846]$; the HPD is $[0.1295, 0.3791]$. The HPD is slightly narrower and excludes the extreme low-density tail on the left.

6.7. Posterior predictive distribution

Having estimated θ , we often want to predict a future observation x_{new} .

Definition 6.5. The **posterior predictive distribution** integrates over parameter uncertainty:

$$p(x_{\text{new}} | D) = \int p(x_{\text{new}} | \theta) p(\theta | D) d\theta.$$

This is the probability distribution for a new observation, having learned θ from data D . No point estimate is needed; we marginalize over the posterior.

Prediction vs plug-in estimation. A naive approach is to plug in the point estimate $\hat{\theta}$ (e.g., the posterior mean):

$$p_{\text{plug-in}}(x_{\text{new}} | D) \approx p(x_{\text{new}} | \hat{\theta}).$$

This ignores **parameter uncertainty**, which can be substantial with sparse data.

Worked example: Poisson rate with zero counts. Suppose we observe zero events in $n = 5$ time intervals: $x_1 = \dots = x_5 = 0$, so $y = \sum_i x_i = 0$.

With a uniform prior $\text{Gamma}(1, 0)$ (improper), the posterior is $\lambda | D \sim \text{Gamma}(1, 5)$.

Plug-in estimate: $\hat{\lambda}_{\text{MLE}} = y/n = 0$, so $p(x_{\text{new}} = 1 | \hat{\lambda}) = 0$. This is unphysical: zero observed events do not mean events are impossible!

Posterior predictive: Integrating over the posterior,

$$p(x_{\text{new}} = 1 \mid D) = \int_0^\infty \frac{\lambda e^{-\lambda}}{1!} \cdot 5e^{-5\lambda} d\lambda = \int_0^\infty 5\lambda e^{-6\lambda} d\lambda = \frac{5}{36} \approx 0.139.$$

Even with zero observed counts, the posterior predicts a 14% chance of seeing an event in the next interval — a reasonable and physically sensible conclusion.

Example 6.4 (*Posterior predictive after a null exposure (exam-style)*)

During an exposure of length T , no events are observed. With a uniform prior on the Poisson rate $\lambda \geq 0$, what is the posterior predictive probability of observing at least one event in a future exposure of the same length T ?

Solution. With the improper uniform prior $p(\lambda) \propto 1$ and zero observed counts, the posterior is

$$p(\lambda \mid D) \propto e^{-\lambda T} \cdot 1 \quad \Rightarrow \quad \lambda \mid D \sim \text{Exp}(T).$$

Conditional on λ , the probability of zero future events in a fresh exposure of length T is $e^{-\lambda T}$. Hence

$$P(X_{\text{new}} = 0 \mid D) = \int_0^\infty e^{-\lambda T} T e^{-T\lambda} d\lambda = \int_0^\infty T e^{-2T\lambda} d\lambda = \frac{1}{2}.$$

Therefore $P(X_{\text{new}} \geq 1 \mid D) = 1/2$. Although no event was observed, the predictive probability of a future event is not zero — residual uncertainty in λ keeps the predictive distribution wide. This is the predictive cousin of the *Rule of Three*.

Example 6.5 (*Predictive probability of m failures (exam-style)*)

With prior $p \sim \text{Beta}(\alpha, \beta)$, k successes are observed in n Bernoulli trials. What is the predictive probability that the next m trials are all failures?

Solution. The posterior is $p \mid D \sim \text{Beta}(\alpha + k, \beta + n - k)$. Conditional on p , the probability of m consecutive failures is $(1 - p)^m$. The predictive probability is

$$P(m \text{ failures} \mid D) = E[(1 - p)^m \mid D].$$

For $p \sim \text{Beta}(a, b)$ one has $E[(1 - p)^m] = B(a, b + m)/B(a, b)$. Therefore

$$P(m \text{ failures} \mid D) = \frac{B(\alpha + k, \beta + n - k + m)}{B(\alpha + k, \beta + n - k)}.$$

This is **not** equal to $(1 - E[p \mid D])^m$ because the predictive integrates over residual uncertainty in p — a feature that the plug-in estimate ignores.

Variance decomposition. The predictive variance splits naturally into two sources:

$$\text{Var}(x_{\text{new}} \mid D) = E_\theta[\text{Var}(x_{\text{new}} \mid \theta) \mid D] + \text{Var}_\theta(E[x_{\text{new}} \mid \theta] \mid D).$$

The first term is **data noise** (aleatoric uncertainty); the second is **parameter uncertainty** (epistemic uncertainty). The full predictive captures both.

6.8. Sequential learning and posterior updating

A key advantage of Bayesian inference is that learning is sequential: today's posterior becomes tomorrow's prior.

Example 6.6 (*Sequential vs batch updating (exam-style)*)

A coin has prior $p \sim \text{Beta}(2, 2)$. In a first run, 3 successes and 1 failure are observed. In a second run, 2 successes and 4 failures are observed. Compute the final posterior both sequentially and in one step.

Solution. Sequentially, after the first run,

$$p \mid D_1 \sim \text{Beta}(2 + 3, 2 + 1) = \text{Beta}(5, 3).$$

Treating that as the new prior and updating on the second run,

$$p \mid D_1, D_2 \sim \text{Beta}(5 + 2, 3 + 4) = \text{Beta}(7, 7).$$

In one step, the combined data have 5 successes and 5 failures, so

$$p \mid D \sim \text{Beta}(2 + 5, 2 + 5) = \text{Beta}(7, 7).$$

The same posterior is obtained either way — conjugate updating is associative, which is just a restatement of the product rule.

Example 6.7

Sequential coin tosses

Start with a $\text{Beta}(1, 1)$ uniform prior. After observing data, we compute the posterior. If we observe more data, the posterior becomes the new prior.

See code 15.18 in Appendix for the implementation.

```
Warning in int_abline(a = a, b = b, h = h, v = v, untf = untf, ...): "alpha" is
not a graphical parameter
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See code 15.19 in Appendix for the implementation.

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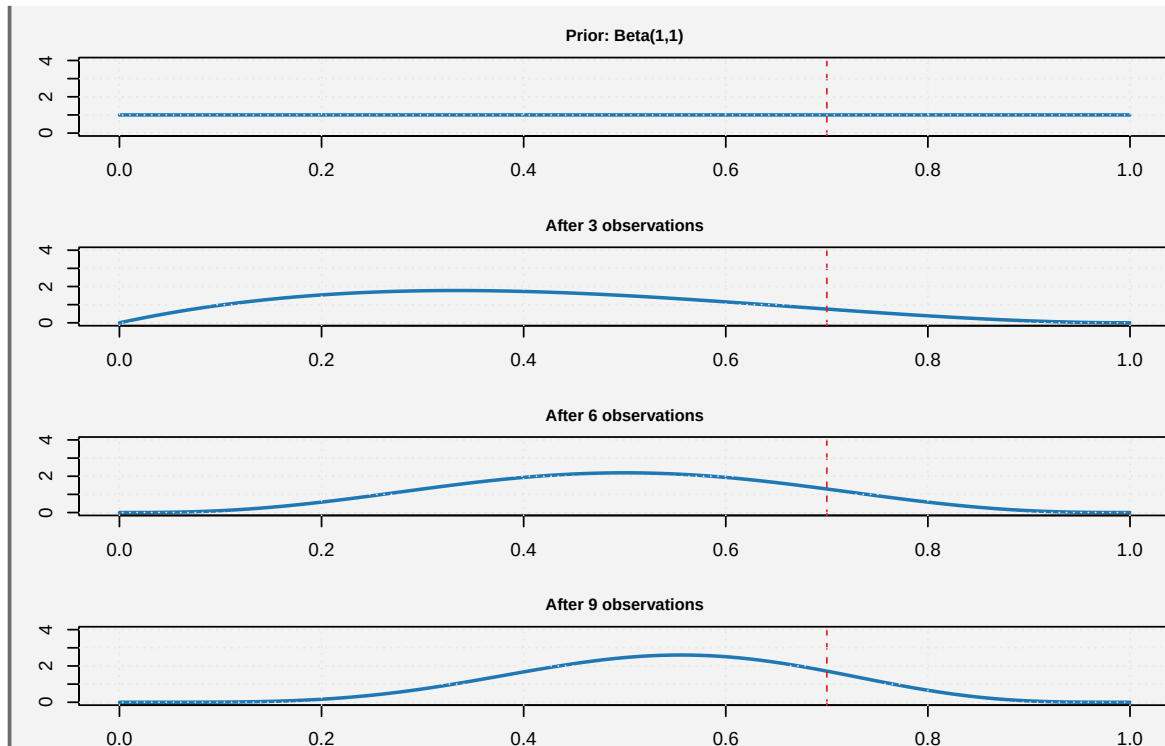


Figure 6.6.: Sequential Bayesian learning for a coin. Top: prior and first observation (1 head, 2 tails). Middle and bottom: posterior becomes new prior as data accumulate. Posterior concentrates around true bias $p=0.7$.

Warning in int_abline(a = a, b = b, h = h, v = v, untf = untf, ...): "alpha" is not a graphical parameter

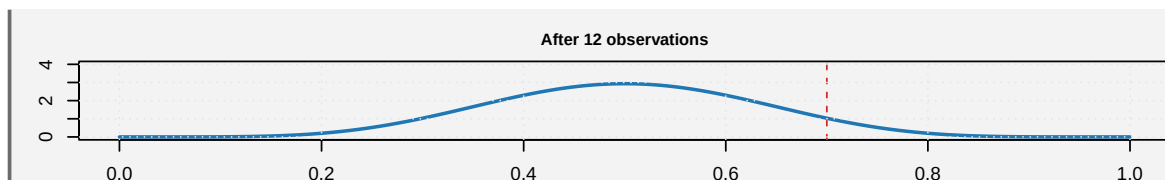


Figure 6.7.: Sequential Bayesian learning for a coin. Top: prior and first observation (1 head, 2 tails). Middle and bottom: posterior becomes new prior as data accumulate. Posterior concentrates around true bias $p=0.7$.

Notice how the posterior tightens and centers on the true value as we accumulate data.

6.9. Frequentist confidence intervals: brief contrast

To highlight what is unique about Bayesian credible intervals, we briefly compare with frequentist confidence intervals.

Definition 6.6. A **frequentist confidence interval** is a random interval $C(X)$ with **coverage probability** $1 - \alpha$:

$$P(\theta \in C(X)) = 1 - \alpha,$$

where the probability is over repeated sampling of data $X \sim p(x | \theta)$ under a **fixed but unknown** θ .

Interpretation: In repeated experiments, $100(1 - \alpha)\%$ of the confidence intervals will contain the true

θ .

Note. **Key differences:**

Aspect	Bayesian Credible Interval	Frequentist Confidence Interval
Probability on	Parameter	Data
Conditioned on	Observed data D	Hypothetical repeated datasets
Interpretation	$P(\theta \in [a, b] \mid D) = \gamma$	In repeated experiments, $100\gamma\%$ of intervals cover θ
Reparametrization	Invariant	Not invariant

Once we observe data, the frequentist interval either contains θ or it does not — there is no probability statement about the observed interval.

Example 6.8 (*Credible vs confidence intervals (exam-style)*)

Define a Bayesian credible interval for a scalar parameter θ . How does its interpretation differ from a frequentist confidence interval?

Solution. A Bayesian credible interval $[a, b]$ at credibility level γ satisfies

$$\int_a^b p(\theta \mid D) d\theta = \gamma.$$

A **central** credible interval excludes equal posterior probability from each tail; a **highest-posterior-density** (HPD) interval is the smallest set containing posterior mass γ (it collects the highest-density region first).

Its interpretation is direct: conditional on the data, model, and prior, the parameter lies in $[a, b]$ with posterior probability γ . A frequentist confidence interval instead has a repeated-sampling coverage interpretation: *before* seeing the data, the random procedure $C(X)$ contains the fixed true θ with probability γ . After conditioning on the observed data, the random interval is fixed and either contains θ or does not — no probability statement is attached to the realised interval.

6.10. One-sided credible intervals

For some problems, we care only about an upper or lower bound on the parameter. A **one-sided credible interval** $[0, \theta^*]$ at level γ satisfies

$$P(\theta \leq \theta^* \mid D) = \gamma.$$

Rule of Three. A famous result in statistics applies to rare-event detection. If you observe zero events in n trials under a Binomial model with unknown rate p , the **95% upper credible bound** (using a uniform Beta(1,1) prior) is approximately

$$p^* \approx \frac{3}{n}.$$

This is the “**Rule of Three**”: if you see zero occurrences in n trials, you can be 95% confident that the true rate is below $3/n$.

Example 6.9**Zero-count drug safety**

If a drug is tested on 100 patients and zero adverse events are observed, the Rule of Three gives a 95% upper bound of $3/100 = 0.03$ or 3%. We can be reasonably confident (in the Bayesian sense, given a weak prior) that the true adverse event rate is below 3%, even though we observed none.

This addresses the zero-count problem: the MLE would say the rate is exactly zero, which is overconfident. The Bayesian posterior hedges appropriately.

6.11. Choosing priors: conjugacy and regularization

Definition 6.7. A prior $\pi(\theta)$ is **conjugate** to a likelihood $p(x | \theta)$ if the posterior is in the same family as the prior:

$$p(\theta | x) \propto p(x | \theta)\pi(\theta) \in \text{same family as } \pi(\theta).$$

Conjugate priors are analytically tractable and allow simple sequential updating. They also naturally encode prior beliefs as “pseudo-data” or “pseudo-observations.”

Two canonical conjugate pairs illustrate the pattern:

- **Beta-Binomial conjugacy.** Prior $p \sim \text{Beta}(\alpha, \beta)$ with density $\pi(p) \propto p^{\alpha-1}(1-p)^{\beta-1}$; likelihood $P(k \text{ heads} | p, n) \propto p^k(1-p)^{n-k}$; posterior $p | k, n \sim \text{Beta}(\alpha + k, \beta + n - k)$. The hyperparameters have an intuitive interpretation: $\alpha - 1$ and $\beta - 1$ are pseudo-successes and pseudo-failures, representing prior pseudo-data.
- **Gamma-Poisson conjugacy.** Prior $\lambda \sim \text{Gamma}(\alpha, \beta)$ with density $\pi(\lambda) \propto \lambda^{\alpha-1}e^{-\beta\lambda}$; likelihood $P(y \text{ counts} | \lambda, n) \propto \lambda^y e^{-n\lambda}$; posterior $\lambda | y, n \sim \text{Gamma}(\alpha + y, \beta + n)$. Here, α is a pseudo-count and β is a pseudo-exposure time.

6.12. Fisher information, Cramér–Rao, and the Jeffreys prior

The theory of parametric estimation rests on a few fundamental objects: the Fisher information matrix, which quantifies the curvature of the likelihood; the Cramér–Rao bound, which sets a lower limit on estimator variance; and the Jeffreys prior, which adapts to the geometry of the parameter space. This section develops these concepts in general, showing how they unify point estimation (Cramér–Rao), prior specification (Jeffreys), and asymptotic posterior behaviour (Bernstein–von Mises).

6.12.1. Score and Fisher Information

Definition 6.8 (Score and Fisher information matrix). Given a model $p(x | \theta)$ and an observation x (or dataset), the **score vector** is

$$s(\theta; x) = \nabla_{\theta} \log p(x | \theta) = \frac{\partial}{\partial \theta} \log p(x | \theta).$$

The **Fisher information matrix** is the expected outer product of the score:

$$\mathcal{I}(\theta) = \mathbb{E}_x[s(\theta; x) s(\theta; x)^{\top}],$$

with (i, j) -entry $\mathcal{I}_{ij}(\theta) = \mathbb{E}[s_i(\theta; x) s_j(\theta; x)]$.

Equivalently, under regularity conditions,

$$\mathcal{J}(\theta) = -\mathbb{E}_x [\nabla_\theta^2 \log p(x | \theta)] = -\mathbb{E}[H(\theta)],$$

where $H(\theta)$ is the Hessian of the log-likelihood.

Proof sketch for the equivalence: Differentiating $\int p(x | \theta) dx = 1$ with respect to θ :

$$\int \frac{\partial}{\partial \theta} p(x | \theta) dx = \int p(x | \theta) \cdot s(\theta; x) dx = \mathbb{E}[s(\theta; x)] = 0.$$

Differentiating again and swapping derivatives and integration (regularity):

$$\mathbb{E} \left[\frac{\partial s(\theta; x)}{\partial \theta} \right] + \mathbb{E}[s(\theta; x)^2] = 0,$$

which gives $\mathbb{E}[s(\theta; x)^2] = -\mathbb{E}[H(\theta)]$.

For an i.i.d. sample of size n , the total score is the sum of individual scores, and the Fisher information scales linearly: $\mathcal{J}_n(\theta) = n\mathcal{J}(\theta)$.

Reparametrization formula. The Fisher information transforms in a particularly clean way under smooth changes of variable.

Proposition 6.2 (Fisher information under reparametrisation). *Suppose we reparameterize via a smooth bijection $\psi = \phi(\theta)$ with Jacobian matrix $J_{ij} = \partial \psi_i / \partial \theta_j$. Then the Fisher information with respect to ψ is*

$$\mathcal{J}_\psi(\psi) = J^\top \mathcal{J}_\theta(\theta) J,$$

where $J = \partial \phi(\theta) / \partial \theta$ evaluated at the appropriate $\theta = \phi^{-1}(\psi)$.

Proof: By the chain rule, $s_\psi(\psi; x) = J^\top s_\theta(\theta; x)$. Hence,

$$\mathcal{J}_\psi(\psi) = \mathbb{E}[J^\top s_\theta s_\theta^\top J] = J^\top \mathcal{J}_\theta(\theta) J.$$

This reparametrization formula is the key to understanding why $\sqrt{\det \mathcal{J}(\theta)}$ behaves like an invariant density element. Under a change of variables $\psi = \phi(\theta)$,

$$\sqrt{\det \mathcal{J}_\psi(\psi)} = \sqrt{\det(J^\top J)} \cdot \sqrt{\det \mathcal{J}_\theta(\theta)} = |\det J| \cdot \sqrt{\det \mathcal{J}_\theta(\theta)},$$

which is precisely the Jacobian rule for transforming a density. This geometric insight motivates the Jeffreys prior.

6.12.2. Cramér–Rao Lower Bound

Theorem 6.3 (Cramér–Rao lower bound). *Let $\hat{\theta}(X)$ be an unbiased estimator of θ from an i.i.d. sample $X = (X_1, \dots, X_n)$. Then*

$$\text{Cov}(\hat{\theta}) \succeq \mathcal{J}_n(\theta)^{-1} = \frac{1}{n} \mathcal{J}(\theta)^{-1},$$

where $\succeq$ denotes the positive semidefinite order (i.e., $\text{Cov}(\hat{\theta}) - \mathcal{J}_n(\theta)^{-1}$ is positive semidefinite).

Proof outline (scalar case): Write the unbiasedness condition as

$$\mathbb{E}_\theta[\hat{\theta} - \theta] = 0.$$

Differentiating with respect to θ and invoking regularity (swapping derivative and expectation):

$$\mathbb{E}_\theta \left[\frac{\partial}{\partial \theta} (\hat{\theta} - \theta) \right] + \mathbb{E}_\theta [s(\theta; X)] = 0.$$

Since $\mathbb{E}[s] = 0$, we have

$$\mathbb{E}[s(\theta; X)] = -1.$$

Wait, I need to be more careful; let me re-phrase. Consider $\mathbb{E}[\hat{\theta} \cdot s(\theta)] = \theta \mathbb{E}[s(\theta)]$. Differentiating the unbiasedness condition $\mathbb{E}[\hat{\theta}] = \theta$:

$$\mathbb{E} \left[\frac{\partial \hat{\theta}}{\partial \theta} \right] = 1 \quad (\text{in scalar case}).$$

By Cauchy–Schwarz applied to $\hat{\theta} \cdot s$ and rearranging, one recovers $\text{Var}(\hat{\theta}) \geq \mathcal{I}(\theta)^{-1}$.

The Cramér–Rao bound sets a fundamental speed limit on estimation: no unbiased estimator can have variance smaller than this bound. Remarkably, the maximum likelihood estimator (MLE) achieves this bound asymptotically.

Asymptotic normality of the MLE. The MLE attains this bound in the large-sample limit.

Theorem 6.4 (Asymptotic normality of the MLE). *Under standard regularity conditions, the MLE $\hat{\theta}_{\text{ML}}$ of θ from a sample of size n satisfies*

$$\sqrt{n} (\hat{\theta}_{\text{ML}} - \theta_*) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta_*)^{-1}),$$

where θ_* is the true parameter and $\mathcal{I}(\theta_*)$ is the Fisher information at the truth.

This result, whose proof relies on Taylor expansions of the score and regularity conditions, shows that for large n the MLE is approximately $\mathcal{N}(\hat{\theta}_{\text{ML}}, \mathcal{I}_n(\theta_*)^{-1})$. Since $\mathcal{I}_n(\theta_*) = n\mathcal{I}(\theta_*)$, the variance shrinks as $1/n$, asymptotically achieving the Cramér–Rao bound. For reference on the detailed conditions and proof, see van der Vaart (1998), Chapter 5.

6.12.3. The Jeffreys Prior

Definition 6.9 (Jeffreys prior). The **Jeffreys prior** is defined as

$$p_J(\theta) \propto \sqrt{\det \mathcal{I}(\theta)},$$

the square root of the determinant of the Fisher information matrix.

Intuitively, the Jeffreys prior allocates more prior mass to regions where the likelihood is less informative (where $\mathcal{I}(\theta)$ is smaller) and less prior mass where the likelihood is highly peaked (where $\mathcal{I}(\theta)$ is large). This gives an *objective* prior that “lets the data speak” in a geometric sense.

Reparametrisation invariance of Jeffreys. The defining property is captured by the following result.

Theorem 6.5 (Jeffreys prior is reparametrisation-invariant). *If $p_J(\theta) \propto \sqrt{\det \mathcal{I}_\theta(\theta)}$ is the Jeffreys prior in the θ parameterization, and we reparametrize via $\psi = \phi(\theta)$ with Jacobian J , then the Jeffreys prior in the ψ parameterization satisfies*

$$p_J(\psi) \propto \sqrt{\det \mathcal{I}_\psi(\psi)} = \sqrt{\det(J^\top J)} \cdot \sqrt{\det \mathcal{I}_\theta(\theta)} = |\det J| \cdot \sqrt{\det \mathcal{I}_\theta(\theta)}.$$

Under the change of variables, the prior density transforms as

$$p_J(\psi) d\psi = p_J(\theta) |\det J| d\theta,$$

which shows that the Jeffreys prior is **invariant under reparametrisation**.

Significance: This is the defining property that distinguishes Jeffreys from a flat prior. A uniform prior $p(\theta) \propto 1$ does *not* remain uniform after reparametrization; its density changes by the Jacobian. In contrast, the Jeffreys prior automatically adapts its form to remain “flat” in an information-geometric sense, independent of how we parameterize the model.

6.12.4. Examples of Jeffreys Priors

Example 6.10 (Bernoulli parameter)

For a Bernoulli(p) model with a single observation $x \in \{0, 1\}$, the log-likelihood is $x \log p + (1 - x) \log(1 - p)$. The Fisher information is

$$\mathcal{J}(p) = \mathbb{E} \left[\left(\frac{\partial}{\partial p} \log p(x | p) \right)^2 \right] = \frac{1}{p(1-p)}.$$

The Jeffreys prior is thus

$$p_J(p) \propto \sqrt{\frac{1}{p(1-p)}} = p^{-1/2}(1-p)^{-1/2},$$

which is the **Beta(1/2, 1/2) distribution**.

Note: the uniform prior is Beta(1, 1), which depends on the choice of the p parameterization. If we instead parameterize via the log-odds $\phi = \log(p/(1-p))$, the uniform prior in ϕ is different from the reparametrized form of Beta(1, 1). But Beta(1/2, 1/2) remains the same (up to the appropriate density form) under any reparametrization.

Example 6.11 (Poisson rate)

For Poisson(λ), the log-likelihood is $x \log \lambda - \lambda$, and the Fisher information is

$$\mathcal{J}(\lambda) = \frac{1}{\lambda}.$$

The Jeffreys prior is

$$p_J(\lambda) \propto \lambda^{-1/2},$$

an improper prior. However, any observed data ($x > 0$) yields a proper posterior, because the likelihood $\lambda^x e^{-\lambda}$ combined with $\lambda^{-1/2}$ gives $\lambda^{x-1/2} e^{-\lambda}$, a Gamma distribution.

Example 6.12 (Jeffreys is reparametrisation-invariant — Poisson with $\phi = \sqrt{\lambda}$ (exam-style))

For a Poisson observation $X \sim \text{Poisson}(\lambda)$, Jeffreys prior is $\pi(\lambda) \propto \lambda^{-1/2}$. Let $\phi = \sqrt{\lambda}$. Find the induced prior on ϕ .

Solution. Since $\lambda = \phi^2$ and $d\lambda/d\phi = 2\phi$, the change-of-variables rule gives

$$\pi_\phi(\phi) = \pi_\lambda(\phi^2) \left| \frac{d\lambda}{d\phi} \right| \propto (\phi^2)^{-1/2} \cdot 2\phi = 2.$$

The Jeffreys prior for the Poisson rate becomes *flat* in $\phi = \sqrt{\lambda}$. This is the geometric content of Jeffreys: in the parameterisation that makes the Fisher information constant, the prior is uniform.

Example 6.13 (*Jeffreys prior for the Lévy scale (exam-style)*)

The Lévy distribution with location 0 and scale $c > 0$ has density

$$f(x | c) = \sqrt{\frac{c}{2\pi}} x^{-3/2} \exp\left(-\frac{c}{2x}\right), \quad x > 0.$$

Useful moments are $E[1/X] = 1/c$ and $E[1/X^2] = 3/c^2$. Compute the Jeffreys prior for c .

Solution. The log-likelihood is

$$\log f(x | c) = \frac{1}{2} \log c - \frac{1}{2} \log(2\pi) - \frac{3}{2} \log x - \frac{c}{2x}.$$

Differentiating with respect to c ,

$$\frac{\partial}{\partial c} \log f(x | c) = \frac{1}{2c} - \frac{1}{2x}.$$

The Fisher information is therefore

$$\mathcal{I}(c) = E\left[\left(\frac{1}{2c} - \frac{1}{2X}\right)^2\right] = \frac{1}{4c^2} - \frac{1}{2c} E\left[\frac{1}{X}\right] + \frac{1}{4} E\left[\frac{1}{X^2}\right] = \frac{1}{4c^2} - \frac{1}{2c^2} + \frac{3}{4c^2} = \frac{1}{2c^2}.$$

Hence

$$\pi_J(c) \propto \sqrt{\mathcal{I}(c)} = \sqrt{\frac{1}{2c^2}} \propto \frac{1}{c}, \quad c > 0.$$

This is the improper scale-invariant prior $p(c) dc \propto d \log c$ — the canonical non-informative choice for a scale parameter.

Example 6.14 (*Gaussian location (known scale)*)

With σ^2 known, the log-likelihood for the mean μ is proportional to $-\frac{1}{2\sigma^2}(x - \mu)^2$. The Fisher information is

$$\mathcal{I}(\mu) = \frac{1}{\sigma^2},$$

a positive constant independent of μ . The Jeffreys prior is thus

$$p_J(\mu) \propto 1,$$

a **uniform (improper) prior**—the standard choice for the unknown mean when the scale is known.

Example 6.15 (*Gaussian scale (known location)*)

With μ known, the log-likelihood is $-\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum (x - \mu)^2$. Differentiating twice with respect to σ^2 gives

$$\mathcal{I}(\sigma^2) = \frac{n}{2(\sigma^2)^2}.$$

The Jeffreys prior is

$$p_J(\sigma^2) \propto \frac{1}{\sigma^2},$$

or equivalently, $p_J(\log \sigma) \propto 1$ (uniform on the log scale). This is the **scale-invariant prior** for variance, classical in Bayesian inference.

Example 6.16 (*Gaussian, joint mean and variance*)

For the Gaussian with both μ and σ^2 unknown, the Fisher information matrix is

$$\mathcal{J}(\mu, \sigma^2) = \begin{pmatrix} 1/\sigma^2 & 0 \\ 0 & 2/\sigma^4 \end{pmatrix},$$

so $\det \mathcal{J} = 2/\sigma^6$, and the Jeffreys prior is

$$p_J(\mu, \sigma^2) \propto \sqrt{\frac{2}{\sigma^6}} = \frac{1}{\sigma^3}.$$

Equivalently, $p_J(\mu, \sigma) \propto 1/\sigma^2$ in terms of the standard deviation.

Subtle point: The *joint* Jeffreys prior $p_J(\mu, \sigma^2) \propto 1/\sigma^3$ does *not* equal the product of the marginal Jeffreys priors for μ (uniform) and σ^2 (proportional to $1/\sigma^2$), which would give $p(\mu) \cdot p(\sigma^2) \propto 1/\sigma^2$. Many practitioners use this product-of-marginals prior and call it “Jeffreys,” but it is technically the **reference prior** or a compromise prior. The true joint Jeffreys prior is $1/\sigma^3$. This distinction matters for hypothesis tests and Bayes factors.

Reference priors. A natural generalisation kicks in when dimension grows.

Remark. When the parameter dimension is high, computing or integrating over the Jeffreys prior becomes cumbersome. **Reference priors** (Bernardo, 1979; Berger, Bernardo & Sun, 2009) generalize the idea: they maximize expected Kullback–Leibler divergence between posterior and prior, thus minimizing the information contained in the prior itself. For one-parameter problems, reference priors coincide with Jeffreys; for multiparameter problems, they offer an alternative construction that often performs better in frequentist coverage. Reference priors are not invariant to reparametrization but are designed to be more “objective” in the sense of information loss.

6.12.5. The Bernstein–von Mises Theorem

Theorem 6.6 (Bernstein–von Mises). *Under standard regularity conditions, for large sample size n , the posterior distribution $p(\theta \mid X_1, \dots, X_n)$ is approximately Gaussian, centered at the MLE $\hat{\theta}_{\text{ML}}$ with covariance given by the inverse Fisher information:*

$$p(\theta \mid X_1, \dots, X_n) \approx \mathcal{N}(\hat{\theta}_{\text{ML}}, \mathcal{J}_n(\theta_*)^{-1}),$$

provided the prior is continuous and positive at θ_ (the true parameter). This holds independently of the choice of prior.*

Interpretation and significance: The Bernstein–von Mises theorem is one of the deepest results in Bayesian asymptotics. It says that, given enough data, the posterior “forgets” the prior (as long as it is not zero at the truth) and concentrates around the MLE with the canonical frequentist covariance $\mathcal{J}_n(\theta_*)^{-1}$. This explains why, asymptotically, Bayesian credible intervals and frequentist confidence intervals coincide: both converge to the same Gaussian limit. In other words, there is no fundamental conflict between Bayesian and frequentist inference in the large-sample regime—they merge. For a detailed proof, see van der Vaart (1998), Chapter 10.

Unifying theme. Fisher information, Cramér–Rao, Jeffreys, and Bernstein–von Mises are tightly interwoven:

- **Cramér–Rao** sets the performance standard for any unbiased estimator.
- **MLE asymptotics** show that the MLE achieves this standard with variance $\mathcal{J}_n(\theta)^{-1}$.
- **Jeffreys prior** is constructed from $\mathcal{J}(\theta)$, ensuring objective prior choice via reparametrization invariance.
- **Bernstein–von Mises** guarantees that the posterior, regardless of the prior, concentrates at the MLE with covariance $\mathcal{J}_n(\theta)^{-1}$.

Together, these results form the backbone of both frequentist and Bayesian asymptotic theory.

6.13. Exercises

Exercise 6.1

(Written test, 15 June 2026 — Q2.) Let $\lambda \mid D \sim \text{Gamma}(a, b)$ with rate parameter b and $a > 1$. The MAP estimator of λ is

$$\hat{\lambda}_{\text{MAP}} = \frac{a-1}{b}.$$

Now define a new parameter $\eta = \log \lambda$. Using the posterior density induced for η , find the MAP estimator $\hat{\eta}_{\text{MAP}}$. Compare $e^{\hat{\eta}_{\text{MAP}}}$ with $\hat{\lambda}_{\text{MAP}}$.

Solution

With $\lambda = e^\eta$ and Jacobian $d\lambda/d\eta = e^\eta$, the induced posterior is

$$p(\eta \mid D) \propto \lambda^{a-1} e^{-b\lambda} \cdot \lambda = e^{a\eta} e^{-be^\eta}.$$

Setting $\partial_\eta \log p = a - be^\eta = 0$ gives

$$\hat{\eta}_{\text{MAP}} = \log(a/b), \quad e^{\hat{\eta}_{\text{MAP}}} = \frac{a}{b} \neq \frac{a-1}{b} = \hat{\lambda}_{\text{MAP}}.$$

The MAP is not invariant under reparametrisation: the Jacobian shifts the mode from $(a-1)/b$ to a/b (the posterior mean of λ).

Exercise 6.2

(Written test, 15 June 2026 — Q3.) A posterior distribution for a scalar parameter is

$$p(\theta \mid D) = 0.9 \mathcal{N}(0, 0.2^2) + 0.1 \mathcal{N}(10, 0.2^2).$$

Which point estimate is favored by squared-error loss? Which point estimate is given by the MAP? Between these two estimates, which is more representative of a typical posterior draw?

Solution

The squared-error (MMSE) estimator is the posterior mean,

$$\hat{\theta}_{\text{LMS}} = 0.9 \cdot 0 + 0.1 \cdot 10 = 1.$$

The MAP is the global mode; since both components have equal width $\sigma = 0.2$ and the first carries

weight $0.9 \gg 0.1$,

$$\hat{\theta}_{\text{MAP}} = 0.$$

The MAP is far more representative: 90% of posterior mass sits near 0, while $\hat{\theta}_{\text{LMS}} = 1$ falls in the low-density gap between the two modes and is unlike any typical draw. This is the classic failure mode of the posterior mean on multimodal posteriors.

Exercise 6.3

(Written test, 15 June 2026 — Q4.) During an exposure of length T , no events are observed. Assume the number of observed events follows a Poisson distribution with mean λT , where λ is the event rate. Use the improper uniform prior $p(\lambda) \propto 1$ with $\lambda \geq 0$. Derive the 99% upper credible bound on λ , then compute it for $T = 50$.

Solution

With zero counts and a flat prior, $p(\lambda | D) \propto e^{-\lambda T}$, so

$$\lambda | D \sim \text{Exp}(T), \quad F(\lambda) = 1 - e^{-T\lambda}.$$

The 99% upper credible bound λ^* solves $F(\lambda^*) = 0.99$, giving

$$\lambda^* = -\frac{\log(0.01)}{T} = \frac{\log 100}{T}.$$

For $T = 50$, $\lambda^* = \log(100)/50 \approx 4.605/50 \approx 0.092$.

Exercise 6.4

A binary outcome is observed n times with s successes. With prior $\theta \sim \text{Uniform}(0, 1)$, show that the posterior is $\text{Beta}(s + 1, n - s + 1)$. Give the posterior mean and the posterior mode.

Solution

Uniform on $(0, 1)$ is $\text{Beta}(1, 1)$, so the likelihood $\theta^s(1 - \theta)^{n-s}$ times the prior 1 gives

$$p(\theta | D) \propto \theta^s(1 - \theta)^{n-s} \implies \theta | D \sim \text{Beta}(s + 1, n - s + 1).$$

Hence

$$\hat{\theta}_{\text{LMS}} = \frac{s + 1}{n + 2}, \quad \hat{\theta}_{\text{MAP}} = \frac{s}{n} \quad (\text{for } 1 \leq s \leq n - 1).$$

The MAP recovers the MLE while the mean is the Laplace-rule-of-succession smoother.

Exercise 6.5

Same Bernoulli likelihood as above. Use prior $\theta \sim \text{Beta}(2, 2)$ and data $n = 12$, $s = 8$. Write the posterior and give its mean and mode.

Solution

Conjugate updating gives

$$\theta | D \sim \text{Beta}(2 + 8, 2 + 4) = \text{Beta}(10, 6).$$

Therefore

$$\hat{\theta}_{\text{LMS}} = \frac{10}{16} = \frac{5}{8}, \quad \hat{\theta}_{\text{MAP}} = \frac{10-1}{10+6-2} = \frac{9}{14}.$$

Exercise 6.6

Conjugate Gamma–Poisson. Observe n Poisson counts with sum s , prior $\lambda \sim \text{Gamma}(\alpha, \beta)$ (rate β). Derive the posterior. Give the posterior mean (MMSE) and the MAP. Verify Bernstein–von Mises numerically by letting $n \rightarrow \infty$ with $s/n \rightarrow \bar{x}$ and checking that the MAP tends to the MLE.

Solution

The likelihood is $\prod_i \lambda^{x_i} e^{-\lambda} / x_i! \propto \lambda^s e^{-n\lambda}$, multiplied by the prior $\lambda^{\alpha-1} e^{-\beta\lambda}$, gives

$$\lambda \mid D \sim \Gamma(\alpha + s, \beta + n).$$

Hence

$$\hat{\lambda}_{\text{LMS}} = \frac{\alpha + s}{\beta + n}, \quad \hat{\lambda}_{\text{MAP}} = \frac{\alpha + s - 1}{\beta + n}.$$

As $n \rightarrow \infty$ with $s/n \rightarrow \bar{x}$, both estimators tend to $\bar{x} = \hat{\lambda}_{\text{ML}}$, and the posterior variance $(\alpha + s)/(\beta + n)^2 \sim \bar{x}/n = \mathcal{I}_n(\lambda)^{-1}$, confirming Bernstein–von Mises.

Exercise 6.7

Exponential model. Observe n i.i.d. samples $x_1, \dots, x_n$ with sum s from $\text{Exp}(\lambda)$, and use a $\text{Gamma}(\alpha, \beta)$ prior on λ . Derive the posterior and the MMSE estimator.

Solution

The likelihood is $\lambda^n e^{-\lambda s}$; multiplying by the prior $\lambda^{\alpha-1} e^{-\beta\lambda}$,

$$\lambda \mid D \sim \Gamma(\alpha + n, \beta + s).$$

The MMSE estimator is the posterior mean

$$\hat{\lambda}_{\text{LMS}} = \frac{\alpha + n}{\beta + s}.$$

As $n, s \rightarrow \infty$ this tends to $n/s = 1/\bar{x} = \hat{\lambda}_{\text{ML}}$.

Exercise 6.8

Reparameterization. For a Bernoulli model with $\text{Beta}(\alpha, \beta)$ posterior on θ , define the odds $\psi = \theta/(1 - \theta)$. Derive the posterior density $p(\psi \mid D)$ and locate its mode. Compare with the mode of $p(\theta \mid D)$ pushed through $\theta \mapsto \theta/(1 - \theta)$.

Solution

With $\theta = \psi/(1 + \psi)$ and $|d\theta/d\psi| = (1 + \psi)^{-2}$,

$$p(\psi \mid D) \propto \frac{\psi^{\alpha-1}}{(1 + \psi)^{\alpha+\beta}}, \quad \psi > 0.$$

Setting the derivative of the log density to zero gives

$$\hat{\psi}_{\text{MAP}} = \frac{\alpha - 1}{\beta + 1}.$$

Meanwhile $\hat{\theta}_{\text{MAP}} = (\alpha - 1)/(\alpha + \beta - 2)$ pushed through the map gives $(\alpha - 1)/(\beta - 1)$, which differs from $(\alpha - 1)/(\beta + 1)$. The discrepancy is exactly the Jacobian $(1 + \psi)^{-2}$ shifting the mode — the MAP is not reparametrisation invariant.

Exercise 6.9

A posterior density is $p(\theta | D) = \frac{3}{2}(1 - \theta^2)$ on $[-1, 1]$. Compute the MMSE estimator, the MAP, and the posterior median.

Solution

The density is symmetric about 0, so by symmetry

$$\hat{\theta}_{\text{LMS}} = E[\theta | D] = 0, \quad \text{median}(\theta | D) = 0.$$

The MAP maximises $1 - \theta^2$, giving

$$\hat{\theta}_{\text{MAP}} = 0.$$

All three coincide at the single peak of this symmetric unimodal posterior.

Exercise 6.10

Skewed posterior. Let $\lambda | D \sim \text{Gamma}(2, 1)$. Compute the posterior mean, the posterior mode, and the median (use the exponential CDF after marginalising).

Solution

With density $\lambda e^{-\lambda}$,

$$\hat{\lambda}_{\text{LMS}} = \frac{a}{b} = 2, \quad \hat{\lambda}_{\text{MAP}} = \frac{a - 1}{b} = 1.$$

The CDF is $F(\lambda) = 1 - (1 + \lambda)e^{-\lambda}$, so the median solves

$$(1 + \lambda)e^{-\lambda} = \frac{1}{2},$$

giving $\text{median}(\lambda | D) \approx 1.678$. Mean $>$ median $>$ mode, the hallmark of a right-skewed Gamma.

Exercise 6.11

Loss functions. Show that the Bayes estimator under the absolute loss $L(a, \theta) = |a - \theta|$ is the posterior median. Then show that under the asymmetric loss

$$L(a, \theta) = c_1 (a - \theta)_+ + c_2 (\theta - a)_+,$$

the optimal estimator is the $c_2/(c_1 + c_2)$ quantile of the posterior.

6. Bayesian parameter estimation, intervals and prediction

Solution

Differentiating $R(a) = E[|a - \theta| | D] = \int_{-\infty}^a (a - \theta)p \, d\theta + \int_a^{\infty} (\theta - a)p \, d\theta$ in a gives

$$R'(a) = P(\theta \leq a | D) - P(\theta > a | D) = 0 \implies P(\theta \leq a | D) = \frac{1}{2},$$

so $\hat{a} = \text{median}(\theta | D)$. For the asymmetric loss the same calculation gives

$$R'(a) = c_1 P(\theta \leq a | D) - c_2 P(\theta > a | D) = 0 \implies P(\theta \leq a | D) = \frac{c_2}{c_1 + c_2},$$

so the optimal $\hat{a}$ is the $c_2/(c_1 + c_2)$ -quantile of the posterior.

Exercise 6.12

Improper flat prior on a location parameter. Let $x | \theta \sim \mathcal{N}(\theta, 1)$, prior $p(\theta) \propto 1$, observe a single value x . Give the posterior and the equal-tailed 95% credible interval (use $z_{0.975} \approx 2$).

Solution

The flat prior leaves the likelihood untouched, so

$$\theta | x \sim \mathcal{N}(x, 1).$$

The equal-tailed 95% credible interval is

$$[\theta^-, \theta^+] = x \pm z_{0.975} \approx [x - 2, x + 2].$$

Exercise 6.13

Sequential updating. Start from prior $\theta \sim \text{Beta}(1, 1)$. After observing a single head update the posterior, then update again after observing a single tail. Compare with the one-shot update using the data (1H, 1T).

Solution

Sequentially,

$$\text{Beta}(1, 1) \xrightarrow{1H} \text{Beta}(2, 1) \xrightarrow{1T} \text{Beta}(2, 2).$$

The one-shot update on the combined data ($s = 1, n = 2$) gives

$$\text{Beta}(1 + 1, 1 + 1) = \text{Beta}(2, 2).$$

Identical posteriors — conjugate updating is associative, as it must be by the product rule.

Exercise 6.14

Posterior predictive. Suppose $\theta | D \sim \text{Beta}(\alpha, \beta)$ and the next observation is $y \sim \text{Bernoulli}(\theta)$. Compute $P(y = 1 | D)$ in closed form.

Solution

By marginalising over θ ,

$$P(y = 1 | D) = E[\theta | D] = \frac{\alpha}{\alpha + \beta}.$$

For Bernoulli predictives the plug-in posterior-mean estimate and the proper posterior-predictive agree, because the model is linear in θ .

Exercise 6.15

One-sided credible bound. Let $\lambda \mid D \sim \text{Gamma}(a, b)$. Express the 95% upper credible bound in terms of the Gamma quantile function. For $a = 1$ (so the posterior is $\text{Exp}(b)$), give it in closed form.

Solution

The 95% upper credible bound solves $P(\lambda \leq \lambda^* \mid D) = 0.95$, so

$$\lambda^* = F_{\Gamma(a,b)}^{-1}(0.95).$$

When $a = 1$ the posterior is $\text{Exp}(b)$ with CDF $1 - e^{-b\lambda}$, hence

$$\lambda^* = -\frac{\log(0.05)}{b} = \frac{\log 20}{b} \approx \frac{2.996}{b}.$$

Exercise 6.16

Jeffreys prior for the binomial parameter $\theta \in (0, 1)$. Starting from the Fisher information $\mathcal{I}(\theta) = 1/[\theta(1 - \theta)]$, derive $p_J(\theta) \propto \theta^{-1/2}(1 - \theta)^{-1/2}$ and identify it as a Beta distribution.

Solution

By definition,

$$p_J(\theta) \propto \sqrt{\mathcal{I}(\theta)} = \frac{1}{\sqrt{\theta(1 - \theta)}} = \theta^{-1/2}(1 - \theta)^{-1/2}.$$

Comparing with the Beta kernel $\theta^{\alpha-1}(1 - \theta)^{\beta-1}$ identifies $\alpha = \beta = 1/2$, so

$$p_J(\theta) = \text{Beta}(\tfrac{1}{2}, \tfrac{1}{2}).$$

This is the arcsine distribution, with U-shaped density placing more mass near the boundaries.

Exercise 6.17

(Define and explain.) Define the posterior mean, MAP, posterior median, and posterior mode. State which of these point estimators are invariant under smooth reparametrisation of θ and which are not, with a one-line justification.

Solution

$\hat{\theta}_{\text{LMS}} = E[\theta \mid D]$ (squared-error optimum), $\hat{\theta}_{\text{med}}$ is the 0.5-quantile (absolute-loss optimum), $\hat{\theta}_{\text{MAP}} = \hat{\theta}_{\text{mode}} = \arg \max_{\theta} p(\theta \mid D)$ (0-1 loss optimum). Under $\psi = g(\theta)$:

$$\hat{\psi}_{\text{med}} = g(\hat{\theta}_{\text{med}}), \quad \hat{\psi}_{\text{LMS}} \neq g(\hat{\theta}_{\text{LMS}}), \quad \hat{\psi}_{\text{MAP}} \neq g(\hat{\theta}_{\text{MAP}}).$$

Only the median is fully invariant (quantiles transform with monotone maps); the mean is integral-based so $E[g(\theta)] \neq g(E\theta)$ generally; the MAP picks up the Jacobian $|d\theta/d\psi|$ which shifts the location of the peak.

Exercise 6.18

(Compare and contrast.) In one paragraph each, describe the operational meaning of a 95% Bayesian credible interval and of a 95% frequentist confidence interval for a Gaussian mean. Highlight where the randomness lives in each.

Solution

Bayesian. Conditional on the observed sample mean $\bar{x}$, prior, and model, $P(\mu \in [\bar{x} - 1.96\sigma/\sqrt{n}, \bar{x} + 1.96\sigma/\sqrt{n}] \mid D) = 0.95$. Randomness lives in the *parameter* μ , treated as a random variable with the posterior distribution; the interval is fixed once the data are seen.

Frequentist. The random interval $C(X) = [\bar{X} - 1.96\sigma/\sqrt{n}, \bar{X} + 1.96\sigma/\sqrt{n}]$ satisfies $P_\mu(\mu \in C(X)) = 0.95$ for every fixed μ . Randomness lives in the *data* X across hypothetical replications; the parameter is fixed but unknown. After observing the data the realised interval either contains μ or not, with no probability attached.

Exercise 6.19

(Define and explain.) Define the Jeffreys prior for a one-dimensional parameter θ . State and prove the property that distinguishes it from a flat prior. Give one concrete one-dimensional example.

Solution

The Jeffreys prior is $p_J(\theta) \propto \sqrt{\mathcal{I}(\theta)}$. It is **reparametrisation invariant**: under $\psi = \phi(\theta)$ with Jacobian $J = d\psi/d\theta$, $\mathcal{I}_\psi(\psi) = \mathcal{I}_\theta(\theta)/J^2$, so

$$p_J(\psi) = \sqrt{\mathcal{I}_\psi(\psi)} = \sqrt{\mathcal{I}_\theta(\theta)} |J|^{-1} = p_J(\theta) |d\theta/d\psi|,$$

which is exactly the change-of-variables formula for a density. A flat prior $p(\theta) \propto 1$ does *not* transform this way and so depends on the choice of parametrisation. **Example:** for $\text{Poisson}(\lambda)$, $\mathcal{I}(\lambda) = 1/\lambda$ gives $p_J(\lambda) \propto \lambda^{-1/2}$.

Exercise 6.20

(Show that.) Show that the posterior predictive variance is strictly larger than the plug-in predictive variance unless the posterior is a point mass. Use the law of total variance.

Solution

By the law of total variance,

$$\text{Var}(x_{\text{new}} \mid D) = \underbrace{E_\theta[\text{Var}(x_{\text{new}} \mid \theta) \mid D]}_{\text{aleatoric}} + \underbrace{\text{Var}_\theta(E[x_{\text{new}} \mid \theta] \mid D)}_{\text{epistemic}}.$$

The plug-in variance evaluates only the first term at $\hat{\theta}$, dropping the epistemic piece. Since $\text{Var}_\theta(\cdot) \geq 0$ with equality iff $E[x_{\text{new}} \mid \theta]$ is constant on the support of the posterior — equivalently iff the posterior is a point mass — the posterior predictive variance strictly exceeds the plug-in variance whenever there is residual parameter uncertainty.

6.14. Sources

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7. Bayesian inference for the Gaussian

7.1. Introduction

The Gaussian (or normal) distribution is the most frequently encountered model in statistical inference. Its ubiquity stems from the Central Limit Theorem, which guarantees that sample averages tend toward normality under weak conditions, and from its mathematical tractability—the Gaussian family is closed under many operations and admits conjugate priors. This chapter develops both Bayesian and frequentist approaches to inference of the mean μ and variance σ^2 of a Gaussian model, emphasizing the conjugate prior structure that yields closed-form posterior distributions.

We begin with the simplest case: inference of the mean with known variance. We then extend to variance estimation with known mean, and finally treat joint inference of both parameters using the Normal-Inverse-Gamma conjugate family. Throughout, we highlight credible intervals, hypothesis testing via the Bayes factor, and the Bayesian t -test, which naturally marginalizes over the unknown variance.

7.2. Gaussian Sampling Model and Sufficient Statistics

Definition 7.1. Let $D = \{x_1, \dots, x_n\}$ be n independent observations from a Gaussian distribution with pdf

$$p(x_i \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right).$$

The likelihood function for the data is the product

$$p(D \mid \mu, \sigma^2) = \prod_{i=1}^n p(x_i \mid \mu, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right).$$

Using the key identity

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2 = S + n(\bar{x} - \mu)^2,$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $S = \sum_{i=1}^n (x_i - \bar{x})^2$, the likelihood becomes

$$p(D \mid \mu, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left(-\frac{S + n(\bar{x} - \mu)^2}{2\sigma^2}\right).$$

Definition 7.2. The statistics $\bar{x}$ and S are **sufficient** for (μ, σ^2) : they capture all information in the data relevant to parameter inference. Any inference on (μ, σ^2) depends on the data only through these two summaries.

7.3. Inference of the Mean with Known Variance

Assume σ^2 is a known constant. The likelihood as a function of μ is

$$p(D \mid \mu, \sigma^2) \propto \exp\left(-\frac{n(\bar{x} - \mu)^2}{2\sigma^2}\right),$$

a Gaussian in μ centered at the sample mean $\bar{x}$. Two priors are natural here.

- **Uniform (Jeffreys) prior.** Assign $p(\mu) \propto 1$. This is also the Jeffreys prior for μ , since the Fisher information $\mathcal{I}(\mu) = n/\sigma^2$ does not depend on μ . By Bayes' theorem,

$$p(\mu \mid D, \sigma^2) \propto p(D \mid \mu, \sigma^2) \cdot p(\mu) \propto \exp\left(-\frac{n(\bar{x} - \mu)^2}{2\sigma^2}\right).$$

Theorem 7.1. Under the flat prior, the posterior distribution of μ is

$$\mu \mid D, \sigma^2 \sim \mathcal{N}\left(\bar{x}, \frac{\sigma^2}{n}\right).$$

The posterior mean equals the maximum likelihood estimate (MLE) $\hat{\mu}_{\text{MLE}} = \bar{x}$, and the posterior variance shrinks as $1/n$, reflecting increased confidence with larger sample size.

- **Conjugate Normal–Normal update.** To incorporate prior knowledge, use a Gaussian prior $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$. Multiplying likelihood by prior and completing the square gives

$$p(\mu \mid D, \sigma^2) \propto \exp\left(-\frac{n(\bar{x} - \mu)^2}{2\sigma^2}\right) \cdot \exp\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right) \propto \exp\left(-\frac{\kappa_n(\mu - \mu_n)^2}{2}\right).$$

Definition 7.3. Normal–Normal Conjugate Update:

$$\kappa_n = \frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}, \tag{7.1}$$

$$\mu_n = \frac{\mu_0/\sigma_0^2 + n\bar{x}/\sigma^2}{\kappa_n} = \frac{\mu_0/\sigma_0^2 + n\bar{x}/\sigma^2}{1/\sigma_0^2 + n/\sigma^2}. \tag{7.2}$$

The posterior is $\mu \mid D, \sigma^2 \sim \mathcal{N}(\mu_n, \kappa_n^{-1})$. **Interpretation:** precisions add; the posterior mean is a precision-weighted average of the prior mean and the sample mean. As $n \rightarrow \infty$ the posterior is dominated by the data ($\mu_n \rightarrow \bar{x}$); for a weak prior ($\sigma_0^2 \rightarrow \infty$), $\kappa_n \rightarrow n/\sigma^2$ and $\mu_n \rightarrow \bar{x}$, recovering the uniform-prior result.

Example 7.1 (Normal–Normal update: small sample with informative prior (exam-style))

Measurements have known variance $\sigma^2 = 4$. The sample mean from $n = 9$ measurements is $\bar{x} = 3$. A prior for the mean is $\mu \sim \mathcal{N}(0, 1)$. Compute the posterior mean and variance, and compare with the flat-prior posterior mean.

Solution. With $\sigma_0^2 = 1$ and $n/\sigma^2 = 9/4$, the conjugate posterior precision and variance are

$$\kappa_n = \frac{1}{\sigma_0^2} + \frac{n}{\sigma^2} = 1 + \frac{9}{4} = \frac{13}{4}, \quad V_n = \kappa_n^{-1} = \frac{4}{13}.$$

The posterior mean is the precision-weighted average

$$\mu_n = V_n \left(\frac{\mu_0}{\sigma_0^2} + \frac{n\bar{x}}{\sigma^2} \right) = \frac{4}{13} \left(0 + \frac{9 \cdot 3}{4} \right) = \frac{27}{13} \approx 2.08.$$

With the flat (improper) prior the posterior would be $\mu \mid D \sim \mathcal{N}(\bar{x}, \sigma^2/n) = \mathcal{N}(3, 4/9)$, so the posterior mean is just $\bar{x} = 3$. The proper prior shrinks the estimate from 3 toward the prior mean 0, by a factor controlled by the ratio of prior to data precision. With only $n = 9$ samples the prior carries non-trivial weight; with $n \rightarrow \infty$ the shrinkage vanishes.

7.4. Inference of the Variance with Known Mean

Now assume μ is known and infer σ^2 . Setting $\nu = \sum_i (x_i - \mu)^2$, the likelihood is

$$p(D \mid \mu, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left(-\frac{\nu}{2\sigma^2}\right).$$

- **Jeffreys prior for a scale parameter.** Variance is a **scale parameter**: if we rescale $x \rightarrow ax$, then $\sigma^2 \rightarrow a^2\sigma^2$. As shown in Chapter 5 (§ Fisher information, Cramér–Rao, and the Jeffreys prior), the Jeffreys prior $p_J(\theta) \propto \sqrt{\det \mathcal{I}(\theta)}$ is reparametrization-invariant. For the Gaussian scale parameter σ^2 with known mean, the Fisher information is $\mathcal{I}(\sigma^2) = n/[2(\sigma^2)^2]$, yielding $p_J(\sigma^2) \propto 1/\sigma^2$ (or equivalently $p_J(\log \sigma^2) \propto 1$, uniform on log-scale). This prior ensures invariance under rescaling — critical for Bayes factors. Chapter 5 gives the extended example and invariance proof; here we apply it directly.
- **Inverse-Gamma conjugate family.** The Inverse-Gamma distribution is conjugate for variance estimation.

Definition 7.4. If $X \sim \text{Gamma}(\alpha, \beta)$ with pdf $f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$, then $1/X \sim \text{Inv-Gamma}(\alpha, \beta)$ with pdf

$$f(z) = \frac{\beta^\alpha}{\Gamma(\alpha)} (1/z)^{\alpha+1} \exp(-\beta/z).$$

A prior $\sigma^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0)$ combined with the likelihood yields the conjugate posterior.

Theorem 7.2. Inverse-Gamma Conjugate Update:

$$\sigma^2 \mid D, \mu \sim \text{Inv-Gamma}\left(\alpha_0 + \frac{n}{2}, \beta_0 + \frac{\nu}{2}\right),$$

where $\nu = \sum_i (x_i - \mu)^2$.

The Jeffreys prior corresponds to $\alpha_0 = 0$ and $\beta_0 = 0$ (or $\alpha_0 = -1/2$ to ensure integrability). The posterior mean is $\mathbb{E}[\sigma^2 \mid D] = \beta_n/(\alpha_n - 1)$ for $\alpha_n > 1$.

7.5. Joint Inference: Normal-Inverse-Gamma Model

The most realistic case has both μ and σ^2 unknown. Use a factorized prior

$$p(\mu, \sigma^2) = p(\mu \mid \sigma^2)p(\sigma^2),$$

with $\mu \mid \sigma^2 \sim \mathcal{N}(\mu_0, \sigma^2/\kappa_0)$ and $\sigma^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0)$. This is the **Normal-Inverse-Gamma (NIG)** conjugate family with hyperparameters $(\mu_0, \kappa_0, \alpha_0, \beta_0)$.

- **Posterior update.** Combining likelihood with the NIG prior yields a posterior of the same form, with updated hyperparameters.

Definition 7.5. Normal-Inverse-Gamma Conjugate Update:

$$\kappa_n = \kappa_0 + n, \quad (7.3)$$

$$\mu_n = \frac{\kappa_0 \mu_0 + n \bar{x}}{\kappa_n}, \quad (7.4)$$

$$\alpha_n = \alpha_0 + \frac{n}{2}, \quad (7.5)$$

$$\beta_n = \beta_0 + \frac{S}{2} + \frac{\kappa_0 n}{2(\kappa_0 + n)} (\bar{x} - \mu_0)^2, \quad (7.6)$$

where $S = \sum_i (x_i - \bar{x})^2$.

The joint posterior factorizes as

$$p(\mu, \sigma^2 \mid D) \propto (\sigma^2)^{-(\alpha_n+1)} \exp \left(-\frac{\beta_n}{\sigma^2} - \frac{\kappa_n (\mu - \mu_n)^2}{2\sigma^2} \right).$$

- **Marginal posterior of the mean.** Marginalizing σ^2 yields a Student- t posterior for μ .

Theorem 7.3. *The posterior marginal distribution of μ given D is*

$$\mu \mid D \sim t_{2\alpha_n} \left(\mu_n, \frac{\beta_n}{\alpha_n \kappa_n} \right),$$

a Student- t distribution with $\nu = 2\alpha_n$ degrees of freedom, location μ_n , and scale parameter $\sqrt{\beta_n/(\alpha_n \kappa_n)}$.

Interpretation: the heavy tails of the t -distribution reflect additional uncertainty from not knowing σ^2 . For large n , $\nu = 2\alpha_n \approx n \rightarrow \infty$, so the t -distribution converges to a Gaussian, recovering asymptotic normality.

7.6. Credible Intervals and Regions

- **Central credible interval for the mean.** A central $(1 - \alpha)\%$ credible interval is

$$\mu_n \pm t_{2\alpha_n, 1-\alpha/2} \sqrt{\frac{\beta_n}{\alpha_n \kappa_n}},$$

where $t_{\nu, p}$ denotes the p -quantile of a Student- t distribution with ν degrees of freedom. **Frequentist analogue:** in the large-sample limit, this recovers the familiar confidence interval $\hat{\mu} \pm z_{1-\alpha/2} \sqrt{\hat{\sigma}^2/n}$.

- **Joint credible region.** A joint $(1 - \alpha)\%$ credible region for $\theta = (\mu, \sigma^2)$ is the ellipse

$$(\theta - \hat{\theta})^T \mathcal{J}_n(\hat{\theta})(\theta - \hat{\theta}) \leq \chi_{2, 1-\alpha}^2,$$

where $\mathcal{J}_n(\hat{\theta})$ is the Fisher information matrix evaluated at the MLE and $\chi_{2, 1-\alpha}^2$ is the $(1 - \alpha)$ quantile of the chi-squared distribution with 2 degrees of freedom.

The posterior asymptotically factorizes: the estimated correlation between μ and σ^2 vanishes as $n \rightarrow \infty$. This convergence to Gaussian form and factorization is guaranteed by the **Bernstein–von Mises theorem** (Chapter 5, § Fisher information, Cramér–Rao, and the Jeffreys prior), which states that the posterior becomes approximately $\mathcal{N}(\hat{\theta}_{\text{ML}}, \mathcal{I}_n(\theta_*)^{-1})$ for large n , independently of the prior. This is why Bayesian credible intervals and frequentist confidence intervals asymptotically agree.

7.7. Bayesian Hypothesis Testing

- **One-sided test for the mean.** Consider $H_0 : \mu \leq \mu^*$ vs $H_1 : \mu > \mu^*$. The posterior probability of H_1 is

$$P(H_1 | D) = 1 - F_t \left(\frac{\mu^* - \mu_n}{\sqrt{\beta_n / (\alpha_n \kappa_n)}} \right),$$

where F_t is the CDF of a Student- t distribution with $2\alpha_n$ degrees of freedom. Decision rule: reject H_0 if $P(H_1 | D) > 1 - \alpha$. Equivalently, reject if μ^* lies below the lower bound of the $(1 - \alpha)\%$ credible interval.

- **Two-sided test and Bayes factor.** For $H_0 : \mu = \mu^*$ vs $H_1 : \mu \neq \mu^*$, compare the marginal likelihoods.

Definition 7.6. The **Bayes factor** of H_1 over H_0 is

$$\text{BF}_{10} = \frac{p(D | H_1)}{p(D | H_0)} = \frac{\int p(D | \mu, \sigma^2) p(\mu, \sigma^2 | H_1) d\mu d\sigma^2}{\int_{\mu=\mu^*} p(D | \mu^*, \sigma^2) p(\sigma^2 | H_0) d\sigma^2}.$$

Interpretation: $\text{BF}_{10} > 1$ favors H_1 (mean differs from μ^*); $\text{BF}_{10} < 1$ favors H_0 ; the magnitude quantifies the strength of evidence. Unlike frequentist p -values, the Bayes factor is invariant under the choice of scale (the Jeffreys prior ensures this).

- **Bayesian t -test.** The Bayesian t -test compares the observed $t = (\bar{x} - \mu^*) / \sqrt{S / (n(n-1))}$ to its posterior predictive distribution under H_0 . With the NIG conjugate prior, the marginal likelihood under H_0 is a t -distribution, naturally accounting for variance uncertainty without ad-hoc degrees-of-freedom corrections.

7.8. Posterior Predictive Distribution

Given posterior $p(\mu, \sigma^2 | D)$, the posterior predictive distribution for a new observation $\tilde{x}$ is

$$p(\tilde{x} | D) = \int p(\tilde{x} | \mu, \sigma^2) p(\mu, \sigma^2 | D) d\mu d\sigma^2.$$

For the NIG model, this marginalizes to a Student- t distribution with the same degrees of freedom and location as the posterior for μ , but with inflated scale (reflecting both parameter and observational uncertainty):

$$\tilde{x} | D \sim t_{2\alpha_n} \left(\mu_n, \frac{\beta_n}{\alpha_n} \left(1 + \frac{1}{\kappa_n} \right) \right).$$

The posterior predictive intervals are wider than credible intervals for the mean, appropriately quantifying full predictive uncertainty.

7.9. Frequentist Contrast

The frequentist approach uses the **sample t -statistic**

$$t = \frac{\bar{x} - \mu}{\sqrt{S/(n(n-1))}},$$

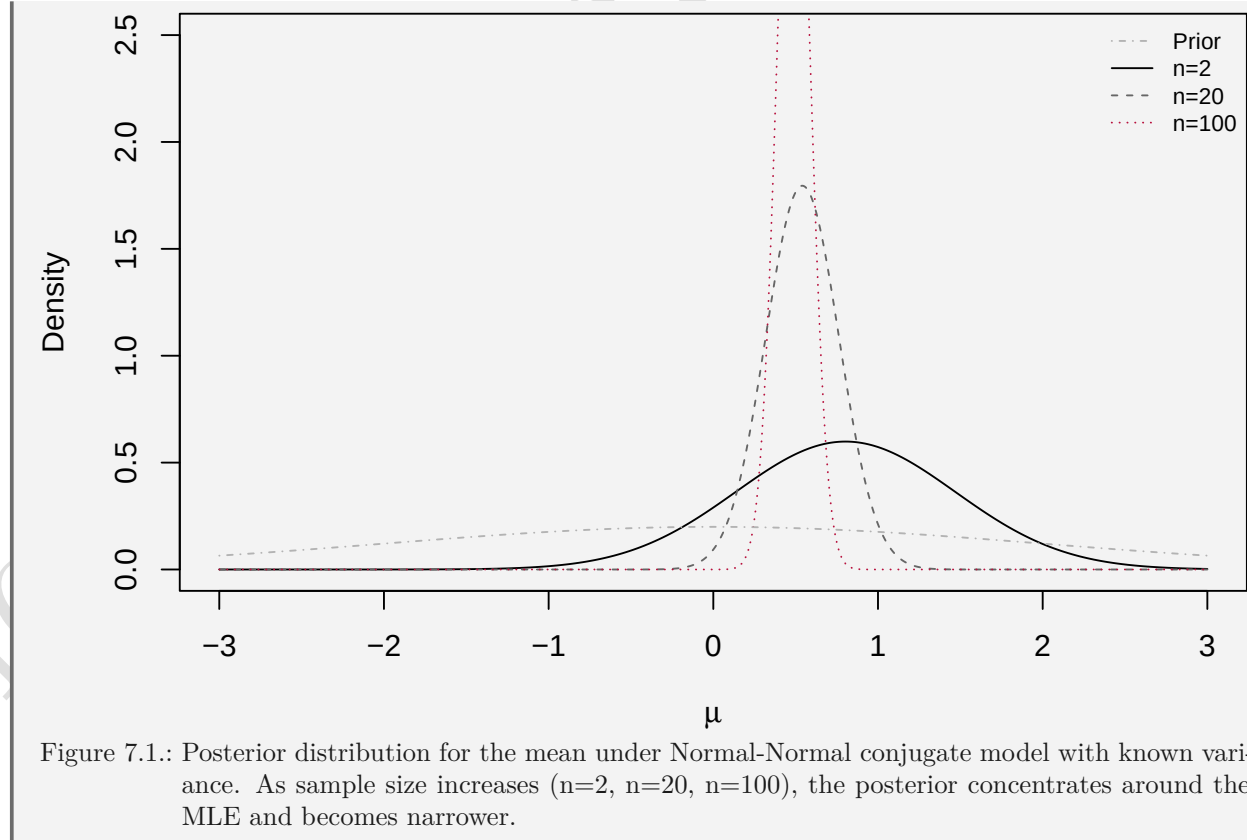
which under $H_0 : \mu = \mu^*$ follows t_{n-1} . Rejection is based on p -value: $p = P(|T| \geq |t_{\text{obs}}| \mid H_0)$.

Key differences from Bayes:

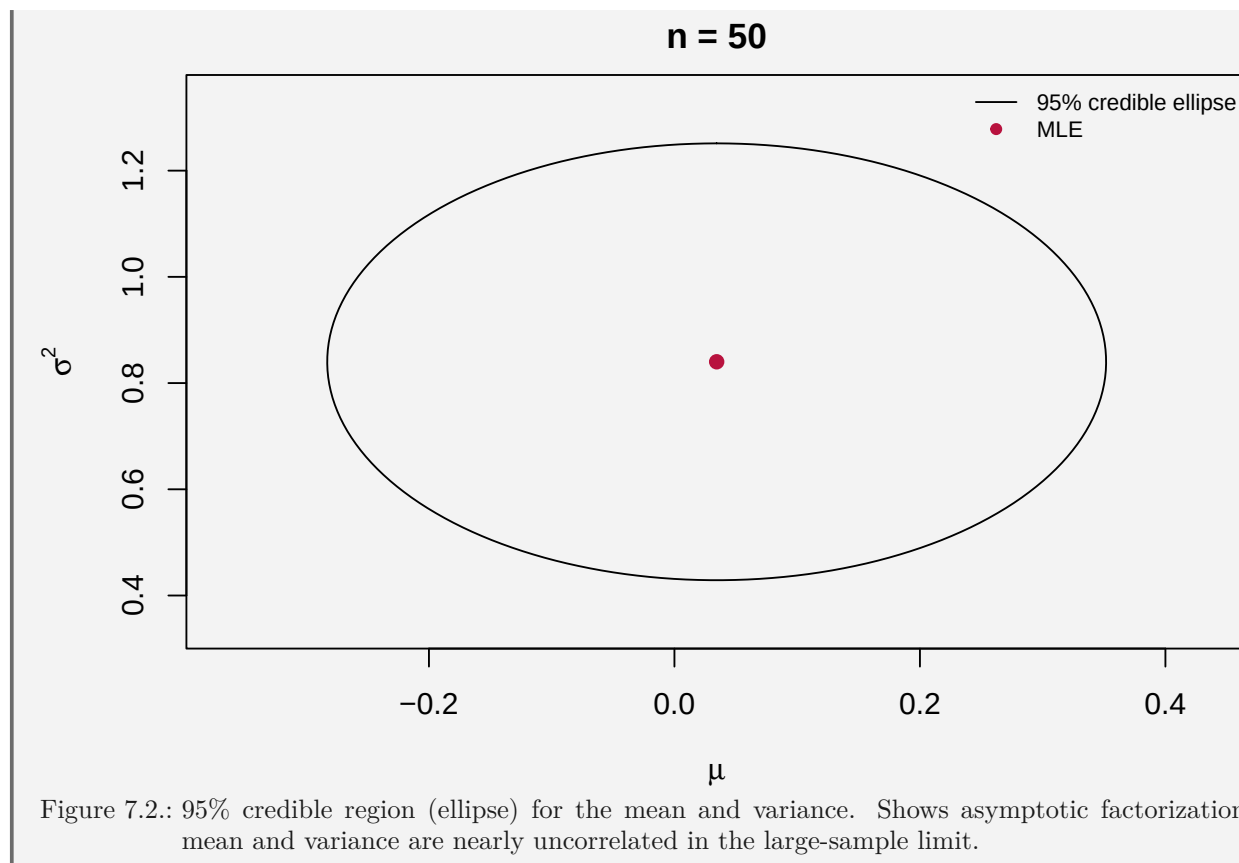
1. **Frequentist t -test:** assumes μ^* exactly true under H_0 ; p -value is the probability of observing data as extreme or more extreme under repeated sampling.
2. **Bayesian t -test:** posterior probability that $\mu > \mu^*$ (or any specified region); updates beliefs given observed data; allows flexible loss functions for decision-making.
3. **Sample size:** Frequentist p -values can remain small for minor effects with large n . Bayes factors adapt: as data accumulate under H_1 , they grow; but if data weakly support H_1 relative to prediction, BF remains modest.
4. **Asymptotic equivalence:** For fixed effect size and large n , frequentist confidence intervals and Bayesian credible intervals converge; the posterior becomes approximately Gaussian by Bernstein–von Mises.

7.10. Illustrative Examples

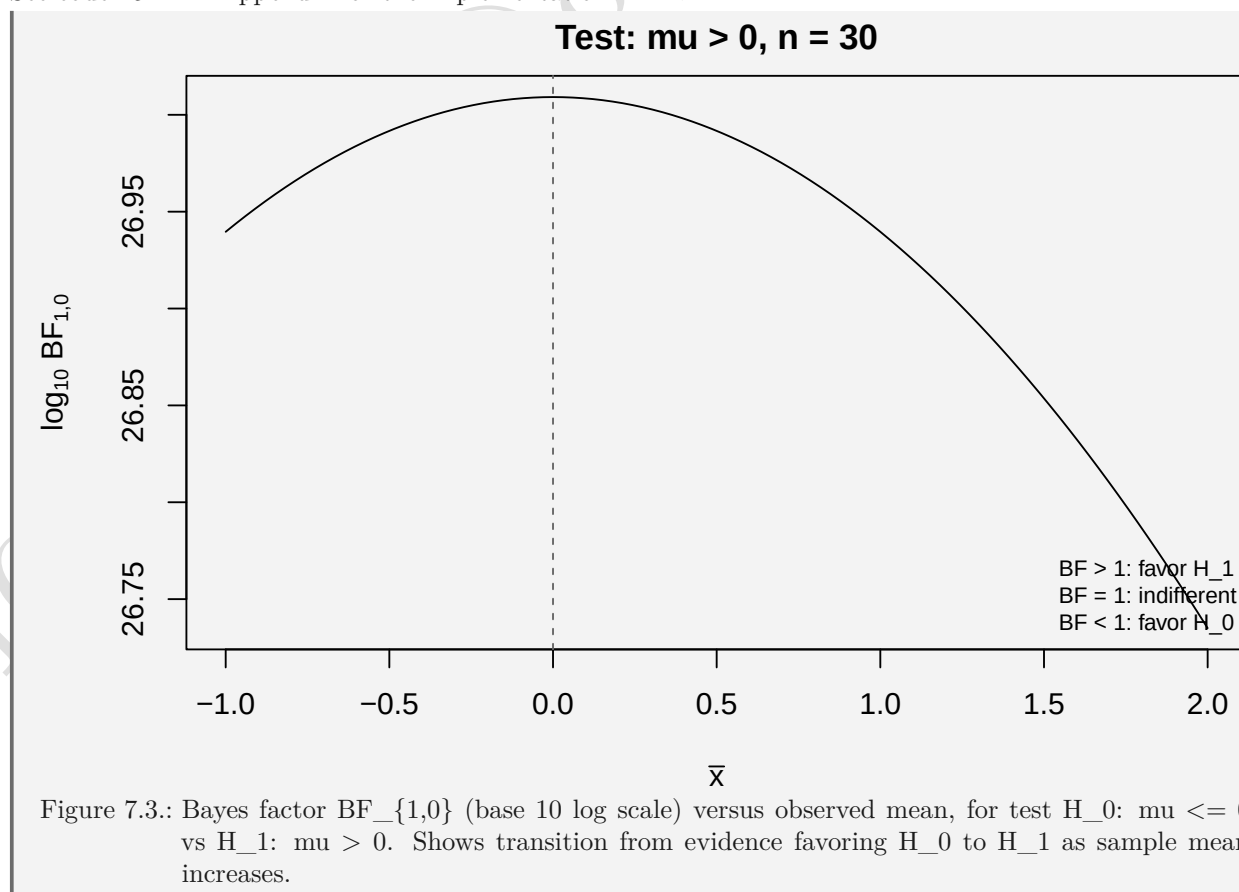
See code 15.20 in Appendix for the implementation.



See code 15.21 in Appendix for the implementation.



See code 15.22 in Appendix for the implementation.



7.11. Exercises

Exercise 7.1

Conjugate Normal–Normal model with known variance. Observe $n = 4$ measurements with sample mean $\bar{x} = 2$ from $\mathcal{N}(\mu, \sigma^2 = 1)$, with prior $\mu \sim \mathcal{N}(0, 1)$. Compute the posterior mean and variance.

Solution

Precisions add: $\kappa_n = 1/\sigma_0^2 + n/\sigma^2 = 1 + 4 = 5$, so $V_n = \kappa_n^{-1} = 1/5$. The posterior mean is the precision-weighted average

$$\mu_n = V_n \left(\frac{\mu_0}{\sigma_0^2} + \frac{n\bar{x}}{\sigma^2} \right) = \frac{1}{5} (0 + 4 \cdot 2) = \frac{8}{5}.$$

Hence $\mu \mid D \sim \mathcal{N}(8/5, 1/5)$.

Exercise 7.2

Same model as above. As $n \rightarrow \infty$ with $\bar{x} \rightarrow \mu_*$ fixed, what does the posterior tend to? State the result in terms of μ_* and σ^2 .

Solution

The data precision n/σ^2 dominates the prior precision $1/\sigma_0^2$, so $\kappa_n \rightarrow n/\sigma^2$ and $\mu_n \rightarrow \bar{x} \rightarrow \mu_*$. Therefore

$$\mu \mid D \xrightarrow{n \rightarrow \infty} \mathcal{N}\left(\mu_*, \frac{\sigma^2}{n}\right),$$

which collapses to a point mass at μ_* and matches the flat-prior result, illustrating Bernstein–von Mises.

Exercise 7.3

Inference of the variance with known mean. Observe n i.i.d. samples from $\mathcal{N}(0, \sigma^2)$ with sum of squares $S = \sum_i x_i^2$. Use the conjugate Inverse-Gamma prior $\sigma^2 \sim \text{IG}(\alpha_0, \beta_0)$ and derive the posterior.

Solution

The likelihood is $p(D \mid \sigma^2) \propto (\sigma^2)^{-n/2} \exp(-S/(2\sigma^2))$, and the prior is $p(\sigma^2) \propto (\sigma^2)^{-(\alpha_0+1)} \exp(-\beta_0/\sigma^2)$. Multiplying gives an IG kernel with

$$\sigma^2 \mid D \sim \text{IG}\left(\alpha_0 + \frac{n}{2}, \beta_0 + \frac{S}{2}\right) = \text{IG}(\alpha_n, \beta_n).$$

Exercise 7.4

Show that $(\bar{x}, s^2) = (\frac{1}{n} \sum_i x_i, \frac{1}{n-1} \sum_i (x_i - \bar{x})^2)$ are sufficient statistics for (μ, σ^2) in the i.i.d. $\mathcal{N}(\mu, \sigma^2)$ model, using the factorisation theorem.

Solution

Using $\sum_i (x_i - \mu)^2 = (n-1)s^2 + n(\bar{x} - \mu)^2$, the joint density factorises as

$$p(D \mid \mu, \sigma^2) = \underbrace{(2\pi\sigma^2)^{-n/2} \exp\left(-\frac{(n-1)s^2 + n(\bar{x} - \mu)^2}{2\sigma^2}\right)}_{g(\bar{x}, s^2; \mu, \sigma^2)} \cdot \frac{1}{h(D)}.$$

Since the data enter only through $(\bar{x}, s^2)$, the Fisher–Neyman factorisation theorem gives sufficiency.

Exercise 7.5

Conjugate Normal–Inverse–Gamma joint posterior. Under prior $(\mu, \sigma^2) \sim \text{NIG}(\mu_0, \kappa_0, \alpha_0, \beta_0)$, write the posterior parameters $(\mu_n, \kappa_n, \alpha_n, \beta_n)$ after observing n samples with sample mean $\bar{x}$ and sum of squares from $\bar{x}$ equal to S_{xx} .

Solution

Conjugacy preserves the NIG form with updates

$$\kappa_n = \kappa_0 + n, \quad \mu_n = \frac{\kappa_0 \mu_0 + n\bar{x}}{\kappa_n}, \quad (7.7)$$

$$\alpha_n = \alpha_0 + \frac{n}{2}, \quad \beta_n = \beta_0 + \frac{S_{xx}}{2} + \frac{\kappa_0 n}{2(\kappa_0 + n)} (\bar{x} - \mu_0)^2. \quad (7.8)$$

The last term in β_n penalises prior–data disagreement between μ_0 and $\bar{x}$.

Exercise 7.6

For the NIG conjugate model, derive the marginal posterior $p(\mu \mid D)$ and identify it as a Student- t distribution. State its location, scale, and degrees of freedom.

Solution

Integrating σ^2 out of the joint NIG posterior gives an IG integral in σ^2 with shape $\alpha_n + 1/2$ and rate $\beta_n + \kappa_n(\mu - \mu_n)^2/2$, leading to

$$p(\mu \mid D) \propto \left[1 + \frac{\kappa_n(\mu - \mu_n)^2}{2\beta_n} \right]^{-(\alpha_n + 1/2)},$$

i.e. a Student- t with $\nu = 2\alpha_n$ degrees of freedom, location μ_n , and squared scale $\beta_n/(\alpha_n \kappa_n)$.

Exercise 7.7

Posterior predictive in the conjugate Normal model with known σ^2 . Derive $p(x_{\text{new}} \mid D)$ when $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$ and the data have sample mean $\bar{x}$.

Solution

With posterior $\mu \mid D \sim \mathcal{N}(\mu_n, V_n)$ where $V_n = \kappa_n^{-1}$, and $x_{\text{new}} \mid \mu \sim \mathcal{N}(\mu, \sigma^2)$, the sum of independent Gaussians gives

$$x_{\text{new}} \mid D \sim \mathcal{N}(\mu_n, V_n + \sigma^2).$$

Parameter uncertainty contributes V_n and irreducible noise contributes σ^2 .

Exercise 7.8

For the conjugate model with known σ^2 , derive the 95% equal-tailed credible interval for μ using the conjugate posterior parameters (μ_n, V_n) . Use $z_{0.975} \approx 2$.

Solution

Since $\mu \mid D \sim \mathcal{N}(\mu_n, V_n)$, the equal-tailed 95% interval is

$$\mu_n \pm z_{0.975} \sqrt{V_n} \approx \mu_n \pm 2\sqrt{V_n}.$$

Exercise 7.9

Sequential updating. With known σ^2 and prior $\mu \sim \mathcal{N}(0, \tau^2)$, you observe data in two batches with sample means $\bar{x}_1, \bar{x}_2$ and counts n_1, n_2 . Show that updating sequentially gives the same posterior as updating in one batch.

Solution

After batch 1: $\kappa_1 = 1/\tau^2 + n_1/\sigma^2$ and $\mu_1 = \kappa_1^{-1}(n_1\bar{x}_1/\sigma^2)$. Using this as prior for batch 2:

$$\kappa_n = \kappa_1 + n_2/\sigma^2 = \frac{1}{\tau^2} + \frac{n_1 + n_2}{\sigma^2}, \quad \mu_n = \kappa_n^{-1} \left(\kappa_1 \mu_1 + \frac{n_2 \bar{x}_2}{\sigma^2} \right) = \kappa_n^{-1} \left(\frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{\sigma^2} \right),$$

which is identical to the one-shot update with pooled mean $(n_1\bar{x}_1 + n_2\bar{x}_2)/(n_1 + n_2)$ and count $n_1 + n_2$. Precisions and precision-weighted means simply concatenate.

Exercise 7.10

Posterior predictive variance decomposition. Show that the posterior predictive variance in the conjugate Normal model is $V_n + \sigma^2$, and identify which term is parameter uncertainty and which is irreducible noise.

Solution

By the law of total variance,

$$\text{Var}(x_{\text{new}} \mid D) = \mathbb{E}_\mu[\text{Var}(x_{\text{new}} \mid \mu)] + \text{Var}_\mu[\mathbb{E}(x_{\text{new}} \mid \mu)] = \sigma^2 + V_n.$$

Here σ^2 is the irreducible observation noise and $V_n = \kappa_n^{-1}$ is the posterior (parameter) uncertainty about μ .

Exercise 7.11

Multivariate conjugate update. For $\mathbf{x} \sim \mathcal{N}(\mu, \Sigma)$ with Σ known and prior $\mu \sim \mathcal{N}(\mu_0, \Sigma_0)$, write the posterior $p(\mu \mid D)$ after observing one sample $\mathbf{x}$.

Solution

Precision matrices add and the posterior mean is precision-weighted:

$$\Sigma_n^{-1} = \Sigma_0^{-1} + \Sigma^{-1}, \quad \mu_n = \Sigma_n (\Sigma_0^{-1} \mu_0 + \Sigma^{-1} \mathbf{x}),$$

so $\mu \mid D \sim \mathcal{N}(\mu_n, \Sigma_n)$.

Exercise 7.12

Bayesian z -test. With prior $\mu \sim \mathcal{N}(0, \tau^2)$ and known σ^2 , compute the Bayes factor between $H_0 : \mu = 0$ and $H_1 : \mu \sim \mathcal{N}(0, \tau^2)$ as a function of the standardised sample mean $z = \bar{x}\sqrt{n}/\sigma$. (You can reuse the worked example “Closed-form Bayes factor for a Gaussian mean” in Chapter 4.)

Solution

Let $g = n\tau^2/\sigma^2$ measure prior relative to data precision. The marginal likelihood under H_1 inflates the data variance by the prior, while H_0 uses just σ^2/n . Standard Gaussian marginalisation (Ch. 4) gives

$$\text{BF}_{10} = \frac{1}{\sqrt{1+g}} \exp\left(\frac{g}{1+g} \cdot \frac{z^2}{2}\right).$$

Large $|z|$ exponentially favours H_1 ; the prefactor $(1+g)^{-1/2}$ is the Ockham penalty for the broader alternative.

Exercise 7.13

Improper Jeffreys prior for the variance. Take $p(\sigma^2) \propto 1/\sigma^2$. Show that the posterior for σ^2 given n i.i.d. $\mathcal{N}(0, \sigma^2)$ samples is Inverse-Gamma, and identify its parameters.

Solution

The Jeffreys prior is the formal limit $\alpha_0 \rightarrow 0$, $\beta_0 \rightarrow 0$ of $\text{IG}(\alpha_0, \beta_0)$. Multiplying by the likelihood $\propto (\sigma^2)^{-n/2} \exp(-S/(2\sigma^2))$ with $S = \sum_i x_i^2$ gives

$$\sigma^2 \mid D \sim \text{IG}(\alpha_n = \frac{n}{2}, \beta_n = \frac{S}{2}),$$

which is proper for $n \geq 1$.

Exercise 7.14

Posterior comparison. With $n = 9$, $\sigma^2 = 4$, $\bar{x} = 3$: compute the posterior mean for μ under the flat prior, the $\mathcal{N}(0, 1)$ prior, and the $\mathcal{N}(0, 100)$ prior. Comment on how the posterior depends on prior width.

Solution

Data precision is $n/\sigma^2 = 9/4 = 2.25$. The precision-weighted-mean form $\mu_n = (n\bar{x}/\sigma^2)/(1/\sigma_0^2 + n/\sigma^2)$ gives

$$\mu_n^{\text{flat}} = \bar{x} = 3, \quad \mu_n^{(\sigma_0^2=1)} = \frac{27/4}{1 + 9/4} = \frac{27}{13} \approx 2.08, \quad \mu_n^{(\sigma_0^2=100)} = \frac{27/4}{0.01 + 9/4} \approx 2.99.$$

A tight prior ($\sigma_0^2 = 1$) shrinks the estimate sharply toward $\mu_0 = 0$; a vague prior ($\sigma_0^2 = 100$) recovers the flat-prior MLE to within 1%. Posterior dependence on prior width vanishes as $\sigma_0^2 \rightarrow \infty$.

Exercise 7.15

Sample size and prior dominance. For the conjugate Normal–Normal model with prior precision τ_0 and likelihood precision n/σ^2 , find the sample size n^* such that the posterior is half-determined by the prior and half by the data.

Solution

Equal weighting means prior precision equals data precision: $\tau_0 = n^*/\sigma^2$, so

$$n^* = \tau_0 \sigma^2.$$

For $n \ll n^*$ the prior dominates, while for $n \gg n^*$ the likelihood dominates and the posterior approaches the flat-prior result.

7.12. Sources

Sivia, D.S. & Skilling, J. (2006). *Data Analysis: A Bayesian Tutorial* (2nd ed.). Oxford University Press. Ch. 3 (Gaussian estimation).

van der Linden, W.J. (2016). *Bayesian Assessment and Reduction of Uncertainty in Measurement*. Springer. Ch. 17 (Bayesian z-test, t-test).

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Part III.

PART III: MONTE CARLO METHODS

8. Monte Carlo methods

8.1. Motivation and the curse of dimensionality

In Bayesian inference, the central computational challenge is to evaluate expectations of the form

$$\mathbb{E}_\pi[f(\theta)] = \int f(\theta)\pi(\theta | D) d\theta,$$

where $\pi(\theta | D)$ is the posterior distribution. These integrals appear everywhere: computing the evidence, credible intervals, point estimates, and predictions. Most realistic models lack analytical solutions, especially in high dimensions.

Classical numerical integration methods fail catastrophically in high dimensions. Consider a grid-based quadrature with n points per dimension in a d -dimensional space. The total number of grid points grows as $N = n^d$ —exponential in dimension. This is the **curse of dimensionality** for deterministic methods.

Example 8.1

Consider the integral over a unit hypercube $[0, 1]^d$ with an indicator function that counts volume inside a ball of radius $r = 1/2$ centered at the origin:

$$I_d = \int_{[0,1]^d} \mathbb{1}_{\|x\| < 1/2} dx.$$

The volume of the region is proportional to r^d while the hypercube has volume 1. As d grows, nearly all of the hypercube's volume lies outside the ball (since $V_{\text{ball}}/V_{\text{cube}} \sim (1/2)^d \rightarrow 0$). A uniform grid wastes almost all points. Even with $N = 10^6$ samples in dimension $d = 20$, we expect essentially zero hits, making the variance unbounded and estimation impossible.

Monte Carlo to the rescue. Monte Carlo methods replace high-dimensional integration with **averaging over random samples**. The fundamental principle is

$$\mathbb{E}_p[f(X)] \approx \frac{1}{N} \sum_{i=1}^N f(x_i), \quad x_i \sim p(x).$$

By the law of large numbers, this average converges to the true expectation as $N \rightarrow \infty$. Crucially, the error does **not depend explicitly on dimension**.

Theorem 8.1. *Let $x_1, \dots, x_N \sim p(x)$ be i.i.d. samples. The Monte Carlo estimator*

$$\hat{I}_N = \frac{1}{N} \sum_{i=1}^N f(x_i)$$

satisfies: - **Unbiasedness:** $\mathbb{E}[\hat{I}_N] = \mathbb{E}_p[f(X)]$ - **Variance:** $\text{Var}(\hat{I}_N) = \frac{1}{N} \text{Var}_p(f)$ - **Convergence:** $\hat{I}_N - \mathbb{E}_p[f] = \mathcal{O}(N^{-1/2})$

The error scales universally as $N^{-1/2}$ regardless of dimension d .

8. Monte Carlo methods

However, there is a catch: while the error rate is independent of dimension, the variance $\text{Var}_p(f)$ itself can grow exponentially with d . For the hypercube example, the variance approaches zero as $d \rightarrow \infty$ (since $I_d \rightarrow 0$), but the *relative* error

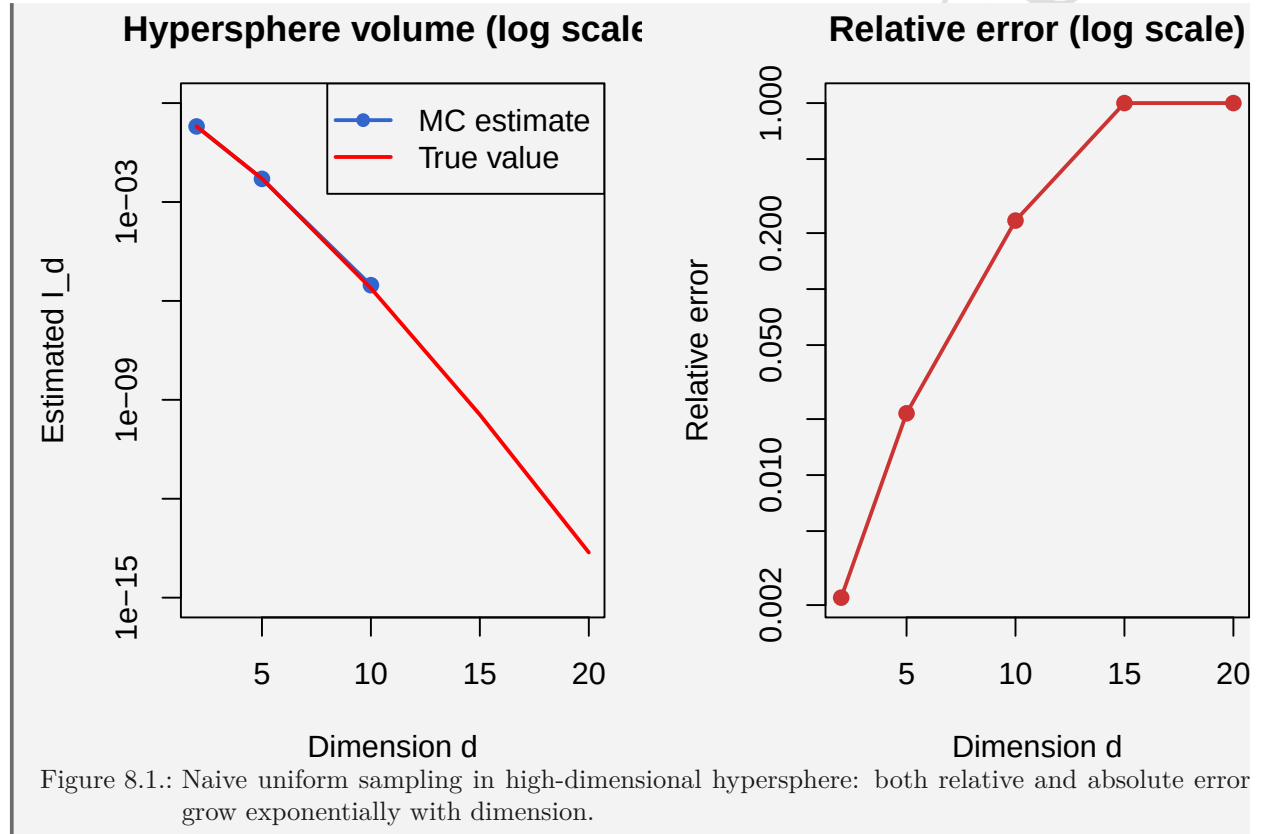
$$\frac{\sqrt{\text{Var}(\hat{I}_N)}}{\mathbb{E}[\hat{I}_N]} \sim \frac{1}{\sqrt{NI_d}} \rightarrow \infty$$

explodes exponentially. We need $N \sim 1/I_d$ to maintain constant relative error—which is infeasible for rare events. Techniques like **rejection sampling** and **importance sampling** exist precisely to mitigate this degeneracy by concentrating samples where the integrand is large.

See code 15.23 in Appendix for the implementation.

```
Warning in xy.coords(x, y, xlabel, ylabel, log): 2 y values <= 0 omitted from
logarithmic plot
```

See code 15.24 in Appendix for the implementation.



Example 8.2 (Why Monte Carlo in Bayesian inference (exam-style))

Explain why Monte Carlo methods are central in Bayesian inference. What is the difference between using independent posterior samples and samples generated by a Markov chain?

Solution. Bayesian inference is built on posterior expectations,

$$E[f(\theta) \mid D] = \int f(\theta) p(\theta \mid D) d\theta.$$

In moderate to high dimensions, deterministic grid integration becomes impractical because the number of grid points grows exponentially with the parameter dimension (curse of dimensionality).

Monte Carlo replaces the integral by an average over samples:

$$E[f(\theta) \mid D] \approx \frac{1}{N} \sum_{i=1}^N f(\theta_i), \quad \theta_i \sim p(\theta \mid D).$$

If the samples θ_i are *independent*, the variance of the estimator scales as $\text{Var}(f)/N$. With **MCMC** the samples are correlated by construction, so the variance is inflated by the integrated autocorrelation time τ_{int} :

$$\text{Var}(\hat{I}_N) \approx \frac{\tau_{\text{int}}}{N} \text{Var}_{\pi}(f).$$

The relevant sample size is therefore the **effective sample size** $N_{\text{eff}} \approx N/\tau_{\text{int}}$, not the raw chain length. MCMC remains valid after burn-in and convergence to the stationary distribution, but reporting N_{eff} — not N — is essential.

8.2. Direct sampling: inverse transform method

The most straightforward sampling approach is the **inverse transform method** (or inverse CDF method). It is exact and works whenever the CDF can be computed and inverted analytically.

Definition 8.1. Let $F(x) = P(X \leq x) = \int_{-\infty}^x p(x') dx'$ be the cumulative distribution function of a continuous random variable $X \sim p(x)$. The **inverse transform method** generates samples by: 1. Compute $F(x)$ and its inverse $F^{-1}(u)$ (the quantile function). 2. Sample $U \sim \text{Uniform}(0, 1)$. 3. Set $X = F^{-1}(U)$.

Then $X \sim p(x)$.

Proof: By the probability integral transform,

$$P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x),$$

so the distribution of X is $p(x)$ as required.

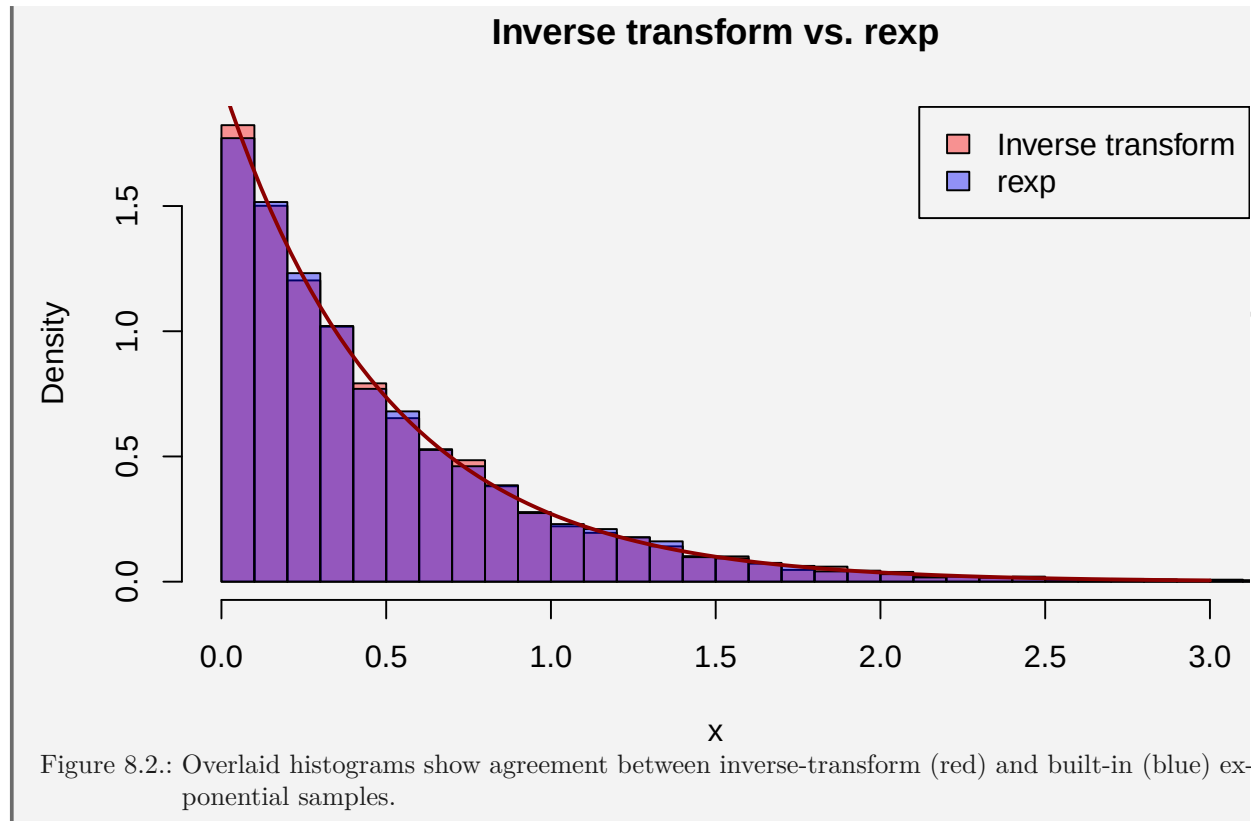
Example: exponential distribution. For $p(x) = \lambda e^{-\lambda x}$ with $x \geq 0$, the CDF is $F(x) = 1 - e^{-\lambda x}$. Inverting $u = 1 - e^{-\lambda x}$ gives

$$x = -\frac{1}{\lambda} \log(1 - u).$$

Since $1 - U \sim \text{Uniform}(0, 1)$ when $U \sim \text{Uniform}(0, 1)$, we have

$$X = -\frac{1}{\lambda} \log(U).$$

See code 15.25 in Appendix for the implementation.



The inverse transform method is **exact**, **simple**, and **efficient in one dimension**. However, it requires analytical inversion of F , which is rarely available for complex distributions (e.g., posteriors) or in high dimensions.

Example 8.3 (*Inverse-CDF sampling of the exponential (exam-style)*)

Let $U \sim \text{Uniform}(0, 1)$ and define $X = -\lambda^{-1} \log(1 - U)$ with $\lambda > 0$. Find the density of X and explain the Monte Carlo interpretation.

Solution. For $x \geq 0$,

$$P(X \leq x) = P(-\lambda^{-1} \log(1 - U) \leq x) = P(U \leq 1 - e^{-\lambda x}) = 1 - e^{-\lambda x}.$$

Differentiating gives $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Thus the inverse-CDF construction transforms uniform random numbers into exponential waiting times with rate λ . This is the simplest non-trivial Monte Carlo recipe and underlies almost every random-number generator in Poisson process simulation.

8.3. Rejection sampling

When the target density $\pi(x)$ is difficult to sample from directly, **rejection sampling** uses a proposal distribution $q(x)$ from which sampling is easy, and accepts or rejects each proposal to asymptotically recover the target.

Definition 8.2. Suppose we want to sample from a target $\pi(x) = \tilde{\pi}(x)/Z$ where $\tilde{\pi}(x)$ is the unnormalized density and Z is unknown. Choose a proposal density $q(x)$ with the same support. If there exists $M > 0$ such that

$$\tilde{\pi}(x) \leq M q(x) \quad \forall x,$$

then the following algorithm produces samples from $\pi(x)$:

Rejection Sampling Algorithm: 1. Sample $x \sim q(x)$. 2. Sample $u \sim \text{Uniform}(0, 1)$. 3. Accept x if $u \leq \tilde{\pi}(x)/(Mq(x))$; else reject and repeat.

The accepted samples are i.i.d. from $\pi(x)$.

Correctness: The joint density of proposing and accepting is

$$p(x, \text{accept}) = q(x) \cdot \frac{\tilde{\pi}(x)}{Mq(x)} = \frac{\tilde{\pi}(x)}{M}.$$

The marginal acceptance probability is

$$P(\text{accept}) = \int \frac{\tilde{\pi}(x)}{M} dx = \frac{Z}{M},$$

and conditioning on acceptance yields

$$p(x \mid \text{accept}) = \frac{\tilde{\pi}(x)/M}{Z/M} = \frac{\tilde{\pi}(x)}{Z} = \pi(x).$$

Acceptance ratio and efficiency. The acceptance probability is $P(\text{accept}) = Z/M$. For **efficiency**, we want M as small as possible (ideally $M^* = \max_x [\tilde{\pi}(x)/q(x)]$). When acceptance probability is low, most computational effort is wasted on rejected proposals.

In high dimensions, rejection sampling becomes pathologically inefficient. Consider sampling from a Gaussian target in dimension d . A reasonably-scaled proposal will have acceptance probability that typically decays as $e^{-\alpha d}$ for some constant $\alpha > 0$. The algorithm wastes an exponentially growing fraction of samples, making it impractical for large d .

8.4. Importance sampling

Importance sampling avoids the rejection step by using weighted samples. Every sample is retained, but weighted by how much the proposal underestimates the target.

Definition 8.3. To compute $\mathbb{E}_\pi[f(\theta)] = \int f(\theta)\pi(\theta) d\theta$, rewrite the integral in terms of a proposal $q(\theta)$ with the same support:

$$\mathbb{E}_\pi[f] = \int f(\theta)\pi(\theta) d\theta = \int f(\theta) \frac{\pi(\theta)}{q(\theta)} q(\theta) d\theta = \mathbb{E}_q[f(\theta)w(\theta)],$$

where the **importance weight** is

$$w(\theta) = \frac{\pi(\theta)}{q(\theta)}.$$

Sample $\theta_i \sim q(\theta)$ i.i.d. and estimate

$$\hat{I} = \frac{1}{N} \sum_{i=1}^N f(\theta_i)w(\theta_i).$$

This is unbiased when π is known up to a normalizing constant.

Self-normalized importance sampling. In practice, the target $\pi(\theta)$ is often known only up to normalization: $\pi(\theta) = \tilde{\pi}(\theta)/Z$ where Z is unknown (as in Bayesian posteriors). Use **unnormalized weights**

8. Monte Carlo methods

$\tilde{w}_i = \tilde{\pi}(\theta_i)/q(\theta_i)$ and estimate via the ratio

$$\hat{I} = \frac{\sum_{i=1}^N f(\theta_i) \tilde{w}_i}{\sum_{i=1}^N \tilde{w}_i}.$$

This estimator is biased but **consistent** (converges to the true value as $N \rightarrow \infty$).

The efficiency of importance sampling depends critically on how well the weights match the target distribution. If the proposal q is far from the target π , most weight concentrates on a few samples, and the effective sample size collapses.

Definition 8.4. The **effective sample size (ESS)** quantifies this degeneracy:

$$\text{ESS} = \frac{(\sum_i \tilde{w}_i)^2}{\sum_i \tilde{w}_i^2}.$$

Intuitively, if all weights are equal ($\tilde{w}_i = \tilde{w}^*$), then $\text{ESS} = N$. If one weight dominates, $\text{ESS} \ll N$, signaling **weight degeneracy**. As a rule of thumb, $\text{ESS} < N/10$ indicates serious efficiency loss.

Optimal proposal. The variance of the importance-sampling estimator is proportional to $\mathbb{E}_q[(f(X)w(X))^2]$. By calculus of variations, the variance-minimizing proposal is

$$q^*(x) \propto |f(x)|\pi(x).$$

However, this optimal proposal is itself as difficult to sample from as π , so it is rarely useful in practice. Practical guidance: match the shape and tail behavior of π , ensure heavy tails to avoid undersampling the tail regions, and capture correlations.

High-dimensional limitations. In very high dimensions, any suboptimal proposal leads to catastrophic weight degeneracy. The mass of π concentrates in thin, high-curvature regions (e.g., the typical set of a high-dimensional Gaussian). Unless q places comparable mass in those regions, the ESS shrinks exponentially with dimension. This is a **fundamental curse of dimensionality** for importance sampling, which motivates the turn to **Markov chain methods**.

8.5. Variance reduction

Before moving to MCMC, two classical variance-reduction tricks deserve mention.

Note. Antithetic sampling: Use correlated pairs to reduce variance. If $U \sim \text{Uniform}(0, 1)$, sample both U and $1 - U$ (which is also uniform). Since $f(F^{-1}(U))$ and $f(F^{-1}(1 - U))$ are negatively correlated for many smooth functions, averaging them reduces variance.

Control variates: If you know the expectation of a related function $g(x)$ under π , use

$$\hat{I}_{\text{cv}} = \frac{1}{N} \sum_i [f(x_i) - c(g(x_i) - \mathbb{E}[g])]$$

for a suitably chosen c . The term $c(g(x_i) - \mathbb{E}[g])$ has zero mean but can have large negative correlation with $f(x_i) - \mathbb{E}[f]$, reducing variance.

8.6. Markov chains and stationarity

When neither inverse sampling, rejection, nor importance sampling are practical, we turn to **Markov chain Monte Carlo (MCMC)**. The key idea is to construct a stochastic dynamics whose long-time distribution equals the target $\pi(x)$.

Definition 8.5. A **Markov chain** is a discrete-time stochastic process $\{X_t\}_{t \geq 0}$ where the next state depends only on the current state (the Markov property):

$$P(X_{t+1} = x' \mid X_t = x, X_{t-1}, \dots, X_0) = P(X_{t+1} = x' \mid X_t = x) \equiv T(x \rightarrow x').$$

The transition operator $T(x \rightarrow x')$ is the core object; it specifies how the chain evolves in a single step.

Stationarity: A distribution $\pi(x)$ is stationary with respect to T if

$$\pi(x) = \sum_{x'} T(x' \rightarrow x) \pi(x'),$$

or in integral form,

$$\pi(x) = \int T(x' \rightarrow x) \pi(x') dx'.$$

Under one transition step, the distribution remains unchanged.

Convergence: For the chain to be useful, we need it to converge to π starting from any initial distribution p_0 . This requires: - **Irreducibility:** all states (or regions) are mutually reachable with positive probability. - **Aperiodicity:** the chain does not cycle with a fixed period.

Under these conditions, the distribution evolves as

$$p_t(x) = T^t p_0(x) \xrightarrow{t \rightarrow \infty} \pi(x).$$

8.7. Detailed balance and reversibility

A sufficient condition for stationarity is **detailed balance**:

$$\pi(x) T(x \rightarrow x') = \pi(x') T(x' \rightarrow x) \quad \forall x, x'.$$

This says that the probability current from x to x' balances the current from x' to x at equilibrium. While not necessary for stationarity (some stationary distributions have non-zero net currents), detailed balance is mathematically convenient and physically intuitive—it ensures the chain is **reversible**.

Example 8.4 (Two-state Markov chain — stationarity and detailed balance (exam-style))

A Markov chain on states $\{0, 1\}$ has transition probabilities $P(0 \rightarrow 1) = a$ and $P(1 \rightarrow 0) = b$, with $a, b \in (0, 1)$. Find the stationary distribution and verify detailed balance.

Solution. Let the stationary distribution be (π_0, π_1) . Stationarity requires the probability flow out of 0 to equal the flow in:

$$\pi_0 a = \pi_1 b, \quad \pi_0 + \pi_1 = 1.$$

Solving,

$$\pi_0 = \frac{b}{a+b}, \quad \pi_1 = \frac{a}{a+b}.$$

Detailed balance asks $\pi_0 P(0 \rightarrow 1) = \pi_1 P(1 \rightarrow 0)$. Substituting,

$$\frac{b}{a+b} a = \frac{a}{a+b} b,$$

which holds trivially. In two-state chains stationarity and detailed balance are equivalent — any stationary distribution is automatically reversible.

8.8. Metropolis–Hastings algorithm

The **Metropolis–Hastings (MH) algorithm** constructs a transition operator satisfying detailed balance by combining a **proposal** $q(x' | x)$ with an **accept/reject step**.

Definition 8.6. Given a target $\pi(x)$ (known up to normalization), the MH transition is

$$T(x \rightarrow x') = q(x' | x) \alpha(x \rightarrow x'),$$

where the **acceptance probability** is

$$\alpha(x \rightarrow x') = \min \left(1, \frac{\pi(x')q(x | x')}{\pi(x)q(x' | x)} \right).$$

This is the **MH ratio** or **Hastings ratio**.

Proof of detailed balance: Define the ratio $r(x, x') = \frac{\pi(x')q(x | x')}{\pi(x)q(x' | x)}$.

Case 1: $r \geq 1$. Then $\alpha(x \rightarrow x') = 1$ and $\alpha(x' \rightarrow x) = 1/r$, so

$$\pi(x)q(x' | x)\alpha(x \rightarrow x') = \pi(x)q(x' | x) = \pi(x')q(x | x') \cdot \frac{1}{r} = \pi(x')q(x | x')\alpha(x' \rightarrow x).$$

Case 2: $r < 1$. Then $\alpha(x \rightarrow x') = r$ and $\alpha(x' \rightarrow x) = 1$, giving the same identity.

Thus detailed balance holds, and π is stationary. $\square$

Practical MH algorithm. One step from the current state x_t :

1. **Propose** $x' \sim q(x' | x_t)$.
2. **Compute the ratio** $r = \pi(x')q(x_t | x') / [\pi(x_t)q(x' | x_t)]$.
3. **Draw** $u \sim \text{Uniform}(0, 1)$.
4. **Accept** if $u < \min(1, r)$ and set $x_{t+1} = x'$; **reject** otherwise and set $x_{t+1} = x_t$.

For a **symmetric proposal** $q(x' | x) = q(x | x')$ (e.g., a Gaussian random walk), the acceptance probability simplifies to

$$\alpha(x \rightarrow x') = \min \left(1, \frac{\pi(x')}{\pi(x)} \right).$$

8.9. Convergence and ergodicity

After an initial **burn-in** period during which the chain forgets its starting point, the samples x_t are approximately distributed as $\pi(x)$. Estimates are formed as time averages:

$$\hat{I} = \frac{1}{N} \sum_{t=1}^N f(x_t) \approx \mathbb{E}_\pi[f].$$

However, successive samples are **correlated**. The integrated autocorrelation time

$$\tau_{\text{int}} = 1 + 2 \sum_{k=1}^{\infty} \rho_f(k)$$

quantifies this correlation, where $\rho_f(k)$ is the normalized autocorrelation at lag k . The effective number of independent samples is

$$N_{\text{eff}} \approx \frac{N}{2\tau_{\text{int}}},$$

which can be much smaller than the total number of draws.

The decay of autocorrelation is controlled by the **spectral gap** of the transition operator—the inverse of the largest non-zero eigenvalue. A small gap indicates slow mixing and long memory, forcing either longer chains or better proposals.

Example 8.5 (τ_{int} and ESS from a geometric autocorrelation (exam-style))

A single MCMC chain has $N = 20\,000$ stored samples. The estimated normalised autocorrelation is approximately $\rho(\tau) = 0.8^\tau$ for $\tau \geq 1$. Estimate τ_{int} and N_{eff} using

$$\tau_{\text{int}} = 1 + 2 \sum_{\tau=1}^{\infty} \rho(\tau).$$

Solution. The tail is a geometric series:

$$\sum_{\tau=1}^{\infty} 0.8^\tau = \frac{0.8}{1-0.8} = 4.$$

Hence

$$\tau_{\text{int}} = 1 + 2 \cdot 4 = 9, \quad N_{\text{eff}} \approx \frac{N}{\tau_{\text{int}}} = \frac{20\,000}{9} \approx 2\,222.$$

The chain contains far less information than its raw length suggests — long-memory autocorrelation discounts every additional sample.

Example 8.6 (Total ESS from M parallel chains (exam-style))

For a stationary Markov chain $\{x_t\}_{t=1}^N$ and an observable $A(x)$, define the autocorrelation function, the normalised autocorrelation function, and the integrated autocorrelation time. If $M = 5$ independent chains each have length $N = 10^4$ and the integrated autocorrelation time is $\tau_{\text{int}} = 250$, estimate the total effective sample size. Is it acceptable or should the proposal be improved?

Solution. The autocorrelation function of A is

$$C_A(\tau) = E[A(x_t) A(x_{t+\tau})] - E[A]^2,$$

the normalised version is $\rho_A(\tau) = C_A(\tau)/C_A(0)$, and

$$\tau_{\text{int}} = 1 + 2 \sum_{\tau=1}^{\infty} \rho_A(\tau).$$

With $M = 5$ chains of length $N = 10^4$, the total number of stored samples is $N_{\text{tot}} = MN = 5 \cdot 10^4$. Inflating the variance by τ_{int} ,

$$N_{\text{eff}} \approx \frac{N_{\text{tot}}}{\tau_{\text{int}}} = \frac{5 \cdot 10^4}{250} = 200.$$

Five correlated chains carry the same information as roughly $2 \cdot 10^2$ independent draws. The ratio $N/\tau_{\text{int}} = 40$ per chain shows that most moves are wasteful — the proposal is poorly matched to the target geometry, and either a better proposal (HMC, adapted scale) or a reparametrisation is needed.

See code 15.26 in Appendix for the implementation.

Acceptance rate: 0.5342107

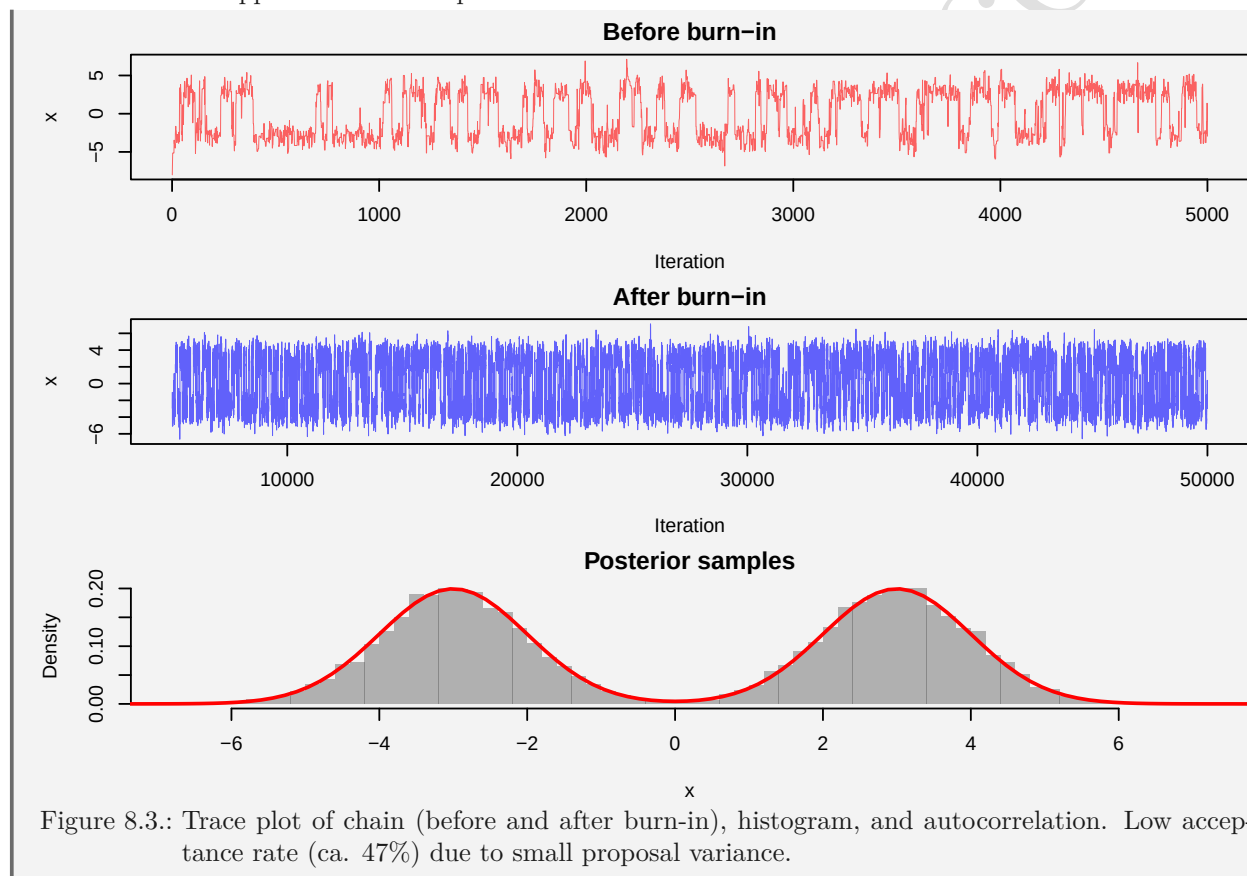
See code 15.27 in Appendix for the implementation.

Integrated autocorr time: 36.05

See code 15.28 in Appendix for the implementation.

N_eff: 624

See code 15.29 in Appendix for the implementation.



See code 15.30 in Appendix for the implementation.

8.10. Summary and roadmap

This chapter has covered the foundational sampling methods:

- **Direct sampling** (inverse transform) is exact when the CDF is invertible, but unavailable for complex posteriors.
- **Rejection sampling** corrects for sampling from the wrong density but wastes samples in high dimensions.

- **Importance sampling** avoids rejection but suffers from weight degeneracy when the proposal is misspecified.
- **Markov chain Monte Carlo** constructs a dynamics whose stationary distribution is the target, enabling asymptotically exact sampling even in high dimensions.

The next chapters develop practical refinements and extensions:

- **Chapter 8** explores advanced MCMC algorithms: Hamiltonian Monte Carlo (which uses gradient information to reduce autocorrelation), adaptive proposal tuning, and parallel tempering for multimodal distributions.
- **Chapter 9** covers **variational inference**—a fast, approximate alternative that trades accuracy for speed via optimization.
- **Chapter 10** introduces **particle methods**: sequential Monte Carlo and resampling strategies for filtering and online inference.

8.11. Exercises

Exercise 8.1

(Written test, 15 June 2026 — Q5.) A target density is $f(x) = 3x^2$ with $0 < x \leq 1$. A proposal density is $g(x) = 1$ with $0 < x \leq 1$. Find the smallest constant M so that g can be used in a rejection sampling algorithm to sample f . Give the expected acceptance probability.

Solution

We need $M \geq \sup_{x \in (0,1]} f(x)/g(x) = \sup 3x^2 = 3$, so the smallest admissible constant is

$$M = 3.$$

The expected acceptance probability is

$$P(\text{accept}) = \frac{1}{M} = \frac{1}{3}.$$

Exercise 8.2

Inverse transform. Show that if $U \sim \text{Uniform}(0,1)$ and $\lambda > 0$, then $X = -\log(U)/\lambda$ has the $\text{Exp}(\lambda)$ distribution.

Solution

For $x \geq 0$, using that U and $1 - U$ are both uniform on $(0,1)$,

$$P(X \leq x) = P(-\log U/\lambda \leq x) = P(U \geq e^{-\lambda x}) = 1 - e^{-\lambda x}.$$

Differentiating yields the density $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, i.e. $X \sim \text{Exp}(\lambda)$.

Exercise 8.3

Inverse transform for the Pareto distribution with CDF $F(x) = 1 - (x_m/x)^\alpha$ for $x \geq x_m$. Derive the sampling formula in terms of $U \sim \text{Uniform}(0,1)$.

Solution

Setting $U = F(X) = 1 - (x_m/X)^\alpha$ and solving for X ,

$$(x_m/X)^\alpha = 1 - U \quad \Rightarrow \quad X = \frac{x_m}{(1 - U)^{1/\alpha}}.$$

Since $1 - U \stackrel{d}{=} U$, one may equivalently use $X = x_m U^{-1/\alpha}$.

Exercise 8.4

Inverse transform for the geometric distribution on $\{1, 2, 3, \dots\}$ with parameter p . Derive the smallest integer X such that $1 - (1 - p)^X \geq U$, with $U \sim \text{Uniform}(0, 1)$.

Solution

The CDF is $F(k) = 1 - (1 - p)^k$ for $k \geq 1$. Requiring $F(X) \geq U$ gives $(1 - p)^X \leq 1 - U$, i.e.

$$X \geq \frac{\log(1 - U)}{\log(1 - p)}.$$

The smallest such integer is

$$X = \left\lceil \frac{\log(1 - U)}{\log(1 - p)} \right\rceil,$$

which (using $1 - U \stackrel{d}{=} U$) is the standard geometric sampler.

Exercise 8.5

Rejection from $\text{Gamma}(2, 1)$ with density $f(x) = x e^{-x}$ for $x > 0$, using the proposal $g(x) = e^{-x}$ (i.e. $\text{Exp}(1)$). Find the smallest constant M such that $f(x) \leq M g(x)$. Give the expected acceptance probability.

Solution

The ratio $f(x)/g(x) = x$ is unbounded on $(0, \infty)$, so no finite M exists and rejection from $\text{Exp}(1)$ fails. Choosing instead a lighter exponential $g_\lambda(x) = \lambda e^{-\lambda x}$ with $\lambda < 1$ gives a bounded ratio

$$\frac{f(x)}{g_\lambda(x)} = \frac{x e^{-(1-\lambda)x}}{\lambda},$$

maximised at $x^* = 1/(1 - \lambda)$, yielding $M(\lambda) = 1/[\lambda e(1 - \lambda)]$, minimised at $\lambda = 1/2$ with $M = 4/e$ and acceptance probability

$$P(\text{accept}) = \frac{1}{M} = \frac{e}{4}.$$

Exercise 8.6

Rejection sampling from the half-normal $f(x) \propto e^{-x^2/2}$ on $x \geq 0$ using the exponential proposal $g(x) = e^{-x}$. Find the smallest M and the acceptance probability.

Solution

Using the normalised half-normal $f(x) = \sqrt{2/\pi} e^{-x^2/2}$, the ratio is

$$\frac{f(x)}{g(x)} = \sqrt{\frac{2}{\pi}} e^{x-x^2/2},$$

which is maximised at $x^* = 1$, giving

$$M = \sqrt{\frac{2e}{\pi}}, \quad P(\text{accept}) = \frac{1}{M} = \sqrt{\frac{\pi}{2e}} \approx 0.76.$$

Exercise 8.7

Show that in rejection sampling the overall acceptance probability is exactly $1/M$, where $M = \sup_x f(x)/g(x)$. State the implication for the expected number of proposals needed to obtain one sample.

Solution

Marginalising over the proposal,

$$P(\text{accept}) = \int g(x) \frac{f(x)}{M g(x)} dx = \frac{1}{M} \int f(x) dx = \frac{1}{M}.$$

Hence the number of proposals until the first acceptance is geometric with mean

$$E[N_{\text{prop}}] = M,$$

so large M (poor proposal) makes rejection sampling prohibitively expensive.

Exercise 8.8

Importance sampling. Estimate $\mathbb{E}_p[X^2]$ for $p = \mathcal{N}(0, 1)$ using the proposal $q = \mathcal{N}(0, 4)$. Write the importance weights $w(x) = p(x)/q(x)$ explicitly and the unbiased estimator based on N samples.

Solution

The weight ratio of two zero-mean Gaussians is

$$w(x) = \frac{p(x)}{q(x)} = \frac{(2\pi)^{-1/2} e^{-x^2/2}}{(2\pi \cdot 4)^{-1/2} e^{-x^2/8}} = 2 \exp\left(-\frac{3x^2}{8}\right).$$

With $x_i \sim q$ i.i.d., the unbiased importance-sampling estimator is

$$\hat{I} = \frac{1}{N} \sum_{i=1}^N x_i^2 w(x_i),$$

which converges to $E_p[X^2] = 1$.

Exercise 8.9

Importance sampling for a tail probability. Estimate $\mathbb{P}_p(X > 3)$ for $p = \mathcal{N}(0, 1)$ using the shifted proposal $q = \mathcal{N}(3, 1)$. Write the estimator and explain why the proposal beats naïve Monte Carlo.

Solution

With $f(x) = \mathbb{1}_{x>3}$ and weights

$$w(x) = \frac{p(x)}{q(x)} = \exp\left(-\frac{1}{2}x^2 + \frac{1}{2}(x-3)^2\right) = e^{-3x+9/2},$$

the estimator with $x_i \sim \mathcal{N}(3, 1)$ is

$$\widehat{P} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}_{x_i>3} e^{-3x_i+9/2}.$$

Naïve sampling from p produces a hit only with probability $\Phi^c(3) \approx 1.35 \cdot 10^{-3}$, so its relative error blows up; the shifted proposal places roughly half the samples in the rare region, making the variance of order $1/N$ instead of $1/(N \Phi^c(3))$.

Exercise 8.10

Effective sample size. For three normalised importance weights $(w_1, w_2, w_3) = (0.7, 0.2, 0.1)$ compute $\text{ESS} = 1 / \sum_i w_i^2$.

Solution

Computing the sum of squares,

$$\sum_i w_i^2 = 0.49 + 0.04 + 0.01 = 0.54,$$

so

$$\text{ESS} = \frac{1}{0.54} = \frac{50}{27} \approx 1.85.$$

Three samples carry less than two effective ones — the first weight dominates and signals severe degeneracy.

Exercise 8.11

Antithetic variates. Let $U \sim \text{Uniform}(0, 1)$ and consider the estimator $\frac{1}{2}[g(U) + g(1-U)]$. Show that when g is monotone, $\text{Cov}(g(U), g(1-U)) \leq 0$ and the estimator has variance no larger than the single-sample estimator.

Solution

If g is monotone, $g(U)$ and $g(1-U)$ are monotone in U in opposite senses, so by the Chebyshev/FKG correlation inequality

$$\text{Cov}(g(U), g(1-U)) \leq 0.$$

With $\sigma^2 = \text{Var } g(U) = \text{Var } g(1-U)$,

$$\text{Var}\left(\frac{1}{2}[g(U) + g(1-U)]\right) = \frac{1}{4}[2\sigma^2 + 2\text{Cov}] = \frac{\sigma^2}{2} + \frac{1}{2}\text{Cov} \leq \frac{\sigma^2}{2},$$

which is at most the variance $\sigma^2/2$ of the average of two independent draws of $g(U)$, and strictly smaller when the covariance is negative.

Exercise 8.12

Control variates. Estimate $\theta = \mathbb{E}[h(X)]$ with $X \sim \text{Uniform}(0, 1)$ and $h(x) = e^x$, using X itself as a control variate (its mean is $1/2$). Derive the optimal coefficient $c^* = -\text{Cov}(h(X), X)/\text{Var}(X)$ and write the control-variate estimator.

Solution

For $X \sim \text{Uniform}(0, 1)$, $\text{Var}(X) = 1/12$, and integrating by parts

$$E[Xe^X] = e - (e - 1) = 1, \quad \text{Cov}(e^X, X) = 1 - \frac{1}{2}(e - 1) = \frac{3-e}{2}.$$

Hence

$$c^* = -\frac{\text{Cov}(e^X, X)}{\text{Var}(X)} = -12 \cdot \frac{3-e}{2} = 6(e-3),$$

and the control-variate estimator is

$$\hat{\theta}_{\text{cv}} = \frac{1}{N} \sum_{i=1}^N [e^{X_i} + c^*(X_i - \frac{1}{2})].$$

Exercise 8.13

Two-state Markov chain. The transition matrix is

$$T = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}.$$

Find the stationary distribution π .

Solution

With $a = T_{12} = 0.3$ and $b = T_{21} = 0.4$, balancing the off-diagonal flows $\pi_1 a = \pi_2 b$ together with $\pi_1 + \pi_2 = 1$ gives

$$\pi_1 = \frac{b}{a+b} = \frac{0.4}{0.7} = \frac{4}{7}, \quad \pi_2 = \frac{a}{a+b} = \frac{0.3}{0.7} = \frac{3}{7}.$$

Exercise 8.14

Detailed balance check. With T as above, verify whether the candidate stationary distribution $\pi = (4/7, 3/7)$ satisfies $\pi_i T_{ij} = \pi_j T_{ji}$.

Solution

Comparing the two flows,

$$\pi_1 T_{12} = \frac{4}{7} \cdot 0.3 = \frac{1.2}{7}, \quad \pi_2 T_{21} = \frac{3}{7} \cdot 0.4 = \frac{1.2}{7}.$$

The two are equal, so detailed balance holds and the chain is reversible with stationary law $\pi = (4/7, 3/7)$.

Exercise 8.15

Metropolis acceptance. For target $\pi(x) \propto e^{-x^2/2}$ on $\mathbb{R}$ with a symmetric proposal $q(x' | x) = q(x | x')$, write the acceptance probability $\alpha(x \rightarrow x')$ in closed form and identify when it equals 1.

Solution

Symmetry of q cancels the proposal ratio, leaving

$$\alpha(x \rightarrow x') = \min\left(1, \frac{\pi(x')}{\pi(x)}\right) = \min\left(1, \exp\left[\frac{1}{2}(x^2 - x'^2)\right]\right).$$

Acceptance equals 1 whenever $x'^2 \leq x^2$, i.e. the proposal moves closer to the mode at the origin; otherwise it accepts with probability $\exp[(x^2 - x'^2)/2] < 1$.

Exercise 8.16

(Define and explain.) For a stationary Markov chain $\{x_t\}_{t=1}^N$ and observable $A(x)$, define the autocorrelation function $C_A(\tau)$, the normalised autocorrelation $\rho_A(\tau)$, and the integrated autocorrelation time τ_{int} . State the formula for the effective sample size N_{eff} .

Solution

The autocorrelation, its normalised version, and the integrated time are

$$C_A(\tau) = E[A(x_t)A(x_{t+\tau})] - E[A]^2, \quad \rho_A(\tau) = \frac{C_A(\tau)}{C_A(0)}, \quad \tau_{\text{int}} = 1 + 2 \sum_{\tau=1}^{\infty} \rho_A(\tau).$$

The effective sample size after N correlated draws is

$$N_{\text{eff}} \approx \frac{N}{\tau_{\text{int}}},$$

i.e. the chain carries N_{eff} independent draws' worth of information about $E_{\pi}[A]$.

Exercise 8.17

(Define and explain.) Explain in one paragraph why Monte Carlo methods are central in Bayesian inference. Contrast with grid-based numerical integration in d dimensions. What is the difference between using independent posterior samples and samples from a Markov chain?

Solution

Bayesian inference reduces to posterior expectations $E[f(\theta) \mid D] = \int f(\theta)\pi(\theta \mid D) d\theta$, which Monte Carlo replaces by $\hat{I}_N = N^{-1} \sum_i f(\theta_i)$. The Monte Carlo error scales as $\mathcal{O}(N^{-1/2})$ regardless of dimension, whereas a grid with n points per axis costs n^d evaluations — exponentially worse. Independent samples give $\text{Var}(\hat{I}_N) = \text{Var}_{\pi}(f)/N$, while an MCMC chain inflates this by the integrated autocorrelation time,

$$\text{Var}(\hat{I}_N) \approx \frac{\tau_{\text{int}}}{N} \text{Var}_{\pi}(f),$$

so the relevant figure of merit is the effective sample size $N_{\text{eff}} = N/\tau_{\text{int}}$, not the raw number of stored draws.

Exercise 8.18

(Compare and contrast.) Define rejection sampling and importance sampling. State one situation where each one is preferable over the other. What is the cost of a bad proposal in each method?

Solution

Rejection sampling draws $x \sim q$ and accepts with probability $\tilde{\pi}(x)/[Mq(x)]$, producing exact i.i.d. samples from π ; **importance sampling** keeps every draw $x_i \sim q$ and weights it by $w(x_i) = \pi(x_i)/q(x_i)$ in $\hat{I} = N^{-1} \sum_i f(x_i)w(x_i)$. Rejection is preferable when downstream code requires unweighted samples (e.g. visualisation, plugging into a black-box estimator); importance sampling is preferable for rare events or tail probabilities where q can be shifted into the region of interest. The cost of a bad proposal is: in rejection, a vanishing acceptance rate $1/M$ and roughly M proposals per sample; in importance sampling, weight degeneracy quantified by $\text{ESS} = 1/\sum_i w_i^2 \ll N$ and unbounded estimator variance.

Exercise 8.19

(Define and explain.) State the conditions for a Markov chain to converge to its stationary distribution. Define detailed balance and explain why it is sufficient (but not necessary) for stationarity.

Solution

A Markov chain with transition $T(x, y)$ converges to a unique stationary law π provided it is **irreducible** (any state reachable from any other in finite time), **aperiodic** (no cyclic class), and admits π as an invariant measure satisfying

$$\pi(y) = \int T(x, y) \pi(x) dx.$$

Detailed balance is the local-equilibrium condition

$$\pi(x) T(x, y) = \pi(y) T(y, x).$$

Integrating both sides over x gives the global balance equation above, so detailed balance implies stationarity. It is *not* necessary: chains can be stationary with non-zero probability currents that close up globally without balancing pairwise (e.g. deterministic cyclic chains).

8.12. Sources

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9. Independence sampler and Gibbs sampling

9.1. Introduction

The Metropolis–Hastings algorithm, introduced in the previous chapter, is a general framework for constructing Markov chains with a target stationary distribution. However, the choice of proposal distribution critically affects both computational efficiency and the quality of samples. In this chapter, we explore two important special cases that reveal a fundamental trade-off in MCMC design: global vs. local exploration, and the role of acceptance rates in controlling mixing.

The **independence sampler** uses a fixed global proposal, independent of the current state. This allows long-distance jumps but requires the proposal to approximate the target well. The **Gibbs sampler**, by contrast, updates coordinates sequentially using exact full conditional distributions, guaranteeing acceptance but potentially moving slowly through high-dimensional spaces when parameters are correlated. Both methods illustrate how MCMC designers must balance the competing demands of efficient exploration and computational tractability.

9.2. The independence sampler: global proposals

In a random-walk Metropolis algorithm, the proposal distribution depends on the current state:

$$q(x' | x) \propto \exp\left(-\frac{\|x' - x\|^2}{2\sigma^2}\right).$$

This locality allows the chain to take small, tentative steps and gradually explore the target space. However, if the proposal variance is too small, the chain mixes slowly; if too large, it wastes computation on distant, low-probability regions.

The **independence sampler** takes a different approach: the proposal is **independent of the current state**,

$$q(x') = q(x' | x),$$

depending only on x' . This allows the chain to jump from anywhere to anywhere in a single step, potentially escaping local modes and exploring the target space more globally.

- **Acceptance probability.** For a generic Metropolis–Hastings step the acceptance probability is

$$\alpha(x \rightarrow x') = \min\left(1, \frac{\pi(x') q(x | x')}{\pi(x) q(x' | x)}\right).$$

For the independence sampler, $q(x | x') = q(x)$ and $q(x' | x) = q(x')$, so

$$\alpha(x \rightarrow x') = \min\left(1, \frac{\pi(x') q(x)}{\pi(x) q(x')}\right).$$

Rearranging in terms of **importance weights** $w(x) = \pi(x)/q(x)$:

$$\alpha(x \rightarrow x') = \min\left(1, \frac{w(x')}{w(x)}\right).$$

Note. The acceptance probability compares the importance weights at the proposed and current states. If $w(x')$ is larger, the proposal is always accepted. If $w(x')$ is smaller, it is accepted with probability $w(x')/w(x)$. When the proposal closely matches the target, weights are nearly constant and acceptance is high.

- **Connection to importance sampling and ESS.** The independence sampler inherits its theoretical motivation from **importance sampling**. In importance sampling, we approximate an expectation

$$E_{\pi}[f(x)] = \int f(x) \pi(x) dx$$

by sampling from a simpler proposal q and weighting:

$$\widehat{E}_{\pi}[f(x)] = \frac{\sum_{i=1}^N w(x_i) f(x_i)}{\sum_{i=1}^N w(x_i)}, \quad w(x_i) = \frac{\pi(x_i)}{q(x_i)}.$$

The **effective sample size (ESS)** reflects how much information the weighted samples contain. If weights are unequal—as they are when q deviates from π —the ESS is smaller than N :

$$\text{ESS} = \frac{1}{\sum_i (w_i / \bar{w})^2}.$$

When $w(x)$ varies widely, many samples receive vanishingly small weight, and the variance of the estimator becomes very large. The independence sampler's acceptance rate is directly related to the relative variation in weights: a narrow or poorly-matched proposal produces highly variable weights, leading to low acceptance and slow exploration. This is the **narrow-proposal failure mode** — if the proposal is too concentrated, the chain accepts rarely and gets stuck.

- **When it works well.** The key requirement is that q **must approximate π globally**. A sufficient ergodicity condition is

$$\pi(x) \leq C q(x) \quad \forall x$$

for some finite constant $C > 0$ (analogous to the envelope condition in rejection sampling), which ensures the chain does not get trapped in regions of arbitrarily high relative weight. The independence sampler is best when we have a good global approximation to the posterior — e.g., from a **Laplace approximation**, **variational inference**, or other fast approximate methods — and when the target is unimodal or has well-separated modes the proposal can accommodate. It fails when the proposal underestimates the tails or misses modes, so that states with large $\pi(x)/q(x)$ are rare and hard to reach.

- **Curse of dimensionality.** A profound limitation emerges in high dimensions. Suppose $\log w(x) = \log \pi(x) - \log q(x)$ decomposes as a sum of roughly independent contributions:

$$\log w(x) \approx \sum_{i=1}^d \Delta_i(x_i),$$

where each Δ_i represents the local mismatch between π and q in coordinate i . Even if each Δ_i is small (order $O(1)$), the total deviation is $\log w(x) = O(d)$, so $w(x) = \exp(O(d))$. The weights vary **exponentially** in dimension, and the acceptance rate typically decays like $P(A) \sim \exp(-cd)$.

Definition 9.1. Concentration of measure: In high dimensions, probability mass concentrates on thin regions. For a standard Gaussian $\pi = N(0, I_d)$, the norm $\|x\|$ concentrates around $\sqrt{d}$. Even a slightly mis-scaled proposal q will place its probability mass in regions irrelevant to π , causing acceptance to decay exponentially with d .

This is the motivation for **local proposals** (random-walk Metropolis), **gradient-based methods** (Hamiltonian MC), and **coordinate-wise samplers** (Gibbs sampling), which we turn to next.

9.3. The Gibbs sampler: coordinate-wise updates

In many Bayesian models, jointly sampling from $p(\theta \mid D)$ where $\theta = (\theta_1, \dots, \theta_d)$ is intractable, but sampling from each **full conditional distribution** is simple.

Definition 9.2. The **full conditional distribution** of θ_i is

$$p(\theta_i \mid \theta_{-i}, D),$$

where $\theta_{-i} = (\theta_1, \dots, \theta_{i-1}, \theta_{i+1}, \dots, \theta_d)$ denotes all other parameters. It is obtained by treating θ_{-i} as fixed and proportional to the joint posterior:

$$p(\theta_i \mid \theta_{-i}, D) \propto p(\theta_i, \theta_{-i} \mid D).$$

9.3.1. Algorithm: deterministic and random scan

Gibbs sampling iteratively samples each parameter from its full conditional.

Theorem 9.1. Gibbs Sampler Algorithm

Initialize $\theta^{(0)} = (\theta_1^{(0)}, \dots, \theta_d^{(0)})$.

For $t = 1, 2, \dots$:

1. $\theta_1^{(t)} \sim p(\theta_1 \mid \theta_2^{(t-1)}, \dots, \theta_d^{(t-1)}, D)$
2. $\theta_2^{(t)} \sim p(\theta_2 \mid \theta_1^{(t)}, \theta_3^{(t-1)}, \dots, \theta_d^{(t-1)}, D)$
3. $\vdots$
4. $\theta_d^{(t)} \sim p(\theta_d \mid \theta_1^{(t)}, \dots, \theta_{d-1}^{(t)}, D)$

One complete cycle of all d updates constitutes one iteration of the Gibbs sampler.

Variants: The **random-scan Gibbs sampler** updates coordinates in random order instead of a fixed sequence. For convergence, both deterministic and random scans are ergodic under mild conditions.

- **Gibbs as a special case of Metropolis–Hastings.** Gibbs sampling can be viewed as Metropolis–Hastings with the proposal being the full conditional itself. At step t , we propose a new value for θ_i by sampling from

$$q(\theta_i' \mid \theta_i^{(t-1)}, \theta_{-i}^{(t)}) = p(\theta_i' \mid \theta_{-i}^{(t)}, D).$$

The acceptance probability is

$$\alpha(\theta^{(t-1)} \rightarrow \theta^{(t)}) = \min \left(1, \frac{p(\theta_i^{(t)}, \theta_{-i}^{(t)} \mid D) p(\theta_i^{(t-1)} \mid \theta_{-i}^{(t)}, D)}{p(\theta_i^{(t-1)}, \theta_{-i}^{(t-1)} \mid D) p(\theta_i^{(t)} \mid \theta_{-i}^{(t)}, D)} \right).$$

Using the factorization $p(\theta_i, \theta_{-i} \mid D) = p(\theta_i \mid \theta_{-i}, D) p(\theta_{-i} \mid D)$, the ratio simplifies:

$$\frac{p(\theta_i^{(t)} \mid \theta_{-i}^{(t)}, D) p(\theta_{-i}^{(t)} \mid D) p(\theta_i^{(t-1)} \mid \theta_{-i}^{(t)}, D)}{p(\theta_i^{(t-1)} \mid \theta_{-i}^{(t-1)}, D) p(\theta_{-i}^{(t-1)} \mid D) p(\theta_i^{(t)} \mid \theta_{-i}^{(t)}, D)} = 1.$$

Therefore, $\alpha = 1$: **every Gibbs step is accepted.**

Note. Acceptance is guaranteed, but samples are still **correlated**. The chain does not jump to a new state at each step; instead, it updates coordinates sequentially, and autocorrelation arises because coordinates are strongly dependent.

9.3.2. Detailed balance for Gibbs updates

For the full Gibbs update (one complete sweep of all d coordinates), detailed balance holds with respect to the joint posterior. We verify that

$$p(\theta \mid D) T(\theta \rightarrow \theta') = p(\theta' \mid D) T(\theta' \rightarrow \theta),$$

where T is the transition kernel of one complete sweep.

For a single coordinate update T_i applied to coordinate i , left-multiply by $p(\theta \mid D)$ and expand using the factorization:

$$p(\theta \mid D) T_i(\theta \rightarrow \theta') = p(\theta_i, \theta_{-i} \mid D) p(\theta'_i \mid \theta_{-i}, D) \delta(\theta'_{-i} - \theta_{-i}) \quad (9.1)$$

$$= p(\theta_i \mid \theta_{-i}, D) p(\theta_{-i} \mid D) p(\theta'_i \mid \theta_{-i}, D) \delta(\theta'_{-i} - \theta_{-i}). \quad (9.2)$$

Permuting terms and using $p(\theta_i \mid \theta_{-i}, D) = p(\theta_i \mid \theta'_{-i}, D)$ (since $\theta_{-i} = \theta'_{-i}$):

$$= p(\theta'_i \mid \theta_{-i}, D) p(\theta_{-i} \mid D) p(\theta_i \mid \theta_{-i}, D) \delta(\theta'_{-i} - \theta_{-i}) = p(\theta' \mid D) T_i(\theta' \rightarrow \theta).$$

Thus each coordinate-wise update preserves the posterior. Since the full sweep is a composition of such updates, it also satisfies detailed balance.

- **Mixing and slow mixing from correlation.** Unlike the independence sampler, Gibbs always accepts. However, this does not imply fast mixing. When parameters are strongly correlated — as in high-dimensional hierarchical models or when the posterior is elongated along a narrow ridge — coordinate-wise updates must take many small steps to explore the space.

Example 9.1

Bivariate Gaussian with high correlation

Consider a bivariate normal distribution:

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right).$$

The full conditionals are:

$$X \mid Y = y \sim N(\rho y, 1 - \rho^2), \quad Y \mid X = x \sim N(\rho x, 1 - \rho^2).$$

When $\rho = 0.9$, the posterior concentrates on a narrow diagonal ridge. A Gibbs step first moves horizontally (X-update), then vertically (Y-update), taking many iterations to traverse the ridge. In contrast, updating both coordinates jointly (e.g., via a rotated-coordinate Gibbs or HMC) moves along the ridge in fewer steps.

- **Metropolis-within-Gibbs and block Gibbs.** When the full conditional for a component is intractable, we can replace the Gibbs step with a Metropolis–Hastings step; when several coordinates are strongly correlated, we can group them and update them jointly.

Definition 9.3. Metropolis-within-Gibbs: Replace the Gibbs step for θ_i with a Metropolis–Hastings move using a local proposal $q_i(\theta'_i | \theta_i)$. The acceptance probability is

$$\alpha_i(\theta_i | \theta_{-i}) = \min \left(1, \frac{p(\theta'_i | \theta_{-i}, D) q_i(\theta_i | \theta'_i)}{p(\theta_i | \theta_{-i}, D) q_i(\theta'_i | \theta_i)} \right).$$

The chain remains valid (it still has $p(\theta | D)$ as stationary distribution), but some steps are rejected.

Block Gibbs sampling updates multiple coordinates jointly,

$$(\theta_1, \dots, \theta_k) \sim p(\theta_1, \dots, \theta_k | \theta_{k+1}, \dots, \theta_d, D),$$

which is effective when joint conditionals are tractable and dramatically reduces autocorrelation in highly correlated subsets.

9.4. Worked example: Gibbs sampling for a bivariate Gaussian

We now demonstrate Gibbs sampling on the bivariate Gaussian with $\rho = 0.9$ to illustrate slow mixing due to correlation, and show how a simple coordinate rotation accelerates mixing.

9.4.1. Slow mixing at high correlation

See code 15.31 in Appendix for the implementation.

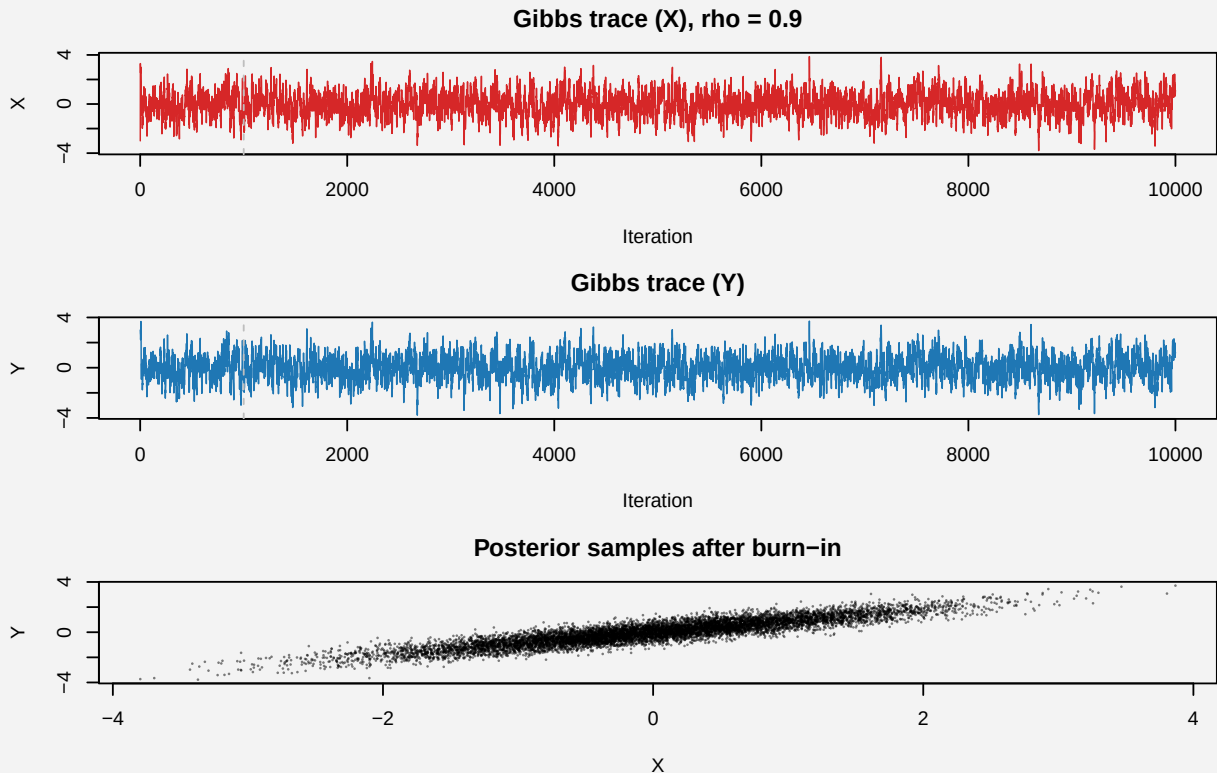


Figure 9.1.: Gibbs sampling trace plots for a bivariate Gaussian with $\rho = 0.9$. The X-trace (top) shows high autocorrelation; the Y-trace (middle) is even slower due to dependence on X. The scatter plot (bottom) reveals the narrow diagonal structure.

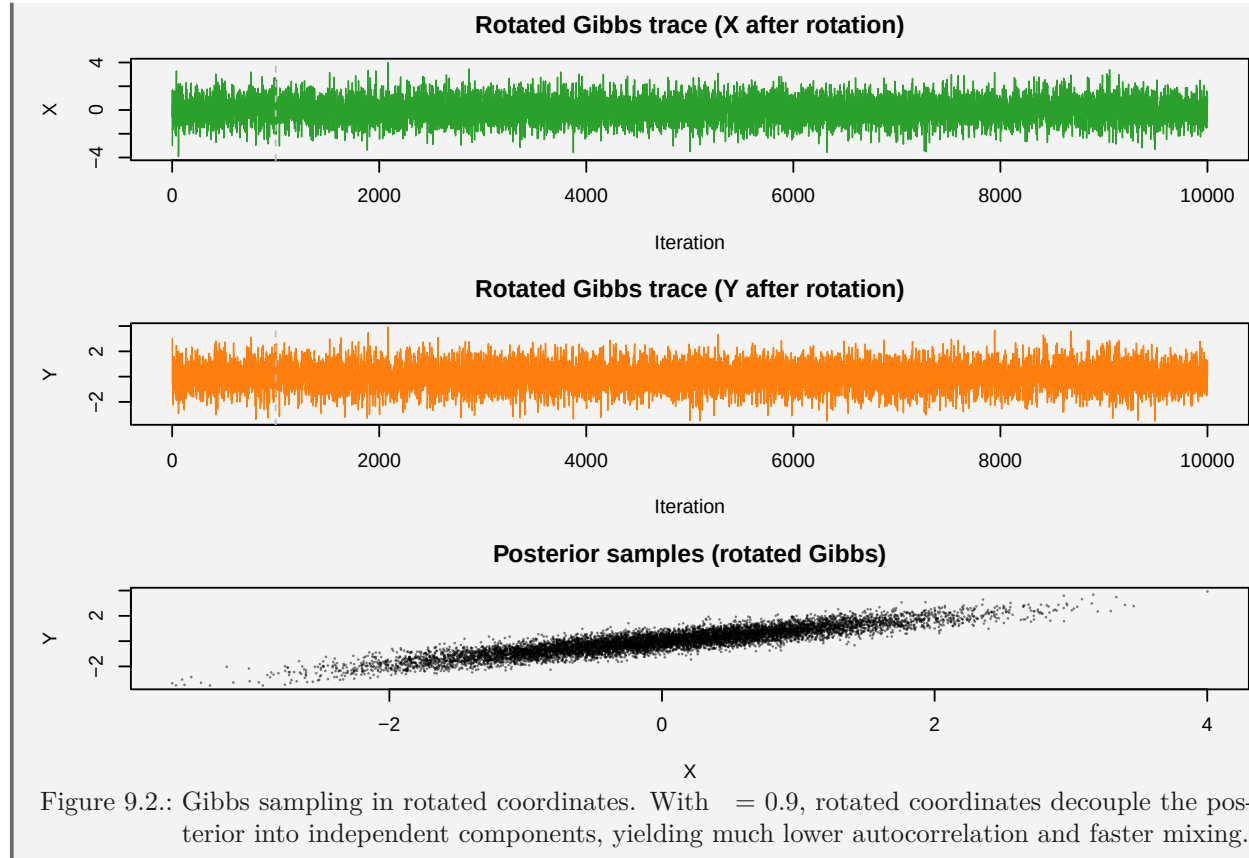
9.4.2. Rotated coordinates for faster mixing

The slow mixing arises because the Gibbs updates align with the coordinate axes, while the posterior concentrates along the diagonal. We can decouple the correlation by rotating coordinates. Let $u = \frac{x+y}{\sqrt{2}}$ and $v = \frac{x-y}{\sqrt{2}}$. For a standard bivariate Gaussian, the joint distribution of (u, v) is:

$$u \sim N(0, (1 + \rho)), \quad v \sim N(0, (1 - \rho)),$$

and u and v are **independent**.

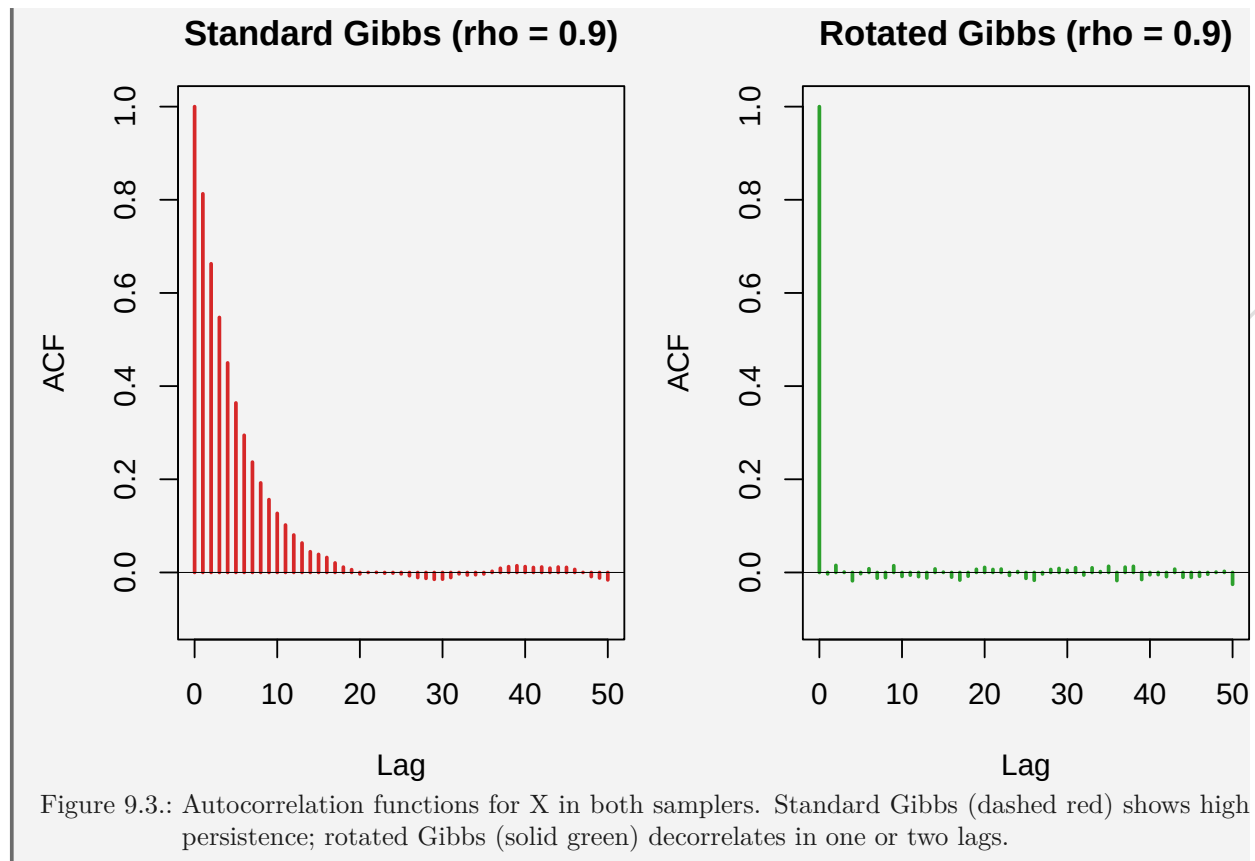
See code 15.32 in Appendix for the implementation.



See code 15.33 in Appendix for the implementation.

9.4.3. Autocorrelation comparison

See code 15.34 in Appendix for the implementation.



See code 15.35 in Appendix for the implementation.

9.5. Comparison: independence sampler vs. Gibbs sampler vs. random-walk Metropolis

The three samplers embody different philosophies and trade-offs:

Aspect	Independence Sampler	Gibbs Sampler	Random-Walk MH
Proposal	$q(x')$ (global, fixed)	$p(\theta_i \theta_{-i}, D)$	$x' \sim N(x, \Sigma)$
Move type	Global jumps anywhere	Coordinate-wise updates	Local steps
Acceptance	MH ratio (often low)	Always 1 (deterministic)	Variable (tunable)
Main requirement	Good global approximation	Tractable conditionals	Reasonable scale estimate
Fails when	q poorly matches π	Strong parameter correlations	Proposal scale wrong
Autocorrelation	Low if $q \approx \pi$	High if correlated structure	Moderate (tunable)
Dimension scaling	Exponential decay $\sim \exp(-cd)$	Polynomial (depends on correlation)	Exponential decay
Typical use	Approximate posteriors	Conjugate/hierarchical models	Flexible fallback

Remark. In practice, these are rarely used in isolation. Modern samplers often combine ideas:

- **Metropolis-within-Gibbs:** Use Gibbs for tractable coordinates, MH for intractable ones.
- **Block Gibbs:** Group correlated coordinates and sample jointly to reduce autocorrelation.
- **Adaptive proposals:** Learn a proposal from preliminary runs (adaptive MCMC).
- **HMC within Gibbs:** Combine gradient-based efficient exploration with block structure.

The choice depends on the model structure, the availability of efficient samplers for conditionals, and computational constraints.

9.6. Summary

The **independence sampler** represents a global exploration strategy: it can jump anywhere but only works well if the proposal closely approximates the target. Its acceptance probability directly reflects the importance weights, making it a bridge between importance sampling and MCMC. However, the exponential degradation of weights in high dimensions—a manifestation of concentration of measure—limits its applicability to high-dimensional problems.

The **Gibbs sampler** exploits model structure by updating parameters from their full conditionals. Every step is accepted, eliminating the rejection waste of Metropolis algorithms. However, when parameters are strongly correlated, the chain mixes slowly because coordinate-wise moves struggle to explore elongated posterior geometries efficiently. Rotation to decorrelated coordinates, block updates, or auxiliary variable methods can dramatically improve performance.

Both methods highlight the central tension in MCMC: acceptance rates, exploration speed, and computational tractability are not independent. A low acceptance rate signals a poorly matched proposal; a high rate (even 100%) does not guarantee fast mixing if the effective step size is small. Successful MCMC practice requires diagnosing mixing, adapting proposals, and sometimes redesigning the parameterization to align with the posterior structure.

9.7. Exercises

Exercise 9.1

(Written test, 15 June 2026 — Q6.) A discrete-time Markov chain has state space $\{0, 1, 2\}$ and transition matrix

$$T = \begin{pmatrix} 1-a-b & a & b \\ b & 1-a-b & a \\ a & b & 1-a-b \end{pmatrix},$$

where $a > 0$, $b > 0$, and $a + b \leq 1$.

- Find the stationary distribution $\pi = (\pi_0, \pi_1, \pi_2)$.
- State the detailed-balance equations for this chain.
- Derive the condition on a and b under which detailed balance can hold.

Solution

- The chain is doubly stochastic (every column also sums to 1), so the uniform vector is invariant:

$$\pi = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

b. Detailed balance requires $\pi_i T(i, j) = \pi_j T(j, i)$ for all $i \neq j$; with $\pi_i = 1/3$ this reduces to

$$T(0, 1) = T(1, 0), \quad T(1, 2) = T(2, 1), \quad T(0, 2) = T(2, 0).$$

c. Substituting the entries gives $a = b$ in each equation, so detailed balance holds **iff** $a = b$. When $a \neq b$ the chain is stationary but only **cyclically** so (clockwise vs. counter-clockwise probability flux differs by $a - b$).

Exercise 9.2

Two-state Markov chain with $T(0 \rightarrow 1) = \alpha$ and $T(1 \rightarrow 0) = \beta$, where $\alpha, \beta \in (0, 1)$. Find the stationary distribution π in closed form and verify detailed balance.

Solution

Stationarity $\pi T = \pi$ yields $\pi_0 \alpha = \pi_1 \beta$, and together with $\pi_0 + \pi_1 = 1$,

$$\pi_0 = \frac{\beta}{\alpha + \beta}, \quad \pi_1 = \frac{\alpha}{\alpha + \beta}.$$

Detailed balance reads $\pi_0 T(0 \rightarrow 1) = \pi_1 T(1 \rightarrow 0)$, i.e. $\frac{\beta \alpha}{\alpha + \beta} = \frac{\alpha \beta}{\alpha + \beta}$, which holds identically. Every two-state irreducible chain is reversible.

Exercise 9.3

Three-state birth-death chain on $\{0, 1, 2\}$ with $T(i \rightarrow i+1) = p$ and $T(i \rightarrow i-1) = q$ (where defined), self-loops chosen so each row sums to one. Find the stationary distribution in terms of the ratio $r = p/q$.

Solution

Birth-death chains are reversible; detailed balance $\pi_i T(i, i+1) = \pi_{i+1} T(i+1, i)$ gives the recursion

$$\pi_{i+1} = \frac{p}{q} \pi_i = r \pi_i,$$

so $\pi_1 = r \pi_0$ and $\pi_2 = r^2 \pi_0$. Normalising,

$$\pi = \frac{1}{1 + r + r^2} (1, r, r^2).$$

Exercise 9.4

Construct a 3-state Markov chain that admits a stationary distribution but does *not* satisfy detailed balance. (Hint: try a deterministic cycle $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$.) Verify both claims explicitly.

Solution

Take $T(i, i+1 \bmod 3) = 1$, all other entries 0. This is doubly stochastic so $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is stationary: $(\pi T)_j = \sum_i \pi_i T(i, j) = \frac{1}{3} \cdot 1 = \pi_j$. But

$$\pi_0 T(0, 1) = \frac{1}{3} \cdot 1 = \frac{1}{3}, \quad \pi_1 T(1, 0) = \frac{1}{3} \cdot 0 = 0,$$

so detailed balance fails. The chain is stationary by *cyclic* probability flux, not by pairwise balance.

Exercise 9.5

Independence sampler with target $\pi(x) = \text{Beta}(2, 2)$ on $(0, 1)$ and proposal $q(x) = \text{Uniform}(0, 1)$. Write the Metropolis–Hastings acceptance probability $\alpha(x \rightarrow x')$ in terms of x, x' , and identify the smallest constant M that would make every proposal accepted in a *rejection* sampler.

Solution

Here $\pi(x) = 6x(1-x)$ and $q(x) = 1$, so $w(x) = 6x(1-x)$ and

$$\alpha(x \rightarrow x') = \min\left\{1, \frac{x'(1-x')}{x(1-x)}\right\}.$$

The weight is maximised at $x = 1/2$ with $w(1/2) = 3/2$, so the smallest envelope constant for rejection sampling is

$$M = \sup_x \frac{\pi(x)}{q(x)} = \frac{3}{2}.$$

Exercise 9.6

Independence sampler with target $\pi = \text{Exp}(1)$ and proposal $q = \text{Exp}(2)$. Compute the importance weight $w(x) = \pi(x)/q(x)$ and the acceptance ratio $\alpha(x \rightarrow x') = \min\{1, w(x')/w(x)\}$.

Solution

With $\pi(x) = e^{-x}$ and $q(x) = 2e^{-2x}$ on $x \geq 0$,

$$w(x) = \frac{e^{-x}}{2e^{-2x}} = \frac{1}{2} e^x.$$

Hence

$$\alpha(x \rightarrow x') = \min\left\{1, \frac{w(x')}{w(x)}\right\} = \min\{1, e^{x'-x}\}.$$

The weight grows unboundedly with x , so w is **not bounded** — the proposal has lighter tails than the target and the chain will tend to stall whenever it reaches large x .

Exercise 9.7

Show that an independence sampler with proposal $q = \pi$ accepts every move and therefore generates i.i.d. samples from π .

Solution

The importance weight is constant: $w(x) = \pi(x)/q(x) = 1$. Therefore

$$\alpha(x \rightarrow x') = \min\left\{1, \frac{w(x')}{w(x)}\right\} = 1$$

for every x, x' . Every proposal is accepted, and since the proposal is independent of the current state and equal to π , the successive draws are i.i.d. samples from π .

Exercise 9.8

Explain qualitatively why an independence sampler whose proposal q has lighter tails than the target π tends to get stuck in the tails of π . Express your answer in terms of the importance weight $w(x)$.

Solution

Light-tailed q means $q(x) \ll \pi(x)$ as $|x| \rightarrow \infty$, so the weight

$$w(x) = \frac{\pi(x)}{q(x)} \rightarrow \infty.$$

If the chain ever reaches a tail state x with very large $w(x)$, the acceptance probability $\alpha(x \rightarrow x') = \min\{1, w(x')/w(x)\}$ is tiny for any typical proposal x' from q (which lives near the mode and has $w(x') \ll w(x)$). The chain rejects almost every move and remains stuck at x , producing a “sticking” pathology and an unbounded weight ratio (no $C < \infty$ in $\pi \leq Cq$).

Exercise 9.9

Gibbs sampler for a bivariate Gaussian $(X, Y) \sim \mathcal{N}(0, \Sigma)$ with $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$. Derive the full conditionals $p(x | y)$ and $p(y | x)$ and give a Gibbs sweep update rule.

Solution

For a centred Gaussian the conditional mean and variance follow from $E[X | Y = y] = \Sigma_{XY} \Sigma_Y^{-1} y = \rho y$ and $\text{Var}(X | Y) = 1 - \rho^2$. Hence

$$X | Y = y \sim \mathcal{N}(\rho y, 1 - \rho^2), \quad Y | X = x \sim \mathcal{N}(\rho x, 1 - \rho^2).$$

One Gibbs sweep at step t is

$$x^{(t)} \sim \mathcal{N}(\rho y^{(t-1)}, 1 - \rho^2), \quad y^{(t)} \sim \mathcal{N}(\rho x^{(t)}, 1 - \rho^2).$$

Exercise 9.10

Beta–Binomial hierarchical model: $x | \theta \sim \text{Binomial}(n, \theta)$, $\theta \sim \text{Beta}(\alpha, \beta)$. Derive the full conditional $p(\theta | x, \alpha, \beta)$ and identify it as a Beta distribution.

Solution

By Bayes’ rule,

$$p(\theta | x, \alpha, \beta) \propto p(x | \theta) p(\theta) \propto \theta^x (1 - \theta)^{n-x} \cdot \theta^{\alpha-1} (1 - \theta)^{\beta-1} = \theta^{\alpha+x-1} (1 - \theta)^{\beta+n-x-1}.$$

This is the kernel of a Beta distribution, so

$$\theta | x, \alpha, \beta \sim \text{Beta}(\alpha + x, \beta + n - x).$$

Conjugacy makes the Gibbs update a closed-form Beta draw.

Exercise 9.11

Two-component Gaussian mixture: $X_i | Z_i = k \sim \mathcal{N}(\mu_k, 1)$ with $Z_i \in \{1, 2\}$ and $P(Z_i = 1) = w$. Given μ_1, μ_2, w and the data, write the full conditional $p(z_i = 1 | x_i)$ (the E-step of EM, but in the Gibbs context).

Solution

By Bayes' rule, with $\phi(x; \mu) = \frac{1}{\sqrt{2\pi}} \exp\{-\frac{1}{2}(x - \mu)^2\}$,

$$p(z_i = 1 \mid x_i) = \frac{w \phi(x_i; \mu_1)}{w \phi(x_i; \mu_1) + (1 - w) \phi(x_i; \mu_2)}.$$

Cancelling the common $1/\sqrt{2\pi}$ factor gives the explicit form

$$p(z_i = 1 \mid x_i) = \frac{w \exp\{-\frac{1}{2}(x_i - \mu_1)^2\}}{w \exp\{-\frac{1}{2}(x_i - \mu_1)^2\} + (1 - w) \exp\{-\frac{1}{2}(x_i - \mu_2)^2\}}.$$

The Gibbs update samples z_i from this Bernoulli; the EM E-step uses the same expression as a soft responsibility.

Exercise 9.12

Show that the Gibbs sampler is a special case of Metropolis–Hastings in which the proposal for component k is the full conditional $\pi(x_k \mid x_{-k})$, and that the corresponding acceptance probability is always 1.

Solution

Take the MH proposal $q(x'_k \mid x_k, x_{-k}) = \pi(x'_k \mid x_{-k})$. The MH ratio factorises as

$$\frac{\pi(x'_k, x_{-k}) \pi(x_k \mid x_{-k})}{\pi(x_k, x_{-k}) \pi(x'_k \mid x_{-k})} = \frac{\pi(x'_k \mid x_{-k}) \pi(x_{-k}) \pi(x_k \mid x_{-k})}{\pi(x_k \mid x_{-k}) \pi(x_{-k}) \pi(x'_k \mid x_{-k})} = 1,$$

using $\pi(x_k, x_{-k}) = \pi(x_k \mid x_{-k}) \pi(x_{-k})$. Hence $\alpha(x \rightarrow x') = \min\{1, 1\} = 1$ for every Gibbs step.

Exercise 9.13

Block Gibbs. Suppose (θ_1, θ_2) have posterior correlation ρ close to 1. Sketch (no calculator) why coordinate-wise Gibbs mixes slowly and why updating (θ_1, θ_2) jointly fixes the problem.

Solution

When $\rho \approx 1$ the posterior concentrates on a narrow diagonal ridge. Each axis-aligned Gibbs step has conditional variance $1 - \rho^2 \ll 1$, so the chain inches along the ridge by small horizontal/vertical moves and the effective step *along* the ridge is $O(\sqrt{1 - \rho^2})$. The block update

$$(\theta_1, \theta_2) \sim p(\theta_1, \theta_2 \mid \text{rest}, D)$$

draws from the joint conditional, which has full variance along the ridge direction. One block step accomplishes what hundreds of coordinate-wise steps would require, eliminating the dependence on ρ .

Exercise 9.14

Reversibility from detailed balance. Verify that if a Markov chain satisfies $\pi(x)T(x, y) = \pi(y)T(y, x)$ for all states x, y , then π is stationary. Is the converse true?

Solution

Summing detailed balance over x :

$$\sum_x \pi(x) T(x, y) = \sum_x \pi(y) T(y, x) = \pi(y) \sum_x T(y, x) = \pi(y),$$

so $(\pi T)(y) = \pi(y)$ and π is stationary. The converse is **false**: the deterministic cyclic chain $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$ has $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ stationary but violates detailed balance.

Exercise 9.15

Trivial chain. A Markov chain has transition kernel $T(x, y) = \pi(y)$ for every x (i.e. each step is an independent draw from π). Verify detailed balance and compute the integrated autocorrelation time of any well-behaved test function.

Solution

Detailed balance: $\pi(x) T(x, y) = \pi(x) \pi(y) = \pi(y) \pi(x) = \pi(y) T(y, x)$. Successive states are i.i.d. from π , so for any test function f with finite variance the lag- k autocorrelation is $\rho_k = 0$ for all $k \geq 1$, and

$$\tau_{\text{int}} = \frac{1}{2} + \sum_{k=1}^{\infty} \rho_k = \frac{1}{2}.$$

This is the minimum possible value — i.i.d. sampling is the gold standard.

Exercise 9.16

(Define and explain.) Define the independence sampler. Write its acceptance probability in terms of the importance weights $w(x) = \pi(x)/q(x)$. Explain its connection to importance sampling.

Solution

The **independence sampler** is a Metropolis–Hastings algorithm whose proposal $q(x' | x) = q(x')$ does not depend on the current state. The MH ratio collapses to

$$\alpha(x \rightarrow x') = \min \left\{ 1, \frac{\pi(x') q(x)}{\pi(x) q(x')} \right\} = \min \left\{ 1, \frac{w(x')}{w(x)} \right\},$$

the ratio of importance weights $w(x) = \pi(x)/q(x)$. It is the MCMC analogue of importance sampling: q plays the role of a global proposal that should approximate π , and the acceptance step (rather than weighting) corrects the mismatch. Ergodicity requires the envelope bound $w(x) \leq C < \infty$.

Exercise 9.17

(Define and explain.) Define the Gibbs sampler. Show that every Gibbs step is accepted, i.e. the Metropolis–Hastings acceptance ratio equals 1 when the proposal is the full conditional.

Solution

The **Gibbs sampler** updates parameters $\theta = (\theta_1, \dots, \theta_d)$ cyclically by sampling each coordinate from its full conditional

$$\theta_i^{(t)} \sim p(\theta_i | \theta_{-i}, D).$$

Viewed as MH with proposal $q(\theta'_i | \theta_{-i}) = p(\theta'_i | \theta_{-i}, D)$, the acceptance ratio factorises via $p(\theta_i, \theta_{-i}) =$

$p(\theta_i | \theta_{-i})p(\theta_{-i})$:

$$\frac{p(\theta'_i, \theta_{-i}) p(\theta_i | \theta_{-i})}{p(\theta_i, \theta_{-i}) p(\theta'_i | \theta_{-i})} = \frac{p(\theta'_i | \theta_{-i}) p(\theta_i | \theta_{-i})}{p(\theta_i | \theta_{-i}) p(\theta'_i | \theta_{-i})} = 1.$$

Hence $\alpha = 1$ — every Gibbs proposal is accepted.

Exercise 9.18

(Compare and contrast.) For a target with strongly correlated coordinates, compare the mixing behaviour of coordinate-wise Gibbs, block Gibbs, and random-walk Metropolis. State which one you would prefer in this situation and why.

Solution

- **Coordinate-wise Gibbs**: always accepts but takes $O(\frac{1}{1-\rho^2})$ steps to cross the posterior ridge — slow when $|\rho| \rightarrow 1$.
- **Random-walk MH with proposal covariance** Σ_{prop} : also slow unless Σ_{prop} is tuned to match the elongated posterior covariance Σ ; an isotropic proposal wastes effort across the ridge.
- **Block Gibbs**: samples (θ_1, θ_2) jointly from the exact joint conditional, removing the dependence on ρ entirely.

Preferred: block Gibbs, because it eliminates the bottleneck at its source. If the joint conditional is intractable, the next best option is random-walk MH with a preconditioned proposal $\Sigma_{\text{prop}} \propto \Sigma$ (or HMC).

Exercise 9.19

(Define and explain.) Define integrated autocorrelation time. Explain in one sentence why grouping (blocking) strongly correlated parameters in a Gibbs sampler lowers it.

Solution

For a stationary chain and test function f with lag- k autocorrelation ρ_k ,

$$\tau_{\text{int}} = \frac{1}{2} + \sum_{k=1}^{\infty} \rho_k,$$

so that the effective sample size is $N_{\text{eff}} = N/(2\tau_{\text{int}})$. Blocking strongly correlated coordinates lets one update move along the posterior ridge in a single step (instead of many small axis-aligned moves), which collapses the slow tail of ρ_k and therefore reduces τ_{int} .

9.8. Sources

- Robert, C. P., & Casella, G. (2004). *Monte Carlo Statistical Methods* (2nd ed., Springer Texts in Statistics, Chapters 7 & 10).
- Falasco, G. (2026). *Advanced statistics for physics analysis* (Sections 6.4–6.5, lecture notes).

10. Hamiltonian Monte Carlo

10.1. Introduction

Markov chain Monte Carlo (MCMC) methods form the backbone of modern Bayesian inference, but they face a fundamental challenge: as dimension and posterior curvature increase, standard samplers become inefficient. Random-walk Metropolis–Hastings requires ever-smaller step sizes to maintain acceptance, leading to slow exploration and highly autocorrelated samples. Independence samplers, which propose large moves, suffer equally in high dimensions as their proposals become increasingly unlikely.

Hamiltonian Monte Carlo (HMC) breaks this curse of dimensionality by exploiting the geometry of the target distribution. Instead of proposing moves blindly, it uses gradient information to propose distant, geometrically informed steps that remain probable under the posterior. By augmenting the state space with auxiliary momentum variables, HMC transforms random exploration into ballistic motion guided by the log-posterior surface.

This chapter synthesizes the mechanics, practical implementation, and automatic tuning of HMC. We cover the theoretical foundation (Hamiltonian dynamics and energy conservation), the discrete leapfrog integrator, the Metropolis correction for discretization error, and modern variants like the No-U-Turn Sampler (NUTS) that remove manual tuning. We also discuss when HMC is preferable to alternatives and highlight its role in packages like Stan.

10.2. Why random-walk Metropolis fails in high dimensions

Remark. Standard random-walk Metropolis proposes local moves:

$$x' = x + \eta, \quad \eta \sim \mathcal{N}(0, \Sigma_{\text{prop}}).$$

Acceptance probability is $\alpha = \min(1, e^{-\Delta \log \pi})$, where $\Delta \log \pi = \log \pi(x') - \log \pi(x)$.

The curse: In d dimensions, for a fixed acceptance rate (typically ~ 0.23 for the Metropolis algorithm in the limit), the optimal step size scales as $O(d^{-1/2})$. This means: - To maintain constant acceptance as d grows, step sizes must shrink like $d^{-1/2}$. - Distance explored per iteration scales as step size $\times \sqrt{d} \sim O(1)$, regardless of dimension. - **Autocorrelation grows linearly with d** , making the effective sample size collapse.

For correlated posteriors with high curvature (e.g., a narrow ridge), the problem is even worse: proposals must shrink to match the local geometry, requiring $O(d)$ or $O(d^2)$ steps to traverse the parameter space.

Example 10.1

Bivariate Gaussian with strong correlation

Consider a target $\pi(x) = \mathcal{N}(0, \Sigma)$ with

$$\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad \rho = 0.95.$$

The ridge is elongated; random-walk MH must take tiny steps perpendicular to the ridge to avoid rejection, then many small steps along the ridge. For a $d = 100$ -dimensional elongated Gaussian, RW-MH becomes prohibitively slow. HMC, by following the natural flow of the posterior, explores this space in far fewer iterations.

10.3. Hamiltonian mechanics: the continuous story

Hamiltonian dynamics govern many physical systems. We repurpose them for sampling by augmenting the parameter space with momentum variables, then exploiting the conservation laws of the resulting flow.

- **Augmented state space.** Let $\mathbf{q} \in \mathbb{R}^d$ be the parameters of interest (confusingly called “position” in physics). We introduce **momentum** $\mathbf{p} \in \mathbb{R}^d$, an auxiliary variable sampled from a Gaussian with covariance matrix M (the “mass matrix”).

Definition 10.1. The **augmented target distribution** is

$$\pi(\mathbf{q}, \mathbf{p}) = \pi(\mathbf{q}) f(\mathbf{p} \mid M),$$

where $f(\mathbf{p} \mid M) = \mathcal{N}(\mathbf{0}, M)$ is the momentum density.

Crucial observation: Marginalizing over $\mathbf{p}$ recovers the original target:

$$\int \pi(\mathbf{q}, \mathbf{p}) d\mathbf{p} = \pi(\mathbf{q}) \int \mathcal{N}(\mathbf{0}, M) d\mathbf{p} = \pi(\mathbf{q}).$$

Thus, sampling $(\mathbf{q}, \mathbf{p})$ and discarding $\mathbf{p}$ gives samples from $\pi(\mathbf{q})$.

- **The Hamiltonian function.** The joint density factorizes as

$$\pi(\mathbf{q}, \mathbf{p}) = \pi(\mathbf{q}) \mathcal{N}(\mathbf{0}, M) \propto \exp[-U(\mathbf{q}) - K(\mathbf{p})],$$

where we define the **Hamiltonian** $H(\mathbf{q}, \mathbf{p}) = U(\mathbf{q}) + K(\mathbf{p})$.

Definition 10.2. • **Potential energy:** $U(\mathbf{q}) = -\log \pi(\mathbf{q})$.

- **Kinetic energy:** $K(\mathbf{p}) = \frac{1}{2} \mathbf{p}^T M^{-1} \mathbf{p}$.

The **Hamiltonian** is $H(\mathbf{q}, \mathbf{p}) = U(\mathbf{q}) + K(\mathbf{p})$.

The joint density becomes $\pi(\mathbf{q}, \mathbf{p}) \propto \exp[-H(\mathbf{q}, \mathbf{p})]$.

The mass matrix M controls the geometry: a diagonal M with larger entries allows larger steps in those directions. The default is $M = I$ (identity), but adaptive mass matrices can be tuned to match posterior covariance.

- **Hamilton’s equations.** The dynamics of a Hamiltonian system are governed by the canonical equations of motion.

Theorem 10.1. *Hamilton’s equations of motion:*

$$\frac{d\mathbf{q}}{dt} = \nabla_{\mathbf{p}} H = M^{-1} \mathbf{p},$$

$$\frac{d\mathbf{p}}{dt} = -\nabla_{\mathbf{q}}H = -\nabla_{\mathbf{q}}U(\mathbf{q}) = \nabla_{\mathbf{q}}\log\pi(\mathbf{q}).$$

Interpretation: - Position evolves along the momentum direction, scaled by M^{-1} . - Momentum evolves in the direction of the gradient (natural gradient of log-posterior). - **Key property:** These equations naturally navigate toward high-probability regions via $\nabla\log\pi$.

- **Energy conservation and volume preservation.** An exact Hamiltonian trajectory conserves total energy $H(\mathbf{q}(t), \mathbf{p}(t)) = \text{const}$, with profound consequences for sampling.

Proposition 10.2. Energy conservation: Along an exact Hamiltonian trajectory, $\frac{dH}{dt} = 0$.

Proof:

$$\frac{dH}{dt} = \dot{\mathbf{q}}^T \nabla_{\mathbf{q}}H + \dot{\mathbf{p}}^T \nabla_{\mathbf{p}}H = (\nabla_{\mathbf{p}}H)^T \nabla_{\mathbf{q}}H + (-\nabla_{\mathbf{q}}H)^T \nabla_{\mathbf{p}}H = 0.$$

Exact trajectories move along **level sets** of the Hamiltonian, staying on surfaces of constant H . Since $\pi(\mathbf{q}, \mathbf{p}) \propto e^{-H}$, exact trajectories move along level sets of the joint density — they are ideal proposals! Additionally, Hamiltonian flow is **volume-preserving** (symplectic) in phase space.

Proposition 10.3. Liouville's theorem: The divergence of the Hamiltonian velocity field is zero:

$$\nabla \cdot \begin{pmatrix} \dot{\mathbf{q}} \\ \dot{\mathbf{p}} \end{pmatrix} = \nabla_{\mathbf{q}} \cdot (M^{-1}\mathbf{p}) + \nabla_{\mathbf{p}} \cdot (-\nabla_{\mathbf{q}}U) = 0 + 0 = 0.$$

Thus Hamiltonian flow preserves volume in phase space and leaves $\pi(\mathbf{q}, \mathbf{p})$ invariant.

Together, energy conservation and volume preservation imply that Hamiltonian flow preserves $\pi(\mathbf{q}, \mathbf{p})$. But running the dynamics alone would never leave the level set determined by the initial condition: to explore the whole space we must periodically jump between level sets of H by **refreshing** $\mathbf{p}$ from its Gaussian marginal.

10.4. The leapfrog integrator

We cannot simulate Hamiltonian dynamics exactly on a computer. Instead, we discretize time using the **leapfrog integrator**, which maintains approximate energy conservation and reversibility while being computationally efficient.

- **Leapfrog discretization.** Given current state $(\mathbf{q}_0, \mathbf{p}_0)$, step size $\varepsilon > 0$, and number of steps L , one leapfrog step is defined below. The name “leapfrog” reflects the staggered half-steps: momentum is updated at half-steps, position at full steps.

Note. **One leapfrog step** (from $(\mathbf{q}_t, \mathbf{p}_t)$ to $(\mathbf{q}_{t+1}, \mathbf{p}_{t+1})$):

$$\begin{aligned} \mathbf{p}_{t+\varepsilon/2} &= \mathbf{p}_t - \frac{\varepsilon}{2} \nabla U(\mathbf{q}_t), \\ \mathbf{q}_{t+\varepsilon} &= \mathbf{q}_t + \varepsilon M^{-1} \mathbf{p}_{t+\varepsilon/2}, \\ \mathbf{p}_{t+\varepsilon} &= \mathbf{p}_{t+\varepsilon/2} - \frac{\varepsilon}{2} \nabla U(\mathbf{q}_{t+\varepsilon}). \end{aligned}$$

Repeating L times produces a trajectory of length $T = L\varepsilon$.

- **Why leapfrog?** The integrator has three essential properties that make it ideal for HMC.

Theorem 10.4. *The leapfrog integrator is:*

1. **Reversible:** *If we run the integrator backward (reversing $\varepsilon \rightarrow -\varepsilon$ and flipping momentum at the end), we recover the initial state exactly (up to numerical precision). This ensures the proposal $(\mathbf{q}', \mathbf{p}') = \text{Leapfrog}(\mathbf{q}, \mathbf{p}, \varepsilon, L)$ and its reverse satisfy $q((\mathbf{q}', \mathbf{p}') \rightarrow (\mathbf{q}, \mathbf{p})) = q((\mathbf{q}, \mathbf{p}) \rightarrow (\mathbf{q}', \mathbf{p}'))$, so the forward and reverse probabilities cancel in the Metropolis ratio.*
2. **Volume-preserving:** *The leapfrog map is a composition of volume-preserving transformations and preserves the Jacobian determinant. This reduces the Metropolis acceptance formula.*
3. **Symplectic:** *The integrator is a symplectic integrator, meaning it preserves the symplectic structure $\mathbf{q} \wedge \mathbf{p}$. The error in energy per step is $O(\varepsilon^2)$, so total error over L steps is $O(\varepsilon^2)$.*

For small ε , the discretization error is tiny, energy error $\Delta H \approx O(\varepsilon^2)$ accumulates slowly, and acceptance rates remain high. If ε is too large, energy errors accumulate and the rejection rate increases.

10.4.1. Example: 2D leapfrog trajectory

Let us visualize a leapfrog trajectory in 2D phase space for the bivariate Gaussian example.

See code 15.36 in Appendix for the implementation.

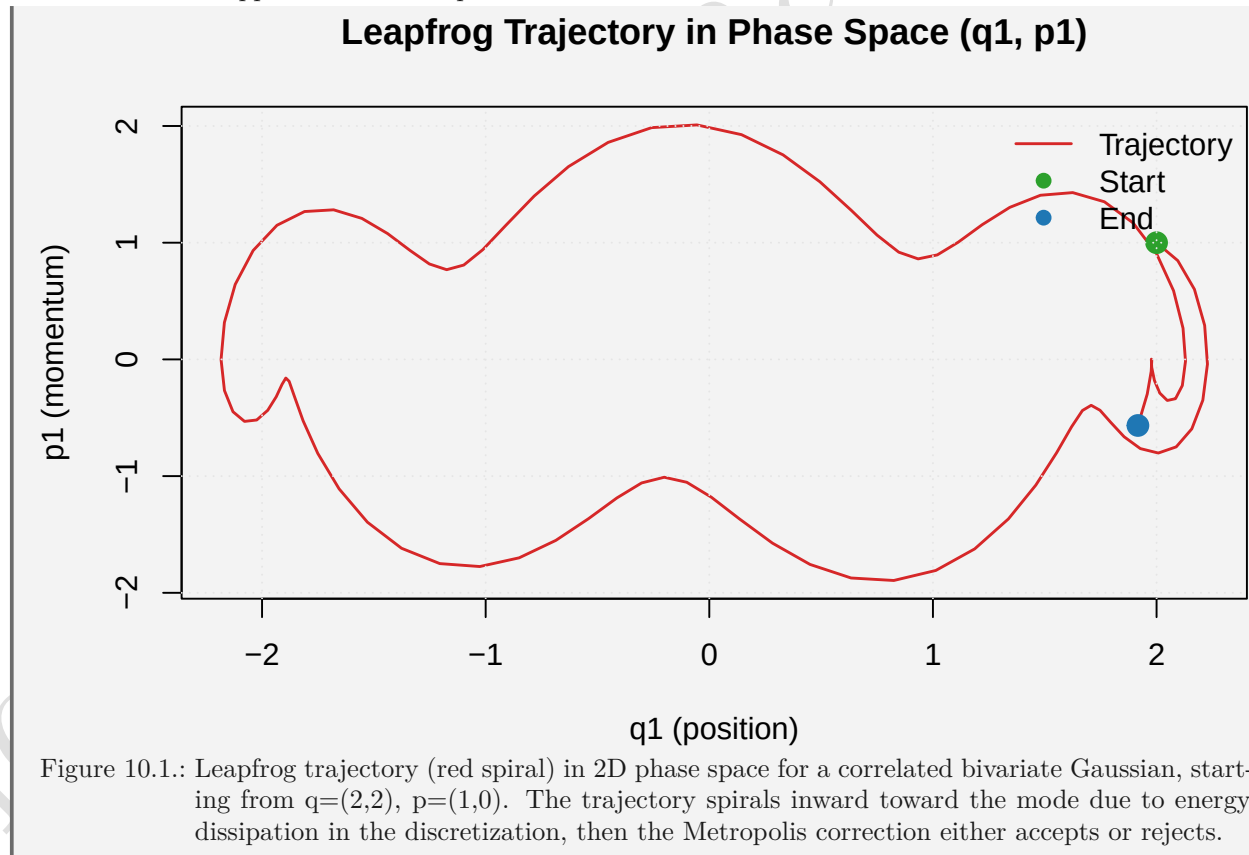


Figure 10.1.: Leapfrog trajectory (red spiral) in 2D phase space for a correlated bivariate Gaussian, starting from $\mathbf{q}=(2,2)$, $\mathbf{p}=(1,0)$. The trajectory spirals inward toward the mode due to energy dissipation in the discretization, then the Metropolis correction either accepts or rejects.

10.5. The HMC algorithm

Combining leapfrog integration with Metropolis acceptance yields the complete HMC algorithm.

Note. HMC Algorithm

Given current position $\mathbf{q}_t$:

1. **Refresh momentum:** Draw $\mathbf{p} \sim \mathcal{N}(\mathbf{0}, M)$.
2. **Store current energy:** Compute $H_{\text{old}} = U(\mathbf{q}_t) + K(\mathbf{p})$.
3. **Integrate:** Run leapfrog for L steps with step size ε :

$$(\mathbf{q}', \mathbf{p}') = \text{Leapfrog}(\mathbf{q}_t, \mathbf{p}, \varepsilon, L).$$

4. **Compute proposal energy:** $H_{\text{new}} = U(\mathbf{q}') + K(\mathbf{p}')$.
5. **Metropolis correction:** Accept with probability

$$\alpha = \min(1, \exp(-\Delta H)), \quad \Delta H = H_{\text{new}} - H_{\text{old}}.$$

6. **Move or reject:**

- If $u \sim \text{Uniform}(0, 1) < \alpha$: set $\mathbf{q}_{t+1} = \mathbf{q}'$.
- Else: set $\mathbf{q}_{t+1} = \mathbf{q}_t$.

- **Why the Metropolis correction?** If we could simulate Hamiltonian dynamics exactly, $\Delta H = 0$ and every proposal would be accepted. However, leapfrog introduces discretization error $\Delta H \approx O(L\varepsilon^2)$. The Metropolis step removes any bias introduced by discretization, restores exact invariance to the joint distribution $\pi(\mathbf{q}, \mathbf{p})$, and ensures that marginalizing over $\mathbf{p}$ gives exact samples from $\pi(\mathbf{q})$. A smaller ε reduces discretization error and increases acceptance; a larger ε reduces computation but risks rejection.
- **Momentum refresh.** In step 1, we draw a fresh momentum independently of the previous momentum. This is crucial: otherwise, the trajectory would continue indefinitely on level sets of H , exploring only one mode. By resampling $\mathbf{p}$ at each iteration, we jump between level sets of H (exploring different energies), allow exploration of the full parameter space, and introduce randomness necessary for ergodicity. Notably, resampling $\mathbf{p} \sim \mathcal{N}(\mathbf{0}, M)$ is **Gibbs sampling** in the (q, p) space.

10.6. Tuning and practical guidance

HMC's efficiency depends critically on three tuning parameters — step size ε , number of leapfrog steps L , and mass matrix M — together with a warm-up phase that adapts them automatically.

- **Step size ε .** This controls discretization granularity, and is the parameter most often tuned automatically.

Definition 10.3. The **step size** ε controls discretization granularity.

- **Too small:** $\varepsilon \rightarrow 0$ gives accurate proposals, high acceptance, but slow exploration (many gradient evaluations).
- **Too large:** ε large causes energy errors to accumulate; acceptance rate α drops.

Typical rule: Target acceptance rate of 0.65–0.80 (varies slightly with d). Adjust ε to match this during warm-up.

- **Trajectory length and number of steps L .** The product $T = L\varepsilon$ is the simulated time and determines how far each proposal travels.

Definition 10.4. The **trajectory length** is $T = L\varepsilon$. A longer trajectory explores farther but at higher computational cost.

- **Too few steps:** Behaves like random-walk MH (local exploration).
- **Too many steps:** Trajectories double back on themselves (waste computation and may reject).

Effective range: Aim for $L \in [20, 100]$ initially; tune via ESS per gradient evaluation.

- **Mass matrix M .** The mass matrix controls how momentum couples to position. With $M = I$ (identity), kinetic energy is $K(\mathbf{p}) = \frac{1}{2}\|\mathbf{p}\|^2$ — isotropic.

Remark. Adaptive mass matrix: If the posterior has vastly different scales or high correlations, an isotropic M is inefficient. Adaptive HMC (as implemented in Stan) estimates the posterior covariance Σ_{post} during warm-up and sets $M = \Sigma_{\text{post}}^{-1}$. This makes the “momentum-space” geometry match the “parameter-space” geometry, enabling larger steps and better mixing.

Equivalently, we can work with a reparametrization or preconditioning. The improvement can be dramatic in correlated or high-dimensional problems.

- **Warm-up and adaptation.** Most HMC implementations (especially Stan) use a **warm-up** (or “burn-in”) phase to (i) find a reasonable starting point and estimate posterior covariance, (ii) adapt ε to target a desired acceptance rate, (iii) adapt M to match posterior structure, and (iv) discard warm-up samples as they are not from the stationary distribution. After warm-up, ε and M are fixed, and the algorithm becomes a standard Markov chain with invariant distribution $\pi(\mathbf{q})$.

10.7. Divergent transitions and energy issues

HMC can fail in subtle ways:

Remark. Divergent transitions occur when the numerical integration goes unstable, typically because:

- Step size ε is too large and energy error $|\Delta H|$ becomes very large, causing rejection.
- Posterior has regions of high curvature that the integrator cannot resolve.
- Posterior has heavy tails or mild multimodality.

Indicator: Many proposals have $\Delta H \gg 10$ (large energy errors).

Solution: Reduce ε , increase warm-up iterations, or use an adaptive mass matrix. Stan prints warnings about divergences; they suggest the sampler is not exploring properly and posterior inferences may be biased.

10.8. NUTS: No-U-Turn Sampler

Modern software like Stan uses the **No-U-Turn Sampler (NUTS)**, which removes the need to manually tune L . Instead of running a fixed number of steps L , NUTS adaptively expands the trajectory until it would begin to double back.

Definition 10.5. Doubling criterion: A trajectory “doubles back” (makes a U-turn) when the “generalized velocity” $(\mathbf{q}' - \mathbf{q}_{\text{left}})^T \mathbf{p}$ becomes negative, indicating backward motion. More precisely, a U-turn

is detected using the dot product:

$$(\mathbf{q}_{\text{right}} - \mathbf{q}_{\text{left}})^T \mathbf{p}_{\text{right}} < 0 \quad \text{or} \quad (\mathbf{q}_{\text{right}} - \mathbf{q}_{\text{left}})^T \mathbf{p}_{\text{left}} < 0.$$

NUTS extends the trajectory by “doubling” (adding steps in the same direction) until a U-turn is detected, then accepts from the entire trajectory.

The advantages of NUTS over fixed- L HMC are:

- **Eliminates manual tuning of L :** The algorithm automatically determines trajectory length.
- **Prevents wasted computation:** Stops before the trajectory begins circling back.
- **Maintains properties:** The randomized doubling ensures detailed balance is preserved (via Metropolis-from-trajectory).
- **Stan implementation:** NUTS is the default sampler in Stan with automatic warm-up and adaptation.

10.9. Comparison: HMC vs random-walk MH vs Gibbs

Proposition 10.5. *Scaling with dimension d :*

<i>Sampler</i>	<i>Acceptance</i>	<i>Auto-correlation</i>	<i>ESS per gradient</i>
<i>RW-MH</i>	$O(1)$	$O(d)$	$O(d^{-2})$
<i>Gibbs</i>	1	$O(d)$	$O(d^{-1})$ if uncorrelated
<i>HMC</i>	0.6–0.9	$O(\sqrt{d})$ or better	$O(1)$ to $O(d^{-1})$

(Rough asymptotics; constants matter.)

Practical implications: - **RW-MH:** Simple to implement, no gradients needed. But prohibitively slow for $d > 20$ unless posterior is nearly isotropic. - **Gibbs:** Requires conditional distributions. Fast if draws from conditionals are cheap. Struggles with correlations. - **HMC:** Requires gradients but dramatically outperforms RW-MH in medium to high dimensions ($d \geq 10$). Excellent when gradients are available (automatic differentiation).

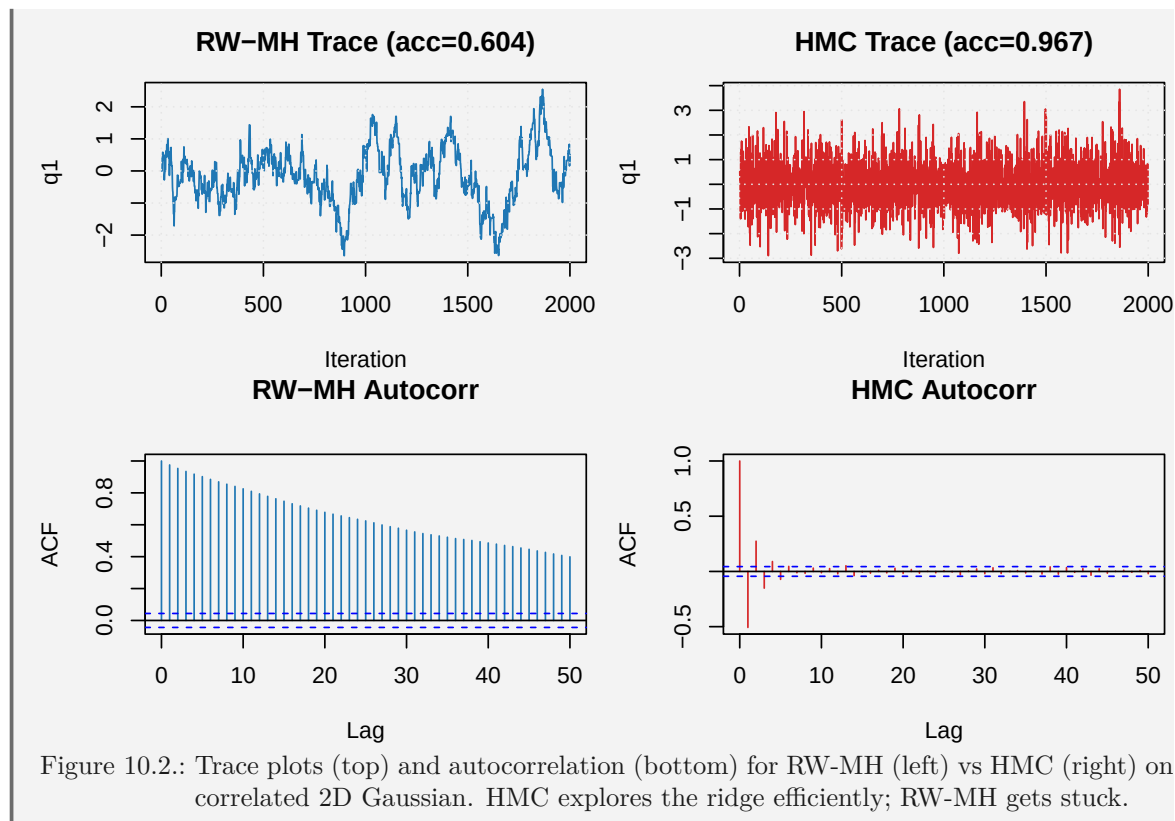
Example 10.2

Comparison on correlated bivariate Gaussian

We sample from $\mathcal{N}(0, \Sigma)$ with $\Sigma = \begin{pmatrix} 1 & 0.95 \\ 0.95 & 1 \end{pmatrix}$ using: 1. Random-walk MH with step = 0.3. 2.

HMC with $\epsilon = 0.15$, $L = 20$.

See code 15.37 in Appendix for the implementation.



RW-MH has acceptance ≈ 0.61 and poor mixing; autocorrelation decays slowly. HMC has acceptance ≈ 0.96 and autocorrelation decays in ~ 10 lags, yielding much larger effective sample size per gradient evaluation.

10.10. Practical recommendations

Remark. When to use HMC: - Posterior dimension $d \geq 10$. - Gradient $\nabla \log \pi$ is available (exact or via automatic differentiation). - Posterior is not highly multimodal with separated modes. - Computational budget permits gradient evaluations (modern autodiff is fast).

When HMC struggles: - Highly multimodal posterior with isolated modes (HMC gets trapped in one mode; consider parallel tempering or other global methods). - Discrete parameters (HMC requires continuous state; marginalize out discrete variables or use other samplers). - Gradient unavailable and too expensive to compute numerically.

When to prefer alternatives: - **Gibbs sampling:** If conditional distributions are easy to sample. Better mixing if conditionals are well-separated; worse if highly correlated. - **Random-walk MH:** Low-dimensional ($d < 10$), no gradients, simple implementation preferred. - **Slice sampling or other adaptive methods:** If gradient unavailable but adaptive exploration is desired.

10.11. Stan and modern practice

The probabilistic programming language **Stan** uses NUTS with automatic warm-up by default. This means:

1. Users specify the log-posterior (via a model statement).

2. Stan automatically differentiates to obtain $\nabla \log \pi$ via the reverse-mode autodiff.
3. During warm-up, step size ε is tuned to target 0.80 acceptance; mass matrix M is adapted.
4. NUTS extends trajectories adaptively without manual specification of L .
5. Convergence diagnostics ($\hat{R}$, effective sample size) are computed automatically.

This makes HMC accessible to practitioners without deep knowledge of tuning. **Stan is now the most widely used tool for general-purpose Bayesian inference**, precisely because it combines HMC's superior performance with ease of use.

10.12. Exercises

Exercise 10.1

For a target $\pi(\theta) \propto \exp(-U(\theta))$ with $U(\theta) = \frac{1}{2}\theta^\top \theta$ (standard Gaussian), write the Hamiltonian $H(\theta, p)$ with a unit mass matrix, the canonical equations of motion, and the analytic solution for $(\theta(t), p(t))$.

Solution

With $M = I$ the Hamiltonian is

$$H(\theta, p) = \frac{1}{2}\theta^\top \theta + \frac{1}{2}p^\top p,$$

and Hamilton's equations give $\dot{\theta} = p$, $\dot{p} = -\theta$. This is the harmonic oscillator, whose closed-form solution is

$$\theta(t) = \theta_0 \cos t + p_0 \sin t, \quad p(t) = -\theta_0 \sin t + p_0 \cos t,$$

i.e. a rigid rotation of phase space, so the trajectory stays exactly on the level set $H = \frac{1}{2}(\|\theta_0\|^2 + \|p_0\|^2)$.

Exercise 10.2

Show that for a separable Hamiltonian $H(\theta, p) = U(\theta) + K(p)$ with $K(p) = \frac{1}{2}p^\top M^{-1}p$, the canonical equations decouple into $\dot{\theta} = M^{-1}p$ and $\dot{p} = -\nabla U(\theta)$.

Solution

Because U depends only on θ and K only on p , the gradients separate: $\nabla_p H = \nabla_p K(p)$ and $\nabla_\theta H = \nabla_\theta U(\theta)$. Hamilton's equations therefore become

$$\dot{\theta} = \nabla_p H = M^{-1}p, \quad \dot{p} = -\nabla_\theta H = -\nabla U(\theta),$$

so position is driven only by p (through M^{-1}) and momentum only by the potential gradient — the coupling between θ and p is gone, which is exactly what lets leapfrog use alternating explicit half-steps.

Exercise 10.3

Energy conservation. Differentiate $H(\theta(t), p(t))$ along an exact trajectory and show that $dH/dt = 0$. State why the leapfrog integrator does not exactly conserve H but conserves a *modified* shadow Hamiltonian.

Solution

Along an exact trajectory,

$$\frac{dH}{dt} = \nabla_{\theta} H \cdot \dot{\theta} + \nabla_p H \cdot \dot{p} = \nabla_{\theta} H \cdot \nabla_p H + \nabla_p H \cdot (-\nabla_{\theta} H) = 0.$$

Leapfrog is a discrete symplectic map, not the exact flow, so H oscillates by $O(\varepsilon^2)$. By backward-error analysis it instead conserves a shadow Hamiltonian $\tilde{H} = H + \varepsilon^2 H_2 + \dots$ exactly, which is why ΔH stays bounded over long trajectories instead of drifting.

Exercise 10.4

Write out one full step of the leapfrog integrator for a separable Hamiltonian with step size ε . Show that the update is symplectic by checking that the Jacobian determinant equals one.

Solution

The three sub-steps are

$$p_{1/2} = p_0 - \frac{\varepsilon}{2} \nabla U(\theta_0), \quad \theta_1 = \theta_0 + \varepsilon M^{-1} p_{1/2}, \quad p_1 = p_{1/2} - \frac{\varepsilon}{2} \nabla U(\theta_1).$$

Each sub-step is a shear in either θ or p , so its Jacobian is upper- or lower-triangular with unit diagonal, e.g. the half kick has Jacobian $\begin{pmatrix} I & 0 \\ -\frac{\varepsilon}{2} \nabla^2 U & I \end{pmatrix}$ with determinant 1. The full step is the composition, hence $\det J = 1 \cdot 1 \cdot 1 = 1$, so leapfrog preserves phase-space volume and the Metropolis ratio reduces to $\exp(-\Delta H)$.

Exercise 10.5

Why random-walk Metropolis fails in high dimensions. Argue that for an isotropic Gaussian target in d dimensions, the optimal step size scales as $\varepsilon \sim d^{-1/2}$ and the number of steps to traverse the typical set scales as $d^{1/2}$, giving a *quadratic* cost in d . Contrast with HMC's $d^{1/4}$ cost.

Solution

A proposal $\theta' = \theta + \varepsilon \xi$ with $\xi \sim \mathcal{N}(0, I)$ produces $\Delta \log \pi$ with mean $-\frac{1}{2} d \varepsilon^2$ and variance $\sim d \varepsilon^2$, so keeping a non-vanishing acceptance forces $d \varepsilon^2 = O(1)$, i.e. $\varepsilon \sim d^{-1/2}$. The displacement per accepted step is then $\|\Delta \theta\| \sim \varepsilon \sqrt{d} \sim 1$, and crossing the typical set of radius $\sqrt{d}$ as a diffusion requires

$$N_{\text{RW-MH}} \sim d, \quad \text{i.e. total cost} \propto d,$$

which becomes d^2 once correlation length is included. HMC instead tolerates $\varepsilon \sim d^{-1/4}$ (the leapfrog energy error is $O(\varepsilon^2)$ summed coherently), so trajectories of length $T = O(1)$ need $L \sim d^{1/4}$ gradient evaluations — exponentially better in high d .

Exercise 10.6

HMC acceptance probability. Show that the acceptance probability of a proposed move $(\theta_0, p_0) \rightarrow (\theta_L, p_L)$ generated by leapfrog is

$$\alpha = \min\{1, \exp(-[H(\theta_L, p_L) - H(\theta_0, p_0)])\}.$$

Solution

The augmented target is $\pi(\theta, p) \propto e^{-H(\theta, p)}$. Leapfrog combined with the final momentum flip $p \mapsto -p$ gives a deterministic, volume-preserving, self-inverse proposal, so the forward and reverse proposal densities are equal and their Jacobian is 1. The Metropolis–Hastings ratio collapses to

$$\alpha = \min\left\{1, \frac{e^{-H(\theta_L, p_L)}}{e^{-H(\theta_0, p_0)}}\right\} = \min\{1, \exp(-\Delta H)\}.$$

Because exact dynamics would give $\Delta H = 0$, the only source of rejection is leapfrog’s discretization error.

Exercise 10.7

Optimal acceptance rate. State the result (Beskos et al., 2013) that the optimal HMC acceptance for isotropic Gaussian targets is approximately 0.65. Sketch the argument in words: too-low acceptance wastes proposals, too-high acceptance means ε is too small.

Solution

For separable Gaussian targets the leapfrog energy error is asymptotically Gaussian, $\Delta H \sim \mathcal{N}(\mu, 2\mu)$ with $\mu \propto d\varepsilon^4$, so the expected squared jumping distance per gradient evaluation is maximised at

$$\alpha^* = \mathbb{E}[\min\{1, e^{-\Delta H}\}] \approx 0.651.$$

Heuristically, very low acceptance means most leapfrog work is thrown away on rejected trajectories, while very high acceptance means ε is so small that each accepted trajectory barely moves; the sweet spot trades these costs.

Exercise 10.8

Mass-matrix reparametrisation. Show that running HMC with mass matrix M on parameter θ is equivalent to running HMC with identity mass matrix on the reparametrised variable $\tilde{\theta} = M^{1/2}\theta$. State the practical implication for posteriors with correlated parameters.

Solution

Let $\tilde{\theta} = M^{1/2}\theta$ and $\tilde{p} = M^{-1/2}p$. Then the kinetic energy becomes

$$K(p) = \frac{1}{2}p^\top M^{-1}p = \frac{1}{2}\tilde{p}^\top \tilde{p},$$

so in $(\tilde{\theta}, \tilde{p})$ coordinates the kinetic energy uses the identity mass matrix, while the Jacobian $|\det M^{1/2}|$ is constant and cancels in the acceptance ratio. Practically, choosing $M \approx \Sigma_{\text{post}}^{-1}$ whitens correlated posteriors: HMC sees an approximately isotropic problem in $\tilde{\theta}$, so a single step size works in all directions and step length need not shrink to the smallest eigendirection.

Exercise 10.9

Divergent transitions. Describe when leapfrog energy errors blow up. Give one example posterior geometry (e.g. a “funnel” in hierarchical models) where divergences are common, and the standard remedy (non-centered reparametrisation).

Solution

Leapfrog energy error scales as $\Delta H = O(\varepsilon^2 \|\nabla^2 U\|)$, so wherever the local curvature spikes, $|\Delta H|$ explodes and Stan flags a *divergence* (typically $|\Delta H| \gtrsim 10^3$). Neal's funnel $\theta_i \mid \tau \sim \mathcal{N}(0, e^\tau)$, $\tau \sim \mathcal{N}(0, 9)$ is the canonical example: near $\tau \ll 0$ the funnel neck is so narrow that no single ε works for both the neck and the mouth. The remedy is the non-centred reparametrisation

$$\theta_i = e^{\tau/2} z_i, \quad z_i \sim \mathcal{N}(0, 1),$$

which decouples scale from location and gives an approximately isotropic posterior in (z, τ) .

Exercise 10.10

NUTS termination criterion. Describe Hoffman & Gelman's no-U-turn condition: stop doubling the trajectory once the proposal starts curling back on itself. Why does this remove the need to tune the trajectory length L manually?

Solution

NUTS keeps doubling the trajectory in random forward/backward directions and terminates once the displacement vector $\Delta\theta = \theta_{\text{right}} - \theta_{\text{left}}$ stops aligning with the momentum at either endpoint, i.e.

$$\Delta\theta^\top p_{\text{right}} < 0 \quad \text{or} \quad \Delta\theta^\top p_{\text{left}} < 0.$$

At that instant any further integration would curve the trajectory back on itself, wasting gradient evaluations. Because the criterion is *local* and computed on the fly, the user no longer has to pick a fixed L : NUTS adapts the trajectory length per iteration to the local geometry while preserving detailed balance through a slice/multinomial sampler over the built tree.

Exercise 10.11

Time reversibility. Show that the leapfrog integrator is reversible: if we apply it for L steps to go from (θ_0, p_0) to (θ_L, p_L) , then a single negation $p_L \mapsto -p_L$ followed by L leapfrog steps returns to $(\theta_0, -p_0)$.

Solution

The kinetic energy is even in p , so under $p \mapsto -p$ the equations $\dot{\theta} = M^{-1}p$, $\dot{p} = -\nabla U$ become $\dot{\theta} = -M^{-1}p$, $\dot{p} = \nabla U$ — exactly the time-reversed dynamics. For one leapfrog step, applying the three sub-steps with negated momentum literally undoes the original substeps in reverse order:

$$\Phi_\varepsilon \circ R \circ \Phi_\varepsilon = R, \quad R : (\theta, p) \mapsto (\theta, -p).$$

Composing L such involutions gives $\Phi_\varepsilon^L \circ R \circ \Phi_\varepsilon^L(\theta_0, p_0) = (\theta_0, -p_0)$, which is what makes the leapfrog proposal symmetric so the MH ratio depends only on ΔH .

Exercise 10.12

Effect of step size on energy error. For leapfrog applied to a quadratic Hamiltonian, show that the per-step energy error scales as $O(\varepsilon^3)$ but the total error over a trajectory of fixed length scales as $O(\varepsilon^2)$. Why does this argue for keeping ε moderately small but not vanishingly small?

Solution

Leapfrog is a second-order symplectic method, so its *local* truncation error in H per step is $O(\varepsilon^3)$. Over a trajectory of fixed length $T = L\varepsilon$, the number of steps is $L = T/\varepsilon$, so the global energy error scales as

$$\Delta H_{\text{total}} = L \cdot O(\varepsilon^3) = O(\varepsilon^2).$$

Shrinking ε further keeps ΔH tiny and acceptance near 1, but doubles the number of gradient evaluations: cost scales like $1/\varepsilon$ while ESS gain is sub-linear, so the sweet spot is a moderate ε targeting ~65–80% acceptance, not $\varepsilon \rightarrow 0$.

Exercise 10.13

HMC vs random-walk Metropolis. Sketch on a single plot: target trace from random-walk MH with poorly tuned step size, target trace from HMC with reasonable (ε, L) . Identify the integrated autocorrelation time of each visually and explain the gap.

Solution

The RW-MH trace looks like a slow diffusive walk with long flat plateaus (frequent rejections) and visible drift over many iterations, so its integrated autocorrelation time τ_{RW} is large; the HMC trace looks like nearly independent draws hopping across the support, with $\tau_{\text{HMC}} \approx 1\text{--}5$. The ESS ratio scales as

$$\frac{\text{ESS}_{\text{HMC}}}{\text{ESS}_{\text{RW}}} = \frac{\tau_{\text{RW}}}{\tau_{\text{HMC}}} \sim \frac{d}{d^{1/2}} = d^{1/2},$$

because each HMC trajectory traverses an $O(1)$ fraction of the typical set in one move, while RW-MH only takes a step of size $O(d^{-1/2})$.

Exercise 10.14

Define the *typical set* of a d -dimensional posterior. Argue why HMC, which conserves energy and follows gradient flow, stays inside the typical set, while random-walk MH wastes most of its time outside it in high dimensions.

Solution

The typical set $\mathcal{T}_\delta$ is the thin shell of states whose log-density equals its expectation up to δ :

$$\mathcal{T}_\delta = \{\theta : |\log \pi(\theta) - \mathbb{E}_\pi[\log \pi]| < \delta\},$$

which in high d concentrates near a $(d-1)$ -dimensional surface far from the mode (e.g. a thin shell of radius $\sqrt{d}$ for $\mathcal{N}(0, I)$). HMC trajectories conserve $H = U + K$, and on average $U(\theta) \approx \mathbb{E}_\pi[U]$, so the dynamics flow *along* this level set — they orbit inside $\mathcal{T}_\delta$ by construction. RW-MH proposes isotropically in $\mathbb{R}^d$, where almost all directions point *off* the thin shell, so most proposals leave the typical set, get rejected, and the chain barely moves.

Exercise 10.15

Practical tuning. A Stan run reports $\hat{R} = 1.01$, ESS = 200 per chain, divergences = 12 out of 4000. Decide: is this acceptable? If not, list the standard remedies in order of priority.

10.13. Sources

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- **Hoffman, M. D., & Gelman, A. (2014).** The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15, 1351–1381. [Foundational paper on NUTS.]
- **Carpenter, B., et al. (2017).** Stan: A probabilistic programming language. *Journal of Statistical Software*, 76(1). [Stan implementation and practical guidance.]

11. Parallel tempering

11.1. Introduction: The multimodality trap

Standard Metropolis–Hastings and Hamiltonian Monte Carlo samplers work well when the target posterior is unimodal or has modes close together. However, when the posterior has several well-separated modes, a single MCMC chain becomes trapped near the mode it encounters first. It cannot escape because crossing the energy barrier between modes occurs with probability decaying exponentially in the barrier height.

Consider two modes separated by a barrier of height ΔE . The probability of escaping is approximately

$$p \approx \exp\left(-\frac{\Delta E}{T}\right)$$

by the Arrhenius formula, where we write the posterior as $\pi(\theta) = \exp[-E(\theta)]$. When $\Delta E \gg 1$, this probability is negligibly small, and the chain explores only a tiny fraction of the parameter space.

Simulated annealing and parallel tempering both tackle this problem by introducing a temperature parameter. They exploit a key observation: at high temperature T , the posterior is flattened, and energy barriers become easier to cross. The two methods differ in how they use this idea.

11.2. Tempered distributions and temperatures

For any target posterior $\pi(\theta)$ with energy representation $\pi(\theta) = \exp[-E(\theta)]$, we define the **tempered posterior** at inverse temperature $\beta \in (0, 1]$ by

$$\pi_\beta(\theta) \propto \pi(\theta)^\beta = \exp[-\beta E(\theta)].$$

Equivalently, using temperature $T = 1/\beta$,

$$\pi_T(\theta) \propto \pi(\theta)^{1/T} = \exp\left[-\frac{E(\theta)}{T}\right].$$

The two limits play distinct roles:

- **Cold limit** ($\beta \rightarrow 1$ or $T \rightarrow 1$). $\pi_\beta(\theta) \rightarrow \pi(\theta)$, the original posterior.
- **Hot limit** ($\beta \rightarrow 0$ or $T \rightarrow \infty$). $\pi_\beta(\theta)$ becomes nearly flat, and the chain explores the entire space freely.
- **Concentrated limit** ($T \rightarrow 0$, used in optimization). As the temperature decreases to zero, $\pi_T(\theta)$ concentrates its mass on the global minima of $E(\theta)$.

At high temperature, the contrast between modes vanishes, making it easy for the chain to move between them. As the temperature decreases, the modes become increasingly separated again.

11.3. Simulated annealing

Simulated annealing is an optimization method that cools a single chain to find the maximum a posteriori (MAP) estimator $\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \pi(\theta)$. It generates a sequence of increasingly concentrated distributions by running a cooling schedule.

Simulated annealing algorithm:

1. Initialize θ_0 and temperature $T_0 > 0$.
2. For $k = 0, 1, \dots, K$:
 - Propose $\theta' \sim q(\theta' | \theta_k)$.
 - Accept with probability $\alpha_T(\theta_k \rightarrow \theta') = \min\{1, \exp[-\beta_k(E(\theta') - E(\theta_k))]\}$ where $\beta_k = 1/T_k$.
 - Update $T_{k+1} = \gamma T_k$ for some $0 < \gamma < 1$ (e.g., geometric cooling) or use $T_k = c/\log k$ (logarithmic cooling).
3. Return the state θ with highest $\pi(\theta)$ encountered.

The choice of cooling schedule controls both correctness and cost:

- **Logarithmic cooling and barrier crossing.** With $T_k = c/\log k$, the chain crosses any fixed energy barrier infinitely often (with probability 1) as $k \rightarrow \infty$, because the cumulative escape probability

$$\sum_{k=1}^{\infty} p_k = \sum_{k=1}^{\infty} \exp\left(-\frac{\Delta E \cdot \log k}{c}\right) = \sum_{k=1}^{\infty} k^{-\Delta E/c}$$

diverges if and only if $\Delta E/c \leq 1$, i.e., $c \geq \Delta E$. Simulated annealing is therefore theoretically guaranteed to find the global optimum almost surely with sufficiently slow cooling.

- **Geometric cooling in practice.** Slower cooling requires more iterations, so the geometric schedule $T_{k+1} = \gamma T_k$ is usually preferred for computational speed, accepting that the optimality guarantee becomes heuristic.
- **What is returned.** The output is the state with the highest $\pi(\theta)$ encountered (or the final low-temperature state), not a sample from π — annealing answers an optimization question, not a sampling one.

11.4. Parallel tempering (replica exchange)

Parallel tempering runs K **chains in parallel**, each at a different inverse temperature $\beta_1 = 1 > \beta_2 > \dots > \beta_K > 0$. The chain at $\beta_1 = 1$ is the “cold chain” that targets the posterior of interest. Chains at smaller β are “hot chains” that explore more freely. The key innovation is **swaps**: occasionally, we exchange the states of neighbouring chains. This allows the cold chain to benefit from discoveries made by hot chains, enabling it to escape local modes without the cold chain ever needing to climb the barrier alone.

The algorithm is built from two pieces — a joint target and two move types:

- **Joint target.** Let θ_l denote the state of chain l , write $\Theta = (\theta_1, \dots, \theta_K)$, and define the **product target**

$$\Pi(\Theta) = \prod_{l=1}^K \pi_{\beta_l}(\theta_l) \propto \prod_{l=1}^K \pi(\theta_l)^{\beta_l}.$$

- **Within-temperature moves.** Each chain runs a standard Metropolis–Hastings update with the tempered acceptance probability

$$\alpha_l(\theta_l \rightarrow \theta'_l) = \min \left\{ 1, \left(\frac{\pi(\theta'_l)}{\pi(\theta_l)} \right)^{\beta_l} \frac{q(\theta_l | \theta'_l)}{q(\theta'_l | \theta_l)} \right\}.$$

Hot chains (small β_l) accept uphill moves more readily, allowing them to traverse the landscape freely.

Note. The marginal distribution of θ_1 (the cold chain) is exactly

$$\pi_{\beta_1}(\theta_1) = \pi(\theta_1),$$

since $\beta_1 = 1$. Samples from the cold chain are valid posterior samples, and expectations computed from the cold chain are unbiased estimates of posterior expectations.

11.4.1. Swap moves and detailed balance

The second type of move exchanges states between neighbouring chains l and $l+1$:

$$(\theta_l, \theta_{l+1}) \rightarrow (\theta_{l+1}, \theta_l).$$

The Metropolis acceptance probability for this move is

$$\alpha_{\text{swap}}(\theta_l, \theta_{l+1}) = \min \left\{ 1, \frac{\pi(\theta_{l+1})^{\beta_l} \pi(\theta_l)^{\beta_{l+1}}}{\pi(\theta_l)^{\beta_l} \pi(\theta_{l+1})^{\beta_{l+1}}} \right\}.$$

In log form (preferred numerically):

$$\log \alpha_{\text{swap}} = \min\{0, (\beta_l - \beta_{l+1})[E(\theta_l) - E(\theta_{l+1})]\}.$$

The normalizing constants cancel, making the swap probability independent of the partition function.

Theorem 11.1. *The combined algorithm of within-temperature MH updates and swap moves satisfies detailed balance with respect to the product target $\Pi(\Theta)$. Therefore, the cold chain's marginal converges to π .*

Proof. Within-temperature moves leave $\pi_{\beta_l}(\theta_l)$ invariant at each temperature. Swap moves satisfy detailed balance for the product density: the probability of being in state (θ_l, θ_{l+1}) and swapping to (θ_{l+1}, θ_l) equals the probability of being in (θ_{l+1}, θ_l) and swapping back (by the Metropolis ratio). The product of these two reversible operations preserves $\Pi(\Theta)$. $\square$

The mechanism that makes the cold chain mix can now be read off the construction:

- **Hot chains explore freely.** Because β_l is small for hot chains, they traverse the landscape easily, finding new modes.
- **Swaps transport information.** When a hot chain discovers a mode, a swap can move it to a colder chain, bringing the discovery closer to the target.
- **Cold chain remains on target.** The cold chain never leaves the target distribution π , yet it benefits from the hot chains' discoveries via successful swaps.

This is the crucial difference from simulated annealing: the temperature is fixed, not decreasing. The algorithm is homogeneous, run indefinitely, and all cold-chain samples are valid posterior samples.

11.5. Temperature ladder design

The spacing of temperatures is critical. If β values are too far apart, swap acceptance rates become vanishingly small, and information cannot flow between chains. If they are too close, many chains are needed.

11. Parallel tempering

Three design choices control the trade-off:

- **Geometric spacing.** A simple and effective choice is

$$\beta_l = \beta_1 r^{l-1}, \quad 0 < r < 1,$$

which keeps temperatures evenly spaced in a logarithmic sense. Under the standard assumption that the energy variance scales linearly with the dimension, geometric spacing approximately equalises swap acceptance between every adjacent pair.

- **Target swap acceptance.** A practical rule of thumb is to set the ladder so that the swap acceptance probability between adjacent chains is in the range 20–40%. Below this range information stops flowing along the ladder; above it the ladder is denser than necessary and one is paying for redundant chains.
- **All-pairs vs. neighbour-only swaps.** The basic algorithm swaps only neighbouring pairs. Allowing swaps between arbitrary chains (“all-pairs”) can improve mixing but at higher computational cost. For most problems, neighbour-only swaps are sufficient if the ladder is carefully tuned.

11.6. Worked example: bimodal 1D mixture

Example 11.1

Consider a bimodal posterior that is a mixture of two Gaussians:

$$\pi(\theta) \propto 0.35 N(\theta; -4, 1) + 0.65 N(\theta; 3, 0.7^2).$$

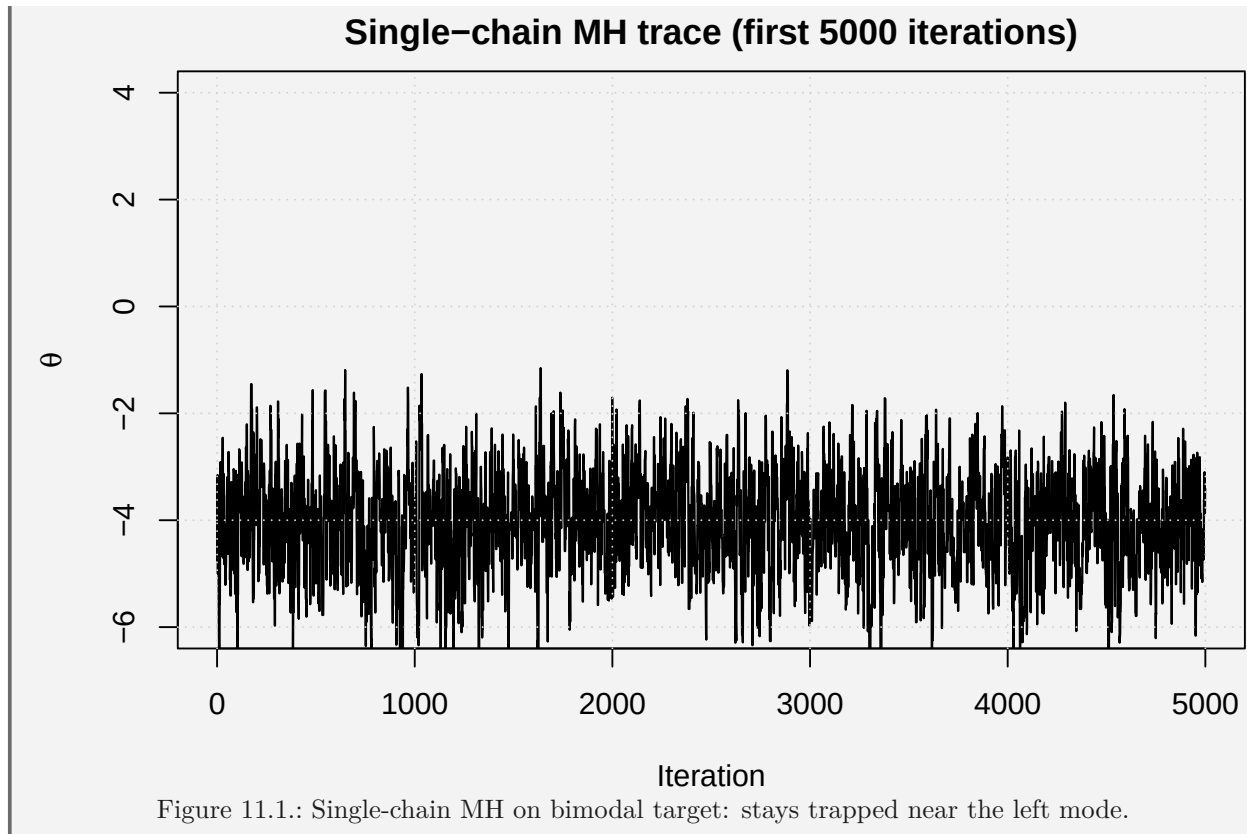
The right mode is higher. We will show that a single-chain MH sampler gets trapped, while parallel tempering explores both modes.

See code 15.38 in Appendix for the implementation.

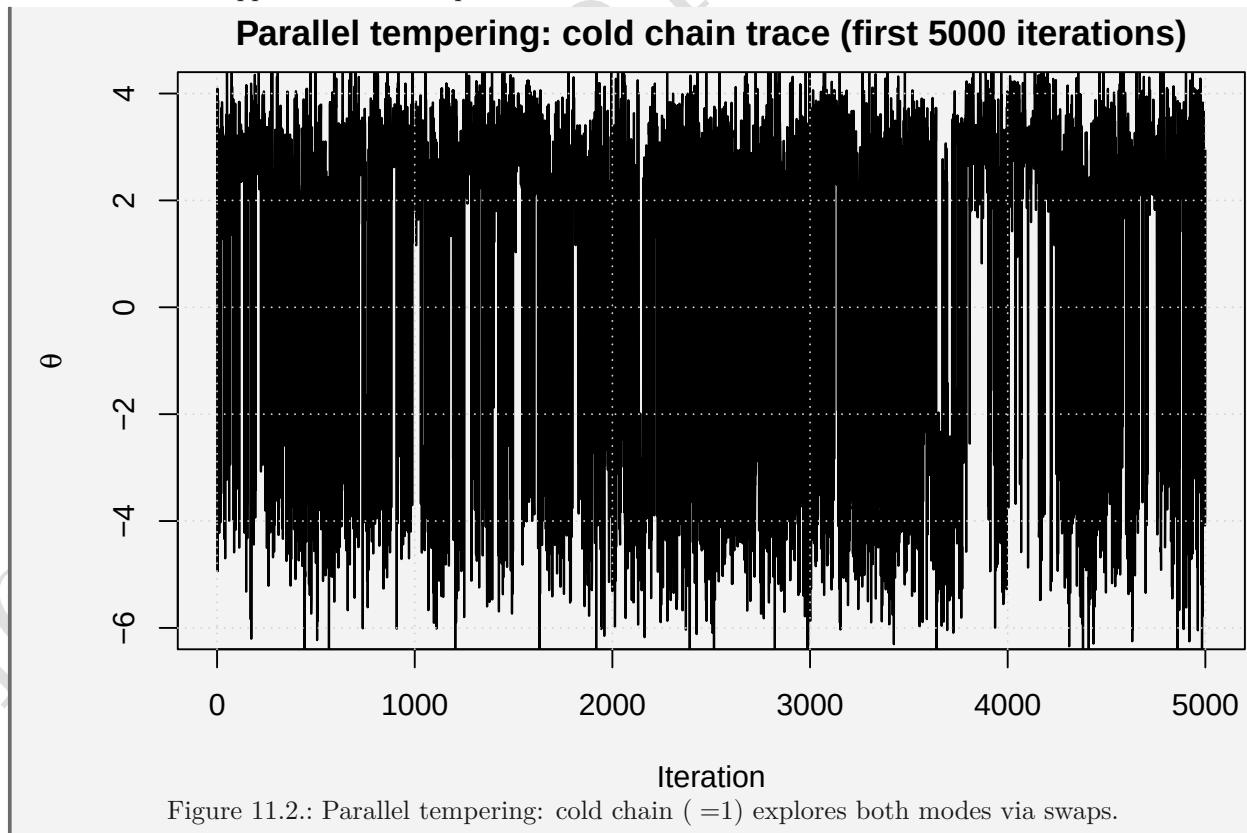
See code 15.39 in Appendix for the implementation.

See code 15.40 in Appendix for the implementation.

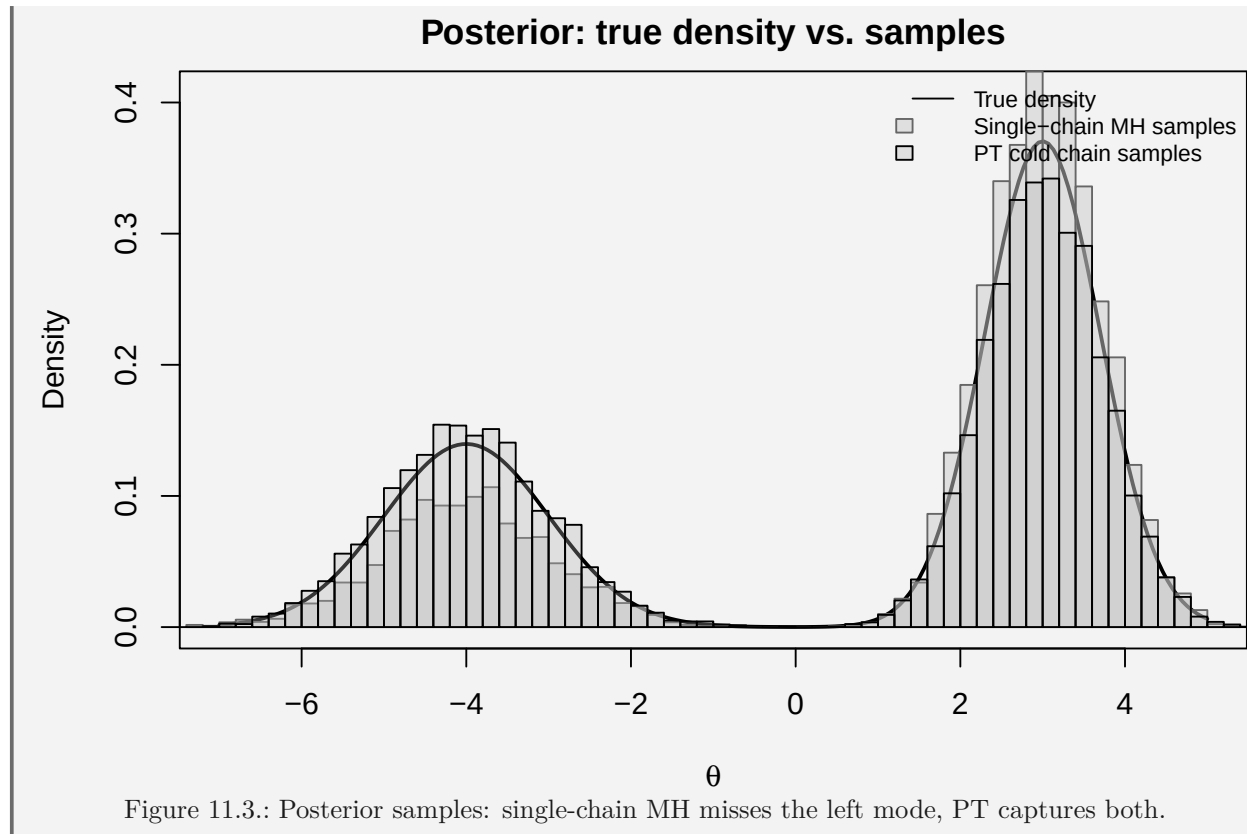
See code 15.41 in Appendix for the implementation.



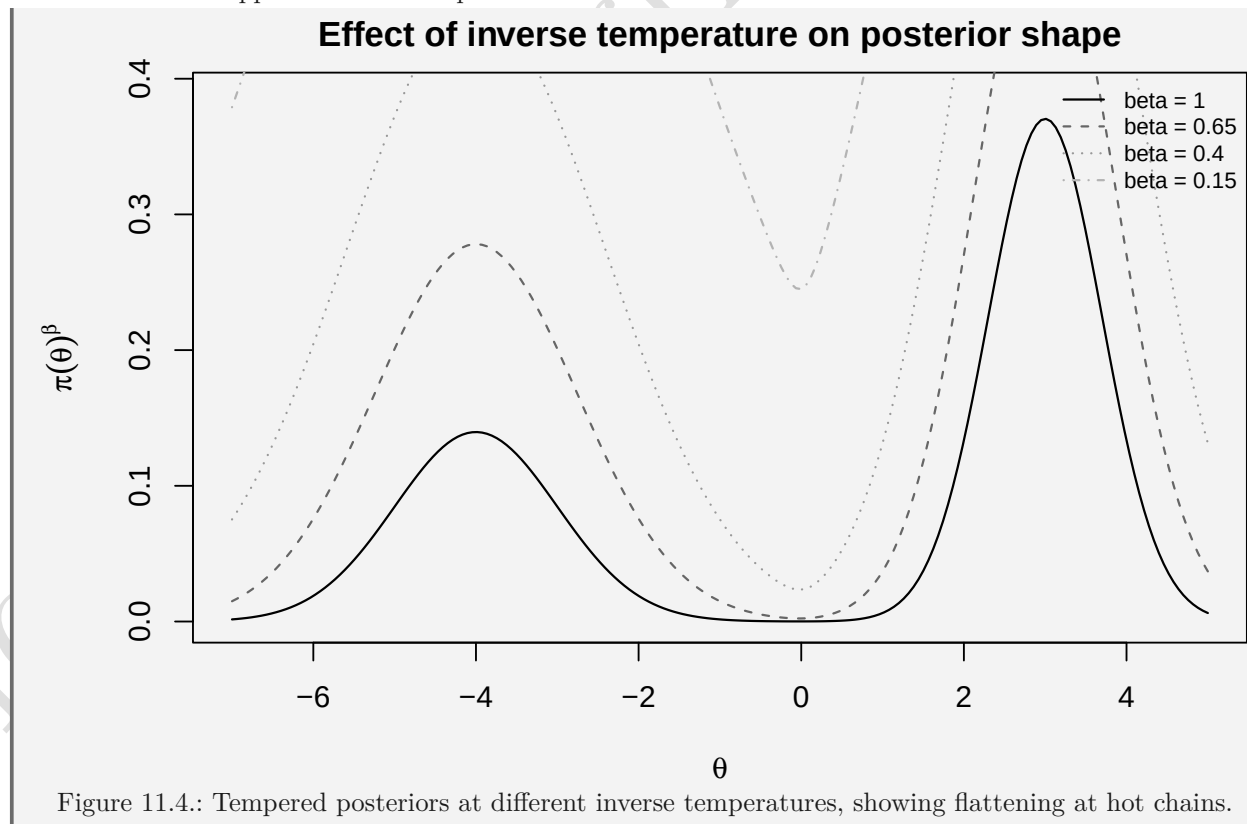
See code 15.42 in Appendix for the implementation.



See code 15.43 in Appendix for the implementation.



See code 15.44 in Appendix for the implementation.



The figures above illustrate the fundamental problem and solution:

- **Figure 1** shows a single chain trapped in the left mode.

- **Figure 2** shows the parallel tempering cold chain (at $\beta = 1$) successfully switching between both modes.
- **Figure 3** shows the histogram of samples: single-chain MH misses the left mode entirely, while PT recovers the true bimodal structure.
- **Figure 4** visualizes how higher temperature ($\beta \rightarrow 0$) flattens the posterior, enabling free movement.

11.7. Practical considerations

Choosing whether to deploy parallel tempering — and what to compare it against — comes down to the question being asked and the resources available:

- **Use parallel tempering when** the posterior is multimodal with well-separated modes, standard MCMC chains get trapped, the interest is in accurate posterior sampling and uncertainty quantification, or the problem involves phase transitions or critical phenomena (common in physics).
- **Use simulated annealing when** the goal is to find the MAP estimate or a point estimate, optimization (not sampling) is the primary objective, and you need fast identification of the best fit, not uncertainty.
- **Prefer HMC/NUTS when** the posterior is unimodal and high-dimensional, gradient information is available, or computational resources are limited and you cannot afford many chains. HMC and NUTS excel in high dimensions because they use gradient information to make large, informed moves, but they fail on multimodal targets because a single trajectory cannot jump between modes.
- **Computational cost.** Running K chains costs roughly K times as much as a single chain. However, this is often money well spent: the mixing time of the cold chain typically decreases dramatically when the temperature ladder is well-designed.

11.8. Summary

Definition 11.1. Tempered distributions scale the exponent of the posterior: $\pi_\beta(\theta) \propto \pi(\theta)^\beta$. At $\beta = 1$ we recover the target; at $\beta \rightarrow 0$ the distribution flattens.

Simulated annealing is a single-chain method with a time-varying cooling schedule; it targets optimization (MAP estimation) and asymptotically concentrates on the global maximum.

Parallel tempering (replica exchange MCMC) runs K chains at fixed inverse temperatures $1 = \beta_1 > \beta_2 > \dots > \beta_K$ and periodically swaps states between neighbours. The cold chain remains on the target distribution throughout and escapes local modes via information shared from hotter chains.

Detailed balance is satisfied: within-temperature moves are standard MH, and swap moves satisfy detailed balance with respect to the product target $\Pi(\Theta) = \prod_{l=1}^K \pi_{\beta_l}(\theta_l)$. Thus the cold chain's marginal is exactly π .

Temperature ladder design is critical: geometric spacing in inverse temperature and target swap acceptance rates of 20–40% are practical guidelines.

The worked example shows that parallel tempering succeeds where single-chain MH fails on a bimodal target, recovering the true posterior over both modes.

11.9. Exercises

Exercise 11.1

Define the tempered distribution $\pi_\beta(x) \propto \pi(x)^\beta$ for $\beta \in [0, 1]$. Identify the two limiting cases $\beta = 0$ and $\beta = 1$.

Solution

At $\beta = 1$ we recover the target,

$$\pi_1(x) \propto \pi(x),$$

while at $\beta = 0$ we have $\pi(x)^0 = 1$, so

$$\pi_0(x) \propto 1,$$

i.e. the uniform (or prior, in a Bayesian setting) reference distribution on the support of π .

Exercise 11.2

Let $\pi(x) \propto \exp(-U(x))$. Show that $\pi_\beta(x) \propto \exp(-\beta U(x))$. Explain why lowering β flattens the landscape and lowers energy barriers.

Solution

Raising the density to the power β gives

$$\pi_\beta(x) \propto \pi(x)^\beta = [\exp(-U(x))]^\beta = \exp(-\beta U(x)).$$

The effective energy becomes $\beta U(x)$, so all energy differences (including barrier heights ΔU) are rescaled by β . For $\beta < 1$ this shrinks the contrast between modes and valleys, so the chain crosses barriers with probability $\sim e^{-\beta \Delta U}$ instead of $e^{-\Delta U}$.

Exercise 11.3

Bimodal target. Consider $\pi(x) = \frac{1}{2}\mathcal{N}(-3, 1) + \frac{1}{2}\mathcal{N}(3, 1)$. Sketch (qualitatively) the tempered density π_β for $\beta = 1, 0.5, 0.1$ and explain why a random-walk Metropolis chain on π_1 has trouble crossing the mode gap.

Solution

At $\beta = 1$ the density has two sharp peaks at ± 3 separated by a deep valley near 0; at $\beta = 0.5$ the peaks broaden and the valley between them rises; at $\beta = 0.1$ the density is nearly flat across $[-5, 5]$. Random-walk MH must traverse the low-density gap by a sequence of small steps, each accepted with probability $\sim e^{-\Delta U}$ where $\Delta U \approx U(0) - U(\pm 3) \approx \frac{9}{2}$, giving an exponentially small per-step crossing probability and effective trapping in one mode.

Exercise 11.4

Swap acceptance probability. For replicas at inverse temperatures β_i and β_j holding states x_i, x_j , derive the Metropolis–Hastings acceptance probability

$$\alpha_{ij} = \min\{1, \exp[(\beta_i - \beta_j)(U(x_i) - U(x_j))]\}.$$

Solution

The product target is $\Pi(x_i, x_j) \propto e^{-\beta_i U(x_i) - \beta_j U(x_j)}$, and the swap proposal $(x_i, x_j) \rightarrow (x_j, x_i)$ is deterministic and symmetric. The MH ratio is therefore

$$\frac{\Pi(x_j, x_i)}{\Pi(x_i, x_j)} = \frac{e^{-\beta_i U(x_j) - \beta_j U(x_i)}}{e^{-\beta_i U(x_i) - \beta_j U(x_j)}} = \exp[(\beta_i - \beta_j)(U(x_i) - U(x_j))],$$

so $\alpha_{ij} = \min\{1, \exp[(\beta_i - \beta_j)(U(x_i) - U(x_j))]\}$.

Exercise 11.5

Show that the swap move described above satisfies detailed balance with respect to the product target $\Pi(x_1, \dots, x_K) = \prod_l \pi_{\beta_l}(x_l)$. Use this to argue that the marginal of the cold chain is exactly π .

Solution

For the swap of replicas i, j with symmetric proposal probability q , the MH rule gives

$$\Pi(\dots, x_i, \dots, x_j, \dots) q \alpha_{ij} = \Pi(\dots, x_j, \dots, x_i, \dots) q \alpha_{ji},$$

which is exactly the standard MH detailed balance identity using $\alpha_{ij} = \min\{1, \Pi'/\Pi\}$ (all factors at other replicas cancel). Since both within-temperature moves and swaps preserve Π , the stationary distribution is Π , and the marginal over x_1 at $\beta_1 = 1$ is $\pi_{\beta_1} = \pi$.

Exercise 11.6

Geometric ladder. Show that with a geometric temperature schedule $\beta_{l+1}/\beta_l = r$ (constant), the expected swap acceptance between adjacent replicas becomes approximately independent of l when the energy variance scales linearly with the dimension.

Solution

The log swap ratio between replicas $l, l+1$ is $\Delta_l = (\beta_l - \beta_{l+1})(U_l - U_{l+1}) = \beta_l(1-r)(U_l - U_{l+1})$. Standard scaling arguments give $\text{Var}_{\pi_\beta}[U] \approx d c(\beta)/\beta^2$ for some smooth $c(\beta)$, so

$$\text{Var}(\Delta_l) \approx \beta_l^2(1-r)^2 \cdot 2 \cdot \frac{d c(\beta_l)}{\beta_l^2} = 2d c(\beta_l) (1-r)^2,$$

which is independent of l up to the slow variation of $c(\beta)$. The expected acceptance, governed by the distribution of Δ_l , is therefore nearly constant along the ladder.

Exercise 11.7

Number of replicas. For a target in d dimensions with average energy gap ΔU between cold and hot ends, estimate the number of replicas K needed to keep adjacent-pair swap acceptance around 25%. Express your answer in terms of ΔU and the desired swap rate.

Solution

For a target acceptance a , each adjacent pair should produce a typical log-ratio of order $|\log a|$, i.e. $(\beta_l - \beta_{l+1}) \Delta U_{\text{adj}} \sim |\log a|$, where the per-pair energy gap on a geometric ladder is $\Delta U_{\text{adj}} \approx$

11. Parallel tempering

$\Delta U/(K-1)$. Setting $(1)(\Delta U/(K-1)) \approx |\log a|$ gives

$$K \approx 1 + \frac{\Delta U}{|\log a|},$$

so for $a \approx 0.25$, $|\log a| \approx 1.4$ and $K \sim 1 + \Delta U/1.4$. (For a target whose energy variance grows like d , the same argument with $\sigma_U \sim \sqrt{d}$ rescales this to $K \sim 1 + \sqrt{d} \Delta U/|\log a|$.)

Exercise 11.8

Simulated annealing vs parallel tempering. Compare the two algorithms in two sentences each: what each one tries to achieve, and why PT generates valid posterior samples while annealing typically does not.

Solution

Simulated annealing runs a single chain whose temperature $T_k \rightarrow 0$ on a schedule; it is an *optimisation* method that concentrates on $\arg \max \pi$, and the resulting trajectory has a non-stationary, time-inhomogeneous law so its iterates are not samples from any fixed distribution. Parallel tempering, by contrast, runs K chains at *fixed* inverse temperatures with $\beta_1 = 1$ and swaps states between neighbours; the joint stationary distribution is $\Pi = \prod_l \pi_{\beta_l}$, so the cold chain marginal is exactly π and its iterates are valid posterior samples.

Exercise 11.9

Thermodynamic integration. Show that

$$\log Z = - \int_0^1 E_{\pi_\beta}[U] d\beta,$$

and explain how this identity allows the marginal likelihood (evidence) to be estimated by sampling at multiple temperatures.

Solution

Define $Z(\beta) = \int e^{-\beta U(x)} dx$, with $Z(0) = |\mathcal{X}|$ (or 1 for a normalised reference) and $Z(1) = Z$. Differentiating under the integral,

$$\frac{d \log Z(\beta)}{d\beta} = \frac{- \int U(x) e^{-\beta U(x)} dx}{Z(\beta)} = -E_{\pi_\beta}[U],$$

so integrating from 0 to 1 gives $\log Z = \log Z(1) - \log Z(0) = - \int_0^1 E_{\pi_\beta}[U] d\beta$ (taking $Z(0) = 1$). In practice one estimates $\hat{U}(\beta_l) \approx E_{\pi_{\beta_l}}[U]$ from the replica at β_l and applies a quadrature rule (trapezoid/Simpson) over the temperature ladder.

Exercise 11.10

Optimal acceptance rate for swap moves. Discuss why empirical guidance places the optimal swap acceptance in the range 20–40%. What goes wrong below and above this range?

Solution

The swap rate measures the rate at which information (replicas) flows along the ladder. Below $\sim 20\%$ the energy histograms of adjacent replicas barely overlap, swaps are rare and the cold chain receives very few hot discoveries (information bottleneck). Above $\sim 40\%$ adjacent temperatures are nearly identical: each accepted swap conveys little new information per chain and one is paying for redundant replicas, so the wall-clock cost per ESS rises. The empirical sweet spot 20–40% balances “swaps happen often enough” against “each swap actually changes the energy”.

Exercise 11.11

Effective sample size after a PT run. With K replicas, only the cold chain produces posterior samples. Explain why the total wall-clock cost scales with K but the ESS only grows by the swap mixing improvement, and why the trade-off is still typically favourable for multimodal targets.

Solution

Each PT iteration costs $\sim K$ likelihood evaluations (one per replica) but only the β_1 chain contributes to the posterior estimate, so cost-per-cold-iterate scales as K . The benefit is a reduction of the cold-chain integrated autocorrelation τ : roughly $\text{ESS} \sim N/\tau_{\text{PT}}$ vs. N/τ_{single} . For unimodal targets $\tau_{\text{PT}} \approx \tau_{\text{single}}$ and the factor K is pure overhead, but for multimodal targets a single chain has $\tau_{\text{single}} \sim e^{\Delta U}$ (exponential in barrier height) while PT gives $\tau_{\text{PT}} \sim K^2$ (polynomial round-trip time), so the speedup $\tau_{\text{single}}/(K \tau_{\text{PT}}) \sim e^{\Delta U}/K^3$ is enormous.

Exercise 11.12

Energy histogram overlap. Show qualitatively that adjacent replicas must have overlapping energy distributions for swaps to be accepted with reasonable probability. Sketch the case of (a) too few replicas and (b) too many.

Solution

The swap log-ratio is $(\beta_l - \beta_{l+1})(U_l - U_{l+1})$, which is appreciable only when U_l and U_{l+1} are drawn from overlapping distributions; if the histograms of U at β_l and β_{l+1} are disjoint then $U_l - U_{l+1}$ is systematically large and negative, making $\alpha \approx 0$.

- (a) ****Too few replicas:**** the histograms at adjacent β_l are widely separated bell curves with negligible overlap, so almost every swap is rejected.
- (b) ****Too many replicas:**** the histograms are nearly identical and almost fully overlap; swaps are accepted essentially every time but exchange very little energy, so information barely propagates per chain.

Good design has adjacent histograms whose tails overlap noticeably but whose means differ.

Exercise 11.13

Replica exchange for a discrete target. A finite chain has stationary distribution π on $\{0, 1, \dots, K-1\}$. Construct the tempered transition kernel at inverse temperature β and write the swap acceptance with a neighbour at β' .

Solution

Write $\pi(x) \propto e^{-U(x)}$ with $U(x) = -\log \pi(x)$, and define the tempered target on the same state space by

$$\pi_\beta(x) \propto \pi(x)^\beta = e^{-\beta U(x)}.$$

A simple within-temperature kernel is Metropolis with a symmetric neighbour proposal $q(x \rightarrow y)$ accepted with probability

$$\alpha_\beta(x \rightarrow y) = \min \left\{ 1, \left(\frac{\pi(y)}{\pi(x)} \right)^\beta \right\}.$$

The swap with a neighbour at β' holding state x' uses

$$\alpha_{\text{swap}}(x, x') = \min \{ 1, \exp[(\beta - \beta')(U(x) - U(x'))] \}.$$

Exercise 11.14

Power-posterior interpretation. In Bayesian inference $\pi(\theta | D) \propto L(\theta)^1 p(\theta)$. State the corresponding tempered version $\pi_\beta(\theta | D)$ and identify the prior, the maximum-likelihood-dominated posterior, and the standard posterior as special cases.

Solution

The power posterior is obtained by raising the likelihood to β :

$$\pi_\beta(\theta | D) \propto L(\theta)^\beta p(\theta).$$

Then $\beta = 0$ gives $\pi_0 \propto p(\theta)$ (the **prior**); $\beta = 1$ gives $\pi_1 \propto L(\theta)p(\theta)$ (the **standard posterior**); and $\beta \rightarrow \infty$ concentrates on $\arg \max_\theta L(\theta)$, the **MLE/likelihood-dominated** limit. Intermediate $\beta \in (0, 1)$ interpolates between prior and posterior and underlies the thermodynamic-integration estimate of the evidence $\log Z = \int_0^1 E_{\pi_\beta}[\log L] d\beta$.

Exercise 11.15

When parallel tempering is overkill. Give two situations where PT is *not* worth the extra computational cost (i.e. a single-chain HMC or random-walk Metropolis is preferable).

Solution

- (i) **Unimodal, well-conditioned targets in moderate-to-high dimension.** Here a single HMC/NUTS chain already mixes in $O(1)$ effective steps; the $K \times$ overhead of PT buys essentially no extra ESS. (ii) **Tight computational budget for a quick exploratory fit**, especially when the posterior is unimodal or only mildly multimodal — running K replicas multiplies the per-iteration cost while a single tuned chain (perhaps with multiple random restarts) gives faster posterior estimates with adequate diagnostics.

11.10. Sources

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Part IV.

PART IV: APPLIED BAYESIAN MODELLING

12. Bayesian regression

12.1. Introduction

Regression analysis seeks to learn a functional relationship between input variables x and observed response variables y from data. In the Bayesian framework, we construct a probabilistic model that combines a generative (noise-free) function $f(x; \theta)$ with a noise model, then use Bayes' theorem to infer the posterior distribution $p(\theta | D)$ of the parameters. For regression models where the parameters enter linearly, exact posterior inference is tractable via conjugate analysis. For nonlinear models or non-Gaussian likelihoods, we rely on Markov chain Monte Carlo (MCMC) sampling. This chapter develops both the theory and practice of Bayesian regression: from the closed-form solution for linear Gaussian regression to the complete MCMC workflow for fitting arbitrary nonlinear regression models and making probabilistic predictions.

12.2. Regression Model: Signal Plus Noise

The basic regression model specifies that each observed response y_i arises from a noise-free mean function plus random noise:

$$y_i = f(x_i; \theta) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

- **Model ingredients.** x_i is the regressor (input, explanatory variable), assumed fixed and measured without error; y_i is the response (dependent, outcome variable); $f(x_i; \theta)$ is the signal—the deterministic part; and ε_i is measurement noise or intrinsic scatter, i.i.d. Gaussian with variance σ^2 .
- **Gaussian likelihood.** Each observation has conditional distribution $p(y_i | x_i, \theta) = \mathcal{N}(f(x_i; \theta), \sigma^2)$, so for N independent observations $D = \{(x_i, y_i)\}_{i=1}^N$ the joint likelihood factorizes as $p(D | \theta) = \prod_{i=1}^N \mathcal{N}(f(x_i; \theta), \sigma^2)$.
- **Log-likelihood.** Taking logarithms,

$$\log p(D | \theta) = -N \log \sigma - \frac{1}{2} \log(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^N [y_i - f(x_i; \theta)]^2 + \text{const.}$$

12.3. Linear Models and the Design Matrix

A regression model is **linear** if the parameters enter linearly, even if the features depend nonlinearly on the inputs:

$$y_i = \phi(x_i)^\top \mathbf{w} + \varepsilon_i,$$

where $\phi(x) = [\phi_1(x), \dots, \phi_M(x)]^\top$ is a feature map and $\mathbf{w} = [w_0, \dots, w_{M-1}]^\top$ is the weight vector.

Definition 12.1. Examples of linear regression models: - Straight line: $\phi(x) = (1, x)$, $f(x) = w_0 + w_1 x$. - Polynomial: $\phi(x) = (1, x, x^2, \dots, x^{M-1})$, $f(x) = \sum_{j=0}^{M-1} w_j x^j$. - Fourier: $\phi(x) = (1, \sin(x), \cos(x), \dots)$. - Linear in unknown parameters: If $f(x) = w_0 e^{-\lambda x}$ with λ known, it is

linear in w_0 . - **Not** linear in unknown parameters: $f(x) = w_0 e^{-\lambda x}$ if λ is unknown (exponential, logistic, etc.).

For N data points and M basis functions, define the $N \times M$ **design matrix** (or regressor matrix)

$$\Phi = \begin{bmatrix} \phi_1(x_1) & \phi_2(x_1) & \cdots & \phi_M(x_1) \\ \phi_1(x_2) & \phi_2(x_2) & \cdots & \phi_M(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(x_N) & \phi_2(x_N) & \cdots & \phi_M(x_N) \end{bmatrix}.$$

The linear model becomes

$$\mathbf{y} = \Phi \mathbf{w} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N),$$

and the likelihood is

$$p(\mathbf{y} \mid \Phi, \mathbf{w}, \sigma^2) = \mathcal{N}(\mathbf{y} \mid \Phi \mathbf{w}, \sigma^2 \mathbf{I}_N).$$

12.4. Bayesian Linear Regression: Closed-Form Solution

Assume σ^2 is known. Choose a Gaussian prior on $\mathbf{w}$:

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mathbf{m}_0, \mathbf{S}_0).$$

The likelihood is Gaussian in $\mathbf{w}$; by conjugacy, the posterior is also Gaussian:

Theorem 12.1. Bayesian Linear Regression with Gaussian Prior:

Given likelihood $p(\mathbf{y} \mid \Phi, \mathbf{w}, \sigma^2) = \mathcal{N}(\mathbf{y} \mid \Phi \mathbf{w}, \sigma^2 \mathbf{I}_N)$ and prior $\mathcal{N}(\mathbf{w} \mid \mathbf{m}_0, \mathbf{S}_0)$, the posterior is

$$p(\mathbf{w} \mid D, \sigma^2) = \mathcal{N}(\mathbf{w} \mid \mathbf{m}_N, \mathbf{S}_N),$$

where

$$\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \sigma^{-2} \Phi^\top \Phi, \tag{12.1}$$

$$\mathbf{m}_N = \mathbf{S}_N (\mathbf{S}_0^{-1} \mathbf{m}_0 + \sigma^{-2} \Phi^\top \mathbf{y}). \tag{12.2}$$

- **Interpretation.** Precisions add. The posterior mean is a precision-weighted average of the prior mean and the data-driven estimate. The posterior covariance $\mathbf{S}_N$ shrinks relative to $\mathbf{S}_0$.
- **Weak isotropic prior.** A common choice is $\mathbf{m}_0 = \mathbf{0}$ and $\mathbf{S}_0 = \sigma_0^2 \mathbf{I}_M$, giving $\mathbf{S}_N^{-1} = \sigma_0^{-2} \mathbf{I}_M + \sigma^{-2} \Phi^\top \Phi$ and $\mathbf{m}_N = \sigma^{-2} \mathbf{S}_N \Phi^\top \mathbf{y}$.
- **Connection to ridge regression.** Ordinary least squares minimizes $\|\mathbf{y} - \Phi \mathbf{w}\|^2$; ridge regression adds an L2 penalty $L_{\text{ridge}} = \|\mathbf{y} - \Phi \mathbf{w}\|^2 + \lambda \|\mathbf{w}\|^2$, with estimator $\hat{\mathbf{w}}_{\text{ridge}} = (\Phi^\top \Phi + \lambda \mathbf{I})^{-1} \Phi^\top \mathbf{y}$. Ridge regression is **maximum a posteriori (MAP) estimation** under a Gaussian prior: the prior variance relates to the penalty by $\lambda = \sigma^2 / \sigma_0^2$. The Bayesian approach goes further by quantifying posterior uncertainty via the full covariance $\mathbf{S}_N$.

12.5. Posterior Predictive Distribution

For a new input x_* , we wish to predict y_* . The posterior predictive distribution marginalizes over parameter uncertainty:

$$p(y_* | x_*, D, \sigma^2) = \int p(y_* | x_*, \mathbf{w}, \sigma^2) p(\mathbf{w} | D, \sigma^2) d\mathbf{w}.$$

Since both terms are Gaussian, the integral is tractable:

Theorem 12.2. Posterior Predictive Distribution:

Let $\phi_* = \phi(x_*)$. Then

$$p(y_* | x_*, D, \sigma^2) = \mathcal{N}(y_* | \phi_*^\top \mathbf{m}_N, \sigma^2 + \phi_*^\top \mathbf{S}_N \phi_*).$$

- **Observation noise.** σ^2 (irreducible error in y_*).
- **Parameter uncertainty.** $\phi_*^\top \mathbf{S}_N \phi_*$ (uncertainty in the regression function $f(x_*)$). The posterior predictive band widens away from the data due to increasing parameter uncertainty.

Example 12.1 (Predictive distribution from total variance (exam-style))

In Bayesian linear regression, assume

$$y_* | \mathbf{w}, x_* \sim \mathcal{N}(\phi_*^\top \mathbf{w}, \sigma^2), \quad \mathbf{w} | D \sim \mathcal{N}(\mathbf{m}, \mathbf{S}).$$

Derive the posterior predictive distribution of y_* .

Solution. Linearity of expectation gives

$$E[y_* | D] = E[\phi_*^\top \mathbf{w} | D] = \phi_*^\top \mathbf{m}.$$

The law of total variance splits the predictive variance into a noise term and a parameter term:

$$\text{Var}(y_* | D) = E[\text{Var}(y_* | \mathbf{w}, D)] + \text{Var}(E[y_* | \mathbf{w}, D]) = \sigma^2 + \phi_*^\top \mathbf{S} \phi_*.$$

Since both contributions are Gaussian and we are linearly combining Gaussians, the predictive distribution is itself Gaussian:

$$y_* | D, x_* \sim \mathcal{N}(\phi_*^\top \mathbf{m}, \sigma^2 + \phi_*^\top \mathbf{S} \phi_*).$$

The first variance term is irreducible noise; the second is parameter uncertainty.

Example 12.2 (Predictive credible interval vs regression-function interval (exam-style))

In Bayesian linear regression, define the posterior predictive distribution and the predictive (e.g. 95%) credible interval for a future observation y_* . What is its physical meaning, and how does it differ from a credible interval for the regression curve?

Solution. The posterior predictive distribution for a future observation y_* at input x_* is

$$p(y_* | x_*, D) = \int p(y_* | x_*, \theta) p(\theta | D) d\theta.$$

Prediction is the likelihood of the future observation averaged over posterior uncertainty in the parameters.

A central (equal-tailed) 95% posterior-predictive credible interval is $C_{0.95}(x_*) = [q_{0.025}(x_*), q_{0.975}(x_*)]$, where $q_\alpha(x_*)$ satisfies $\int_{-\infty}^{q_\alpha(x_*)} p(y_* | x_*, D) dy_* = \alpha$. Conditional on the model, prior, and data, a future

measurement at x_* lies in $C_{0.95}(x_*)$ with posterior predictive probability 95%.

The credible interval for the **regression function** concerns the latent response $f(x_*, \theta)$ and is obtained from

$$p(f \mid x_*, D) = \int \delta(f - f(x_*, \theta)) p(\theta \mid D) d\theta.$$

This describes uncertainty about the underlying curve, not the scatter of future observations around it. Consequently the posterior predictive interval is generally **wider** than the regression-function interval — it adds the irreducible noise variance to the parameter uncertainty.

12.6. Unknown Noise Level

In practice, σ^2 is usually unknown and inferred from data. The parameters become $\theta = (\mathbf{w}, \sigma)$, and we use a prior on the noise scale. A standard choice is the **Jeffreys prior** for scale parameters:

$$p(\sigma) \propto \frac{1}{\sigma}, \quad \text{or equivalently} \quad p(\log \sigma) \text{ is uniform.}$$

The joint posterior is

$$p(\mathbf{w}, \sigma \mid D) \propto p(\mathbf{y} \mid \Phi, \mathbf{w}, \sigma) p(\mathbf{w}) p(\sigma).$$

When both $\mathbf{w}$ and σ are unknown, the posterior no longer has a closed form in closed-form distributions, but conjugate analysis can still apply to marginal posteriors or conditional distributions. For computational convenience and to make use of existing MCMC software, we often resort to sampling.

12.7. Nonlinear Regression and MCMC

A model is **nonlinear in parameters** if $f(x; \theta)$ is a nonlinear function of some θ . Typical examples are the exponential $f(x) = ae^{bx}$ (nonlinear in both a and b unless one is known), the logistic $f(x) = L/(1+e^{-k(x-x_0)})$ (nonlinear in location x_0 , scale k , amplitude L), and the power law $f(x) = ax^b$ (nonlinear in exponent b). For such models, the likelihood is no longer Gaussian in the parameters, and conjugate priors do not yield closed-form posteriors. Similarly, generalized linear models (logistic regression for binary data, Poisson regression for counts) have non-Gaussian likelihoods, precluding analytical solutions. In these cases, we must either use **approximations** (Laplace, variational inference) or use **MCMC** or other sampling methods to draw from $p(\theta \mid D) \propto p(D \mid \theta) p(\theta)$. The next section lays out the standard MCMC workflow; the Laplace approximation is treated alongside the worked example below.

12.8. Practical MCMC Fitting Workflow

The following seven-step workflow is the standard approach for fitting regression models via MCMC.

- **Step 1: specify the generative and noise model.** Write down the regression function $f(x; \theta)$, the noise distribution (e.g., $\varepsilon \sim \mathcal{N}(0, \sigma^2)$), and the resulting observation model $y_i \sim \mathcal{N}(f(x_i; \theta), \sigma^2)$.
- **Step 2: choose priors for all parameters.** Specify weakly informative priors for each parameter. For regression coefficients, use broad Gaussians (e.g., $\mathcal{N}(0, 10^2)$). For scale parameters like σ , use a Jeffreys prior $p(\sigma) \propto 1/\sigma$ or truncated exponentials. To avoid boundary issues, sample on the log-scale: $\eta = \log \sigma$.

Note. Reparameterization for Bounded Parameters:

Parameters constrained to positive values (like $\sigma > 0$) should be sampled in transformed space: - Replace σ with $\eta = \log \sigma$, so $\sigma = e^\eta$. - A flat prior on η corresponds to a Jeffreys prior on σ . - The Jacobian $|d\sigma/d\eta| = e^\eta = \sigma$ automatically cancels the $1/\sigma$ prior, yielding a uniform prior in log-space.

For slope parameters, reparameterize via angle: $\beta = \tan(\alpha)$, so all slopes are finite and comparable.

- **Step 3: write log-prior, log-likelihood, and log-posterior functions.** Define functions that compute

$$\log p(\theta) : \text{sum of log-prior densities for all parameters,} \quad (12.3)$$

$$\log p(D | \theta) : \text{log-likelihood, typically } \sum_{i=1}^N \log p(y_i | x_i, \theta), \quad (12.4)$$

$$\log p(\theta | D) = \log p(D | \theta) + \log p(\theta) + \text{const.} \quad (12.5)$$

For MCMC, we only need the unnormalized posterior (up to an additive constant).

- **Step 4: run MCMC and tune the proposal covariance.** Initialize parameters at a reasonable starting point (e.g., a classical fit or MAP estimate). Specify a proposal distribution, typically a multivariate Gaussian with tuned covariance. Run MCMC (e.g., random-walk Metropolis-Hastings, or preferably NUTS via Stan) for a burn-in period, then for additional sampling iterations. Tune the proposal variance to achieve a target acceptance rate (typically 20–30% for random-walk MH, automatic for NUTS).
- **Step 5: diagnose and inspect the chain.** Use trace plots, autocorrelation, effective sample size, and $\hat{R}$ as detailed in the box below.

Remark. Key Diagnostics: - **Trace plots:** Plot θ_j versus iteration. Traces should resemble white noise (good mixing), not slow drifts or stuck chains. - **Autocorrelation function (ACF):** Compute $\text{Corr}(\theta_j^{(k)}, \theta_j^{(k+\ell)})$ as a function of lag ℓ . Rapid decay (small effective sample size if slow decay) indicates good mixing. - **Effective sample size (ESS):** $\text{ESS} = N_{\text{samples}} / (1 + 2 \sum_{\ell=1}^{\infty} \text{Corr}(\ell))$. Rule of thumb: aim for $\text{ESS} > 1000$ per parameter. - **$\hat{R}$ (Gelman-Rubin statistic):** Ratio of between-chain to within-chain variance. $\hat{R} < 1.01$ indicates convergence; $\hat{R} > 1.05$ indicates poor mixing or insufficient burn-in. - **Joint posterior pairs plot:** Scatter plot of all pairs of parameters. Reveals correlations and identifies poorly identified parameters.

- **Step 6: summarize parameters.** Compute posterior summaries from the converged samples (after thinning to reduce autocorrelation if desired): **MAP** (the sample with highest posterior density), **posterior mean** (average of all samples), **posterior covariance** (empirical covariance of samples), and **credible intervals** (2.5th and 97.5th quantiles, or other quantiles, per parameter). Always transform samples back to the original parameterization before computing summaries.
- **Step 7: plot predictions and posterior predictive checks.** Show the **data with error bars** (x_i, y_i) at $\pm \sigma_{\text{MAP}}$ or $\pm$ posterior SD of y_i ; overlay the **fitted regression curve** $\hat{f}(x) = \phi(x)^\top \mathbb{E}[\mathbf{w} | D]$; add a **credible band for the mean curve** by computing posterior quantiles of $f(x; \theta)$ on a grid of x values from each posterior sample (this shows uncertainty in the regression function, not noise); and add a **posterior predictive band** that includes both parameter uncertainty and noise — for each x on the grid and each posterior sample $\theta^{(k)}$, generate $y^{(k)} \sim \mathcal{N}(f(x; \theta^{(k)}), \sigma^{(k)2})$ and take the 2.5th and 97.5th quantiles across samples.

12.9. Example: Linear Regression with MCMC

Consider fitting a straight line $f(x) = b_0 + b_1 x$ to observed data $\{(x_i, y_i)\}$, with unknown noise σ .

Parameters (transformed): $\theta = (b_0, \alpha, \eta)$ where $b_1 = \tan(\alpha)$ and $\eta = \log \sigma$.

Priors: - $b_0 \sim \mathcal{N}(0, 10^2)$. - $\alpha \sim \text{Uniform}(-\pi/2, \pi/2)$ (equivalently, $p(\alpha) \propto 1$). - $\eta \sim \text{Uniform}$ (flat prior on $\log \sigma$, Jeffreys prior on σ).

Log-likelihood:

$$\log p(\mathbf{y} | b_0, \alpha, \eta) = -N\eta - \frac{1}{2} \sum_{i=1}^N \frac{[y_i - (b_0 + \tan(\alpha)x_i)]^2}{e^{2\eta}} + \text{const.}$$

Log-posterior: Sum of log-likelihood and log-prior densities.

Run MCMC (e.g., via random-walk MH or Stan's NUTS sampler) and diagnose convergence.

See code 15.45 in Appendix for the implementation.

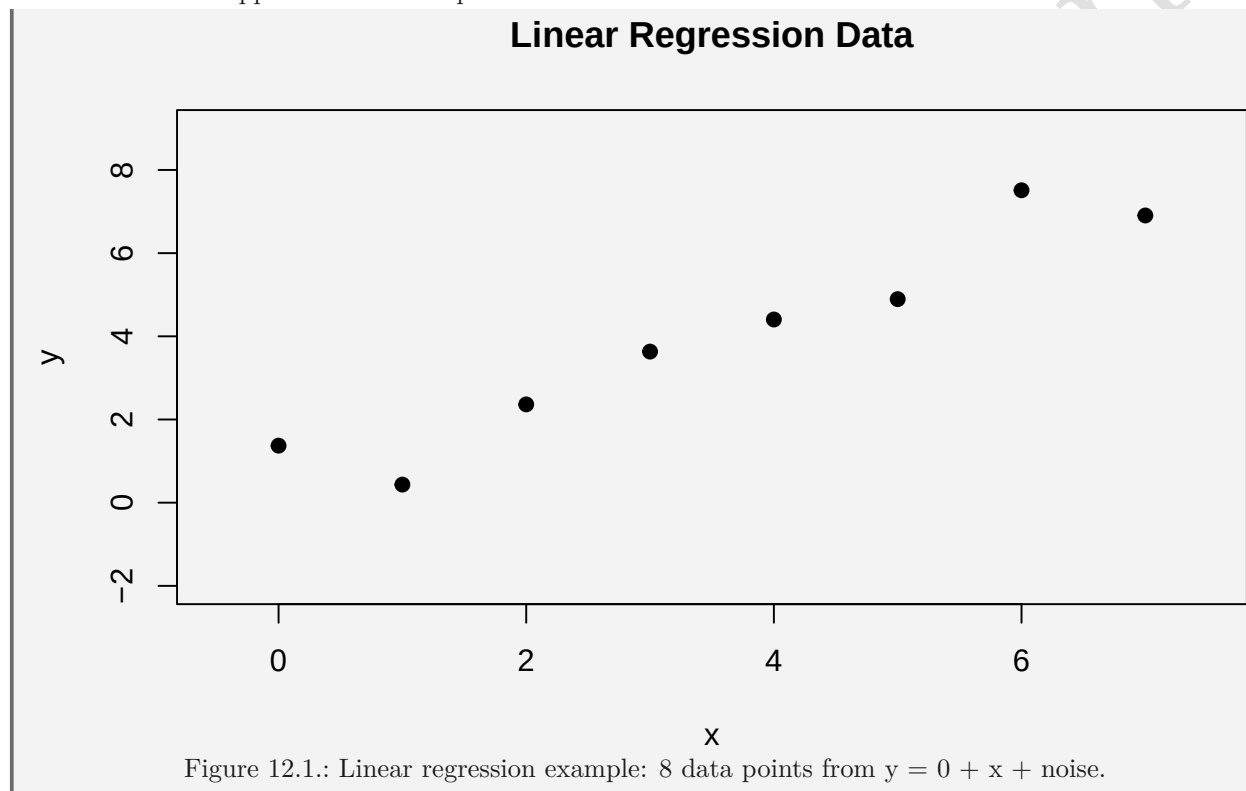


Figure 12.1.: Linear regression example: 8 data points from $y = 0 + x + \text{noise}$.

Now run the MCMC sampler (pseudocode; in practice, use Stan, JAGS, or a hand-coded sampler):

See code 15.46 in Appendix for the implementation.

After running MCMC for 50,000 iterations (with 10,000 burn-in), extract the posterior samples and compute diagnostics:

See code 15.47 in Appendix for the implementation.

After burn-in, summarize the posterior:

See code 15.48 in Appendix for the implementation.

See code 15.49 in Appendix for the implementation.

12.10. Nonlinear Models: Laplace Approximation and MCMC

When $f(x; \theta)$ is nonlinear in θ , the posterior cannot be written in closed form. Two main approaches:

Laplace Approximation: Expand the log-posterior around the MAP estimate $\hat{\theta}$ as a multivariate Gaussian:

$$p(\theta | D) \approx \mathcal{N}(\hat{\theta}, [-\nabla^2 \log p(\theta | D)|_{\hat{\theta}}]^{-1}).$$

This is fast and often accurate for well-behaved posteriors. For multimodal or heavy-tailed posteriors, it can fail.

MCMC: Sample directly from the posterior, handling arbitrary nonlinearity and non-Gaussian shapes. The disadvantage is computational cost; the advantage is generality and accuracy. Modern samplers like NUTS (in Stan) adapt the step size and use Hamiltonian dynamics, making MCMC scalable.

12.11. Posterior Predictive Checks

A fundamental tool in Bayesian regression is **posterior predictive checking**: generate new data under the posterior and compare to observed data.

Definition 12.2. For a regression model, the posterior predictive distribution is

$$p(y_{\text{new}} | x_{\text{new}}, D) = \int p(y_{\text{new}} | x_{\text{new}}, \theta) p(\theta | D) d\theta.$$

The marginal predictive distribution averages over parameter uncertainty, automatically accounting for both the fitted model and the noise in the data.

Posterior predictive checks: 1. For each posterior sample $\theta^{(k)}$, generate $y_{\text{rep}}^{(k)}(x_i) \sim p(y | x_i, \theta^{(k)})$ at each observed x_i . 2. Compare the distribution of simulated y_{rep} to observed y using summary statistics (e.g., mean, variance, quantiles, or max absolute residual). 3. If the posterior predictive distribution is consistent with the observed data, the model is plausible.

12.12. Robustness and Extensions

12.12.1. Student- t Noise for Robustness

Gaussian noise is sensitive to outliers. A robust alternative is **Student- t noise**:

$$y_i \sim t_\nu(f(x_i; \theta), \sigma^2),$$

where ν controls tail heaviness (small ν allows larger outliers). As $\nu \rightarrow \infty$, recovers Gaussian. Often set $\nu = 3$ or $\nu = 4$ for robustness.

The likelihood becomes

$$p(y_i | \dots) \propto \left[1 + \frac{[y_i - f(x_i; \theta)]^2}{\nu \sigma^2} \right]^{-(\nu+1)/2},$$

which is heavier-tailed than Gaussian. MCMC is needed to fit such models.

12.12.2. Errors in All Variables

If both x_i and y_i are measured with error, the problem becomes significantly harder:

$$x_i = x_i^{\text{true}} + \delta_{x,i}, \quad y_i = f(x_i^{\text{true}}; \theta) + \delta_{y,i}.$$

The true x_i^{true} must be treated as latent variables, increasing dimensionality. Bayesian approaches handle this naturally via hierarchical modeling, but require additional priors and tuning.

12.13. Summary

Bayesian regression provides a principled framework for learning functional relationships:

- **Linear models** (parameters enter linearly) admit conjugate closed-form posteriors via Gaussian theory, enabling exact inference and efficient prediction.
- **Posterior prediction** integrates over parameter uncertainty, yielding predictive distributions that account for both fitted function and observation noise.
- **Nonlinear and generalized models** (nonlinear parameters, non-Gaussian likelihoods) require MCMC sampling or approximations.
- **Practical MCMC workflow** (specify model, choose priors, write log-posterior, run sampler, diagnose, summarize, plot) is systematic and reproducible.
- **Posterior predictive checks** verify model adequacy and reveal misfit.
- **Robustness** (heavy-tailed noise) and **latent variable models** (measurement error) extend the framework to real data.

The same logic applies across simple linear regression, polynomial fits, nonlinear curves, and generalized linear models: only the parameterization and likelihood change; the Bayesian template—prior $\times$ likelihood $\propto$ evidence—remains constant.

12.14. Exercises

Exercise 12.1

Conjugate posterior. Observe $y \mid \mathbf{w}, x \sim \mathcal{N}(\phi(x)^\top \mathbf{w}, \sigma^2)$ with known σ^2 and prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$. Derive the posterior parameters $(\mathbf{m}_N, \mathbf{S}_N)$ in terms of the design matrix Φ and the response vector $\mathbf{y}$.

Solution

Completing the square in $\mathbf{w}$ in $\log p(\mathbf{w} \mid D) = -\frac{1}{2\sigma^2} \|\mathbf{y} - \Phi \mathbf{w}\|^2 - \frac{1}{2\tau^2} \|\mathbf{w}\|^2 + \text{const}$ gives a Gaussian posterior $\mathcal{N}(\mathbf{m}_N, \mathbf{S}_N)$ with

$$\mathbf{S}_N^{-1} = \tau^{-2} \mathbf{I} + \sigma^{-2} \Phi^\top \Phi, \quad \mathbf{m}_N = \mathbf{S}_N \sigma^{-2} \Phi^\top \mathbf{y},$$

which is the precision-form update with $\mathbf{S}_0 = \tau^2 \mathbf{I}$ and $\mathbf{m}_0 = \mathbf{0}$.

Exercise 12.2

Show that posterior precisions add: $\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \frac{1}{\sigma^2} \Phi^\top \Phi$. Interpret each term.

Solution

The quadratic in $\mathbf{w}$ in $\log p(\mathbf{w} \mid D)$ is the sum of the prior quadratic $-\frac{1}{2} \mathbf{w}^\top \mathbf{S}_0^{-1} \mathbf{w}$ and the likelihood quadratic $-\frac{1}{2\sigma^2} \mathbf{w}^\top \Phi^\top \Phi \mathbf{w}$, so by matching to $-\frac{1}{2} \mathbf{w}^\top \mathbf{S}_N^{-1} \mathbf{w}$,

$$\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \sigma^{-2} \Phi^\top \Phi.$$

The first term is prior information (precision from the prior); the second is data information (Fisher information per unit noise variance). Precisions, not variances, add.

Exercise 12.3

Single-observation update. Take $\sigma^2 = 1$, $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, $\phi = (1, 0)^\top$, observe $y = 3$. Compute the posterior mean and covariance in closed form.

Solution

With $\mathbf{S}_0 = \mathbf{I}$, $\Phi^\top \Phi = \phi \phi^\top = \text{diag}(1, 0)$ and $\Phi^\top \mathbf{y} = 3 \phi = (3, 0)^\top$, so

$$\mathbf{S}_N^{-1} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{S}_N = \begin{pmatrix} 1/2 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{m}_N = \mathbf{S}_N \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 3/2 \\ 0 \end{pmatrix}.$$

The first weight is pulled halfway to the data; the second remains at its prior because it is unobserved.

Exercise 12.4

Predictive variance away from data. Show that the posterior predictive variance $\sigma^2 + \phi_*^\top \mathbf{S}_N \phi_*$ grows when ϕ_* aligns with directions where $\mathbf{S}_N$ has large eigenvalues. Connect this to “extrapolation uncertainty”.

Solution

Diagonalise $\mathbf{S}_N = \sum_k \lambda_k \mathbf{u}_k \mathbf{u}_k^\top$; then

$$\phi_*^\top \mathbf{S}_N \phi_* = \sum_k \lambda_k (\mathbf{u}_k^\top \phi_*)^2,$$

which is dominated by the eigendirections of largest λ_k (least-constrained weight directions). Since $\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \sigma^{-2} \Phi^\top \Phi$, large λ_k corresponds to directions where $\Phi^\top \Phi$ has little support, i.e. feature directions far from those covered by the training data, so the predictive variance balloons under extrapolation.

Exercise 12.5

Ridge regression as MAP. Show that the MAP under $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$ solves the ridge problem $\arg \min_{\mathbf{w}} [\|\mathbf{y} - \Phi \mathbf{w}\|^2 + (\sigma^2/\tau^2) \|\mathbf{w}\|^2]$.

Solution

The negative log-posterior, up to constants in $\mathbf{w}$, is

$$-\log p(\mathbf{w} | D) = \frac{1}{2\sigma^2} \|\mathbf{y} - \Phi \mathbf{w}\|^2 + \frac{1}{2\tau^2} \|\mathbf{w}\|^2 + \text{const.}$$

Multiplying by $2\sigma^2$ gives the ridge objective with $\lambda = \sigma^2/\tau^2$, and setting the gradient to zero yields the MAP $\hat{\mathbf{w}}_{\text{MAP}} = (\Phi^\top \Phi + \lambda \mathbf{I})^{-1} \Phi^\top \mathbf{y} = \mathbf{m}_N$.

Exercise 12.6

Lasso as MAP. Show that the MAP under a Laplace (double-exponential) prior on each w_j corresponds to an L^1 -penalised least-squares problem (the lasso).

Solution

With independent Laplace priors $p(w_j) = \frac{1}{2b} \exp(-|w_j|/b)$, the negative log-posterior is

$$-\log p(\mathbf{w} \mid D) = \frac{1}{2\sigma^2} \|\mathbf{y} - \Phi \mathbf{w}\|^2 + \frac{1}{b} \sum_j |w_j| + \text{const.}$$

Minimising this is the lasso $\arg \min_{\mathbf{w}} [\|\mathbf{y} - \Phi \mathbf{w}\|^2 + \lambda \|\mathbf{w}\|_1]$ with $\lambda = 2\sigma^2/b$; the non-differentiable ℓ^1 penalty drives some coefficients exactly to zero.

Exercise 12.7

Marginal likelihood (evidence). Derive the marginal likelihood $p(\mathbf{y} \mid \Phi)$ for the Bayesian linear regression with Gaussian prior and Gaussian likelihood. Identify the Occam factor.

Solution

Marginalising the Gaussian likelihood against the Gaussian prior gives a Gaussian on $\mathbf{y}$,

$$p(\mathbf{y} \mid \Phi) = \mathcal{N}(\mathbf{y} \mid \Phi \mathbf{m}_0, \sigma^2 \mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top).$$

Equivalently, $\log p(\mathbf{y} \mid \Phi) = \log p(\mathbf{y} \mid \hat{\mathbf{w}}_{\text{MAP}}) + \frac{1}{2} \log \frac{|\mathbf{S}_N|}{|\mathbf{S}_0|} + \text{const}$; the best-fit likelihood at the MAP times the Occam factor $\sqrt{|\mathbf{S}_N|/|\mathbf{S}_0|}$, which is the ratio of posterior to prior volume in $\mathbf{w}$ -space and penalises models that need to fine-tune their parameters.

Exercise 12.8

Conjugate NIG update. Add an unknown noise scale σ^2 with prior $\sigma^2 \sim \text{IG}(\alpha_0, \beta_0)$ and conditional prior $\mathbf{w} \mid \sigma^2 \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{V}_0)$. Write the posterior parameters $(\mathbf{m}_N, \mathbf{V}_N, \alpha_N, \beta_N)$ after observing $\mathbf{y}, \Phi$.

Solution

The Normal–Inverse–Gamma family is conjugate; absorbing the common σ^2 factor, the posterior is again NIG with

$$\begin{aligned} \mathbf{V}_N^{-1} &= \mathbf{V}_0^{-1} + \Phi^\top \Phi, & \mathbf{m}_N &= \mathbf{V}_N (\mathbf{V}_0^{-1} \mathbf{m}_0 + \Phi^\top \mathbf{y}), \\ \alpha_N &= \alpha_0 + \frac{N}{2}, & \beta_N &= \beta_0 + \frac{1}{2} (\mathbf{y}^\top \mathbf{y} + \mathbf{m}_0^\top \mathbf{V}_0^{-1} \mathbf{m}_0 - \mathbf{m}_N^\top \mathbf{V}_N^{-1} \mathbf{m}_N). \end{aligned}$$

Note the precision-form mean/covariance update is identical to the known-noise case with $\mathbf{S}_0 = \sigma^2 \mathbf{V}_0$, while (α_N, β_N) accumulate N observations and the residual sum of squares.

Exercise 12.9

Equal-tail 95% predictive interval. For a new input x_* , write the interval explicitly in terms of ϕ_* , $\mathbf{m}_N$, $\mathbf{S}_N$, σ^2 , and $z_{0.975} \approx 2$.

Solution

Since $y_* \mid D \sim \mathcal{N}(\phi_*^\top \mathbf{m}_N, \sigma^2 + \phi_*^\top \mathbf{S}_N \phi_*)$, the equal-tail 95% predictive interval is

$$\phi_*^\top \mathbf{m}_N \pm z_{0.975} \sqrt{\sigma^2 + \phi_*^\top \mathbf{S}_N \phi_*} \approx \phi_*^\top \mathbf{m}_N \pm 2\sqrt{\sigma^2 + \phi_*^\top \mathbf{S}_N \phi_*}.$$

Exercise 12.10

Posterior predictive vs plug-in. Argue with one line of algebra why the posterior predictive variance is strictly larger than the plug-in variance unless the posterior is a point mass.

Solution

By the law of total variance,

$$\text{Var}(y_* \mid D) = E[\text{Var}(y_* \mid \mathbf{w})] + \text{Var}(E[y_* \mid \mathbf{w}]) = \sigma^2 + \phi_*^\top \mathbf{S}_N \phi_* \geq \sigma^2,$$

with equality only when $\phi_*^\top \mathbf{S}_N \phi_* = 0$, i.e. the posterior puts all mass on a single $\mathbf{w}$. Hence the plug-in variance σ^2 systematically understates uncertainty.

Exercise 12.11

Effect of prior width. Take a one-feature linear regression with prior $w \sim \mathcal{N}(0, \tau^2)$ and known σ^2 . Compute the posterior mean and discuss its behaviour as $\tau \rightarrow 0$ and $\tau \rightarrow \infty$.

Solution

With $\Phi^\top \Phi = \sum_i x_i^2 \equiv S_{xx}$ and $\Phi^\top \mathbf{y} = \sum_i x_i y_i \equiv S_{xy}$, the precision-form update gives

$$m_N = \frac{S_{xy}/\sigma^2}{\tau^{-2} + S_{xx}/\sigma^2} = \frac{S_{xy}}{S_{xx} + \sigma^2/\tau^2}.$$

As $\tau \rightarrow 0$ the prior dominates and $m_N \rightarrow 0$ (full shrinkage); as $\tau \rightarrow \infty$ the prior is uninformative and $m_N \rightarrow S_{xy}/S_{xx} = \hat{w}_{\text{OLS}}$.

Exercise 12.12

Effective degrees of freedom. For ridge regression with prior precision $\lambda = \sigma^2/\tau^2$, the effective degrees of freedom is $\text{df}(\lambda) = \text{tr}[\Phi(\Phi^\top \Phi + \lambda \mathbf{I})^{-1} \Phi^\top]$. Compute it for a single feature.

Solution

With $\Phi = (x_1, \dots, x_N)^\top$, $\Phi^\top \Phi = S_{xx}$ is a scalar, so

$$\text{df}(\lambda) = \text{tr}[\Phi \Phi^\top] \frac{1}{S_{xx} + \lambda} = \frac{S_{xx}}{S_{xx} + \lambda} \in (0, 1).$$

As $\lambda \rightarrow 0$ we recover 1 degree of freedom (OLS); as $\lambda \rightarrow \infty$, $\text{df} \rightarrow 0$ (full shrinkage).

Exercise 12.13

Sequential update. Show that updating the conjugate posterior on $(\mathbf{y}_1, \Phi_1)$ and then on $(\mathbf{y}_2, \Phi_2)$ gives the same posterior as a single update on the stacked data.

Solution

After the first batch, $\mathbf{S}_1^{-1} = \mathbf{S}_0^{-1} + \sigma^{-2} \Phi_1^\top \Phi_1$ and $\mathbf{S}_1^{-1} \mathbf{m}_1 = \mathbf{S}_0^{-1} \mathbf{m}_0 + \sigma^{-2} \Phi_1^\top \mathbf{y}_1$. Using $(\mathbf{m}_1, \mathbf{S}_1)$ as the new prior and applying the same rule to $(\mathbf{y}_2, \Phi_2)$ yields

$$\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \sigma^{-2} (\Phi_1^\top \Phi_1 + \Phi_2^\top \Phi_2), \quad \mathbf{S}_N^{-1} \mathbf{m}_N = \mathbf{S}_0^{-1} \mathbf{m}_0 + \sigma^{-2} (\Phi_1^\top \mathbf{y}_1 + \Phi_2^\top \mathbf{y}_2),$$

which is exactly the single-step update with stacked $\Phi = [\Phi_1; \Phi_2]$ and $\mathbf{y} = [\mathbf{y}_1; \mathbf{y}_2]$. Bayesian updating is associative.

Exercise 12.14

Posterior mean as shrinkage. Show that the posterior mean in the conjugate model with prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$ is a shrunk version of the OLS estimator $\hat{\mathbf{w}}_{\text{OLS}}$, and identify the shrinkage factor in the eigenbasis of $\Phi^\top \Phi$.

Exercise 12.15

Posterior predictive check. Describe in one sentence what a posterior predictive check is, what diagnostic plots you would draw, and what would indicate a misspecified model.

12.15. Sources

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13. Probabilistic programming: JAGS, Stan, ADVI

13.1. Introduction: from hand-coded to declarative models

In preceding chapters, we implemented Markov chain Monte Carlo by hand: we wrote log-likelihood and log-prior functions, tuned proposal covariance matrices, diagnosed convergence, and iterated. This approach is powerful pedagogically—it reveals exactly what is being sampled and how—but it is labour-intensive and error-prone for realistic models.

Real-world inference often requires many hyperparameters, hierarchical structure, and thousands of latent variables. Writing the full log-posterior by hand for such models is impractical. We need a language in which to *declare* the model in terms of its statistical structure, and delegate the numerical sampling to software.

Probabilistic programming languages (PPLs) solve this. They allow you to write:

$$\text{model} + \text{data} \rightarrow p(\theta \mid \mathbf{y}).$$

Two mainstream PPLs in the applied Bayesian community are **JAGS** and **Stan**. We also cover **variational inference**, an alternative to MCMC that solves the posterior by optimisation, and its automation in **ADVI** (automatic differentiation variational inference).

13.2. The classical limitation: conjugacy

In the first chapters of this course, we exploited conjugate priors: when the prior and likelihood share a specific functional form, the posterior is tractable analytically. For example, a Beta prior on a Bernoulli probability yields a Beta posterior; a Normal prior on a Gaussian mean (with known variance) yields a Normal posterior.

Conjugacy is rare in practice. Most real models are *non-conjugate*: the posterior is not available in closed form. In such cases we have two paths:

1. **MCMC**: Sample from the posterior.
2. **Variational inference**: Approximate the posterior with a tractable family.

Both JAGS and Stan use MCMC. JAGS uses Gibbs sampling or Metropolis-within-Gibbs (no gradient information). Stan uses Hamiltonian Monte Carlo with automatic differentiation (gradient-based), which mixes much faster.

13.3. JAGS: Gibbs sampling with a model syntax

13.3.1. Overview

JAGS stands for *Just Another Gibbs Sampler*. It is a language and engine for specifying Bayesian hierarchical models and sampling from them using Gibbs updates or Metropolis-Hastings steps.

The key idea: you write the model in a declarative syntax that lists the conditional distribution of each variable given its parents. JAGS parses this, determines which variables can be sampled by Gibbs and which need Metropolis-Hastings, and executes the sampler automatically.

13.3.2. Model syntax

A JAGS model comprises three optional blocks: **data**, **model**, and **inits**.

The **data block** declares observed quantities (fixed).

The **model block** specifies the generative structure: for each stochastic node (parameter, latent variable, observation), you write its conditional distribution.

The **inits block** (usually separate in R) provides starting values.

Example: centered quadratic regression.

```
model {
  # Likelihood
  for (i in 1:N) {
    y[i] ~ dnorm(mu[i], tau)
    mu[i] <- a0 + a1 * z[i] + a2 * pow(z[i], 2)
  }

  # Priors
  a0 ~ dnorm(0, 1/100)
  a1 ~ dnorm(0, 1/100)
  a2 ~ dnorm(0, 1/100)

  # Precision parameterization
  tau <- 1 / pow(sigma, 2)
  sigma ~ dunif(0, 20)
}
```

Key syntax notes:

- JAGS uses *precision* $\tau = 1/\sigma^2$, not standard deviation.
- Deterministic transformations use `<-`; stochastic nodes use `~`.
- Distributions are prefixed d: `dnorm`, `dunif`, `dbinom`, etc.
- The prior on σ must be proper.

13.3.3. Workflow in R with rjags

```
library(rjags)
library(coda)

# Prepare data
jags_data <- list(
  N = nrow(obsdata),
  z = obsdata$z,
  y = obsdata$y
)

# Initialize
inits <- function() list(a0=0, a1=0, a2=0, sigma=1)

# Compile
jm <- jags.model(textConnection(model_string),
  data=jags_data,
```

```

        inits=inits,
        n.chains=4)

# Adapt (burn-in)
update(jm, 2000)

# Sample
samp <- coda.samples(jm,
                     variable.names=c("a0","a1","a2","sigma"),
                     n.iter=20000)

# Summarize
summary(samp)
effectiveSize(samp)

```

Advantages of JAGS:

- Simple declarative syntax.
- Automatic Gibbs / MH switching.
- Handles discrete and continuous parameters.

Disadvantages:

- Gibbs sampling mixes slowly for correlated posteriors.
- No gradient information; cannot exploit manifold geometry.
- Precision parameterization less intuitive than standard deviation.

13.4. Stan: Hamiltonian Monte Carlo with gradients

13.4.1. Overview

Stan is a probabilistic programming language that uses **Hamiltonian Monte Carlo** with the **No-U-Turn Sampler** (NUTS) algorithm. By leveraging automatic differentiation to compute gradients of the log-posterior, Stan explores the posterior much more efficiently than Gibbs sampling, especially in high dimensions.

Modern Stan workflows use **cmdstanr**, a lightweight R interface.

13.4.2. Model structure

A Stan program has five optional blocks, executed in order:

1. **data**: Observed quantities, fixed.
2. **transformed data**: Derived quantities computed once.
3. **parameters**: The target of inference.
4. **transformed parameters**: Deterministic functions of parameters.
5. **model**: The log-posterior (likelihood + priors).
6. **generated quantities**: Posterior predictions.

13.4.3. Example: centered quadratic regression

```

data {
  int<lower=1> N;
  vector[N] z;
  vector[N] y;
}

parameters {
  real a0;
  real a1;
  real a2;
  real<lower=0> sigma;
}

model {
  // Likelihood
  y ~ normal(a0 + a1 * z + a2 * square(z), sigma);

  // Priors
  a0 ~ normal(0, 20);
  a1 ~ normal(0, 20);
  a2 ~ normal(0, 20);
  sigma ~ normal(0, 10); // Half-normal (due to lower=0)
}

```

Key syntax notes:

- Type declarations: `real`, `vector[N]`, `int`.
- Constraints are declared inline: `<lower=0>`, `<upper=1>`.
- Stan samples unconstrained internally; constraints are handled via smooth transformations (e.g., log for positive parameters).
- Standard deviation is natural; no precision.
- Vectorized operations (`y ~ normal(...)` broadcasts over the vector) are encouraged.

13.4.4. Workflow in R with cmdstanr

```

library(cmdstanr)

# Prepare data
stan_data <- list(
  N = nrow(obsdata),
  z = obsdata$z,
  y = obsdata$y
)

# Compile
mod <- cmdstan_model("quad_centered.stan")

# Sample
fit <- mod$sample(
  data = stan_data,
  chains = 4,
  parallel_chains = 4,
  iter_warmup = 1000,
  iter_sampling = 2000,
  seed = 123
)

```

```
# Summary
fit$summary(c("a0", "a1", "a2", "sigma"))

# Diagnostics
fit$cmdstan_diagnose()
```

13.4.5. Posterior diagnostics in Stan

Stan reports several convergence statistics:

- $\hat{R}$ (**potential scale reduction factor**): Should be < 1.01 . Values > 1.05 indicate poor convergence.
- **ESS (effective sample size)**: Bulk ESS and tail ESS. Rule of thumb: both $\gtrsim 400$ per chain. Low ESS indicates high autocorrelation.
- **Divergent transitions**: Occur when the leapfrog integrator leaves the high-probability region. May indicate a multimodal or heavy-tailed target. Typical remedy: increase `adapt_delta` (default 0.8) to 0.9 or 0.95.

13.5. Comparison: hand-coded MH vs JAGS vs Stan

Aspect	Hand-coded MH	JAGS	Stan
Model coding	Log-posterior by hand	Declarative (BUGS)	Declarative + blocks
Sampler	Random-walk MH	Gibbs + MH	HMC (NUTS)
Gradients	No	No	Automatic (AD)
Tuning	Manual proposal covariance	Partly automatic	Automatic warmup
Discrete parameters	OK	OK	Require marginalisation
Convergence speed	Often slow	Moderate	Fast (high-dimensional)
Best for	Understanding, learning	Hierarchical models	Efficient inference
Diagnostic burden	High	Moderate	Lower (but still essential)

Note.

13.6. Variational inference: approximate posterior via optimisation

13.6.1. The KL divergence decomposition

MCMC samples exactly (asymptotically) from the posterior $p(\theta \mid \mathbf{y})$. Variational inference instead *optimises* over a family of tractable distributions $\mathcal{Q}$ to find the member $q^*(\theta)$ that best approximates the posterior.

13. Probabilistic programming: JAGS, Stan, ADVI

Define the Kullback–Leibler divergence:

$$\text{KL}(q \parallel p) = \mathbb{E}_q \left[\log \frac{q(\theta)}{p(\theta \mid \mathbf{y})} \right].$$

By the law of total probability,

$$\log p(\mathbf{y}) = \mathbb{E}_q [\log p(\mathbf{y}, \theta) - \log q(\theta)] + \text{KL}(q \parallel p),$$

which rearranges to

$$\boxed{\log p(\mathbf{y}) = \mathcal{L}(q) + \text{KL}(q \parallel p)}.$$

Here $\mathcal{L}(q)$ is the **evidence lower bound (ELBO)**:

$$\mathcal{L}(q) = \mathbb{E}_q [\log p(\mathbf{y}, \theta) - \log q(\theta)].$$

Since $\log p(\mathbf{y})$ is fixed and $\text{KL}(q \parallel p) \geq 0$ always, *maximising the ELBO minimises the KL divergence*. Thus the optimal approximation is

$$q^*(\theta) = \arg \max_{q \in \mathcal{Q}} \mathcal{L}(q).$$

13.6.2. Mean-field approximation

To make the optimisation tractable, we restrict $\mathcal{Q}$ to a *mean-field family*:

$$q(\theta) = \prod_{i=1}^d q_i(\theta_i),$$

i.e., a product of independent marginal distributions (one per parameter). This destroys posterior correlations but is far easier to optimise than a full covariance Gaussian.

For each parameter, q_i is typically a standard distribution (Gaussian, Beta, Gamma, etc.).

13.6.3. The reparameterisation trick

To optimise the ELBO by stochastic gradient descent, we need to differentiate through expectations. The **reparameterisation trick** rewrites an expectation:

$$\mathbb{E}_{q(\theta)} [f(\theta)] = \mathbb{E}_{\epsilon \sim N(0, I)} [f(\mu + \sigma \epsilon)],$$

where μ and σ are the (optimisable) parameters of q , and ϵ is auxiliary randomness. Now we can differentiate w.r.t. μ and σ using the chain rule.

13.7. ADVI: Automatic Differentiation Variational Inference

13.7.1. Automatic transformation to unconstrained space

Many parameters have constraints: $\sigma > 0$, $p \in [0, 1]$, etc. ADVI automatically transforms such parameters to an unconstrained space, optimises a Gaussian approximation there, and back-transforms samples.

For example, $\sigma > 0$ is mapped to $\eta = \log \sigma \in \mathbb{R}$. The variational family becomes $q(\eta)$ (a univariate Gaussian in η), and we sample $\eta \sim N(\mu, \sigma^2)$, then set $\sigma = \exp(\eta)$.

13.7.2. Gaussian variational family

ADVI uses a multivariate Gaussian approximation:

$$q(\theta) = N(\theta \mid \mu, \Sigma),$$

where Σ is either diagonal (mean-field) or full-rank. The parameters (μ, Σ) are optimised by stochastic gradient ascent on the ELBO.

13.7.3. Stan's `vb()` interface

```
# Variational inference in Stan (instead of MCMC)
fit_vb <- mod$variational(
  data = stan_data,
  algorithm = "meanfield", # or "fullrank"
  iter = 10000,
  seed = 123
)

# Approximately posterior samples
draws <- fit_vb$draws()
```

Advantages of ADVI:

- Much faster than MCMC for high-dimensional models.
- Deterministic; no convergence diagnostics needed.
- Useful for exploratory analysis.

Disadvantages:

- Mean-field assumption often too restrictive; underestimates posterior variance.
- No automatic diagnostic (unlike MCMC diagnostics like $\hat{R}$).
- For posterior inference, MCMC is often preferable.

13.8. Practical workflow: from hand-rolled MCMC to PPLs

The progression we have traced reflects a real scientific workflow:

1. **Prototype with hand-coded MCMC** (Chapter 11 onwards): Learn the model, understand the geometry of the posterior, diagnose what works and what doesn't.
2. **Apply parametrisation tricks**: Center and scale predictors, use whitening transforms, exploit posterior covariance structure.
3. **Upgrade to a gradient-based sampler**: Use HMC or NUTS to exploit smoothness of the log-posterior.
4. **Automate with Stan or JAGS**: Declare the model once, let the software handle sampling, diagnostics, and adaptation.
5. **Consider variational inference if**:
 - You need very fast approximate inference for exploration.

- Your model is too large for MCMC.
- You are happy accepting mean-field correlations.

13.9. Failure modes and remedies

13.9.1. Multimodal posteriors

If the target distribution has multiple distinct modes, MCMC may get stuck in one mode and miss others. JAGS and Stan both struggle with multimodality.

Remedies:

- Use parallel tempering (Chapter 10).
- Run many short chains from dispersed starting values.
- Reparametrise to reduce multimodality (e.g., via identifiability constraints).

13.9.2. Discrete parameters

Stan requires all parameters to be continuous. If your model includes discrete latent variables (e.g., mixture component labels), you must marginalise them out analytically (integrate over all possible values) before passing the model to Stan.

JAGS handles discrete nodes naturally.

13.9.3. Heavy-tailed posteriors

If the posterior has heavy tails (e.g., a mixture of normals with small weights), the leapfrog integrator in HMC can take very long steps and diverge. Divergent transitions are a warning.

Remedies:

- Increase `adapt_delta` ($0.8 \rightarrow 0.95$ or higher).
- Increase `max_treedepth`.
- Reparametrise to reduce tail heaviness (e.g., use a Student- t prior to allow more dispersed posteriors).

13.9.4. Weak identifiability

If two parameters affect the likelihood in nearly the same way, the posterior is highly correlated or even degenerate. MCMC mixes slowly; variational inference underestimates uncertainty.

Remedy:

- Include informative priors that break the symmetry.
- Reparametrise (e.g., use a sum-of-effects rather than separate effects).

13.10. What to report

For any Bayesian analysis using JAGS or Stan, report:

1. **Model specification:** Likelihood, priors, parametrisation.
2. **Sampling method:** Algorithm, number of chains, iterations, warmup.
3. **Convergence diagnostics:** Trace plots, $\hat{R}$, ESS, acceptance rate, divergent transitions (Stan only).

4. **Posterior summaries:** Mean or median, 95% credible intervals.
5. **Posterior predictions:** Regression curves with credible intervals; posterior predictive intervals if relevant.

Do not report precise posterior point estimates from a badly mixing chain—they are numerical artifacts, not inference.

13.11. Sources

JAGS: - Martyn Plummer. JAGS: A program for analysis of Bayesian graphical models using Gibbs sampling. In *Proc. 3rd Int. Workshop on Distributed Statistical Computing*, 2003.

Stan: - Stan Development Team. Stan user's guide and reference manual, version 2.33. <https://mc-stan.org>.
- Gelman et al. Bayesian workflow. *ArXiv preprint arXiv:2011.01808*, 2020.

Variational inference: - David M. Blei, Alp Kucukelbir, and Jon D. McInman. Variational inference: A review for statisticians. *Journal of the American Statistical Association*, 112(518):859–877, 2017. - Alp Kucukelbir et al. Automatic differentiation variational inference. *Journal of Machine Learning Research*, 18(1):1–45, 2017.

References cited: - Gianmaria Falasco. Advanced statistics for physics analysis: Efficient MCMC for Bayesian regression: from hand tuning to JAGS and Stan. Lecture slides, May 19, 2026. - Sivia, D.S. & Skilling, J. (2006). *Data Analysis: A Bayesian Tutorial*. Oxford University Press. - Falasco, G. (2026). Lecture on Hamiltonian Monte Carlo. ASPA course notes.

14. Generalized linear models

14.1. Introduction: When linear regression fails

Classical Gaussian linear regression assumes that the response variable y_i is continuous and approximately normally distributed with mean $\mu_i = f(x_i; \beta)$ and constant variance σ^2 . This assumption works well for many physical measurements with small, additive noise.

However, in many experiments we encounter response variables that violate this assumption fundamentally:

- **Count data:** number of detected photons in a time window, decay events in a bin, or success counts in repeated trials. These are non-negative integers with variance proportional to the mean, not constant.
- **Binary outcomes:** detection (yes/no), selection success (pass/fail), or signal presence (present/absent). The mean must lie in $(0, 1)$ to represent a probability.
- **Positive-only data:** time-to-event, energy, or rate parameters. The mean must be strictly positive.

Fitting a standard linear model $y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$ to such data produces:

- **Invalid predictions:** fitted means may fall outside the natural parameter space (e.g., negative counts or probabilities outside $[0, 1]$).
- **Incorrect uncertainty quantification:** constant-variance assumptions badly misrepresent the variability structure of heteroscedastic responses.
- **Biased inference:** parameter estimates become inefficient and confidence intervals lose their intended coverage.

Generalized linear models extend regression to non-Gaussian responses by replacing the normality assumption with a member of the exponential family and introducing a **link function** that maps the linear predictor onto the valid parameter space.

14.2. The three ingredients of a GLM

A generalized linear model comprises three interconnected components:

Definition 14.1. Definition (GLM structure): A generalized linear model for observations $y_1, \dots, y_N$ is specified by:

1. **Random component:** $y_i \mid \theta_i \sim p(y_i; \theta_i)$, where p belongs to an exponential-family distribution parameterized by a natural parameter θ_i .
2. **Systematic component:** a linear predictor

$$\eta_i = \beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p} = \mathbf{x}_i^\top \beta,$$

where β is a vector of regression coefficients and $\mathbf{x}_i$ is the design vector for observation i .

3. **Link function:** a smooth, invertible function $g(\cdot)$ that relates the mean $\mu_i = \mathbb{E}[y_i \mid \mathbf{x}_i]$ to the

linear predictor:

$$g(\mu_i) = \eta_i, \quad \text{equivalently} \quad \mu_i = g^{-1}(\eta_i).$$

The inverse link, g^{-1} , is called the **mean function**.

The link function ensures that even though η_i can take any real value, the predicted mean $\mu_i = g^{-1}(\eta_i)$ respects the constraints of the response distribution. Gaussian linear regression is the special case in which p is Gaussian and g is the identity, so $\mu_i = \eta_i$.

14.3. The exponential family and canonical form

The exponential family is a large class of distributions sharing a common parameterization that unifies likelihood inference for GLMs.

Definition 14.2. Definition (Exponential family canonical form): A probability distribution for y belongs to the exponential family if it can be written as

$$p(y; \theta, \phi) = h(y, \phi) \exp \left(\frac{\theta y - A(\theta)}{\phi} \right),$$

where: - θ is the **natural parameter** (or canonical parameter). - $\phi > 0$ is the **dispersion parameter**, often related to variance. - $A(\theta)$ is the **cumulant function** or **log-partition function**. - $h(y, \phi)$ is the **base measure** (carries y -dependent normalization and ϕ terms not involving θ).

The term $A(\theta)$ ensures the distribution integrates or sums to one.

Theorem 14.1. Theorem (Moment relations): For a distribution in exponential-family form,

$$\mathbb{E}[y] = A'(\theta), \quad \text{Var}(y) = \phi \cdot A''(\theta).$$

Proof sketch: The log-likelihood is $\ell(\theta; y) = \theta y - A(\theta) + \text{const}$. The score is $\frac{\partial \ell}{\partial \theta} = y - A'(\theta)$. Setting $\mathbb{E}[\text{score}] = 0$ gives $\mathbb{E}[y] = A'(\theta)$. The variance follows from the Hessian of ℓ .

These moment relations are powerful: they immediately connect the natural parameter θ to the mean μ and relate the mean to the variance through $A''(\theta)$.

14.4. Canonical links and the link-function table

A **canonical link** is the link function such that $g(\mu) = \theta$. For canonical links, the natural parameter equals the linear predictor directly, which simplifies both interpretation and computation (the sufficient statistics for β are linear in the data).

The following table summarizes the most common GLM families used in practice:

Table 14.1.: **Table: Exponential families and canonical links**

Distribution	$A(\theta)$	$\theta = g(\mu)$ (canonical)	$g^{-1}(\eta)$ (inverse link)	Constraint
Gaussian	$\theta^2/2$	μ	η	$\mu \in \mathbb{R}$
Bernoulli	$\log(1 + e^\theta)$	$\log \frac{\mu}{1-\mu}$ (logit)	$\frac{1}{1+e^{-\eta}}$	$\mu \in (0, 1)$
Poisson	e^θ	$\log \mu$	e^η	$\mu > 0$

Distribution	$A(\theta)$	$\theta = g(\mu)$ (canonical)	$g^{-1}(\eta)$ (inverse link)	Constraint
Gamma	$-\log(-\theta)$	$-1/\mu$	$-1/\eta$	$\mu > 0$

14.5. Logistic regression: Bernoulli response

Logistic regression models a **binary response** $y_i \in \{0, 1\}$ as a Bernoulli trial:

$$y_i \mid \pi_i \sim \text{Bernoulli}(\pi_i),$$

where $\pi_i = \Pr(y_i = 1 \mid \mathbf{x}_i)$ is the success probability.

- **Canonical link.** The Bernoulli distribution has mean $\mu_i = \pi_i$ and variance $\pi_i(1 - \pi_i)$. The cumulant function is $A(\theta) = \log(1 + e^\theta)$, so the canonical link is the **logit**:

$$g(\pi_i) = \log \frac{\pi_i}{1 - \pi_i} = \eta_i = \mathbf{x}_i^\top \beta.$$

The **inverse logit** or **logistic function** recovers the probability:

$$\pi_i = \sigma(\eta_i) = \frac{1}{1 + e^{-\eta_i}}.$$

- **Likelihood.** The likelihood for N independent observations is

$$p(D \mid \beta) = \prod_{i=1}^N \pi_i^{y_i} (1 - \pi_i)^{1 - y_i},$$

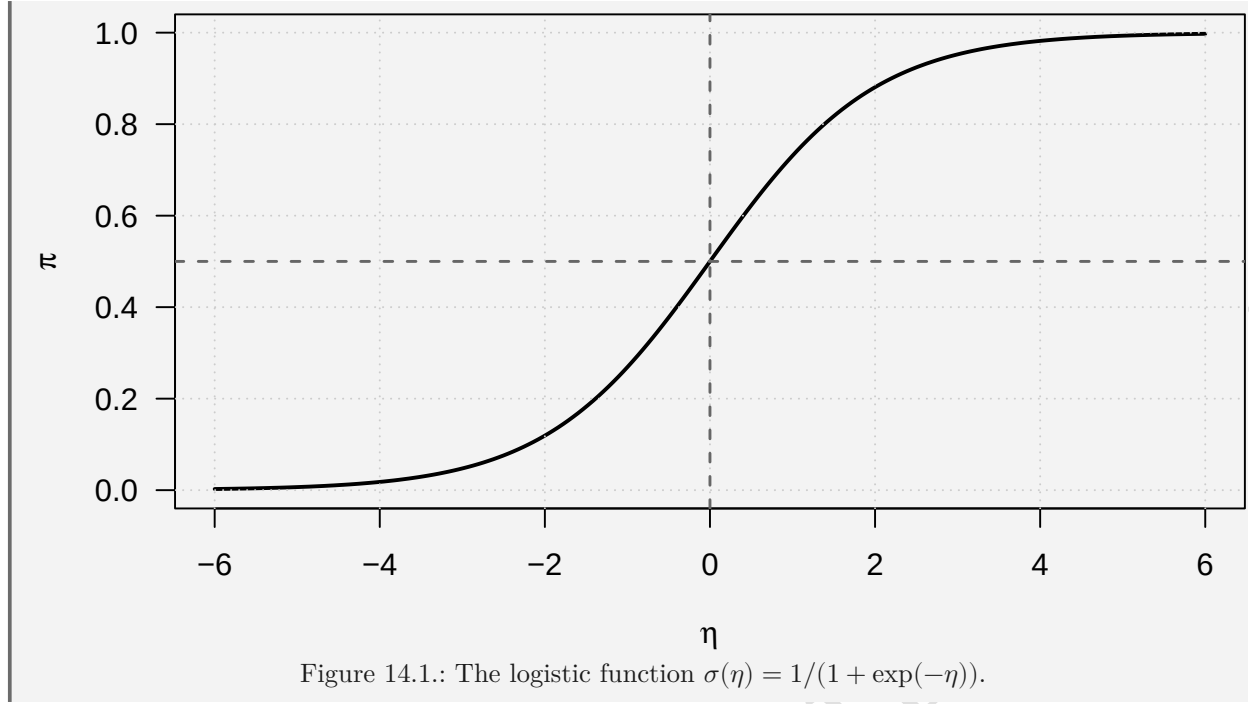
with log-likelihood

$$\ell(\beta; D) = \sum_{i=1}^N [y_i \eta_i - \log(1 + e^{\eta_i})].$$

- **Log-odds and odds ratios.** The logit transforms the probability π into the **log-odds**. The **odds** are $\frac{\pi}{1 - \pi}$, the ratio of success to failure probability. Logistic regression is linear on the log-odds scale: $\log \frac{\pi_i}{1 - \pi_i} = \beta_0 + \beta_1 x_{i,1} + \dots$. A one-unit increase in x_j changes the log-odds by β_j , or equivalently, **multiplies the odds by e^{β_j}** . If $\beta_1 = 0.5$, a one-unit increase in x_1 multiplies the odds of success by $e^{0.5} \approx 1.65$. The change in probability is nonlinear: the same increase in x produces a larger change in π near $\pi = 0.5$ and smaller changes near 0 or 1.
- **Variance is heteroscedastic.** Since $\text{Var}(y_i \mid \pi_i) = \pi_i(1 - \pi_i)$, the noise is largest near $\pi = 1/2$ and shrinks to zero near 0 or 1. Logistic regression estimates a detection probability, not a continuous measured response.

The logistic function below shows how η is squashed into $(0, 1)$.

See code 15.50 in Appendix for the implementation.



14.6. Poisson regression: Count response

Poisson regression models **count data** $y_i \in \{0, 1, 2, \dots\}$ as a Poisson distribution:

$$y_i \mid \lambda_i \sim \text{Poisson}(\lambda_i),$$

where $\lambda_i > 0$ is the expected count or rate. Both mean and variance equal λ_i : $\mathbb{E}[y_i] = \text{Var}(y_i) = \lambda_i$.

- **Canonical link.** The cumulant function is $A(\theta) = e^\theta$, and the canonical link is the **log link**:

$$g(\lambda_i) = \log \lambda_i = \eta_i = \beta_0 + \beta_1 x_{i,1} + \dots,$$

so $\lambda_i = e^{\eta_i}$.

- **Likelihood.** The Poisson likelihood is

$$p(D \mid \beta) = \prod_{i=1}^N \frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!},$$

with log-likelihood

$$\ell(\beta; D) = \sum_{i=1}^N [y_i \log \lambda_i - \lambda_i - \log(y_i!)].$$

- **Exponential effects.** The log link means that coefficients have **exponential interpretation**. A one-unit increase in x_j multiplies the expected count by e^{β_j} :

$$\frac{\lambda(x_j + 1)}{\lambda(x_j)} = e^{\beta_j}.$$

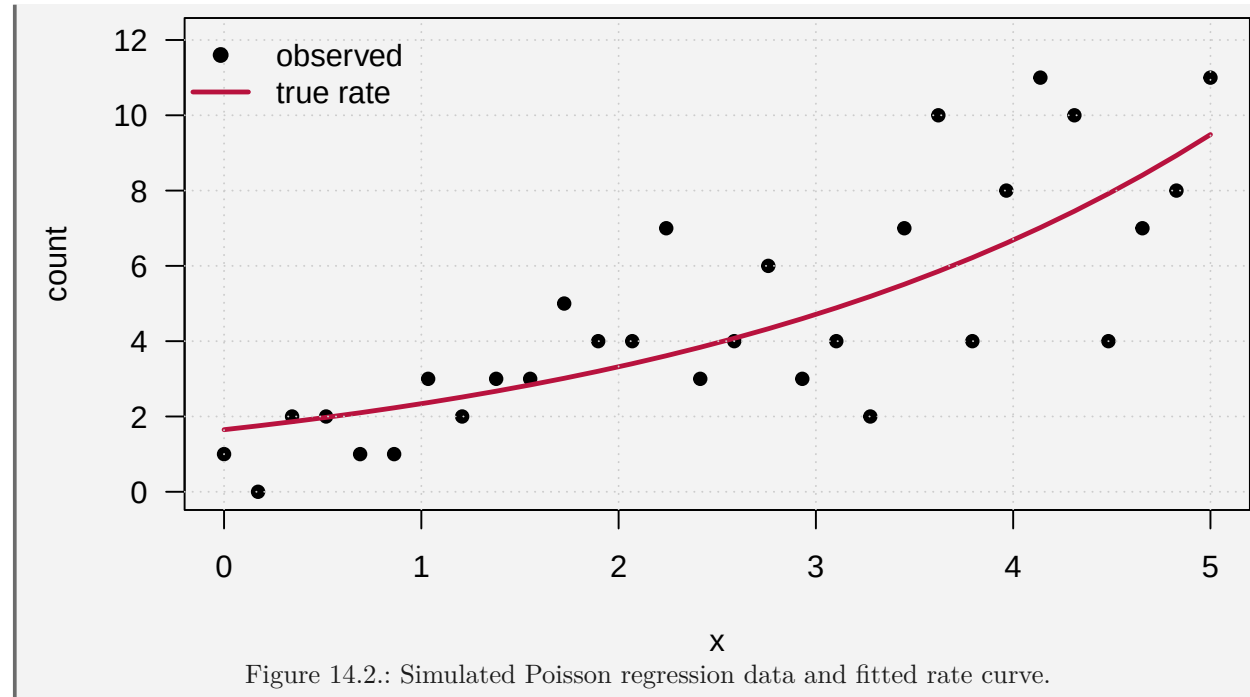
If $\beta_1 = 0.3$, then increasing x_1 by one unit multiplies the expected count by $e^{0.3} \approx 1.35$, a 35% increase. Log-linear models naturally describe rates that change by multiplicative factors.

- **Offsets and exposure.** In many applications, observations have different **exposure times** or **bin widths**. For example, detector data collected over different durations E_i . The expected count scales

with exposure: $\lambda_i = E_i r_i$, where r_i is the event rate. Using the log link, $\log \lambda_i = \log E_i + \log r_i = \log E_i + \eta_i$, where $\eta_i = \beta_0 + \beta_1 x_{i,1} + \dots$. The term $\log E_i$ is a known **offset**, a coefficient fixed at 1. In Stan or R, specify `log(exposure) + w0 + w1*x` in the linear predictor; the software interprets the known contribution separately.

The simulation below shows a Poisson regression with $\log \lambda_i = 0.5 + 0.35x_i$.

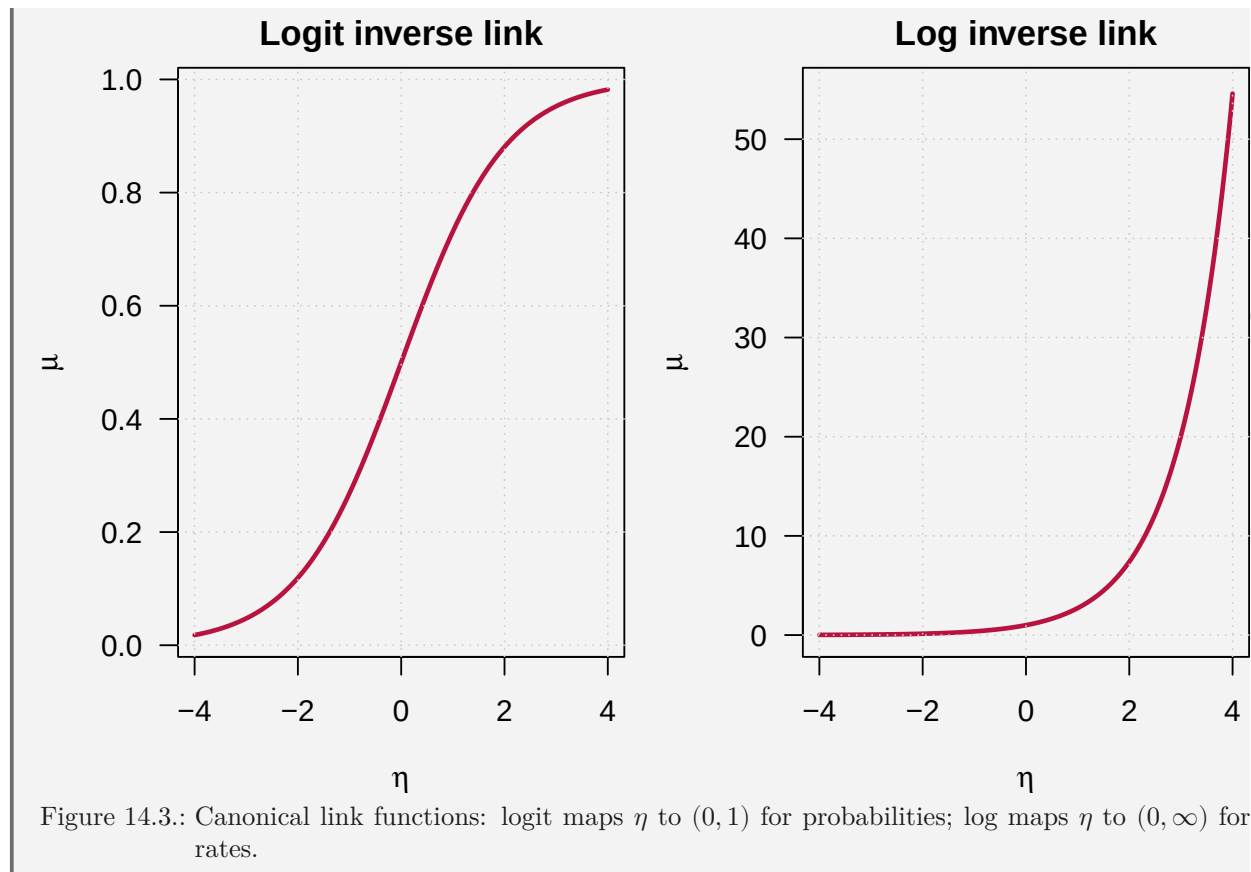
See code 15.51 in Appendix for the implementation.



14.7. Illustration: link functions

The choice of link function transforms the linear predictor onto the appropriate scale. Here we compare logit (for Bernoulli) and log (for Poisson) side by side:

See code 15.52 in Appendix for the implementation.



See code 15.53 in Appendix for the implementation.

14.8. Bayesian inference via MCMC

Frequentist GLMs are typically fitted by maximum likelihood using iteratively reweighted least squares (IRLS). The Bayesian approach places a prior on the coefficients and explores the posterior via Markov chain Monte Carlo. Priors are placed on the coefficient scale; weakly informative normal priors such as $w_k \sim \mathcal{N}(0, s^2)$ are a reasonable default, and standardizing predictors makes a single prior scale meaningful across coefficients.

14.8.1. Logistic regression in Stan

The following Stan model fits logistic regression with weakly informative normal priors:

```
data {
  int<lower=1> N;
  vector[N] x;
  array[N] int<lower=0, upper=1> y;
}
parameters {
  real w0;
  real w1;
}
model {
  w0 ~ normal(0, 5);
  w1 ~ normal(0, 5);
  y ~ bernoulli_logit(w0 + w1 * x);
}
```

```

}
generated quantities {
  array[N] int y_rep;
  for (i in 1:N) {
    y_rep[i] = bernoulli_logit_rng(w0 + w1 * x[i]);
  }
}

```

Stan's `bernoulli_logit` function is numerically stable; it computes the log-likelihood directly without forming π_i explicitly. The fitted curve below shows a logistic regression with $w_0 = -0.4$, $w_1 = 1.2$:

See code 15.54 in Appendix for the implementation.

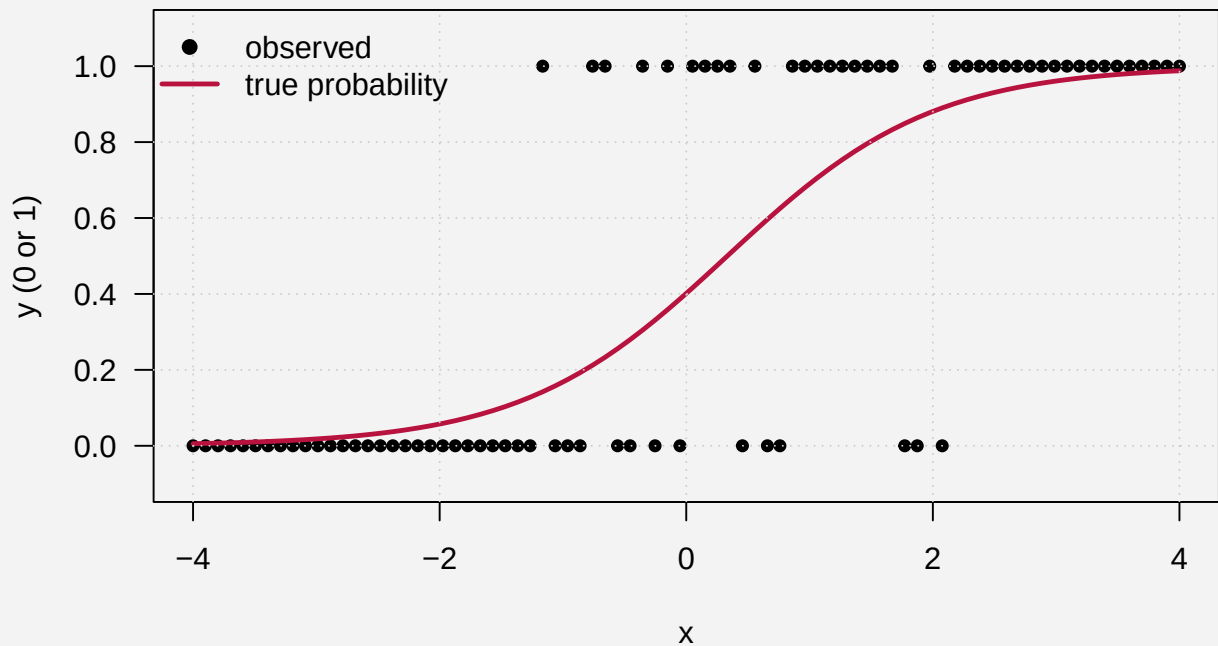


Figure 14.4.: Simulated binary data (0/1 points) with fitted logistic curve.

14.8.2. Poisson regression in Stan

```

data {
  int<lower=1> N;
  vector[N] x;
  array[N] int<lower=0> y;
}
parameters {
  real w0;
  real w1;
}
model {
  w0 ~ normal(0, 5);
  w1 ~ normal(0, 5);
  y ~ poisson_log(w0 + w1 * x);
}
generated quantities {
  array[N] int y_rep;
  for (i in 1:N) {
    y_rep[i] = poisson_log_rng(w0 + w1 * x[i]);
  }
}

```

The `poisson_log` function computes the Poisson likelihood with log link: given $\eta = w_0 + w_1x$, it increments log-probability by $y\eta - e^\eta - \log(y!)$. The generated quantities block produces posterior predictive replicates for model checking. For an exposure-weighted model, replace the sampling statement with `y ~ poisson_log(log(exposure) + w0 + w1 * x)`. JAGS implements the same models with syntax close to the mathematical form (`y[i] ~ dpois(lambda[i])` with `log(lambda[i]) <- w0 + w1*x[i]`), and analogously `dbern` with `logit(pi[i])` for the Bernoulli case); Stan is often more efficient for continuous parameters with strong posterior correlations.

14.9. Residual diagnostics and posterior predictive checks

- **Deviance.** In a fitted GLM, the **deviance** is twice the log-likelihood difference between the saturated model (one parameter per observation) and the fitted model:

$$D = 2 \sum_{i=1}^N [\ell_{\text{sat}}(y_i) - \ell_{\text{fitted}}(\hat{\mu}_i, y_i)].$$

For logistic regression, this simplifies to a sum of log-likelihood comparisons; for Poisson, it involves Kullback–Leibler divergence between observed and fitted counts.

- **Pearson residuals.** These scale differences by the estimated standard deviation:

$$r_i^{\text{Pearson}} = \frac{y_i - \hat{\mu}_i}{\sqrt{\widehat{\text{Var}}(y_i)}}.$$

In well-fitting models, Pearson residuals should be approximately standard normal. Large residuals flag individual observations that the model does not explain well.

- **Posterior predictive checks.** In Bayesian inference, we can generate **replicated data** $y_{\text{rep}}^{(s)}$ from the posterior predictive distribution for each MCMC draw (s). By comparing summaries of observed data y to summaries of replicated data, we can assess whether the model is plausible:
 - **Poisson:** Check the proportion of zeros, the maximum count, the ratio of variance to mean.
 - **Bernoulli:** Check the total number of successes, the success fraction in subgroups, the patterns of runs.

If the observed summary is consistently in the tail of the replicated-data distribution, the model misses important structure.

14.10. Coefficient interpretation and prediction

- **Logistic coefficients.** A one-unit increase in x_j (holding others fixed) changes the **odds** by a factor of e^{β_j} : if $\beta_j = 0.5$, the odds increase by a factor of $e^{0.5} \approx 1.65$ (65% increase); if $\beta_j = -0.3$, the odds decrease by a factor of $e^{-0.3} \approx 0.74$ (26% decrease). The corresponding **probability** change depends on the current probability level. Near $\pi = 0.5$ (maximum sensitivity), a one-unit change in x_j shifts π by approximately $\beta_j/4$. Near extreme probabilities, the shift is much smaller. Always report uncertainty: credible intervals for β_j quantify the precision of the estimate; transform endpoints through e^{β_j} to report odds ratios and their intervals.
- **Poisson coefficients.** β_j is the change in log rate per unit increase in x_j . The expected count is multiplied by e^{β_j} per unit increase: if $\beta_1 = 0.3$, a one-unit increase in x_1 multiplies the expected count by $e^{0.3} \approx 1.35$ (35% increase); if $\beta_2 = -0.2$, the expected count is multiplied by $e^{-0.2} \approx 0.82$ (18% decrease). These are **relative effects**: they describe multiplicative changes in the rate, not absolute changes. In a detector experiment, a 35% increase in count rate might be substantial or negligible

depending on the baseline and the experimental goal.

- **Posterior predictive intervals for logistic regression.** Posterior draws of (w_0, w_1) give posterior samples of $\pi_* = \sigma(w_0 + w_1 x_*)$ at a new location x_* . The credible interval for π_* quantifies uncertainty in the **probability**; a separate predictive interval addresses the binary outcome itself ($y_* \in \{0, 1\}$ with some probability).
- **Posterior predictive intervals for Poisson regression.** The posterior predictive interval for a new count y_* at x_* involves: (1) drawing $(w_0^{(s)}, w_1^{(s)})$ from the posterior, (2) computing $\lambda_*^{(s)} = \exp(w_0^{(s)} + w_1^{(s)} x_*)$, (3) drawing $y_*^{(s)} \sim \text{Poisson}(\lambda_*^{(s)})$, and (4) summarizing the empirical distribution of $y_*^{(s)}$ by quantiles (e.g., 2.5%, 50%, 97.5%). The predictive interval is discrete and generally asymmetric, reflecting the discrete and skewed nature of Poisson noise.

14.11. Overdispersion and the negative binomial

A fundamental property of the Poisson is that **variance equals the mean**. In real count data, the variance often exceeds the mean—a phenomenon called **overdispersion**. Causes include:

- Unobserved heterogeneity: different subgroups have different rates.
- Clustering: observations within groups are not independent.
- Extra-Poisson variability: a genuine excess beyond Poisson noise.

Proposition 14.2. Overdispersion test: Compute $\hat{\lambda}$ from fitted Poisson regression. If the sample variance of residuals $\sum_i (y_i - \hat{\lambda}_i)^2 / (N - p)$ is much larger than the average fitted rate, overdispersion is present.

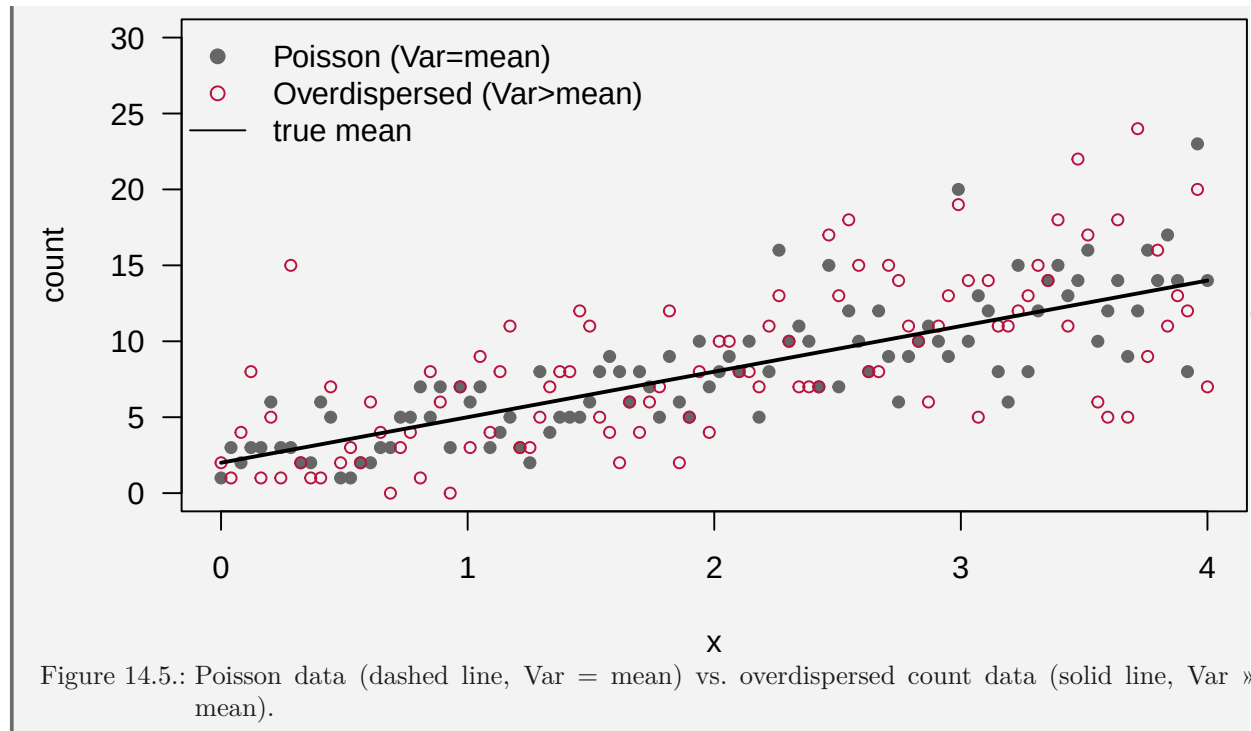
The **negative binomial** distribution extends Poisson regression by adding a shape parameter ϕ that allows variance to exceed the mean:

$$\text{Var}(y_i) = \lambda_i + \frac{\lambda_i^2}{\phi}.$$

As $\phi \rightarrow \infty$, the negative binomial approaches Poisson. A negative-binomial GLM is fitted similarly to Poisson but with an additional dispersion parameter to estimate.

The simulation below contrasts Poisson noise (variance equal to mean) with overdispersed counts:

See code 15.55 in Appendix for the implementation.



14.12. Common pitfalls and best practices

Note. Common mistakes:

1. **Ignoring distributional mismatch:** Fitting Gaussian regression to counts or binary data leads to invalid predictions and misaligned uncertainty.
2. **Forgetting the offset:** In exposure-weighted Poisson models, omitting the offset biases rate estimates and inflates uncertainty.
3. **Misinterpreting coefficients:** Logistic coefficients are log-odds, not probability changes. Poisson coefficients are log-rates, not expected count changes.
4. **Reporting point estimates without credible intervals:** GLM coefficients are estimated quantities; always include uncertainty via posterior quantiles.
5. **Confusing credible and predictive intervals:** The credible interval for μ (or π , λ) reflects estimation uncertainty. The predictive interval for y also includes observation noise and is wider.
6. **Ignoring posterior predictive checks:** A high likelihood value does not guarantee model adequacy. Generate replicated data and compare to observed summaries.
7. **Trusting diagnostics without thought:** High $\hat{R}$ and large ESS are necessary but not sufficient. Check trace plots, residual patterns, and posterior predictive checks for deeper assessment.

Remark. Best practices:

- **Center or standardize predictors** before fitting to reduce collinearity and improve MCMC mixing. A unit increase in a centered x is meaningful and comparable across covariates.
- **Visualize the fitted curve** with uncertainty bands overlaid on the data. This reveals whether the model captures the dominant patterns.
- **Use weakly informative priors** that regularize without imposing strong prior beliefs. For coefficients on a standardized scale, $\text{Normal}(0, 2)$ or $\text{Normal}(0, 3)$ is reasonable.

- **Check trace plots** for all parameters after sampling. Chains should mix well (no trending, not stuck, high autocorrelation).
- **Report both point and interval estimates** and be explicit about the scale (log-odds, log-rate, probability, count).
- **Document the model specification** (link function, prior, offset, if any) to enable reproducibility and interpretation.

14.13. Conclusion

Generalized linear models extend Bayesian regression to count, binary, and other non-Gaussian responses by:

- Specifying a response distribution from the exponential family (Poisson, Bernoulli, etc.).
- Connecting the mean to a linear predictor via a link function that respects parameter constraints.
- Inferring the linear coefficients while propagating uncertainty through the posterior.

The three ingredients—random component, linear predictor, and link function—unify a diverse set of regression tasks under one framework. Logistic regression models probabilities; Poisson regression models rates and counts; gamma and inverse-Gaussian regressions model positive continuous data. MCMC and probabilistic programming allow flexible, fully Bayesian inference, with posterior checks guiding model refinement.

14.14. Exercises

Exercise 14.1

State the three ingredients of a GLM. Write them out for logistic regression: random component, linear predictor, and link function.

Solution

The three ingredients are: a random component from the exponential family, a linear predictor $\eta = \phi^\top \beta$, and a link $g(\mu) = \eta$. For logistic regression

$$y_i | p_i \sim \text{Bernoulli}(p_i), \quad \eta_i = \phi_i^\top \beta, \quad g(p_i) = \log \frac{p_i}{1-p_i} = \eta_i,$$

so that $p_i = \sigma(\beta_0 + \beta_1 x_i)$.

Exercise 14.2

Canonical links. State the canonical link for the binomial, Poisson, Gaussian, gamma, and inverse-Gaussian families. Briefly justify why the link is *canonical* by referring to the natural parameter of the exponential family.

Solution

The canonical link is the function g such that $g(\mu) = \theta$, the natural parameter:

$$\text{binomial: } \text{logit } \log \frac{\mu}{1-\mu}, \quad \text{Poisson: } \log \mu, \quad \text{Gaussian: } \mu, \quad \text{gamma: } -1/\mu, \quad \text{IG: } -1/\mu^2.$$

With the canonical link the score has the clean form $\Phi^\top(\mathbf{y} - \mu)$ and $\eta = \phi^\top \beta$ is exactly the natural

parameter.

Exercise 14.3

Score for logistic regression. Show that the score function (gradient of the log-likelihood) for logistic regression with a single covariate is $\sum_i (y_i - p_i) x_i$, where $p_i = \sigma(\beta_0 + \beta_1 x_i)$ and σ is the logistic function.

Solution

The log-likelihood is $\ell(\beta) = \sum_i [y_i \eta_i - \log(1 + e^{\eta_i})]$ with $\eta_i = \beta_0 + \beta_1 x_i$. Using $\partial \log(1 + e^{\eta_i}) / \partial \beta_1 = \sigma(\eta_i) x_i = p_i x_i$,

$$\frac{\partial \ell}{\partial \beta_1} = \sum_i (y_i - p_i) x_i,$$

and analogously $\partial \ell / \partial \beta_0 = \sum_i (y_i - p_i)$.

Exercise 14.4

Fisher information. For logistic regression with a single feature x_i , derive the Fisher information matrix entry $\mathcal{J}_{11} = \sum_i p_i (1 - p_i) x_i^2$.

Solution

Differentiating the score $\partial \ell / \partial \beta_1 = \sum_i (y_i - p_i) x_i$ with $\partial p_i / \partial \beta_1 = p_i (1 - p_i) x_i$ gives the negative Hessian

$$\mathcal{J}_{11} = -\mathbb{E} \left[\frac{\partial^2 \ell}{\partial \beta_1^2} \right] = \sum_i p_i (1 - p_i) x_i^2.$$

In matrix form $\mathcal{J}(\beta) = \Phi^\top W \Phi$ with $W = \text{diag}(p_i (1 - p_i))$.

Exercise 14.5

Posterior under a Gaussian prior. With prior $\beta \sim \mathcal{N}(0, \tau^2 \mathbf{I})$, write the log-posterior for logistic regression up to a constant. Identify why no closed form exists for the posterior mean.

Solution

Up to an additive constant,

$$\log p(\beta \mid D) = \sum_i [y_i \eta_i - \log(1 + e^{\eta_i})] - \frac{1}{2\tau^2} \|\beta\|^2.$$

The likelihood term $\log(1 + e^{\eta_i})$ is non-conjugate to the Gaussian prior, so the posterior is not Gaussian and the normalising constant has no closed form; one must use IRLS/Laplace or MCMC.

Exercise 14.6

MAP via penalised IRLS. Sketch how iteratively re-weighted least squares (IRLS) reduces to a Newton step on the log-posterior, and how adding a quadratic penalty corresponds to the Gaussian prior. Write down one iteration.

Solution

A Newton step on the penalised log-posterior uses gradient $\Phi^\top(\mathbf{y} - \mathbf{p}) - \beta/\tau^2$ and Hessian $-(\Phi^\top W \Phi + \mathbf{I}/\tau^2)$, yielding

$$\beta^{(t+1)} = (\Phi^\top W^{(t)} \Phi + \tau^{-2} \mathbf{I})^{-1} \Phi^\top W^{(t)} \mathbf{z}^{(t)},$$

with working response $\mathbf{z}^{(t)} = \Phi \beta^{(t)} + W^{(t)-1}(\mathbf{y} - \mathbf{p}^{(t)})$. The Gaussian prior contributes the $\tau^{-2} \mathbf{I}$ ridge term.

Exercise 14.7

Posterior predictive in logistic regression. For a new input x_* with MCMC samples $\{\beta^{(s)}\}_{s=1}^S$, write the posterior predictive probability $P(y_* = 1 \mid x_*, D)$ as a Monte Carlo average.

Solution

Marginalising over the posterior gives

$$P(y_* = 1 \mid x_*, D) = \int \sigma(\beta^\top \phi_*) p(\beta \mid D) d\beta \approx \frac{1}{S} \sum_{s=1}^S \sigma(\beta^{(s)\top} \phi_*).$$

Exercise 14.8

Log-odds interpretation. In logistic regression with coefficient β_1 on feature x_1 , what does a one-unit increase in x_1 do to the odds? Give the formula in closed form.

Solution

Since $\log \frac{p}{1-p} = \beta_0 + \beta_1 x_1$, the odds at $x_1 + 1$ relative to x_1 satisfy

$$\frac{\text{odds}(x_1 + 1)}{\text{odds}(x_1)} = e^{\beta_1},$$

so a unit increase multiplies the odds by e^{β_1} .

Exercise 14.9

Rate-ratio interpretation. In Poisson regression with log link and coefficient β_1 , what does a one-unit increase in x_1 do to the rate? Give the formula and interpret in plain language.

Solution

With $\log \mu = \beta_0 + \beta_1 x_1$,

$$\frac{\mu(x_1 + 1)}{\mu(x_1)} = e^{\beta_1},$$

so a unit increase in x_1 multiplies the expected count (rate) by e^{β_1} ; e.g. $\beta_1 = 0.3$ gives a $\sim 35\%$ increase.

Exercise 14.10

Deviance residuals. State the formula for the deviance residual for a Poisson GLM. Use it to derive the Pearson residual in the small-count limit.

Solution

The Poisson deviance residual is

$$r_i^{\text{dev}} = \text{sign}(y_i - \hat{\mu}_i) \sqrt{2[y_i \log(y_i/\hat{\mu}_i) - (y_i - \hat{\mu}_i)]}.$$

Expanding $y_i \log(y_i/\hat{\mu}_i)$ around $y_i \approx \hat{\mu}_i$ to second order gives $2[\cdot] \approx (y_i - \hat{\mu}_i)^2/\hat{\mu}_i$, recovering the Pearson residual $r_i^P = (y_i - \hat{\mu}_i)/\sqrt{\hat{\mu}_i}$.

Exercise 14.11

Overdispersion. Explain why Poisson regression often fails when the data show $\text{Var}(Y_i) > E[Y_i]$. Identify the standard remedy (negative binomial) and write its variance function $\text{Var}(Y_i) = \mu_i + \mu_i^2/\kappa$.

Solution

Poisson enforces $\text{Var}(Y_i) = \mu_i$, so when data are overdispersed the model underestimates uncertainty and standard errors are too narrow. The negative-binomial GLM adds a shape parameter κ giving

$$\text{Var}(Y_i) = \mu_i + \mu_i^2/\kappa,$$

which reduces to Poisson as $\kappa \rightarrow \infty$.

Exercise 14.12

Identifiability. In logistic regression, what happens to the MLE when the data are perfectly separable (every $y = 1$ point has higher x than every $y = 0$ point)? How does a Gaussian prior restore identifiability?

Solution

Under perfect separation the log-likelihood is monotone in $\|\beta\|$, so the MLE diverges ($\hat{\beta}_1 \rightarrow \infty$) while the likelihood approaches its supremum. A Gaussian prior adds the penalty $-\|\beta\|^2/(2\tau^2)$,

$$\log p(\beta \mid D) = \ell(\beta) - \frac{1}{2\tau^2} \|\beta\|^2,$$

which pulls the posterior mode back to a finite value and restores a proper posterior.

Exercise 14.13

Posterior summary. From an MCMC run on Bayesian logistic regression you obtain posterior samples $\{\beta_1^{(s)}\}$ with mean 0.4 and standard deviation 0.1. Give a 95% equal-tailed credible interval for β_1 and for the odds ratio e^{β_1} , using $z_{0.975} \approx 2$.

Solution

Approximating the posterior as Gaussian, $0.4 \pm 2(0.1)$ gives

$$\beta_1 \in (0.2, 0.6), \quad e^{\beta_1} \in (e^{0.2}, e^{0.6}) \approx (1.22, 1.82).$$

Exercise 14.14

Link choice. For binary data, the probit link uses Φ^{-1} instead of logit. Compare the two: which has heavier tails of the latent process, and how does this change the interpretation of large coefficients?

Solution

The logistic latent distribution has heavier tails than the Gaussian used by probit, so for the same extreme η the logit assigns probabilities closer to 0 or 1 more slowly. Coefficients differ by roughly $\beta_{\text{logit}} \approx 1.6 \beta_{\text{probit}}$, and the logit retains clean odds-ratio interpretation e^{β_1} , while probit coefficients are read as standard-deviation shifts of the latent Gaussian.

Exercise 14.15

Pitfalls. Identify three common pitfalls of GLM modelling (e.g. unmodelled overdispersion, ignored zero-inflation, mis-specified link) and the diagnostic that would reveal each one.

Solution

- (i) Unmodelled overdispersion: detected by a Pearson statistic $\sum_i r_i^{P^2}/(N-p) \gg 1$ or by posterior-predictive variance-to-mean checks. (ii) Zero-inflation: detected by comparing the observed proportion of zeros to its posterior-predictive distribution. (iii) Mis-specified link: detected by plotting deviance/Pearson residuals against the linear predictor η and looking for systematic curvature.

14.15. Sources

- Sivia, D.S., & Skilling, J. (2006). *Data analysis: A Bayesian tutorial for scientists and engineers*. Oxford University Press. [Chapter 8: Model fitting.]
- McCullagh, P., & Nelder, J.A. (1989). *Generalized linear models* (2nd ed.). Chapman & Hall. [Classic reference; foundational theory of GLMs.]
- Gelman, A., Hill, J., Vehtari, A., & Burkner, P.C. (2020). *Regression and other stories*. Cambridge University Press. [Modern Bayesian regression, GLMs, and causal inference.]
- Stan Development Team (2023). *Stan User's Guide*. [Probabilistic programming; chapters on GLMs and efficient computation.]

Part V.

APPENDIX

15. Appendix: Source code

Listing 15.1: Discrete probability mass functions

```
1
2 par(mar = c(4.4, 4.4, 2.2, 1.2), cex = 0.9)
3
4 x_bin <- 0:20
5 pmf_bin <- dbinom(x_bin, size = 10, prob = 0.3)
6
7 x_pois <- 0:20
8 pmf_pois <- dpois(x_pois, lambda = 3)
9
10 x_geom <- 0:20
11 pmf_geom <- dgeom(x_geom, prob = 0.3)
12
13 plot(x_bin, pmf_bin, type = "h", lwd = 1.5, col = "black",
14      xlim = c(0, 20), ylim = c(0, max(pmf_bin, pmf_pois, pmf_geom)),
15      xlab = "k", ylab = "P(X = k)", main = "Discrete distributions")
16 points(x_bin, pmf_bin, pch = 16, col = "black", cex = 0.6)
17
18 lines(x_pois + 0.1, pmf_pois, type = "h", lwd = 1.5, col = "gray40")
19 points(x_pois + 0.1, pmf_pois, pch = 16, col = "gray40", cex = 0.6)
20
21 lines(x_geom - 0.1, pmf_geom, type = "h", lwd = 1.5, col = "#b8123e", lty = 2)
22 points(x_geom - 0.1, pmf_geom, pch = 16, col = "#b8123e", cex = 0.6)
23
24 legend("topright", bty = "n", cex = 0.75,
25       legend = c("Binomial(n=10, p=0.3)", "Poisson(lambda=3)", "Geometric(p=0.3)"),
26       col = c("black", "gray40", "#b8123e"), lwd = c(1.5, 1.5, 1.5), lty = c(1, 1, 2))
```

Listing 15.2: Poisson limit of Binomial

```
1
2 par(mar = c(4.4, 4.4, 2.2, 1.2), cex = 0.9)
3
4 x <- 0:12
5 lambda <- 3
6
7 # Binomial with n=30, p=0.1
8 pmf_bin_30 <- dbinom(x, size = 30, prob = 0.1)
9
10 # Poisson with lambda=3
11 pmf_pois <- dpois(x, lambda = 3)
12
13 plot(x, pmf_bin_30, type = "h", lwd = 1.5, col = "black",
14      xlim = c(-0.5, 12.5), ylim = c(0, max(pmf_bin_30, pmf_pois)),
15      xlab = "k", ylab = "P(X = k)", main = "Poisson limit: Binomial(n=30, p=0.1) vs
16      Poisson(lambda=3)")
17 points(x, pmf_bin_30, pch = 16, col = "black", cex = 0.7)
18
19 lines(x + 0.15, pmf_pois, type = "h", lwd = 1.5, col = "#b8123e")
20 points(x + 0.15, pmf_pois, pch = 16, col = "#b8123e", cex = 0.7)
21
22 legend("topright", bty = "n", cex = 0.75,
23       legend = c("Binomial(n=30, p=0.1)", "Poisson(lambda=3)"),
```

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```
23 col = c("black", "#b8123e"), lwd = c(1.5, 1.5), pch = c(16, 16))
```

Listing 15.3: Overlaid normal distributions showing effect of mean and precision

```
1 par(mar = c(4.4, 4.4, 2.2, 1.2))
2 mu1 <- 0; sigma1 <- 1
3 mu2 <- 2; sigma2 <- 0.5
4 x <- seq(-4, 6, length = 500)
5 plot(x, dnorm(x, mean = mu1, sd = sigma1),
6      type = "l", lwd = 1, ylim = c(0, 1),
7      xlab = "x", ylab = "Density", main = "Normal distributions")
8 lines(x, dnorm(x, mean = mu2, sd = sigma2),
9      lwd = 1, lty = 2, col = "gray40")
10 legend("topright", bty = "n", cex = 0.75,
11       legend = c(expression(N(0, 1)), expression(N(2, 0.25))),
12       lwd = 1, lty = c(1, 2), col = c("black", "gray40"))
```

Listing 15.4: Beta distribution family with various shape parameters

```
1 par(mar = c(4.4, 4.4, 2.2, 1.2))
2 x <- seq(0, 1, length = 200)
3 plot(x, dbeta(x, 2, 5), type = "l", lwd = 1, ylim = c(0, 3.5),
4      xlab = "x", ylab = "Density", main = "Beta distributions")
5 lines(x, dbeta(x, 5, 2), lwd = 1, lty = 2, col = "gray40")
6 lines(x, dbeta(x, 2, 2), lwd = 1, lty = 3, col = "#b8123e")
7 legend("topright", bty = "n", cex = 0.75,
8      legend = c("Beta(2, 5)", "Beta(5, 2)", "Beta(2, 2)"),
9      lwd = 1, lty = c(1, 2, 3), col = c("black", "gray40", "#b8123e"))
```

Listing 15.5: Distribution of sum of two uniform random variables

```
1 set.seed(42)
2 par(mar = c(4.4, 4.4, 2.2, 1.2))
3 n <- 10000
4 x1 <- runif(n)
5 x2 <- runif(n)
6 y <- x1 + x2
7 hist(y, breaks = 50, prob = TRUE, ylim = c(0, 1.2),
8     xlab = "y = X1 + X2", ylab = "Density", main = "Sum of two Uniform(0,1)")
9 yy <- seq(0, 2, length = 200)
10 fy <- ifelse(yy <= 1, yy, 2 - yy)
11 lines(yy, fy, lwd = 1, col = "#b8123e")
12 legend("topright", bty = "n", cex = 0.75,
13      legend = c("Empirical", "Theoretical"),
14      lwd = c(NA, 1), col = c("black", "#b8123e"), fill = c("gray70", NA))
```

Listing 15.6: Law of Large Numbers illustrated by running sample mean

```
1 set.seed(42)
2 par(mar = c(4.4, 4.4, 2.2, 1.2))
3 n <- 10000
4 y <- runif(n, -1, 1)
5 my <- cumsum(y) / (1:n)
6 plot(my, type = "l", lwd = 1, xlab = "n", ylab = "Sample mean",
7     main = "Law of Large Numbers (Uniform on [-1, 1])")
8 abline(h = 0, col = "gray40", lwd = 1, lty = 2)
9 legend("topright", bty = "n", cex = 0.75,
10      legend = c("Sample mean", "True mean = 0"),
11      lwd = 1, lty = c(1, 2), col = c("black", "gray40"))
```

Listing 15.7: Central Limit Theorem with uniform samples

```

1  set.seed(42)
2  par(mar = c(4.4, 4.4, 2.2, 1.2))
3  n <- 50
4  m <- replicate(1000, mean(runif(n)))
5  hist(m, prob = TRUE, breaks = 30, ylim = c(0, 20),
6       xlab = "Sample mean", ylab = "Density",
7       main = "CLT: means of 1000 samples of size 50 from Uniform(0,1)")
8  curve(dnorm(x, 0.5, sd = sqrt(1 / (12 * n))), add = TRUE,
9        col = "#b8123e", lwd = 1)
10
11 legend("topright", bty = "n", cex = 0.75,
12        legend = c("Empirical", "N(0.5, 0.00167)"),
13        fill = c("gray70", NA), lwd = c(NA, 1), col = c(NA, "#b8123e"))

```

Listing 15.8: Bayes factor for coin fairness as a function of prior width on theta under H1.

```

1  # Coin fairness example: n = 250 flips, h = 140 heads
2  # H0: theta = 0.5 (fair coin)
3  # H1: theta ~ Beta(alpha, alpha) with varying alpha
4
5
6  n <- 250
7  h <- 140
8
9  # Likelihood under H0
10 log_likelihood_H0 <- h * log(0.5) + (n - h) * log(0.5) + lchoose(n, h)
11
12 # Marginal likelihood under H1 with Beta(alpha, alpha) prior
13 # For Beta-Binomial model: P(D|H1) = C(n,h) * Beta(h+alpha, n-h+alpha) / Beta(alpha, alpha)
14 bayes_factor_beta_binomial <- function(n, h, alpha) {
15   log_ll_H1 <- lchoose(n, h) +
16     lbeta(h + alpha, n - h + alpha) -
17     lbeta(alpha, alpha)
18   exp(log_ll_H1 - log_likelihood_H0)
19 }
20
21 # Grid of prior concentrations
22 alpha_values <- seq(0.5, 50, by = 0.5)
23 bf_values <- sapply(alpha_values, function(a) bayes_factor_beta_binomial(n, h, a))
24
25 # Three-tone palette
26 col_bf <- "#1f77b4" # Blue for BF curve
27 col_guide <- "#ff7f0e" # Orange for prior-width guide
28 col_accent <- "#2ca02c" # Green for annotations
29
30 par(mar = c(4.4, 4.4, 2.2, 1.2))
31 plot(alpha_values, bf_values, type = "l", lwd = 1,
32      xlab = "Prior concentration (alpha)",
33      ylab = "Bayes Factor (BF_10)",
34      col = col_bf, ylim = c(0, max(bf_values) * 1.1))
35 abline(h = 1, lty = 2, col = col_guide, lwd = 0.8)
36 text(30, 1.05, "BF = 1 (indifference)", col = col_guide, cex = 0.85, adj = 0)
37
38 # Highlight a specific prior width
39 alpha_focus <- 5
40 bf_focus <- bayes_factor_beta_binomial(n, h, alpha_focus)
41 points(alpha_focus, bf_focus, pch = 16, col = col_accent, cex = 1.2)
42 segments(alpha_focus, 0, alpha_focus, bf_focus, lty = 3, col = col_accent, lwd = 0.8)
43 text(alpha_focus - 8, bf_focus + 0.15, sprintf("alpha=%.1f\nBF=%.3f", alpha_focus, bf_focus),
44      col = col_accent, cex = 0.8, adj = 1)

```

Listing 15.9: Growth of posterior odds as data accumulates.

15. Appendix: Source code

```

1
2 # Sequential analysis: build up data gradually
3 n_max <- 250
4 h_max <- 140
5
6 # Track data proportionally
7 flips_seq <- seq(10, n_max, by = 10)
8 heads_seq <- round(flips_seq * (h_max / n_max))
9
10 # Compute Bayes factor at each stage
11 bf_seq <- mapply(function(n_i, h_i) {
12   bayes_factor_beta_binomial(n_i, h_i, alpha = 1)
13 }, flips_seq, heads_seq)
14
15 # Posterior odds with uniform prior on H0 vs H1
16 posterior_odds <- bf_seq
17
18 par(mar = c(4.4, 4.4, 2.2, 1.2))
19 plot(flips_seq, posterior_odds, type = "l", lwd = 1,
20      xlab = "Number of flips",
21      ylab = "Posterior odds ratio (R = BF)",
22      col = col_bf, log = "y")
23 abline(h = 1, lty = 2, col = col_guide, lwd = 0.8)
24 abline(h = 10, lty = 2, col = col_guide, lwd = 0.8)
25 text(20, 0.8, "R = 1 (equal)", col = col_guide, cex = 0.8, adj = 0)
26 text(20, 12, "R = 10 (strong)", col = col_guide, cex = 0.8, adj = 0)
27
28 # Mark final observation
29 points(n_max, posterior_odds[length(posterior_odds)], pch = 16, col = col_accent, cex = 1.2)
30 text(n_max - 30, posterior_odds[length(posterior_odds)] * 0.7,
31      sprintf("n=250\nR=%.2e", posterior_odds[length(posterior_odds)]),
32      col = col_accent, cex = 0.8, adj = 1)

```

Listing 15.10: Illustration of the Occam factor penalty for model complexity.

```

1
2 par(mar = c(4.4, 4.4, 2.2, 1.2))
3
4 # Smooth likelihood function centered at theta_hat = 0.56
5 theta <- seq(0, 1, length.out = 1000)
6 theta_hat <- 0.56
7 delta_theta_eff <- 0.08
8
9 # Likelihood (normalized for display)
10 likelihood <- exp(-((theta - theta_hat)^2) / (2 * delta_theta_eff^2))
11 likelihood <- likelihood / max(likelihood)
12
13 # Prior widths
14 prior_width_simple <- 0.1 # H0 narrow
15 prior_width_complex <- 0.8 # H1 wide
16
17 # Priors
18 prior_simple <- ifelse(theta >= 0.45 & theta <= 0.55, 1 / prior_width_simple, 0)
19 prior_complex <- rep(1 / 1, length(theta)) # Uniform over [0,1]
20
21 plot(theta, likelihood, type = "l", lwd = 2, col = col_bf,
22      xlab = "Parameter theta",
23      ylab = "Density (normalized)",
24      ylim = c(0, 1.1),
25      main = "")
26
27 # Draw prior widths
28 lines(c(0.45, 0.55), c(1.0, 1.0), col = col_accent, lwd = 3)
29 points(c(0.45, 0.55), c(1.0, 1.0), pch = "|", col = col_accent, cex = 2)

```

```

30 text(0.5, 1.05, "H0 prior width", col = col_accent, cex = 0.8, adj = 0.5)
31
32 lines(c(0, 1), c(0.85, 0.85), col = col_guide, lwd = 2)
33 points(c(0, 1), c(0.85, 0.85), pch = "|", col = col_guide, cex = 2)
34 text(0.5, 0.90, "H1 prior width", col = col_guide, cex = 0.8, adj = 0.5)
35
36 # Shade the effective width
37 polygon(c(theta_hat - 2*delta_theta_eff, theta_hat + 2*delta_theta_eff,
38           theta_hat + 2*delta_theta_eff, theta_hat - 2*delta_theta_eff),
39         c(-0.02, -0.02, 0.2, 0.2), col = rgb(0.5, 0.5, 0.5, 0.2), border = NA)
40 text(theta_hat, 0.12, "Effective\nlikelihood\nwidth", cex = 0.7, adj = 0.5)
41
42 legend("topright", c("Likelihood", "H0 prior", "H1 prior"),
43       col = c(col_bf, col_accent, col_guide), lwd = c(2, 3, 2),
44       cex = 0.8, bg = "white")

```

Listing 15.11: Loss functions

```

1
2 a_values <- seq(-3, 3, length.out = 500)
3 theta_true <- 0.7
4
5 par(mfrow = c(1, 3), mar = c(4, 4, 3, 1), cex.main = 0.95)
6
7 # Squared loss
8 L_squared <- (a_values - theta_true)^2
9 plot(a_values, L_squared, type = "l", lwd = 2.5, col = "#d62728",
10      main = "Squared Loss", xlab = expression(a), ylab = expression(L(a, theta)))
11 abline(v = theta_true, lty = 2, col = "grey")
12 grid(col = "grey90")
13
14 # Absolute loss
15 L_abs <- abs(a_values - theta_true)
16 plot(a_values, L_abs, type = "l", lwd = 2.5, col = "#1f77b4",
17      main = "Absolute Loss", xlab = expression(a), ylab = expression(L(a, theta)))
18 abline(v = theta_true, lty = 2, col = "grey")
19 grid(col = "grey90")
20
21 # 0-1 loss
22 L_01 <- ifelse(abs(a_values - theta_true) < 0.2, 0, 1)
23 plot(a_values, L_01, type = "s", lwd = 2.5, col = "#2ca02c",
24      main = "0-1 Loss", xlab = expression(a), ylab = expression(L(a, theta)),
25      ylim = c(-0.1, 1.2))
26 abline(v = theta_true, lty = 2, col = "grey")
27 grid(col = "grey90")

```

Listing 15.12: Asymmetric posterior

```

1
2 p <- seq(0, 1, length.out = 1000)
3 posterior <- dbeta(p, shape1 = 4, shape2 = 8)
4
5 # Compute statistics
6 p_map <- (4 - 1) / (4 + 8 - 2)
7 p_mean <- 4 / (4 + 8)
8 p_median <- qbeta(0.5, 4, 8)
9
10 plot(p, posterior, type = "l", lwd = 2.5, col = "#d62728",
11      main = "Asymmetric Beta(4,8) Posterior",
12      xlab = expression(p), ylab = "Density")
13 abline(v = p_map, col = "#2ca02c", lwd = 2, lty = 2)
14 abline(v = p_median, col = "#1f77b4", lwd = 2, lty = 2)
15 abline(v = p_mean, col = "#9467bd", lwd = 2, lty = 2)
16 legend("topright", legend = c("Posterior", "MAP", "Median", "Mean"),

```

15. Appendix: Source code

```

17     col = c("#d62728", "#2ca02c", "#1f77b4", "#9467bd"),
18     lwd = c(2.5, 2, 2, 2), lty = c(1, 2, 2, 2), bty = "n")
19 grid(col = "grey90")

```

Listing 15.13: Laplace approximation

```

1  par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
2
3  # Good case: many counts
4  alpha_good <- 50
5  beta_good <- 10
6  lambda_grid <- seq(0, 8, length.out = 1000)
7  post_good <- dgamma(lambda_grid, shape = alpha_good, rate = beta_good)
8
9
10 map_good <- (alpha_good - 1) / beta_good
11 var_good <- alpha_good / (beta_good^2)
12 gauss_good <- dnorm(lambda_grid, mean = map_good, sd = sqrt(var_good))
13
14 plot(lambda_grid, post_good, type = "l", lwd = 2.5, col = "#d62728",
15      main = "Well-peaked Gamma(50,10)", xlab = expression(lambda), ylab = "Density")
16 lines(lambda_grid, gauss_good, lwd = 2, col = "black", lty = 2)
17 legend("topright", legend = c("True posterior", "Laplace approx."),
18      col = c("#d62728", "black"), lwd = c(2.5, 2), lty = c(1, 2), bty = "n")
19 grid(col = "grey90")
20
21 # Bad case: no counts
22 alpha_bad <- 1
23 beta_bad <- 5
24 lambda_grid_bad <- seq(0, 2, length.out = 1000)
25 post_bad <- dgamma(lambda_grid_bad, shape = alpha_bad, rate = beta_bad)
26
27 map_bad <- (alpha_bad - 1) / beta_bad
28 var_bad <- alpha_bad / (beta_bad^2)
29 gauss_bad <- dnorm(lambda_grid_bad, mean = map_bad, sd = sqrt(var_bad))
30
31 plot(lambda_grid_bad, post_bad, type = "l", lwd = 2.5, col = "#d62728",
32      main = "Broad Gamma(1,5) [zero counts]", xlab = expression(lambda), ylab = "Density")
33 lines(lambda_grid_bad, gauss_bad, lwd = 2, col = "black", lty = 2)
34 legend("topright", legend = c("True posterior", "Laplace approx."),
35      col = c("#d62728", "black"), lwd = c(2.5, 2), lty = c(1, 2), bty = "n")
36 grid(col = "grey90")

```

Listing 15.14: HpD intervals

```

1  x <- seq(0, 1, length.out = 2000)
2  dens <- 0.5 * dbeta(x, 2, 8) + 0.5 * dbeta(x, 8, 2)
3
4
5  gamma <- 0.80
6  dx <- x[2] - x[1]
7  dens_sorted <- sort(dens, decreasing = TRUE)
8  cum_prob <- cumsum(dens_sorted) * dx
9  k_gamma <- dens_sorted[which(cum_prob >= gamma)[1]]
10
11 # Central interval (for comparison)
12 cum_dens <- cumsum(dens) * dx
13 cum_dens_norm <- cum_dens / max(cum_dens)
14 idx_central <- which(cum_dens_norm <= (1 + gamma)/2 & cum_dens_norm >= (1 - gamma)/2)
15
16 plot(x, dens, type = "l", lwd = 2.5, col = "#1f77b4",
17      main = paste0("HPD vs Central Interval (gamma = ", gamma, ")"),
18      xlab = expression(theta), ylab = "Posterior density")
19

```

```

20 # Shade HPD
21 hpd_idx <- dens >= k_gamma
22 polygon(c(x[hpd_idx], rev(x[hpd_idx])), c(dens[hpd_idx], rep(0, sum(hpd_idx))),
23         col = rgb(1, 0, 0, 0.3), border = NA)
24
25 # Threshold line
26 abline(h = k_gamma, col = "red", lwd = 1.5, lty = 2)
27
28 legend("top", legend = c("Posterior density", "HPD region", expression(k[gamma])),
29       col = c("#1f77b4", rgb(1,0,0,0.3), "red"),
30       lwd = c(2.5, NA, 1.5), lty = c(1, NA, 2), pch = c(NA, 15, NA), bty = "n")
31 grid(col = "grey90")

```

Listing 15.15: Beta credible intervals

```

1
2 p <- seq(0, 1, length.out = 1000)
3 alpha_post <- 6
4 beta_post <- 16
5 posterior <- dbeta(p, alpha_post, beta_post)
6
7 gamma <- 0.95
8
9 # Equal-tail interval
10 p_lower_central <- qbeta((1 - gamma) / 2, alpha_post, beta_post)
11 p_upper_central <- qbeta((1 + gamma) / 2, alpha_post, beta_post)
12
13 # HPD: find threshold
14 dens_sorted <- sort(posterior, decreasing = TRUE)
15 dx <- p[2] - p[1]
16 cum_prob <- cumsum(dens_sorted) * dx
17 k_gamma <- dens_sorted[which(cum_prob >= gamma)[1]]
18 hpd_idx <- posterior >= k_gamma
19
20 # Plot
21 plot(p, posterior, type = "l", lwd = 2.5, col = "#1f77b4",
22      main = "Beta(6,16): Equal-tail vs HPD (95%)",
23      xlab = expression(p), ylab = "Density")
24
25 # Shade equal-tail
26 idx_et <- (p >= p_lower_central) & (p <= p_upper_central)
27 polygon(c(p[idx_et], rev(p[idx_et])), c(posterior[idx_et], rep(0, sum(idx_et))),
28        col = rgb(0, 0, 1, 0.2), border = NA)
29
30 # Shade HPD
31 polygon(c(p[hpd_idx], rev(p[hpd_idx])), c(posterior[hpd_idx], rep(0, sum(hpd_idx))),
32        col = rgb(1, 0, 0, 0.25), border = NA)
33
34 # Mean and mode
35 p_mean <- alpha_post / (alpha_post + beta_post)
36 p_mode <- (alpha_post - 1) / (alpha_post + beta_post - 2)
37 abline(v = p_mean, col = "#9467bd", lwd = 1.5, lty = 3)
38 abline(v = p_mode, col = "#2ca02c", lwd = 1.5, lty = 3)
39
40 legend("topright",
41       legend = c("Posterior", "Equal-tail [95%]", "HPD [95%]", "Mean", "Mode"),
42       col = c("#1f77b4", rgb(0,0,1,0.2), rgb(1,0,0,0.25), "#9467bd", "#2ca02c"),
43       lwd = c(2.5, NA, NA, 1.5, 1.5), lty = c(1, NA, NA, 3, 3),
44       pch = c(NA, 15, 15, NA, NA), bty = "n")
45
46 cat("Equal-tail:  [", sprintf("%.4f", p_lower_central), ",", sprintf("%.4f",
47   p_upper_central), "]\n", sep = "")

```

Listing 15.16: Beta credible intervals

```

1
2 cat("HPD:          [", sprintf("%.4f", min(p[hpd_idx])), ", ", sprintf("%.4f",
   max(p[hpd_idx])), "]\n", sep = "")

```

Listing 15.17: Beta credible intervals

```

1
2 grid(col = "grey90")

```

Listing 15.18: Sequential updating

```

1
2 set.seed(42)
3 p_true <- 0.7
4 n_obs_per_step <- 3
5 n_steps <- 4
6
7 par(mfrow = c(n_steps, 1), mar = c(3, 3, 2, 1), cex.main = 0.85)
8
9 p_grid <- seq(0, 1, length.out = 1000)
10
11 # Initial prior
12 alpha <- 1
13 beta <- 1
14 prior_dens <- dbeta(p_grid, alpha, beta)
15
16 plot(p_grid, prior_dens, type = "l", lwd = 2, col = "#1f77b4",
17      main = "Prior: Beta(1,1)", xlab = expression(p), ylab = "Density",
18      ylim = c(0, 4))
19 abline(v = p_true, col = "#d62728", lwd = 1, lty = 2, alpha = 0.7)

```

Listing 15.19: Sequential updating

```

1
2 grid(col = "grey90")
3
4 # Sequential updates
5 for (step in 1:n_steps) {
6   # Generate new observation
7   obs <- rbinom(n_obs_per_step, 1, p_true)
8   k <- sum(obs)
9   n <- n_obs_per_step
10
11   # Update posterior (which becomes new prior)
12   alpha <- alpha + k
13   beta <- beta + n - k
14   posterior_dens <- dbeta(p_grid, alpha, beta)
15
16   # Plot
17   plot(p_grid, posterior_dens, type = "l", lwd = 2, col = "#1f77b4",
18        main = paste0("After ", step * n_obs_per_step, " observations"),
19        xlab = expression(p), ylab = "Density", ylim = c(0, 4))
20   abline(v = p_true, col = "#d62728", lwd = 1, lty = 2, alpha = 0.7)
21   grid(col = "grey90")
22 }

```

Listing 15.20: Prior-to-posterior updating with increasing sample size

```

1
2 set.seed(42)
3
4 mu_true <- 0.5

```

```

5 sigma_true <- 1.0
6
7 mu0 <- 0
8 sigma0 <- 2
9 n_list <- c(2, 20, 100)
10
11 mu_grid <- seq(-3, 3, length.out = 2000)
12
13 par(mar = c(4.4, 4.4, 2.2, 1.2))
14 plot(NA, xlim = c(-3, 3), ylim = c(0, 2.5), xlab = expression(mu), ylab = "Density", main =
    "")
15
16 for (i in seq_along(n_list)) {
17   n <- n_list[i]
18   x <- rnorm(n, mu_true, sigma_true)
19   xbar <- mean(x)
20
21   kappa_n <- 1/sigma0^2 + n/sigma_true^2
22   mu_n <- (mu0/sigma0^2 + n*xbar/sigma_true^2) / kappa_n
23   sigma_n <- sqrt(1/kappa_n)
24
25   posterior_density <- dnorm(mu_grid, mu_n, sigma_n)
26   col_vec <- c("black", "gray40", "#b8123e")
27   lines(mu_grid, posterior_density, lwd = 1, lty = i, col = col_vec[i])
28 }
29
30 prior_density <- dnorm(mu_grid, mu0, sigma0)
31 lines(mu_grid, prior_density, lwd = 1, lty = 4, col = "gray70")
32
33 legend("topright", legend = c("Prior", "n=2", "n=20", "n=100"),
34       lty = c(4, 1, 2, 3), col = c("gray70", "black", "gray40", "#b8123e"),
35       lwd = 1, bty = "n", cex = 0.75)

```

Listing 15.21: Asymptotic Gaussian posterior for joint mean and variance

```

1
2 set.seed(123)
3
4 n <- 50
5 mu_true <- 0
6 sigma_true <- 1
7
8 x <- rnorm(n, mu_true, sigma_true)
9 xbar <- mean(x)
10 S <- sum((x - xbar)^2)
11
12 mu_hat <- xbar
13 sigma2_hat <- S / n
14
15 theta_hat <- c(mu_hat, sigma2_hat)
16
17 Sigma <- matrix(c(sigma2_hat / n, 0, 0, 2 * sigma2_hat^2 / n), 2, 2)
18
19 alpha_level <- 0.05
20 chi2_val <- qchisq(1 - alpha_level, df = 2)
21
22 eig <- eigen(Sigma)
23 A <- eig$vectors %*% diag(sqrt(eig$values * chi2_val))
24
25 angles <- seq(0, 2*pi, length.out = 200)
26 circle <- rbind(cos(angles), sin(angles))
27 ellipse_mat <- A %*% circle
28 ellipse_df <- data.frame(
29   mu = ellipse_mat[1,] + theta_hat[1],

```

15. Appendix: Source code

```

30   sigma2 = ellipse_mat[2,] + theta_hat[2]
31 )
32
33 par(mar = c(4.4, 4.4, 2.2, 1.2))
34 plot(ellipse_df$mu, ellipse_df$sigma2, type = "l", lwd = 1,
35      xlab = expression(mu), ylab = expression(sigma^2),
36      main = paste("n =", n),
37      xlim = c(theta_hat[1] - 0.4, theta_hat[1] + 0.4),
38      ylim = c(max(0, theta_hat[2] - 0.5), theta_hat[2] + 0.5))
39
40 points(theta_hat[1], theta_hat[2], pch = 19, col = "#b8123e", cex = 1)
41
42 legend("topright", legend = c("95% credible ellipse", "MLE"),
43      lty = c(1, NA), pch = c(NA, 19), col = c("black", "#b8123e"),
44      lwd = 1, bty = "n", cex = 0.75)

```

Listing 15.22: Bayes factor for one-sided Gaussian test

```

1  mu_star <- 0
2  alpha0 <- 1.1
3  beta0 <- 0.5
4  kappa0 <- 0.01
5  mu0 <- 0
6
7
8  n <- 30
9  sigma_true <- 1
10
11 xbar_grid <- seq(-1, 2, length.out = 100)
12
13 log10_bf <- numeric(length(xbar_grid))
14
15 for (j in seq_along(xbar_grid)) {
16   xbar <- xbar_grid[j]
17
18   kappa_n <- kappa0 + n
19   mu_n <- (kappa0*mu0 + n*xbar) / kappa_n
20   alpha_n <- alpha0 + n/2
21
22   S_approx <- 0
23   beta_n <- beta0 + S_approx/2 + (kappa0*n)/(2*(kappa0+n)) * (xbar - mu0)^2
24
25   log_bf <- 0.5*log(kappa0) - 0.5*log(kappa_n) +
26     alpha0*log(beta0) - alpha_n*log(beta_n) +
27     lgamma(alpha_n) - lgamma(alpha0) -
28     0.5*log(pi) + 0.5*log(2*alpha_n) +
29     lgamma(alpha_n - 0.5) - lgamma(alpha0 - 0.5)
30
31   log10_bf[j] <- log_bf / log(10)
32 }
33
34 par(mar = c(4.4, 4.4, 2.2, 1.2))
35 plot(xbar_grid, log10_bf, type = "l", lwd = 1,
36      xlab = expression(bar(x)), ylab = expression(log[10]*" BF"["1*", "*0"]),
37      main = paste("Test: mu > 0, n =", n), col = "black")
38 abline(h = 0, lty = 2, col = "gray70", lwd = 1)
39 abline(v = mu_star, lty = 2, col = "gray40", lwd = 1)
40
41 legend("bottomright", legend = c("BF > 1: favor H_1", "BF = 1: indifferent", "BF < 1: favor
42   H_0"),
43      bty = "n", cex = 0.75)

```

Listing 15.23: Monte Carlo error vs. dimension

```

1
2 # True integral: volume of ball of radius 0.5 in  $[0,1]^d$ 
3 true_val <- function(d) {
4   pi^(d/2) / gamma(d/2 + 1) * 2^(-2*d)
5 }
6
7 # Naive MC
8 dims <- c(2, 5, 10, 15, 20)
9 N <- 1e6
10 results <- data.frame(d = dims, est = NA, rel_err = NA)
11
12 for (i in seq_along(dims)) {
13   d <- dims[i]
14   x <- matrix(runif(N * d), ncol = d)
15   inside <- rowSums(x^2) < 0.25
16   est <- mean(inside)
17   true <- true_val(d)
18   rel_err <- if (true > 0) abs(est - true) / true else NA
19   results$est[i] <- est
20   results$rel_err[i] <- rel_err
21 }
22
23 # Plot
24 par(mfrow = c(1, 2), mar = c(5, 5, 3, 1))
25
26 plot(results$d, results$est, type = "o", pch = 16, cex = 1.2, lwd = 2,
27       col = rgb(0.2, 0.4, 0.8), log = "y",
28       xlab = "Dimension d", ylab = "Estimated I_d",
29       main = "Hypersphere volume (log scale)", ylim = c(1e-15, 1))

```

Listing 15.24: Monte Carlo error vs. dimension

```

1
2 lines(results$d, true_val(results$d), col = "red", lwd = 2)
3 legend("topright", legend = c("MC estimate", "True value"),
4       col = c(rgb(0.2, 0.4, 0.8), "red"), lwd = 2, pch = c(16, NA))
5
6 plot(results$d, results$rel_err, type = "o", pch = 16, cex = 1.2, lwd = 2,
7       col = rgb(0.8, 0.2, 0.2), log = "y",
8       xlab = "Dimension d", ylab = "Relative error",
9       main = "Relative error (log scale)")

```

Listing 15.25: Inverse transform sampling for exponential distribution

```

1
2 N <- 10000
3 u <- runif(N)
4 lambda <- 2
5 x <- -log(u) / lambda
6 x_builtin <- rexp(N, rate = lambda)
7
8 hist(x, probability = TRUE, col = rgb(1, 0, 0, 0.4),
9     breaks = 50, main = "Inverse transform vs. rexp",
10    xlab = "x", xlim = c(0, 3))
11 hist(x_builtin, probability = TRUE, col = rgb(0, 0, 1, 0.4),
12     breaks = 50, add = TRUE)
13 curve(dexp(x, rate = lambda), col = "darkred", lwd = 2, add = TRUE)
14 legend("topright", legend = c("Inverse transform", "rexp"),
15     fill = c(rgb(1, 0, 0, 0.4), rgb(0, 0, 1, 0.4)))

```

Listing 15.26: Metropolis–Hastings on a bimodal target

```

1
2 # Target: mixture of two Gaussians

```

15. Appendix: Source code

```

3 target_density <- function(x) {
4   0.5 * dnorm(x, mean = -3, sd = 1) + 0.5 * dnorm(x, mean = 3, sd = 1)
5 }
6
7 log_target <- function(x) log(target_density(x))
8
9 # MH sampler
10 mh_sample <- function(n_steps, x0, proposal_sd) {
11   x <- numeric(n_steps)
12   x[1] <- x0
13   n_accept <- 0
14   for (t in 2:n_steps) {
15     x_current <- x[t - 1]
16     x_prop <- rnorm(1, mean = x_current, sd = proposal_sd)
17     log_r <- log_target(x_prop) - log_target(x_current)
18     if (log(runif(1)) < log_r) {
19       x[t] <- x_prop
20       n_accept <- n_accept + 1
21     } else {
22       x[t] <- x_current
23     }
24   }
25   list(samples = x, accept_rate = n_accept / (n_steps - 1))
26 }
27
28 # Run sampler
29 n_steps <- 50000
30 burn_in <- 5000
31 proposal_sd <- 2.0
32 mh <- mh_sample(n_steps, x0 = -8, proposal_sd = proposal_sd)
33 samples <- mh$samples
34 samples_burn <- samples[(burn_in + 1):n_steps]
35
36 cat("Acceptance rate:", mh$accept_rate, "\n")

```

Listing 15.27: Metropolis–Hastings on a bimodal target

```

1
2 # Compute autocorrelation
3 lag_max <- 500
4 acf_vals <- acf(samples_burn, lag.max = lag_max, plot = FALSE)
5 lags <- as.numeric(acf_vals$lag)
6 rho <- as.numeric(acf_vals$acf)
7
8 # Integrated autocorrelation time
9 t_int <- 1 + 2 * sum(rho[2:min(which(abs(rho[-1]) < 0.05)[1], length(rho))])
10 N_eff <- length(samples_burn) / (2 * t_int)
11
12 cat("Integrated autocorr time:", round(t_int, 2), "\n")

```

Listing 15.28: Metropolis–Hastings on a bimodal target

```

1
2 cat("N_eff:", round(N_eff, 0), "\n")

```

Listing 15.29: Metropolis–Hastings on a bimodal target

```

1
2 # Plots
3 par(mfrow = c(3, 1), mar = c(4, 4, 2, 1))
4
5 # Trace before and after burn-in
6 plot(1:burn_in, samples[1:burn_in], type = "l", xlab = "Iteration",
7      ylab = "x", main = "Before burn-in", lwd = 0.5, col = rgb(1, 0, 0, 0.6))

```

```

8
9 plot((burn_in + 1):n_steps, samples_burn, type = "l",
10      xlab = "Iteration", ylab = "x", main = "After burn-in",
11      lwd = 0.5, col = rgb(0, 0, 1, 0.6))
12
13 # Histogram + target
14 hist(samples_burn, breaks = 80, probability = TRUE,
15      main = "Posterior samples", xlab = "x",
16      col = rgb(0.3, 0.3, 0.3, 0.4), border = NA)
17 curve(target_density(x), from = -8, to = 8, lwd = 2, col = "red", add = TRUE)

```

Listing 15.30: Metropolis–Hastings on a bimodal target

```

1
2 par(mfrow = c(1, 1))

```

Listing 15.31: Gibbs bivariate high corr

```

1
2 set.seed(42)
3
4 gibbs_bivariate <- function(n_steps, rho, start = c(0, 0)) {
5   x <- numeric(n_steps)
6   y <- numeric(n_steps)
7   x[1] <- start[1]
8   y[1] <- start[2]
9   cond_sd <- sqrt(1 - rho^2)
10
11   for (t in 2:n_steps) {
12     # Update X / Y
13     x[t] <- rnorm(1, mean = rho * y[t - 1], sd = cond_sd)
14     # Update Y / X
15     y[t] <- rnorm(1, mean = rho * x[t], sd = cond_sd)
16   }
17
18   cbind(x = x, y = y)
19 }
20
21 rho_high <- 0.9
22 chain_high <- gibbs_bivariate(n_steps = 10000, rho = rho_high, start = c(-3, 3))
23
24 # Diagnostics
25 burn_in <- 1000
26 samples_high <- chain_high[(burn_in + 1):nrow(chain_high), ]
27
28 par(mfrow = c(3, 1), mar = c(4, 4, 3, 1))
29
30 # Trace plots
31 plot(chain_high[, "x"], type = "l", lwd = 0.8, col = "#d62728",
32      main = paste0("Gibbs trace (X), rho = ", rho_high),
33      xlab = "Iteration", ylab = "X")
34 abline(v = burn_in, col = "grey", lty = 2)
35
36 plot(chain_high[, "y"], type = "l", lwd = 0.8, col = "#1f77b4",
37      main = "Gibbs trace (Y)", xlab = "Iteration", ylab = "Y")
38 abline(v = burn_in, col = "grey", lty = 2)
39
40 # Scatter of posterior samples
41 plot(samples_high[, "x"], samples_high[, "y"], pch = 16, cex = 0.3,
42      col = rgb(0, 0, 0, 0.5), xlab = "X", ylab = "Y",
43      main = "Posterior samples after burn-in")

```

Listing 15.32: Gibbs rotated

15. Appendix: Source code

```

1
2 gibbs_rotated <- function(n_steps, rho, start = c(0, 0)) {
3   # Transform starting point to rotated coordinates
4   u_start <- (start[1] + start[2]) / sqrt(2)
5   v_start <- (start[1] - start[2]) / sqrt(2)
6
7   u <- numeric(n_steps)
8   v <- numeric(n_steps)
9   u[1] <- u_start
10  v[1] <- v_start
11
12  sd_u <- sqrt(1 + rho)
13  sd_v <- sqrt(1 - rho)
14
15  for (t in 2:n_steps) {
16    # u and v are now independent, so Gibbs reduces to independent sampling
17    u[t] <- rnorm(1, mean = 0, sd = sd_u)
18    v[t] <- rnorm(1, mean = 0, sd = sd_v)
19  }
20
21  # Transform back to original coordinates
22  x <- (u + v) / sqrt(2)
23  y <- (u - v) / sqrt(2)
24
25  cbind(x = x, y = y)
26 }
27
28 chain_rotated <- gibbs_rotated(n_steps = 10000, rho = rho_high, start = c(-3, 3))
29 samples_rotated <- chain_rotated[(burn_in + 1):nrow(chain_rotated), ]
30
31 par(mfrow = c(3, 1), mar = c(4, 4, 3, 1))
32
33 # Trace plots for rotated coordinates
34 plot(chain_rotated[, "x"], type = "l", lwd = 0.8, col = "#2ca02c",
35       main = "Rotated Gibbs trace (X after rotation)",
36       xlab = "Iteration", ylab = "X")
37 abline(v = burn_in, col = "grey", lty = 2)
38
39 plot(chain_rotated[, "y"], type = "l", lwd = 0.8, col = "#ff7f0e",
40       main = "Rotated Gibbs trace (Y after rotation)",
41       xlab = "Iteration", ylab = "Y")
42 abline(v = burn_in, col = "grey", lty = 2)
43
44 # Scatter plot
45 plot(samples_rotated[, "x"], samples_rotated[, "y"], pch = 16, cex = 0.3,
46       col = rgb(0, 0, 0, 0.5), xlab = "X", ylab = "Y",
47       main = "Posterior samples (rotated Gibbs)")

```

Listing 15.33: Gibbs rotated

```

1
2 par(mfrow = c(1, 1))

```

Listing 15.34: Gibbs acf comparison

```

1
2 par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
3
4 acf_high <- acf(samples_high[, "x"], lag.max = 50, plot = FALSE)
5 acf_rotated <- acf(samples_rotated[, "x"], lag.max = 50, plot = FALSE)
6
7 plot(acf_high$lag, acf_high$acf, type = "h", lwd = 2, col = "#d62728",
8       main = "Standard Gibbs (rho = 0.9)",
9       xlab = "Lag", ylab = "ACF", ylim = c(-0.1, 1))

```

```

10 abline(h = 0, col = "black", lwd = 0.5)
11
12 plot(acf_rotated$lag, acf_rotated$acf, type = "h", lwd = 2, col = "#2ca02c",
13      main = "Rotated Gibbs (rho = 0.9)",
14      xlab = "Lag", ylab = "ACF", ylim = c(-0.1, 1))
15 abline(h = 0, col = "black", lwd = 0.5)

```

Listing 15.35: Gibbs acf comparison

```

1
2 par(mfrow = c(1, 1))

```

Listing 15.36: Leapfrog trajectory

```

1
2 # Define target (log-posterior and gradient)
3 rho <- 0.95
4 Sigma_inv <- solve(matrix(c(1, rho, rho, 1), 2, 2))
5 log_pi <- function(q) {
6   -0.5 * as.numeric(t(q) %*% Sigma_inv %*% q)
7 }
8 grad_U <- function(q) {
9   Sigma_inv %*% q
10 }
11
12 # Leapfrog integrator
13 leapfrog <- function(q, p, epsilon, L) {
14   p <- p - 0.5 * epsilon * grad_U(q)
15   for (i in 1:L) {
16     q <- q + epsilon * p
17     if (i < L) {
18       p <- p - epsilon * grad_U(q)
19     }
20   }
21   p <- p - 0.5 * epsilon * grad_U(q)
22   list(q = q, p = -p)
23 }
24
25 # Simulate trajectory and track states
26 q0 <- c(2, 2)
27 p0 <- c(1, 0)
28 epsilon <- 0.1
29 L <- 100
30
31 q_curr <- q0
32 p_curr <- p0
33 trajectory_q <- matrix(NA, L+1, 2)
34 trajectory_p <- matrix(NA, L+1, 2)
35 trajectory_q[1,] <- q0
36 trajectory_p[1,] <- p0
37
38 for (step in 1:L) {
39   # Half-step momentum
40   p_curr <- p_curr - 0.5 * epsilon * grad_U(q_curr)
41   # Full-step position
42   q_curr <- q_curr + epsilon * p_curr
43   # Half-step momentum
44   p_curr <- p_curr - 0.5 * epsilon * grad_U(q_curr)
45   trajectory_q[step+1,] <- q_curr
46   trajectory_p[step+1,] <- p_curr
47 }
48
49 # Plot phase space trajectory
50 plot(trajectory_q[,1], trajectory_p[,1],

```

15. Appendix: Source code

```

51     type = "l", lwd = 1.5, col = "#d62728",
52     main = "Leapfrog Trajectory in Phase Space (q1, p1)",
53     xlab = "q1 (position)", ylab = "p1 (momentum)"
54 points(q0[1], p0[1], pch = 19, cex = 1.5, col = "#2ca02c")
55 points(trajectory_q[L+1, 1], trajectory_p[L+1, 1], pch = 19, cex = 1.5, col = "#1f77b4")
56 legend("topright", legend = c("Trajectory", "Start", "End"),
57       col = c("#d62728", "#2ca02c", "#1f77b4"), pch = c(NA, 19, 19),
58       lwd = c(1.5, NA, NA), bty = "n")
59 grid(col = "grey90")

```

Listing 15.37: Hmc vs rwmh

```

1
2 # Setup
3 rho <- 0.95
4 Sigma <- matrix(c(1, rho, rho, 1), 2, 2)
5 Sigma_inv <- solve(Sigma)
6
7 log_pi <- function(q) {
8   -0.5 * as.numeric(t(q) %*% Sigma_inv %*% q)
9 }
10
11 grad_U <- function(q) {
12   Sigma_inv %*% q
13 }
14
15 # RW-MH
16 rw_mh <- function(n_iter, step = 0.3) {
17   q <- c(0, 0)
18   samples <- matrix(NA, n_iter, 2)
19   acc <- 0
20   for (i in 1:n_iter) {
21     q_prop <- q + rnorm(2, 0, step)
22     log_alpha <- log_pi(q_prop) - log_pi(q)
23     if (log(runif(1)) < log_alpha) {
24       q <- q_prop
25       acc <- acc + 1
26     }
27     samples[i, ] <- q
28   }
29   list(samples = samples, acc_rate = acc / n_iter)
30 }
31
32 # Leapfrog for HMC
33 leapfrog <- function(q, p, eps, L) {
34   p <- p - 0.5 * eps * grad_U(q)
35   for (i in 1:L) {
36     q <- q + eps * p
37     if (i < L) p <- p - eps * grad_U(q)
38   }
39   p <- p - 0.5 * eps * grad_U(q)
40   list(q = q, p = -p)
41 }
42
43 # HMC
44 hmc <- function(n_iter, eps = 0.15, L = 20) {
45   q <- c(0, 0)
46   samples <- matrix(NA, n_iter, 2)
47   acc <- 0
48   for (i in 1:n_iter) {
49     p <- rnorm(2)
50     H_old <- 0.5 * as.numeric(t(q) %*% Sigma_inv %*% q) + 0.5 * sum(p^2)
51     prop <- leapfrog(q, p, eps, L)
52     q_prop <- prop$q

```

```

53   p_prop <- prop$p
54   H_new <- 0.5 * as.numeric(t(q_prop) %*% Sigma_inv %*% q_prop) + 0.5 * sum(p_prop^2)
55   log_alpha <- -(H_new - H_old)
56   if (log(runif(1)) < log_alpha) {
57     q <- q_prop
58     acc <- acc + 1
59   }
60   samples[i, ] <- q
61 }
62 list(samples = samples, acc_rate = acc / n_iter)
63 }
64
65 # Run samplers
66 set.seed(42)
67 n_iter <- 2000
68 fit_mh <- rw_mh(n_iter, step = 0.3)
69 fit_hmc <- hmc(n_iter, eps = 0.15, L = 20)
70
71 # Plot traces
72 par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))
73
74 plot(fit_mh$samples[,1], type = "l", lwd = 1, col = "#1f77b4",
75      main = paste0("RW-MH Trace (acc=", round(fit_mh$acc_rate, 3), ")"),
76      xlab = "Iteration", ylab = "q1")
77 grid(col = "grey90")
78
79 plot(fit_hmc$samples[,1], type = "l", lwd = 1, col = "#d62728",
80      main = paste0("HMC Trace (acc=", round(fit_hmc$acc_rate, 3), ")"),
81      xlab = "Iteration", ylab = "q1")
82 grid(col = "grey90")
83
84 # Plot ACF
85 acf(fit_mh$samples[,1], main = "RW-MH Autocorr", lag.max = 50, col = "#1f77b4")
86 acf(fit_hmc$samples[,1], main = "HMC Autocorr", lag.max = 50, col = "#d62728")

```

Listing 15.38: Metropolis update at tempered temperature.

```

1  # Log-posterior for bimodal mixture (minus normalization)
2  log_pi <- function(x) {
3    a <- log(0.35) + dnorm(x, -4, 1, log = TRUE)
4    b <- log(0.65) + dnorm(x, 3, 0.7, log = TRUE)
5    m <- pmax(a, b)
6    m + log(exp(a - m) + exp(b - m))
7  }
8
9
10 # Target density (for plotting)
11 pi_density <- function(x) {
12   0.35 * dnorm(x, -4, 1) + 0.65 * dnorm(x, 3, 0.7)
13 }
14
15 # Single MH step at temperature beta
16 tempered_step <- function(x, beta, step = 1.0) {
17   x_prop <- x + rnorm(1, 0, step)
18   loga <- beta * (log_pi(x_prop) - log_pi(x))
19   if (log(runif(1)) < loga) x_prop else x
20 }

```

Listing 15.39: Single-chain MH sampler on bimodal target.

```

1  single_chain_mh <- function(n_iter = 20000, x0 = -4, step = 1.0) {
2    x <- x0
3    out <- numeric(n_iter)
4

```

15. Appendix: Source code

```

5   for (i in 1:n_iter) {
6     x <- tempered_step(x, beta = 1.0, step = step)
7     out[i] <- x
8   }
9   out
10 }
11
12 set.seed(1)
13 trace_single <- single_chain_mh(n_iter = 20000, x0 = -4, step = 1.2)

```

Listing 15.40: Parallel tempering algorithm with swaps.

```

1
2 parallel_tempering <- function(n_iter = 20000,
3                               betas = c(1.0, 0.65, 0.4, 0.2, 0.08),
4                               step = 1.2) {
5   L <- length(betas)
6   x <- seq(-4, 4, length.out = L) # Initialize chains at different locations
7
8   x_by_temp <- matrix(NA, nrow = n_iter, ncol = L)
9
10  for (i in 1:n_iter) {
11    # Within-temperature MH updates
12    for (l in 1:L) {
13      x[l] <- tempered_step(x[l], betas[l], step = step)
14    }
15
16    # Swap moves between neighbouring chains
17    for (l in sample(1:(L - 1))) {
18      loga <- (betas[l] - betas[l + 1]) * (log_pi(x[l + 1]) - log_pi(x[l]))
19      if (log(runif(1)) < loga) {
20        tmp <- x[l]
21        x[l] <- x[l + 1]
22        x[l + 1] <- tmp
23      }
24    }
25
26    # Store state of cold chain (beta = 1)
27    x_by_temp[i, ] <- x
28  }
29
30  list(x_by_temp = x_by_temp, betas = betas)
31 }
32
33 set.seed(1)
34 res_pt <- parallel_tempering(n_iter = 20000,
35                             betas = c(1.0, 0.65, 0.4, 0.2, 0.08),
36                             step = 1.2)

```

Listing 15.41: Fig single chain trace

```

1
2 par(mar = c(4.4, 4.4, 2.2, 1.2))
3 plot(trace_single[1:5000], type = "l", lwd = 1.2,
4      xlab = "Iteration", ylab = expression(theta),
5      main = "Single-chain MH trace (first 5000 iterations)",
6      col = "black", ylim = c(-6, 4))
7 grid()

```

Listing 15.42: Fig parallel tempering trace

```

1
2 par(mar = c(4.4, 4.4, 2.2, 1.2))
3 cold_trace <- res_pt$x_by_temp[1:5000, 1]

```

```

4 plot(cold_trace, type = "l", lwd = 1.2,
5      xlab = "Iteration", ylab = expression(theta),
6      main = "Parallel tempering: cold chain trace (first 5000 iterations)",
7      col = "black", ylim = c(-6, 4))
8 grid()

```

Listing 15.43: Fig posterior comparison

```

1
2 par(mar = c(4.4, 4.4, 2.2, 1.2))
3
4 # Burn-in
5 burn <- 5000
6 x_single_burn <- trace_single[-(1:burn)]
7 x_pt_burn <- res_pt$x_by_temp[-(1:burn), 1]
8
9 # Plot true density
10 xx <- seq(-7, 5, length.out = 200)
11 plot(xx, pi_density(xx), type = "l", lwd = 2, col = "black",
12      xlab = expression(theta), ylab = "Density",
13      main = "Posterior: true density vs. samples",
14      ylim = c(0, max(pi_density(xx)) * 1.1))
15
16 # Histogram of single-chain samples
17 hist(x_single_burn, breaks = 60, freq = FALSE, col = rgb(0.8, 0.8, 0.8, 0.5),
18      border = "gray40", add = TRUE)
19
20 # Histogram of PT samples
21 hist(x_pt_burn, breaks = 60, freq = FALSE, col = rgb(0.7, 0.7, 0.7, 0.3),
22      border = "black", add = TRUE)
23
24 legend("topright", bty = "n", cex = 0.75,
25      legend = c("True density", "Single-chain MH samples", "PT cold chain samples"),
26      lty = c(1, NA, NA), col = c("black", NA, NA),
27      fill = c(NA, rgb(0.8, 0.8, 0.8, 0.5), rgb(0.7, 0.7, 0.7, 0.3)),
28      border = c(NA, "gray40", "black"))

```

Listing 15.44: Fig temperature ladder

```

1
2 par(mar = c(4.4, 4.4, 2.2, 1.2))
3
4 betas_plot <- c(1.0, 0.65, 0.4, 0.15)
5 xx <- seq(-7, 5, length.out = 200)
6 colors_seq <- c("black", "gray40", "gray60", "gray70")
7 ltys <- c(1, 2, 3, 4)
8
9 plot(xx, pi_density(xx)^betas_plot[1], type = "l", lwd = 1.2, col = colors_seq[1],
10      lty = ltys[1], xlab = expression(theta), ylab = expression(pi(theta)^beta),
11      main = "Effect of inverse temperature on posterior shape",
12      ylim = c(0, max(pi_density(xx)^betas_plot[1]) * 1.05))
13
14 for (i in 2:length(betas_plot)) {
15   lines(xx, pi_density(xx)^betas_plot[i], lwd = 1.2, col = colors_seq[i], lty = ltys[i])
16 }
17
18 legend("topright", bty = "n", cex = 0.75,
19      legend = paste0(expression(beta), " = ", betas_plot),
20      col = colors_seq, lty = ltys, lwd = 1.2)

```

Listing 15.45: Simulate and plot linear regression data.

```

1
2 set.seed(42)

```

15. Appendix: Source code

```
3 N <- 8
4 x <- seq(0, 7, length.out = N)
5 sigma_true <- 1.0
6 y <- 0 + 1 * x + rnorm(N, 0, sigma_true)
7
8 plot(x, y, pch = 19, col = "black", xlab = "x", ylab = "y",
9      xlim = c(-0.5, 7.5), ylim = c(-2, 9),
10     main = "Linear Regression Data")
```

Listing 15.46: Log-posterior for linear regression.

```
1
2 # Log-likelihood for  $y_i \sim N(b_0 + b_1 * x_i, \sigma^2)$ 
3 log_likelihood <- function(b0, b1, sigma, y, x) {
4   mu <- b0 + b1 * x
5   sum(dnorm(y, mu, sigma, log = TRUE))
6 }
7
8 # Log-posterior (unnormalized)
9 log_posterior <- function(theta, data) {
10   b0 <- theta[1]
11   alpha <- theta[2]
12   eta <- theta[3]
13
14   b1 <- tan(alpha)
15   sigma <- exp(eta)
16
17   # Priors
18   log_prior_b0 <- dnorm(b0, 0, 10, log = TRUE)
19   log_prior_alpha <- 0 # Flat in alpha
20   log_prior_eta <- 0 # Flat in eta (Jeffreys for sigma)
21
22   # Likelihood
23   log_like <- log_likelihood(b0, b1, sigma, data$y, data$x)
24
25   log_like + log_prior_b0 + log_prior_alpha + log_prior_eta
26 }
```

Listing 15.47: Trace plots and autocorrelation diagnostics for linear fit.

```
1
2 # This is pseudocode; actual implementation depends on your MCMC sampler.
3 # Assume 'samples' is a matrix with columns: b0, alpha, eta
4
5 # Trace plot
6 par(mfrow = c(3, 1), mar = c(3.5, 3.5, 1, 1))
7 plot(samples[, 1], type = "l", xlab = "iteration", ylab = "b0",
8      col = "black", lwd = 0.5)
9 plot(samples[, 2], type = "l", xlab = "iteration", ylab = "alpha",
10     col = "black", lwd = 0.5)
11 plot(samples[, 3], type = "l", xlab = "iteration", ylab = "log(sigma)",
12     col = "black", lwd = 0.5)
```

Listing 15.48: Posterior summary for linear regression.

```
1
2 # Transform back to original scale
3 b0_samp <- samples[, 1]
4 b1_samp <- tan(samples[, 2])
5 sigma_samp <- exp(samples[, 3])
6
7 # Posterior means and credible intervals
8 cat("Posterior summaries:\n")
9 cat("b0:", mean(b0_samp), "± [", quantile(b0_samp, 0.025), ",",
```

```

10 quantile(b0_samp, 0.975), "]\n")
11 cat("b1:", mean(b1_samp), "± [", quantile(b1_samp, 0.025), ",",
12     quantile(b1_samp, 0.975), "]\n")
13 cat("sigma:", mean(sigma_samp), "± [", quantile(sigma_samp, 0.025), ",",
14     quantile(sigma_samp, 0.975), "]\n")

```

Listing 15.49: Linear regression fit with credible band and posterior predictive band.

```

1  # Prediction grid
2  x_grid <- seq(-0.5, 7.5, length.out = 200)
3
4  # Compute posterior mean and predictive curves
5  f_samples <- matrix(NA, nrow = length(x_grid), ncol = length(b0_samp))
6  y_pred_samples <- matrix(NA, nrow = length(x_grid), ncol = length(b0_samp))
7
8  for (k in seq_along(b0_samp)) {
9    f_samples[, k] <- b0_samp[k] + b1_samp[k] * x_grid
10   y_pred_samples[, k] <- f_samples[, k] + rnorm(length(x_grid), 0, sigma_samp[k])
11 }
12
13 # Compute quantiles
14 f_lo <- apply(f_samples, 1, quantile, 0.025)
15 f_hi <- apply(f_samples, 1, quantile, 0.975)
16 f_med <- apply(f_samples, 1, median)
17
18 y_lo <- apply(y_pred_samples, 1, quantile, 0.025)
19 y_hi <- apply(y_pred_samples, 1, quantile, 0.975)
20
21 # Plot
22 plot(x, y, pch = 19, col = "black", xlab = "x", ylab = "y",
23      xlim = c(-0.5, 7.5), ylim = c(-4, 10),
24      main = "Linear Regression Fit with Uncertainty Bands")
25
26 # Posterior predictive band (wider, includes noise)
27 polygon(c(x_grid, rev(x_grid)), c(y_lo, rev(y_hi)),
28        col = rgb(0.5, 0.5, 0.5, 0.3), border = NA)
29
30 # Credible band for mean (narrower, only parameter uncertainty)
31 polygon(c(x_grid, rev(x_grid)), c(f_lo, rev(f_hi)),
32        col = rgb(0.3, 0.3, 0.3, 0.4), border = NA)
33
34 # Posterior mean fit
35 lines(x_grid, f_med, col = "black", lwd = 2)
36
37 # Legend
38 legend("topleft", c("Data", "Posterior mean", "Credible band", "Predictive band"),
39       pch = c(19, NA, NA, NA), lty = c(NA, 1, NA, NA),
40       col = c("black", "black", "gray30", "gray50"),
41       fill = c(NA, NA, rgb(0.3, 0.3, 0.3, 0.4), rgb(0.5, 0.5, 0.5, 0.3)),
42       border = NA)
43

```

Listing 15.50: Logistic function

```

1  eta <- seq(-6, 6, length.out = 200)
2  pi <- 1 / (1 + exp(-eta))
3
4  plot(eta, pi, type = "l", lwd = 2, col = "black",
5       xlab = expression(eta), ylab = expression(pi),
6       ylim = c(0, 1), las = 1)
7  grid(col = "gray80")
8  abline(h = 0.5, lty = 2, col = "gray40", lwd = 1.5)
9  abline(v = 0, lty = 2, col = "gray40", lwd = 1.5)
10

```

Listing 15.51: Poisson sim

```

1
2 set.seed(101)
3 N <- 30
4 x <- seq(0, 5, length.out = N)
5 w0_true <- 0.5
6 w1_true <- 0.35
7 eta <- w0_true + w1_true * x
8 lambda <- exp(eta)
9 y <- rpois(N, lambda = lambda)
10
11 plot(x, y, pch = 19, col = "black", xlab = "x", ylab = "count",
12       ylim = c(0, max(y) * 1.1), cex = 0.9, las = 1)
13 lines(x, lambda, lwd = 2.5, col = "#b8123e")
14 grid(col = "gray80")
15 legend("topleft", legend = c("observed", "true rate"),
16       pch = c(19, NA), lty = c(NA, 1), lwd = c(NA, 2.5),
17       col = c("black", "#b8123e"), bty = "n")

```

Listing 15.52: Link comparison

```

1
2 eta_seq <- seq(-4, 4, length.out = 200)
3
4 par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
5
6 # Logit (inverse)
7 pi <- 1 / (1 + exp(-eta_seq))
8 plot(eta_seq, pi, type = "l", lwd = 2.5, col = "#b8123e",
9       xlab = expression(eta), ylab = expression(mu),
10      main = "Logit inverse link", las = 1)
11 grid(col = "gray80")
12
13 # Log (inverse)
14 lambda <- exp(eta_seq)
15 plot(eta_seq, lambda, type = "l", lwd = 2.5, col = "#b8123e",
16      xlab = expression(eta), ylab = expression(mu),
17      main = "Log inverse link", ylim = c(0, 55), las = 1)
18 grid(col = "gray80")

```

Listing 15.53: Link comparison

```

1
2 par(mfrow = c(1, 1))

```

Listing 15.54: Logistic fit

```

1
2 set.seed(202)
3 N <- 80
4 x <- seq(-4, 4, length.out = N)
5 w0_true <- -0.4
6 w1_true <- 1.2
7 eta <- w0_true + w1_true * x
8 pi <- plogis(eta) # 1 / (1 + exp(-eta))
9 y <- rbinom(N, size = 1, prob = pi)
10
11 plot(x, y, pch = 19, col = "black", ylim = c(-0.1, 1.1),
12      xlab = "x", ylab = "y (0 or 1)", cex = 0.8, las = 1)
13 lines(x, pi, lwd = 2.5, col = "#b8123e")
14 grid(col = "gray80")
15 legend("topleft", legend = c("observed", "true probability"),
16      pch = c(19, NA), lty = c(NA, 1), lwd = c(NA, 2.5),

```

```
17 | col = c("black", "#b8123e"), bty = "n")
```

Listing 15.55: Overdispersion

```
1 |
2 | set.seed(303)
3 | N <- 100
4 | x <- seq(0, 4, length.out = N)
5 | lambda <- 2 + 3 * x
6 |
7 | # Poisson: variance = mean
8 | y_poisson <- rpois(N, lambda = lambda)
9 |
10 | # Overdispersed: inflate variance by factor of 2
11 | y_overdispersed <- sapply(lambda, function(mu) {
12 |   nb_size <- mu / 1 # size parameter for negative binomial
13 |   rnbinom(1, size = nb_size, mu = mu)
14 | })
15 |
16 | plot(x, y_poisson, pch = 19, col = "gray40", cex = 0.8,
17 |      xlab = "x", ylab = "count", las = 1, ylim = c(0, 30))
18 | points(x, y_overdispersed, pch = 1, col = "#b8123e", cex = 0.8)
19 | lines(x, lambda, lwd = 2, col = "black")
20 | legend("topleft", legend = c("Poisson (Var=mean)", "Overdispersed (Var>mean)", "true mean"),
21 |      pch = c(19, 1, NA), lty = c(NA, NA, 1), col = c("gray40", "#b8123e", "black"),
22 |      bty = "n")
```

This appendix collects the R code used to produce the figures, tables and numerical estimates that appear from Chapter 2 onward. Every labelled listing in this appendix is referenced from the body of the book through `\ref{lst:...}`, with a placeholder of the form “See code ?? in Appendix for the implementation.”

The R-introduction code (Chapter 1) is **not** repeated here: its listings appear inline together with their annotations.

Remark. How the move happens. A Pandoc Lua filter (`move-code.lua`) walks the book and moves every code block that (i) is *not* the R-introduction chapter, (ii) carries an `lst-cap` chunk option, and (iii) does *not* contain `# <1>`-style annotations or the `.no-move` class into this appendix as a `lstlisting` environment.

